

# Floorplan

Synopsys University Program

WEEK 3 Read Design

WEEK 4 Floorplan

WEEK 5 Powerplan

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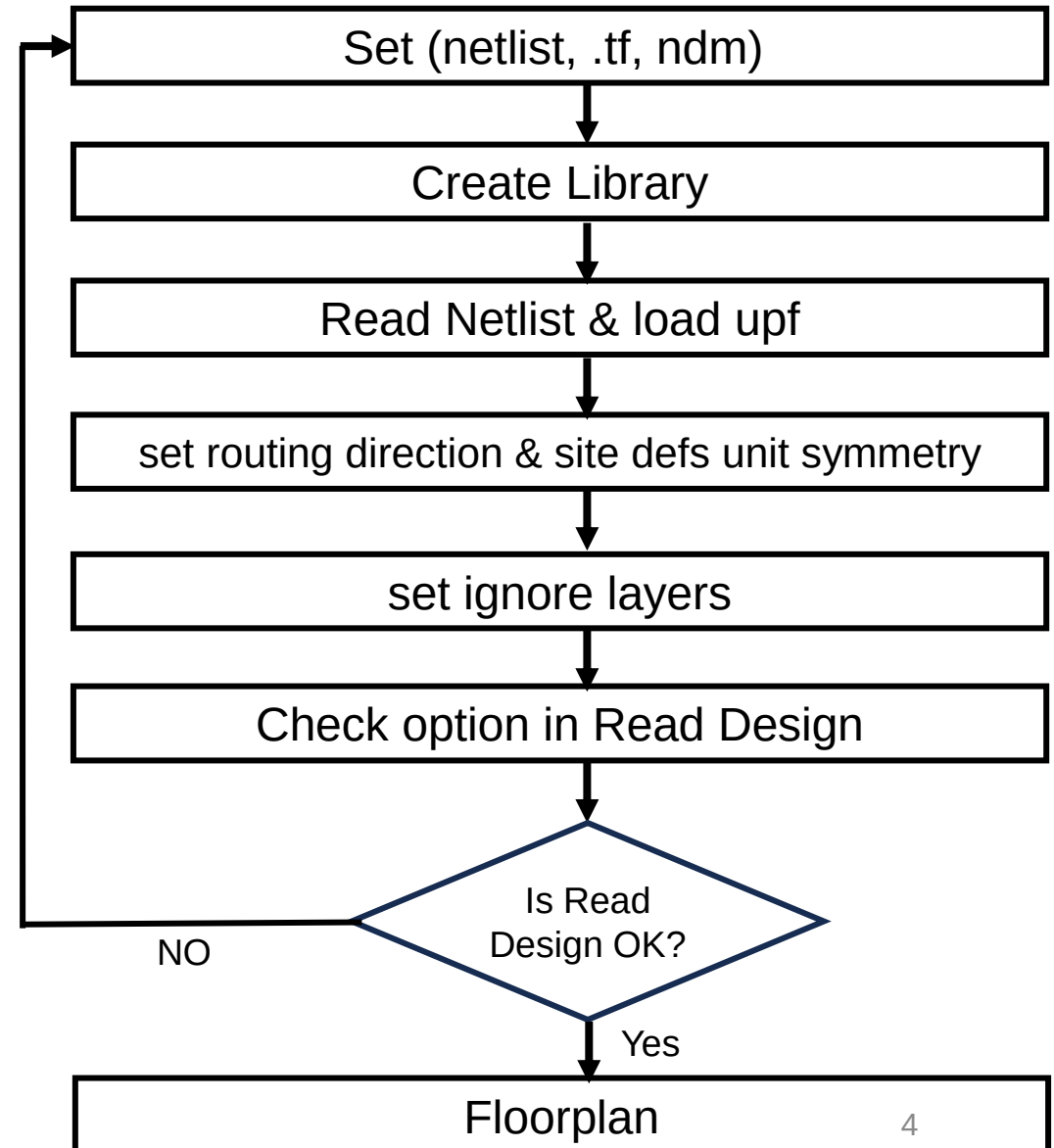
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# WEEK 3 : Read Design in P&R

# Introduction to Read Design

- Read Design Overview

- This module provides an overview of some of the key steps in Read Design :
- **Defining**
  - Tech file (.tf)
  - Netlist
  - Library files (.lib & .lef)
- **Overview to Read Design (directory, input files, script)**
- **Read Design**



# Introduction to Read Design : Input files

- Technology file
  - Defines Units and Design Rules for Layers and Vias as per the Technology
  - Name and Number conventions of Layers and Vias
  - Physical and Electrical parameters of Layers and Vias
  - Direction, Type, Pitch, Width, Offset, Thickness, Resistance, Capacitance, Max Metal Density, Antenna Rule, Blockages, Design rules, Spacing
  - Manufacturing Grid definition
  - Site/Unit Tile definition
  - Tech file has to load before loading other LEF files since it holds the layer information for that particular technology
  - .tf is Synopsys technology file format

# Introduction to Read Design : Input files

- Netlist (.v)
  - Netlist is the combination of sequential elements and their logical connectivity
  - Netlist contains (Std Cell instance – Name & Drive Strength, Macros & MEM instances)
  - Netlist consists of (Ports of STD Cells & Macros, Interconnection details)
- Liberty Timing File (.lib .db)
  - This file is Cell Logical View/ The Timing Library
  - There are ex) Std Cell, Macro, IO, IP in the type of this file
  - Library file contains (Cell type and functionality, Delay Models, Pin/Cell Timings & design rules, PVT conditions, Power Details)
- Library Exchange Format (LEF)
  - Lef file is Cell Abstract View/The Physical Library
  - There are ex) Std Cell, Macro, IO, IP in the type of this file
  - LEF file contains (Cell Name, Shape, Size, Orientation & Class, Port/Pin Name, Direction and Layout Geometries, Obstruction/ Blockages, Antenna Diff, Area)

# WEEK 3 : LAB for Read Design in P&R

# Read Design : introduction to user\_design.tcl

- Review and check the variables in the user\_design.tcl script

```
set WORK_DIR "/slowfs/dgscratch2/asic/ASIC_Project"

# Name of the design to be worked on
set DESIGN_NAME "ORCA"
# Name of the design library, default is ${DESIGN_NAME}.nlib
set DESIGN_LIBRARY "${DESIGN_NAME}.nlib"
#set library directory
set libdir "${WORK_DIR}/2_BACKEND/1_pnr/libdir"

# A technology file; TECH_FILE and TECH_LIB are mutually exclusive ways to specify technology information
set data(tech) "${libdir}/TECH/saed32nm_1p9m_mw.tf

# Set tluplus file (min max)
set data(tech_tlupwc) "${libdir}/TECH/saed32nm_1p9m_Cmin.tluplus"
set data(tech_tlupbc) "${libdir}/TECH/saed32nm_1p9m_Cmax.tluplus"

# Set SDC file
set data(shift_sdc) "${libdir}/SDC/ORCA_shift.sdc"
set data(func_sdc) "${libdir}/SDC/ORCA_func.sdc"
set data(capture_sdc) "${libdir}/SDC/ORCA_capture.sdc"

# set scan_def file
set data(scan_def) "${libdir}/SDC/scan.def"

#std_cell lef,lib,dbs
set data(dbs) [glob -nocomplain ${libdir}/DBS/*.db ]
set data(libs) [glob -nocomplain ${libdir}/LIB/*.lib]
set data(lefs) [glob -nocomplain ${libdir}/LEF/*lef ]

#set mapfile
set data(mapfile) "${libdir}/TECH/saed32nm_tf_itf_tluplus.map"

#set UPF file
set UPF_FILE "${libdir}/UPF/ORCA.upf"

#set NDM directory & file
set NDM_DIR "${WORK_DIR}/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm"
set NDM_FILE [glob -nocomplain $NDM_DIR/*.ndm ]

#A Tcl script placeholder to create your corners, modes, and scenarios,
set TCL_MCMC_SETUP_FILE "${WORK_DIR}/2_BACKEND/1_pnr/rundir/04_place/icc2_common_env/mcmm_ORCA_TOP.tcl"

#Verilog netlist file
set netlist "${libdir}/NETLIST/ORCA.v"

#A list of horizontal routing layers; optional; should be defined if missing in tech file
set HORIZONTAL_ROUTING_LAYER_LIST "M1 M3 M5 M7 M9"
#A list of vertical routing layers; optional; should be defined if missing in tech file
set VERTICAL_ROUTING_LAYER_LIST "M2 M4 M6 M8 MRDL"
#Max routing layer name
set MAX_ROUTE_LAYER "M6"
#Min routing layer name
set MIN_ROUTE_LAYER "M1"
```

# Read Design : Introduction to read\_design.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
< user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/01_read_design/script
< user@linux script >> ls
01_source_design_setup.tcl  03_read_netlist.tcl  05_set_routing_direction_site_defs.tcl  full_script
02_create_lib.tcl          04_load_upf.tcl      06_set_ignored_layers_and_save.tcl
```

- **Review 01\_source\_design\_setup.tcl**

```
# module load
#module load icc2/2021.06-SP5
# open icc2_shell
#icc2_shell -gui | tee -i ./01_read_design.log

# set step
set step read_design

# source design_setup.tcl
source -echo ./icc2_common_env/design_setup.tcl
```

- **Review 02\_create\_lib.tcl**

```
# set step
set step read_design

# create lib
create_lib -technology ${data(tech)} -ref_libs $NDM_FILE ./nlib/${DESIGN_NAME}_${step}.nlib
```

# Read Design : Introduction to read\_design.tcl (about Symmetry)

- Review and check the UPF files before load ;# \${WORK\_DIR}/2\_BACKEND/1\_pnr/libdir/UPF/ORCA.upf
- **Review upf file**
  - **Create\_power\_domain & create\_supply\_net**

```
## UPF EXAM ##

create_power_domain TOP -include_scope

# supply net creation for domain TOP
create_supply_net VDD -domain TOP
create_supply_net VSS -domain TOP
create_supply_net VDDIO -domain TOP
create_supply_net VSSIO -domain TOP
create_supply_net AVDD1 -domain TOP
create_supply_net AVDD2 -domain TOP
create_supply_net DVDD1 -domain TOP
create_supply_net DVDD2 -domain TOP
create_supply_net AVSS -domain TOP
```

Creates or updates a power domain, which provides a power supply distribution network

create\_supply\_net command enables you to create a supply net defined in the specified power domain

- **Set\_domain\_supply\_net & connect\_supply\_net**

```
# set domain supply net
set_domain_supply_net TOP -primary_power_net VDD \
                          -primary_ground_net VSS

connect_supply_net VDD -ports { \
pclk_iopad/VDD \
sys_clk_iopad/VDD \
sdr_clk_iopad/VDD \
```

The set\_domain\_supply\_net command enables sets the primary power net and the primary ground net of a power domain

The connect\_supply\_net command connects a supply net to any supply port or pin

# Read Design : Introduction to read\_design.tcl

- Review and check the variables in each scripts

- **Review 03\_read\_netlist.tcl**

```
# read Netlist
read_verilog -library ${DESIGN_NAME}_${step}.nlib -design $DESIGN_NAME -top $DESIGN_NAME ${netlist}
```

- **Review 04\_load\_upf.tcl** ;# upf for low power design such as multi power domain / isolation / levelshifter

```
#load upf

load_upf $UPF_FILE ;# upf is a power specification file used in -
                  ;# the design cycle to implement low-power design approaches

commit_upf

#connect_pg_net
connect_pg_net -automatic ;# Connects power,ground and tie-off pins to the power & ground nets
```

- **Review 05\_set\_routing\_direction\_site\_defs.tcl**

```
# set routing direction & site_defs unit symmetry
set_attribute [get_layers $HORIZONTAL_ROUTING_LAYER_LIST] routing_direction horizontal ;# {M1 M3 M5 M7 M9}
set_attribute [get_layers $VERTICAL_ROUTING_LAYER_LIST] routing_direction vertical ;# {M2 M4 M6 M8 MRDL}
set_attribute [get_site_defs unit] symmetry Y ;# default is no symmetry

# check site_defs unit
get_attribute [get_site_defs unit] is_default
get_attribute [get_site_defs unit] symmetry

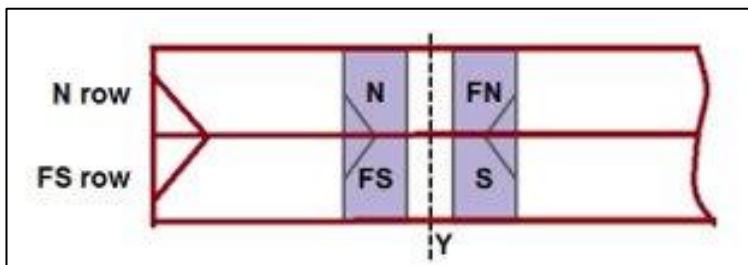
# check routing direction
get_attribute [get_layers M?] routing_direction
```

# Read Design : Introduction to read\_design.tcl (about Symmetry)

- Review and check the variables in each scripts
  - **Review 05\_set\_routing\_direction\_site\_defs.tcl**
    - **get\_site\_defs**  
Returns a collection of site defs from the tech file of current or specified library

```
Tile          "unit" {  
    width      = 0.152  
    height     = 1.672  
}
```

- **-symmetry option**
  - Specifies if the site definition is symmetrical about X/Y/90 degree rotation (By default there is no symmetry)
  - Valid symmetry is a set of combination of the following values
    - X (symmetrical about the X-axis)
    - **Y (symmetrical about the Y-axis)**
    - R90 (symmetrical about the 90 degrees rotation)



- This setting indicates that the site is symmetric about the y-axis, as shown in the following figure
- This means that N and FN sites are equivalent, and S and FS sites are equivalent
- The tool can place N and FN cells in N or FN rows, and S and FS cells in S or FS rows
- Use this setting for standard cells for flipped and abutted horizontal rows

# Read Design : Introduction to read\_design.tcl

- Review and check the variables in each scripts
  - **Review 06\_set\_ignored\_layers\_and\_save.tcl**

```
# report ignored_layers
report_ignored_layers

# set ignore_layers & recheck
set_ignored_layers -min_routing_layer $MIN_ROUTE_LAYER ;# Min routing layer - M1 / Max routing layer - M6
set_ignored_layers -max_routing_layer $MAX_ROUTE_LAYER ;# {routing layers - M1 to M6} {ignore layers P0, M7, M8, M9, MRDL}

#report ignore layers
report_ignored_layers

#####
## Save the library & exit
#####

save_lib -all

exit
```

# Read Design : Use ICC2 (IC Compiler II tool)

Enter the command line by line in 01\_read\_design.tcl and check the result

## 1. Invoke tool icc2\_shell

- **Recommend the icc2/2021.06-SP5 version**

>> icc2\_shell -gui | tee -i ./01\_read\_design.log ;# (| tee -i ./01\_read\_design.log) is the command to output log

```
< user@linux 01_read_design >> icc2_shell -gui | tee -i ./01_read_design.log

                                IC Compiler II (TM)

                                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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                                Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
                                icc2_shell>
```

## 2. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/01\_read\_design/script/01\_source\_design\_script.tcl

- **set step & source design\_setup.tcl**

```
icc2_shell> set step read_design
read_design
icc2_shell> source -echo ./icc2 common env/design_setup.tcl
```

# Read Design : Use ICC2\_shell tool

3. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/02_create_lib.tcl`

- Create library using technology file and NDM (nlib format)

```
icc2_shell> create_lib -technology ${data(tech)} -ref_libs $NDM_FILE ./nlib/${DESIGN_NAME}_${step}.nlib
Warning: Layer 'VIA1' 'onGrid' and 'onWireTrack' coexist. (TECH-025)
Information: Loading technology file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/TECH/saed32nm_1p9m_mw
.tf' (FILE-007)
{ORCA_read_design.nlib}
```

4. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/03_read_netlist.tcl`

- Read gate level Netlist

```
icc2_shell> read_verilog -library ${DESIGN_NAME}_${step}.nlib -design $DESIGN_NAME -top $DESIGN_NAME ${netlist}
Information: Reading Verilog into new design 'ORCA' in library 'ORCA_read_design.nlib'. (VR-012)
Loading verilog file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/NETLIST/ORCA.v'
Number of modules read: 18
Top level ports: 71
Total ports in all modules: 1469
Total nets in all modules: 12051
Total instances in all modules: 9475
Elapsed = 00:00:00.13, CPU = 00:00:00.11
1
icc2_shell> link_block
Using libraries: ORCA_read_design.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32rvt saed32
sram
Linking block ORCA_read_design.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Design 'ORCA' was successfully linked.
1
```

# Read Design : Use ICC2\_shell tool

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/01\_read\_design/script/04\_load\_upf.tcl

- load upf & connect\_pg\_net

1. load\_upf (you must check if an Error occurs during sourcing)

```
icc2_shell> load_upf $UPF_FILE ;# upf is a power specification file used in -
Information: Loading UPF file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/libdir/UPF/ORCA.upf' (FILE-007)
## UPF EXAM ##
create_power_domain TOP -include_scope
# supply net creation for domain TOP
create_supply_net VDD -domain TOP
create_supply_net VSS -domain TOP
create_supply_net VDDIO -domain TOP
create_supply_net VSSIO -domain TOP
create_supply_net AVDD -domain TOP
create_supply_net DVDD -domain TOP
create_supply_net AVSS -domain TOP
# supply port creation for domain TOP
create_supply_port VDD -domain TOP
create_supply_port VSS -domain TOP
create_supply_port VDDIO -domain TOP
create_supply_port VSSIO -domain TOP
create_supply_port AVDD -domain TOP
create_supply_port DVDD -domain TOP
create_supply_port AVSS -domain TOP
# connect supply net and port
connect_supply_net VDD -port VDD
connect_supply_net VSS -port VSS
connect_supply_net VDDIO -port VDDIO
connect_supply_net VSSIO -port VSSIO
connect_supply_net AVDD -port AVDD
connect_supply_net DVDD -port DVDD
connect_supply_net AVSS -port AVSS
# set domain supply net
set_domain_supply_net TOP -primary_power_net VDD -primary_ground_net VSS
```

```
connect_supply_net AVDD -ports { I_CLOCK_GEN/I_PLL_PCI/AVDD U_IOPOWER/DVDD_PLLA/VDDPAD }
connect_supply_net AVDD -ports { I_CLOCK_GEN/I_PLL_SD/AVDD U_IOPOWER/DVDD_PLLB/VDDPAD }
connect_supply_net DVDD -ports { U_IOPOWER/DVDD_PLL1/VDDPAD I_CLOCK_GEN/I_PLL_PCI/DVDD }
connect_supply_net DVDD -ports { U_IOPOWER/DVDD_PLL2/VDDPAD I_CLOCK_GEN/I_PLL_SD/DVDD }
connect_supply_net AVSS -ports { I_CLOCK_GEN/I_PLL_PCI/VSS I_CLOCK_GEN/I_PLL_SD/VSS U_IOPOWER/DVSS_PLLA/VSSPAD U_IOPOWER/DVSS_PLLB/VSSPAD }
```

If sourcing is completed without an Error, you can check output 1 as shown below

# Read Design : Use ICC2\_shell tool

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/04_load_upf.tcl`

- **load upf & connect\_pg\_net**

2. `commit_upf` (you must check if an Error occurs during sourcing upf file)

```
icc2_shell> commit_upf
Information: Supply net 'AVSS' is inferred as 'ground' by name. (MV-204)
Warning: Pre-existing standard cell I_CLOCK_GEN/I_PLL_PCI cannot be connected to pad cell pclk_iopad: standard cells may not be inserted between a pad and the outside world. is_pad attribute on pad pin pclk_iopad/DOUT will be ignored. (MV-143)
Warning: Pre-existing standard cell DFTC_27 cannot be connected to pad cell sys_clk_iopad: standard cells may not be inserted between a pad and the outside world. is_pad attribute on pad pin sys_clk_iopad/DOUT will be ignored. (MV-143)
Warning: Pre-existing standard cell I_CLOCK_GEN/I_PLL_SD cannot be connected to pad cell sdr_clk_iopad: standard cells may not be inserted between a pad and the outside world. is_pad attribute on pad pin sdr_clk_iopad/DOUT will be ignored. (MV-143)
Warning: Pre-existing standard cell I_ORCA_TOP/I_BLENDER/U350 cannot be connected to pad cell test_mode_iopad: standard cells may not be inserted between a pad and the outside world. is_pad attribute on pad pin test_mode_iopad/DOUT will be ignored. (MV-143)
```

•  
•  
•

```
Information: Related supplies are not explicitly specified on 10 port(s) and primary supplies (VDD, VSS) of top power domain will be assumed as the related supply. (UPF-467)
Information: Power intent has been successfully committed. (UPF-072)
Information: Total 0 isolation cell(s) in the design. (MV-021)
Information: Total 0 level shifter cell(s) in the design. (MV-021)
Information: Total 0 enable level shifter cell(s) in the design. (MV-021)
Information: Total 0 repeater cell(s) in the design. (MV-021)
Information: Total 0 retention cell(s) in the design. (MV-021)
Information: Total 0 power switch cell(s) in the design. (MV-021)
Information: Total 4 netlist change(s) have been made to resolve conflicts between power intent and PG netlist. (UPF-073)
```

1

If sourcing is completed without an Error, you can check output 1 as shown above

# Read Design : Use ICC2\_shell tool

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/04_load_upf.tcl`

- `load upf & connect_pg_net`

3. `connect_pg_net -automatic` (Connects power, ground, and tie-off pins to the power and ground automatically)

```
connect_pg_net -automatic ;# Connects power,ground and tie-off pins to the power & ground nets
*****
Report : Power/Ground Connection Summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Oct  4 13:59:41 2023
*****
P/G net name                P/G pin count(previous/current)
-----
Power net AVDD               4/4
Power net DVDD               4/4
Power net VDD                8990/8990
Power net VDDIO              105/105
Ground net AVSS              4/4
Ground net VSS               8990/8990
Ground net VSSIO             105/105
-----
Information: connections of 0 power/ground pin(s) are created or changed.
```

# Read Design : Use ICC2\_shell tool

6. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/05_set_routing_direction_site_defs.tcl`

- Set Routing Direction (Horizontal & Vertical) and site\_defs unit symmetry
- Check Routing Direction and site\_defs unit symmetry

```
icc2_shell> set_attribute [get_layers "M1 M3 M5 M7 M9"] routing_direction horizontal
Information: The design specific attribute override for layer 'M1' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M3' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M5' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M7' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M9' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
{M1 M3 M5 M7 M9}
icc2_shell> set_attribute [get_layers "M2 M4 M6 M8 MRDL"] routing_direction vertical
Information: The design specific attribute override for layer 'M2' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M4' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M6' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'M8' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for layer 'MRDL' is set in the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
{M2 M4 M6 M8 MRDL}
icc2_shell> set_attribute [get_site_defs unit] symmetry Y
{unit}
icc2_shell> get_attribute [get_site_defs unit] is_default
true
icc2_shell> get_attribute [get_site_defs unit] symmetry
Y
icc2_shell> get_attribute [get_layers M?] routing_direction
Information: The returned value for layer 'M1' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M2' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M3' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M4' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M5' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M6' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M7' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M8' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
Information: The returned value for layer 'M9' is the design specific attribute override defined within the current block 'ORCA'. (ATTR-11)
horizontal vertical horizontal vertical horizontal vertical horizontal
```

# Read Design : Use ICC2\_shell tool

7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/06_set_ignored_layers_and_save.tcl`

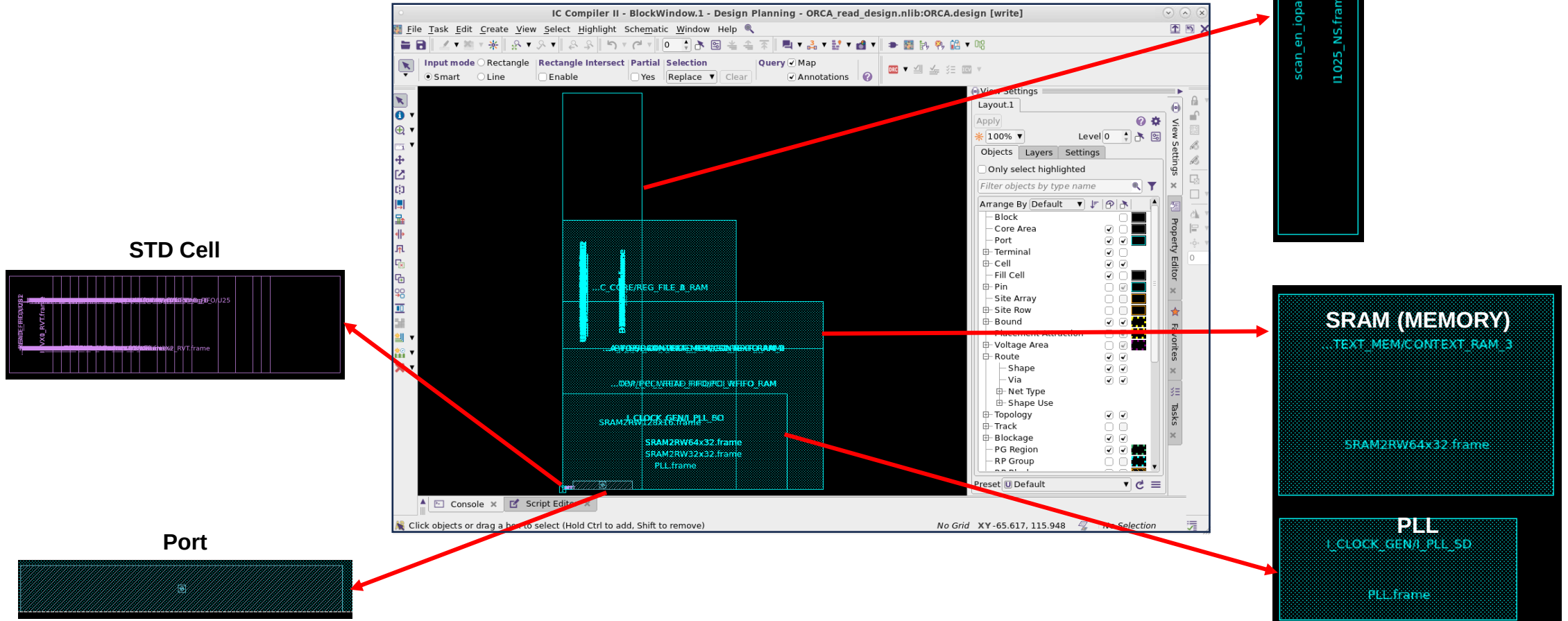
- Set Ignore layers & check Ignore layers
- Save lib (.nlib format)

```
icc2_shell> report_ignored_layers
*****
Report : Ignored Layers
Design : ORCA
Version: S-2021.06-SP5
Date   : Thu Jul 27 23:30:58 2023
*****
Layer Attribute          Value
-----
Min Routing Layer        (none given)
Max Routing Layer        (none given)
RC Estimation Ignored Layers (none given)
1
icc2_shell> set_ignored_layers -min_routing_layer $MIN_ROUTE_LAYER
1
icc2_shell> set_ignored_layers -max_routing_layer $MAX_ROUTE_LAYER
1
icc2_shell>
icc2_shell> report_ignored_layers
*****
Report : Ignored Layers
Design : ORCA
Version: S-2021.06-SP5
Date   : Thu Jul 27 23:31:02 2023
*****
Layer Attribute          Value
-----
Min Routing Layer        M1
Max Routing Layer        M6
RC Estimation Ignored Layers PO M7 M8 M9 MRDL
1
```

# Read Design : Use ICC2 tool

## 7. Review Read Design at ICC2 gui window

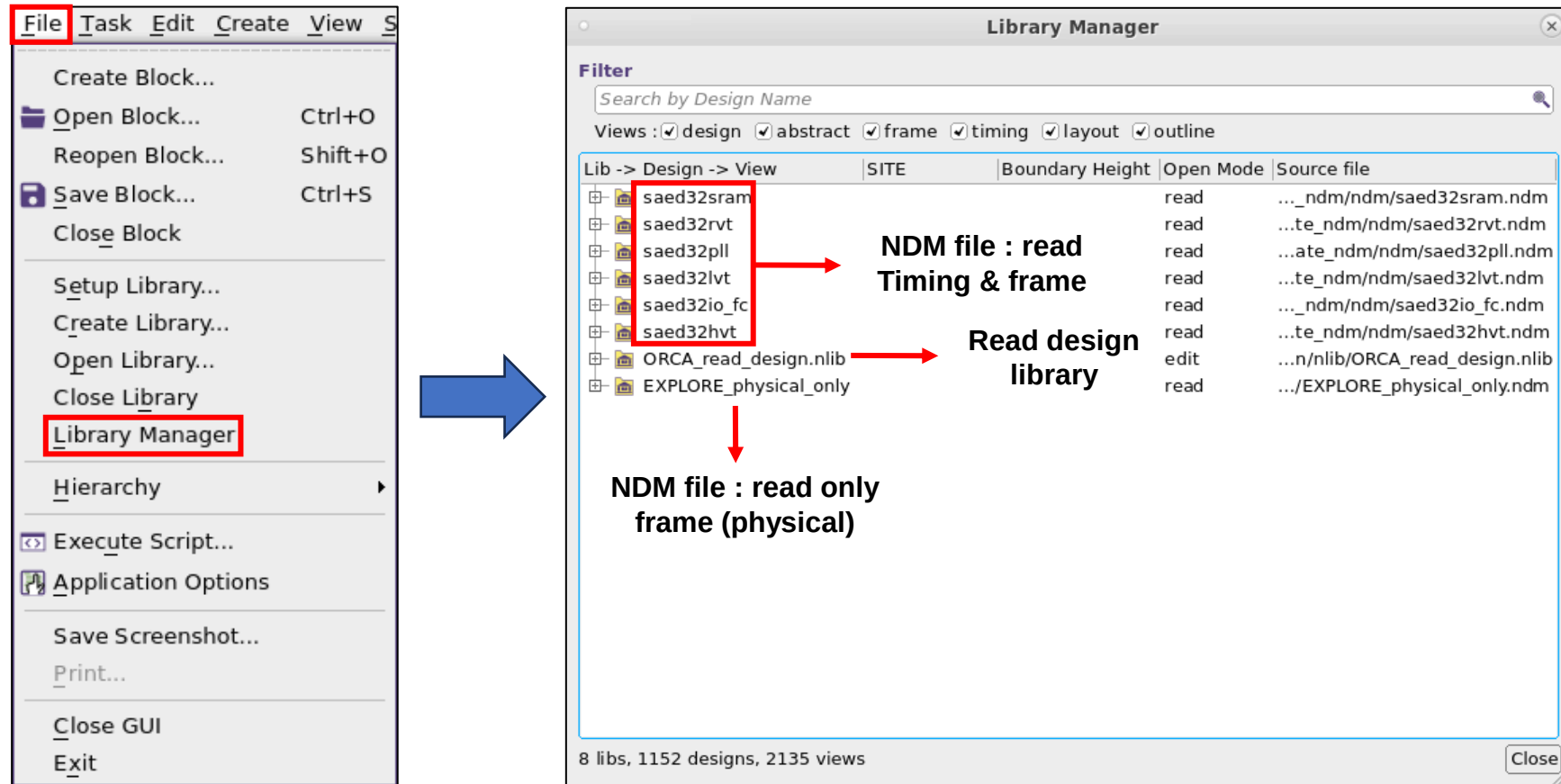
- Review the (IO PAD, STD Cell, SRAM, PLL, Ports) are properly read
- Cells of std/macro/IO is overlapped



# Read Design : Use ICC2 tool

## 7. Review Read\_Design at ICC2 gui window

- Review that the NDM is created and applied correctly (File -> Library Manager)

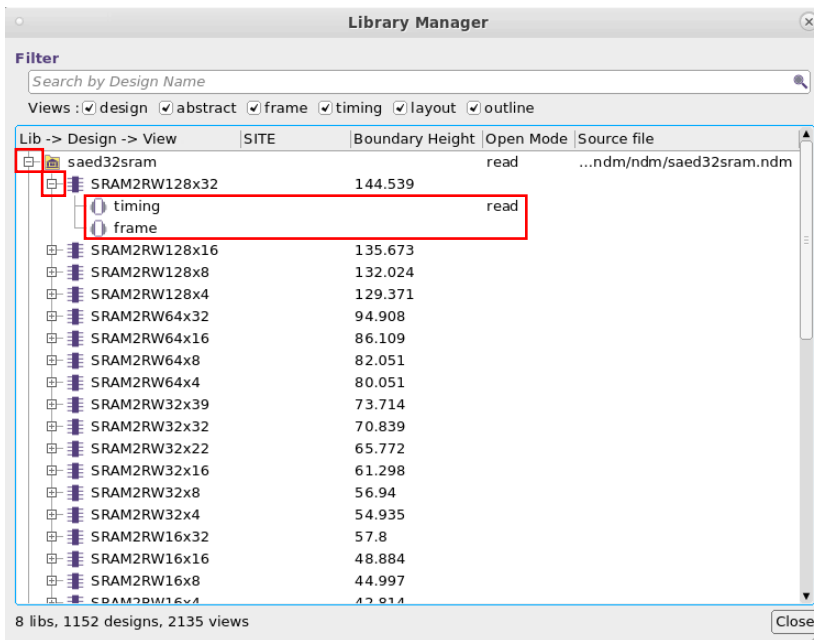


# Read Design : Use ICC2 tool

## 7. Review Read\_Design at ICC2 gui window

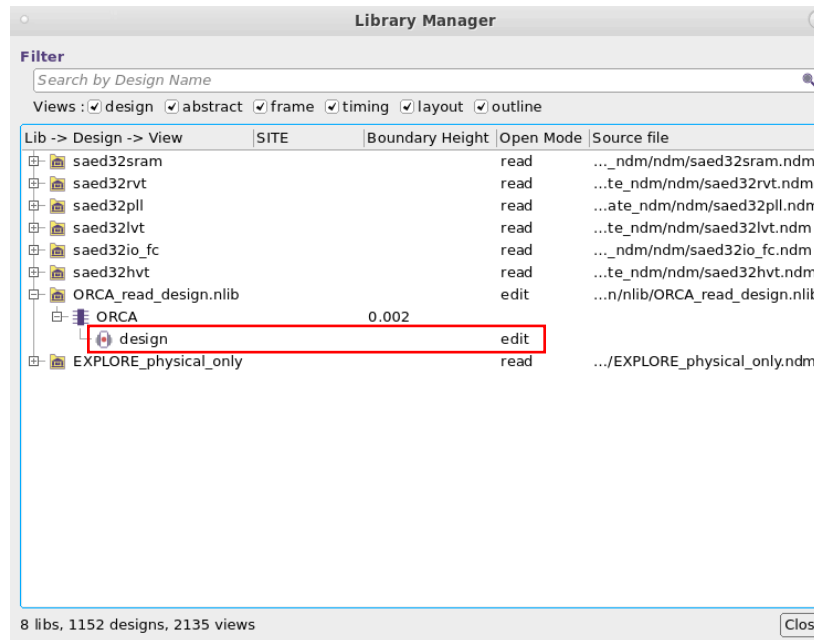
- Review that the NDM is created and applied correctly (File -> Library Manager)
- You can check if each NDM file is properly generated and read by clicking on the  icon under the Lib

### <Std Cell, PLL, IO PAD, SRAM ndm>



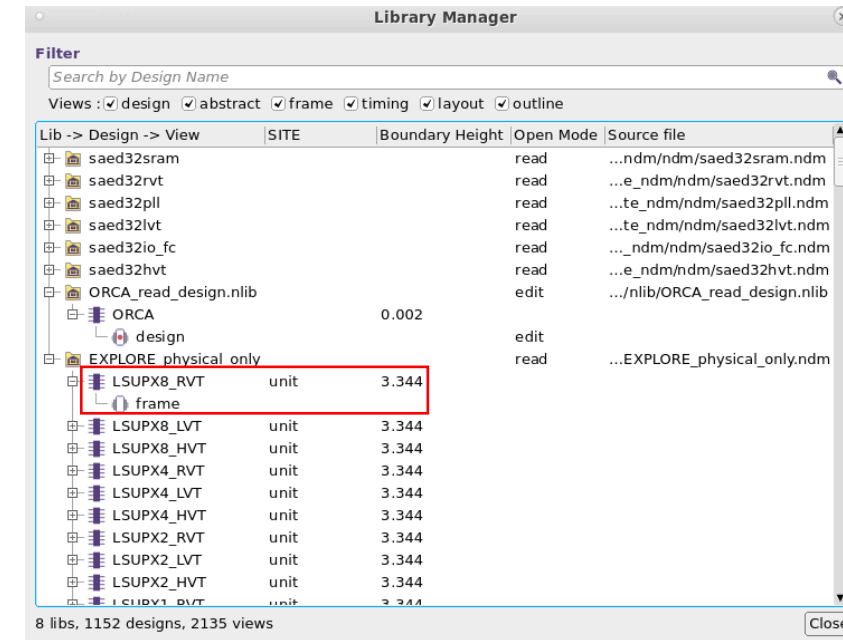
Check to see if you have read the information about the timing and frame

### <ORCA\_read\_design.nlib>



Check ORCA\_read\_design.nlib that you created, and check this lib open mode is edit

### <EXPLORE\_physical\_only ndm>

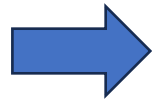
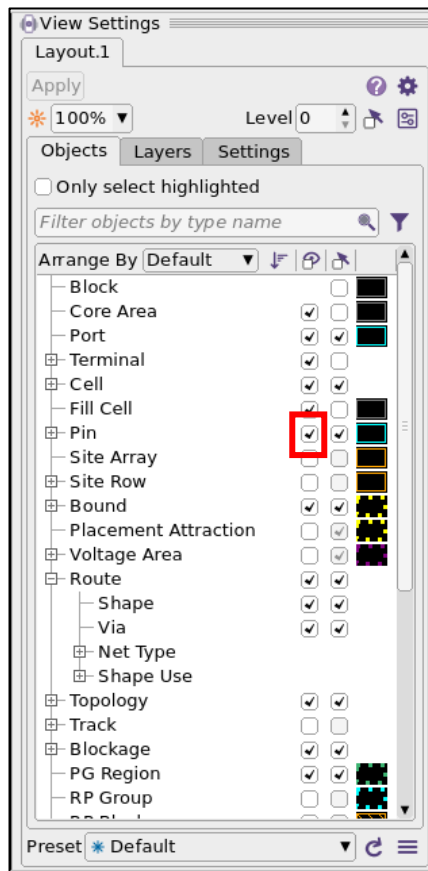


Check to see if you have read the information about only frame (physical\_only : No Timing)

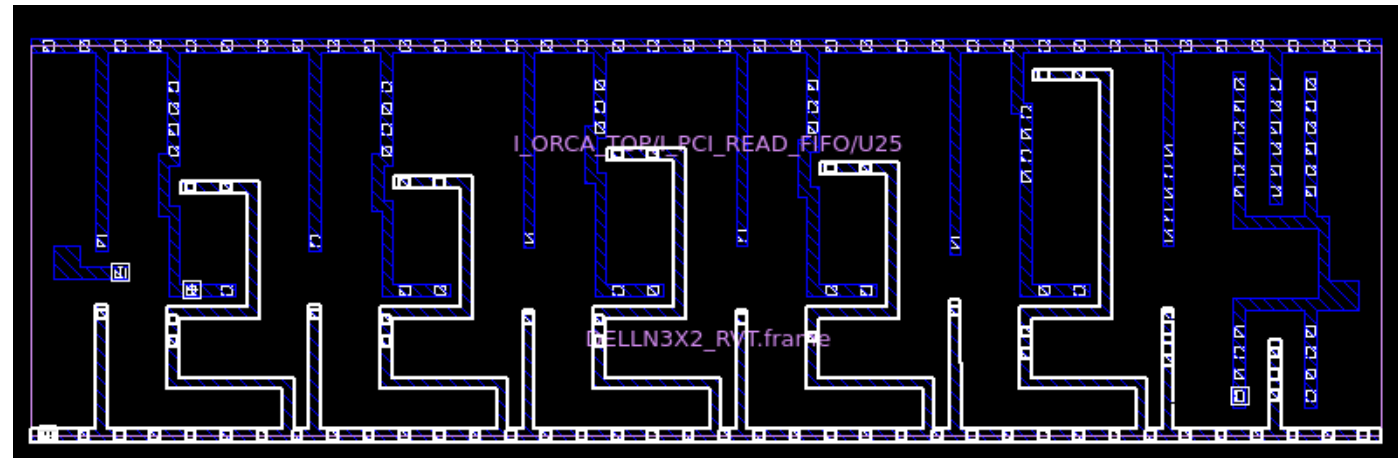
# Read Design : Use ICC2 tool

## 8. Check the internal structure of the cell in ICC2

- You can move the object by pressing a Shortcut (M) key and selecting the object you want to move
- Move the object you want to review into the empty space, and then check the internal structure of the cell in ICC2
- Check Pins in View settings



You can see Pin structure in Cell



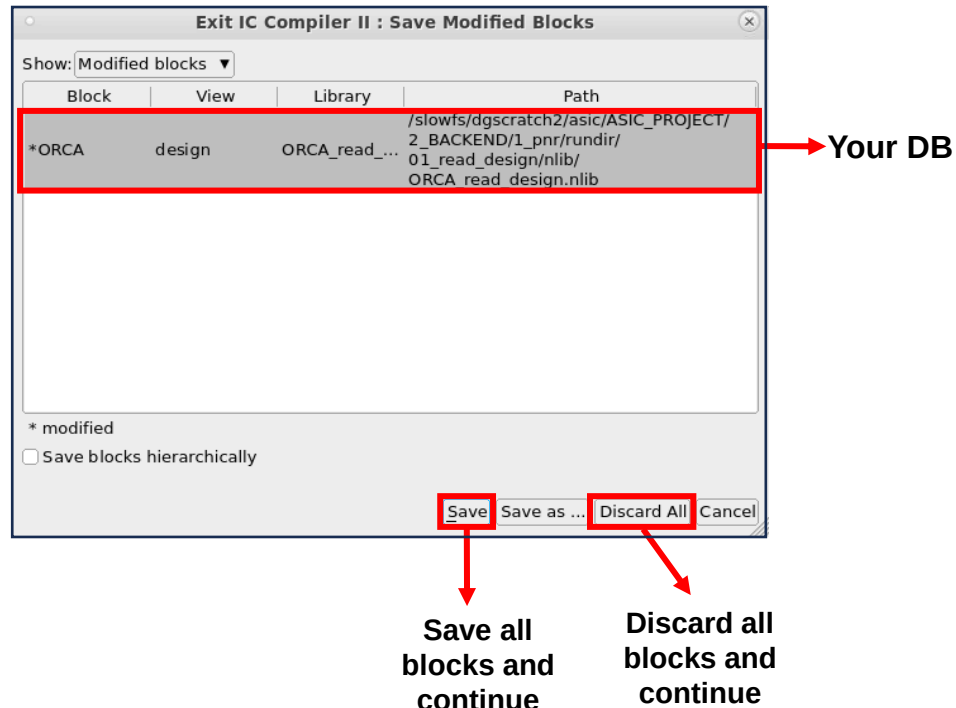
ex) STD Cell pin structure (M1)

# Read Design: Use ICC2 tool

## 9. Exit ICC2

- When the read design is finished, exit from ICC2
- After the read design, Select (Discard all) to exit ICC2 (Because there are no changes in read\_design.nlib)

icc2 shell> exit



- Check that the .nlib is saved correctly

```
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/01_read_design/nlib  
< user@linux nlib >> ls  
ORCA_read_design.nlib
```

# WEEK 4 : introduction to Floorplan

# Introduction to Floorplan : Overview to Floorplan

- What is Floorplan

- Floorplan is one the critical & important step in Physical design
- Quality of your Chip / Design implementation depends on how good is the Floorplan
- A good floorplan can be make implementation process (place, cts, route & timing closure) cake walk
- On similar lines a bad floorplan can create all kind issues in the design (congestion, timing, noise, IR, routing issues)
- A bad floorplan will blow up the area, power & affects reliability, life of the IC and also it can increase overall IC cost (more effort to closure, more LVTs/ULVTs).

- Goals of Floorplan

- Minimize the area
- Minimize the Timing
- Reduce the wire length
- Making routing easy
- Reduce IR drop

# Introduction to Floorplan : Overview to Floorplan

- Floorplan Qualification

- No I/O ports short
- All I/O ports should be placed in routing grid
- All Macros in placement grid
- No Macros overlapping
- All the Macros should be placed at the boundary
- There should not be any notches. If unavoidable, proper blockages has to be added
- Remove all unnecessary placement blockages & routing blockages (which might be put during floorplan & pre-place)

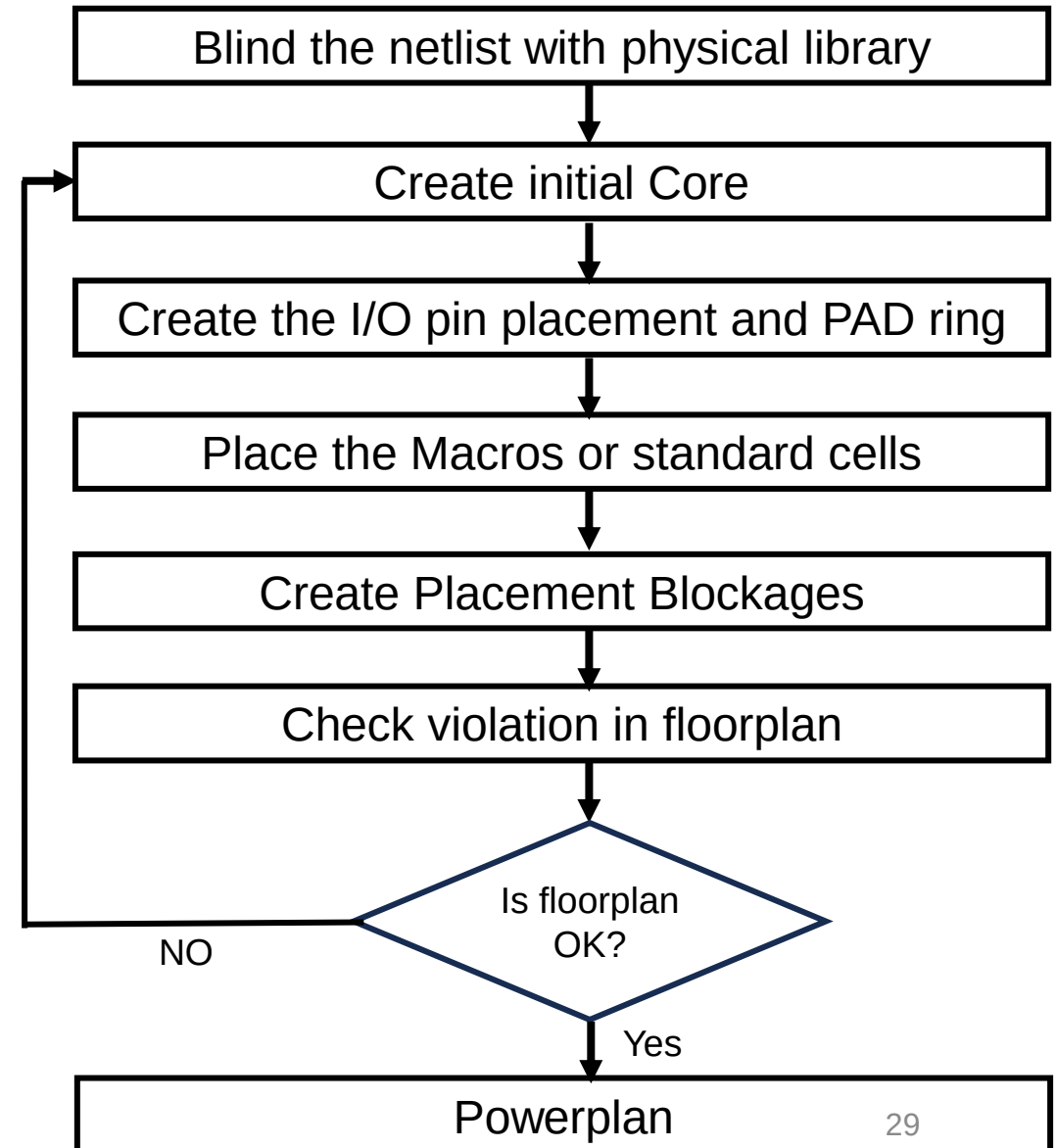
- Floorplan outputs

- IO ports placed
- cell rows created
- macro placement final
- core boundary and area
- pin position
- floorplan def

# Introduction to Floorplan

- **Floorplan Overview**

- This module provides an overview of some of the key steps in Floorplan :
- **Defining**
  - Tech file (.tf)
  - Netlist
  - Library files (.lib & .lef)
  - Power requirements
  - Physical partitioning information
  - IO placement & Macro placement
- **Overview to Floorplan (directory, input files, script)**
- **Floorplan**



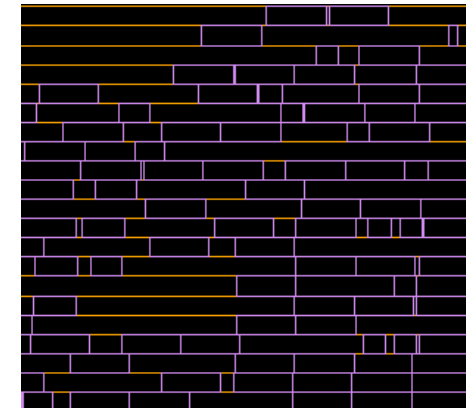
# Introduction to Floorplan : Terminologies & Definitions

- **Utilization**

- Area of the core that is used by placed Standard Cells and Macros expressed in percentage
- Utilization dictates how densely user wants cells to be placed within the block
- Increasing utilization will reduce the core area

$$\text{Utilization} = \frac{\text{Total Standard Cell Area} + \text{Macro Area}}{\text{Total Core Area}} \times 100 \%$$

- A typical starting utilization might be 70% (default)
- High utilization can make it difficult to close a design
  - Routing congestion
  - Negative impact during optimization legalization stages
- Local congestion
  - Can occur with pin-dense cells like multiplexers, so utilization is not completely sufficient for determining die size
  - Run a quick trial route to check for routing congestion
  - Refine the synthesis or increase routing resources



Low STD cell utilization

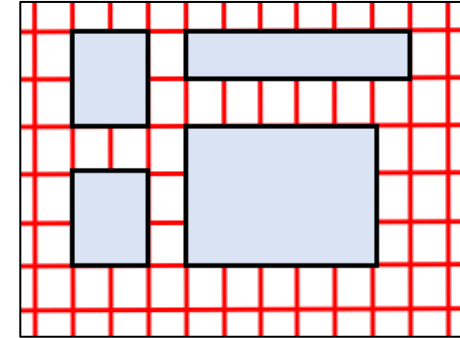


High STD cell utilization

# Introduction to Floorplan : Terminologies & Definitions

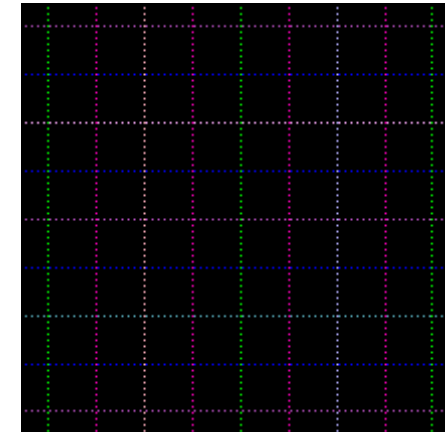
- Placement Grid

- Placement Grid is made up of Standard Cell Site
- Its always a multiple of Manufacturing Grid
- Placement Grid is made up of the Rows which are composed of Sites



- Routing Grid and Routing Track

- Horizontal and Vertical line drawn on the layout area which will guide for making interconnections
- The Routing Grid is made up of the Routing Tracks
- Routing Tracks can be Grid-based, Gridless based or Subgrid-based



- Data-Flow-Flyline

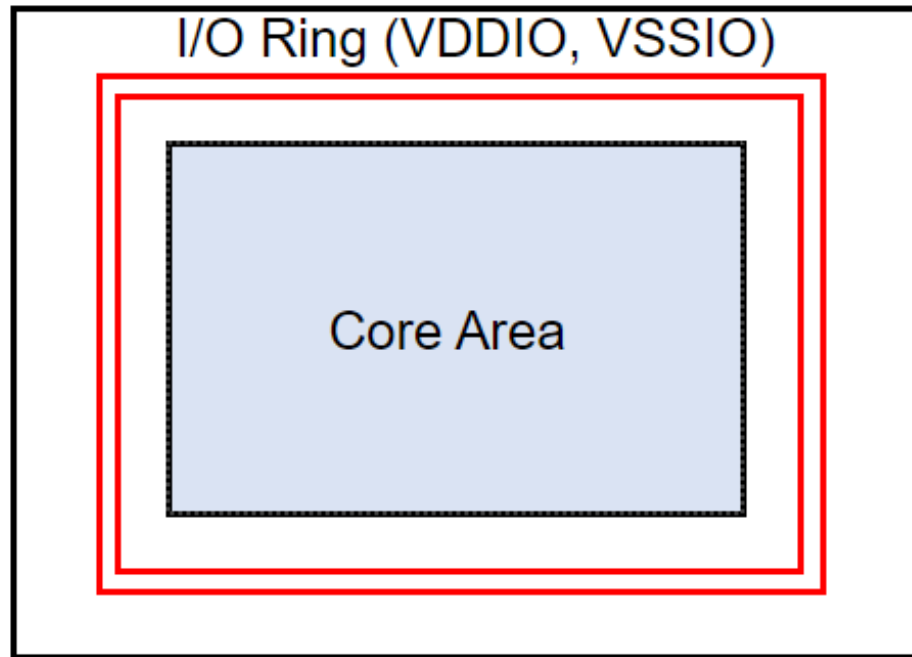
- Virtual connection between Macros and Macro or Macros and IOs

# Introduction to Floorplan : Terminologies & Definitions

- Standard Cell site/Standard Cell Placement Tile/Unit Tile
  - The minimum Width and Height a Cell that can occupy in the design
  - The Standard Cell Site will have the same height as Standard Cells, but the width will be as small as your smallest Filler Cell
  - It's one Vertical Routing Track and the Standard Cell Height
  - All Standard Cells must be multiple of Unit Tile
- Standard Cell Rows
  - Rows are actually the Standard Cell Sites abut side by side and then Standard Cells are placed on these Rows
  - Cells with the equal no. of Track definition will have same height
- Macro
  - Any instances other than Standard Cell and is as loaded as black box to the design is Macro
  - Intellectual Property (IP) e.g. RAM, ROM, PLL, Analog Designs etc.
  - Hard Macro: IP with Layout implemented
  - Soft Macro: IP without Layout implemented (HDL)

# Introduction to Floorplan : Terminologies & Definitions

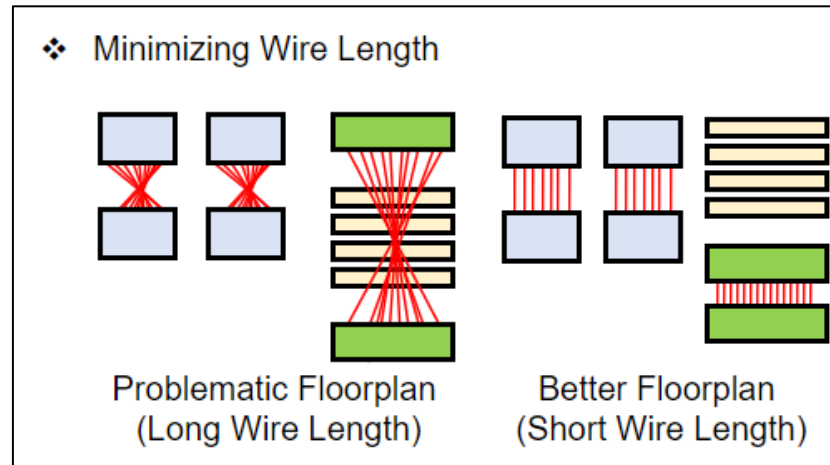
- IO Ring
  - The pinout is often decided by FE designers, with input from physical design and packaging engineers
  - IOs are not only needed for connecting signals to the outside world, but also to provide power to the chip
  - Therefore, IO planning is a critical and very central stage in Floorplanning the chip



# Introduction to Floorplan : Macro Placement

- Macro placement is done based on connectivity information of Macro to IO cell and Macro to Macro and is very critical for congestion and timing and should result in uniform std cell area
  - Fly-line Analysis (For Connectivity information)
  - Macro keep-out (For Uniform Standard Cell Region)
  - Channel Calculation (Critical for Congestion and Timing)
  - Avoid odd shaped area for Standard Cells
  - Funnel shaped Macro Placements are preferred
  - Fix the Macro locations, so that tool won't alter during Optimization
  - Spacing between Macro:

$$\frac{\text{Pitch of the Routing Layers} \times \text{No. of pins to be Routed}}{\text{Available Routing Layers in the preferred direction}} + \text{Buffer Spacing}$$



# Introduction to Floorplan : Tips of Macro Placement

- Place macros around chip periphery, so that core area will be clustered
- Consider connections to fixed cells when placing Macros
- In advanced Technology Nodes Macro Orientation is fixed since the Poly Orientation can't vary, so there will be restrictions in Macro Orientation
- Reserve enough room around Macros for IO Routing
- Reduce open fields as much as possible
- Provide necessary Blockages around the Macro

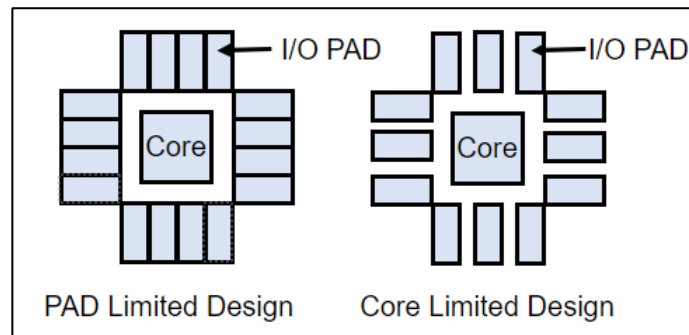
# Introduction to Floorplan : IO Design

- Pad Limited design

- The area of Pad limits the size of Die
- No. Of IO Pads are more or larger in size (technology dependent)
- Pad limited designs pose several challenges for design implementation and to the backend designers, if Die area is a constraint
- The Solution would be to use Flip Chip or Staggered IO placement techniques

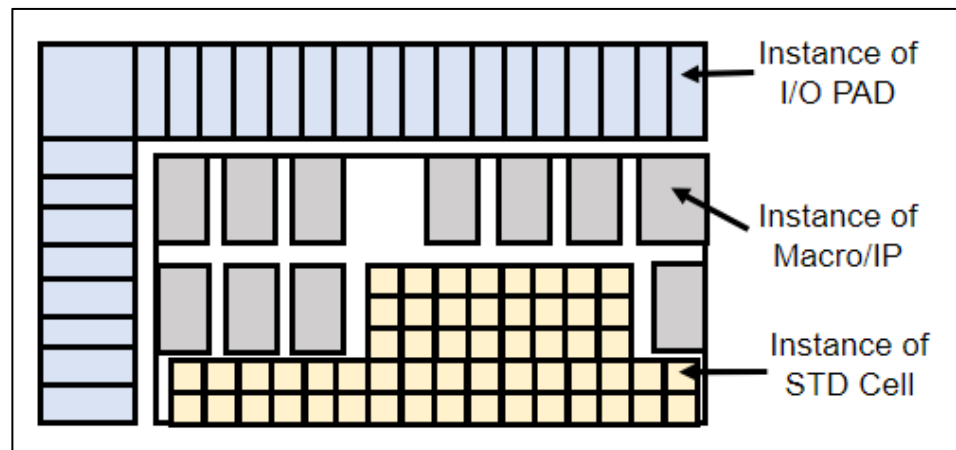
- Core Limited Design

- The area of Core limits the size of Die
- No. of IO Pads are lesser
- In these designs inline IOs will be used
- It can be either due to large no. of Macros the design or due to larger logic



# Introduction to Floorplan : IO Placement

- Chip Level its IO Pads and Block Level its IO Pins
- Pin is a logical entity and is a property of a Port
- Port is a physical entity and a Port have only 1 Pin associated with it
- Netlist will have Pins and Layout will have Ports
- Unplaced Port is not represented in the Layout
- Different types of IOs (Signal, Core Power, IO Power PAD/Pins)
- Corner Pads (Doesn't hold any logic, provides IO Pad Ring connectivity)
- Filler Cells(PAD) (Fill the gaps between IO pads to get the Ring Connectivity)
- Physical-only pads that are not part of the input Gate level Netlist need to be inserted prior to reading IO constraints
- IO Pads enables the design to operate at different voltages with the help of Level Shifters, Pre-Drivers (at Core Voltage)Post-Drivers (at IO Voltage)

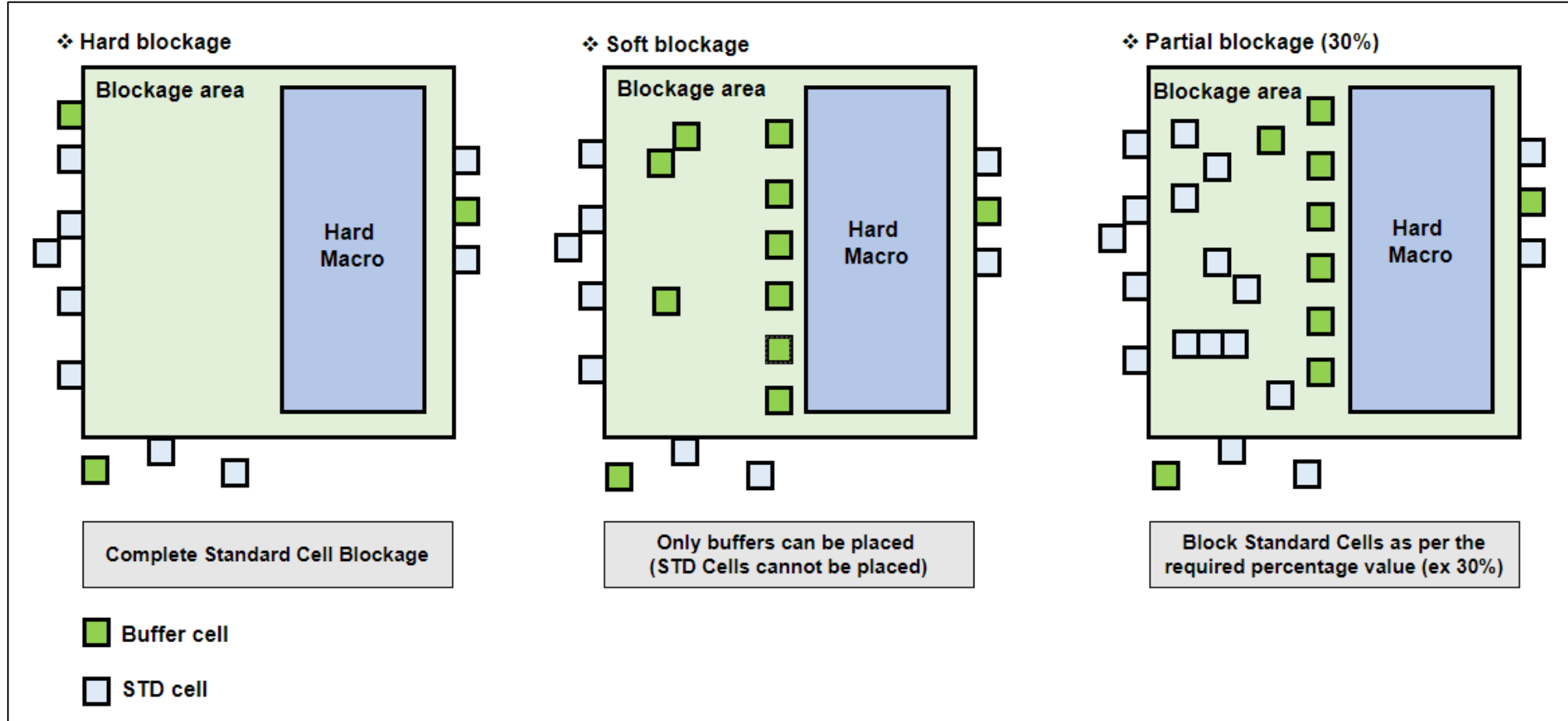


# Introduction to Floorplan : Blockages

- Blockage
  - blockages are specific location where placing of cells are blocked
  - blockages acts like guidelines for placement of std cells
  - blockages will not be guiding the tool to place the std cells at some particular area, but it won't allow the tool to place the std cell in the blocked area
- Type of Blockages (Placement Blockages)
  - Both of the Blockages can be classified as (Hard, Soft and Partial Blockages)
    - **Hard Blockage**
      - Complete Standard Cell Blockage
      - Std cell blockages are mostly used to
        - avoid routing congestion at macro corners
        - Restrict std cells to certain regions in the design
        - control power rail generation at macro cells
    - **Soft Blockage**
      - Non-Buffering Blockage, only buffers can be placed and std cells cannot be placed
    - **Partial Blockage**
      - Partial Standard Cell Blockage and is used to avoid congestion
      - We can Block Standard Cells as per the required percentage value

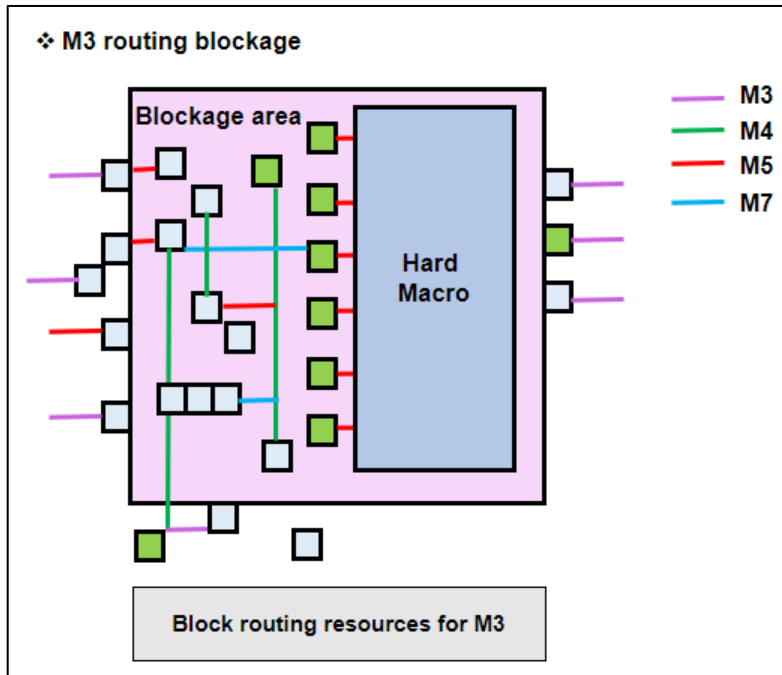
# Introduction to Floorplan : Blockages

- Type of Blockages (Placement Blockages)



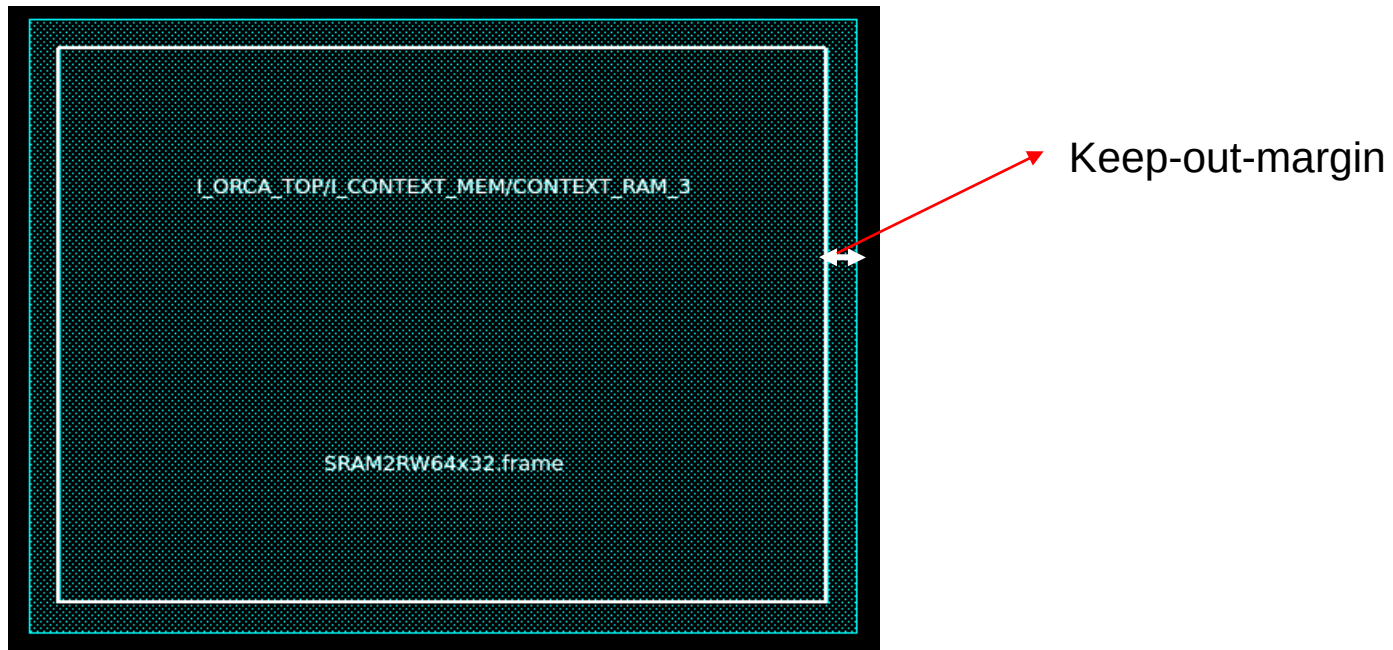
# Introduction to Floorplan : Blockages

- Type of Blockages (Routing Blockages)
  - **Routing Blockage**
    - Routing blockages are block routing resources on one or more layers and it can be created any location in design flow
    - Typically, placement blockages are created at floorplan step, but routing blockages are created before using any routers
    - It acts like guidelines for placement of standard cells  
(Because it affects routing, it also affects cell placement)

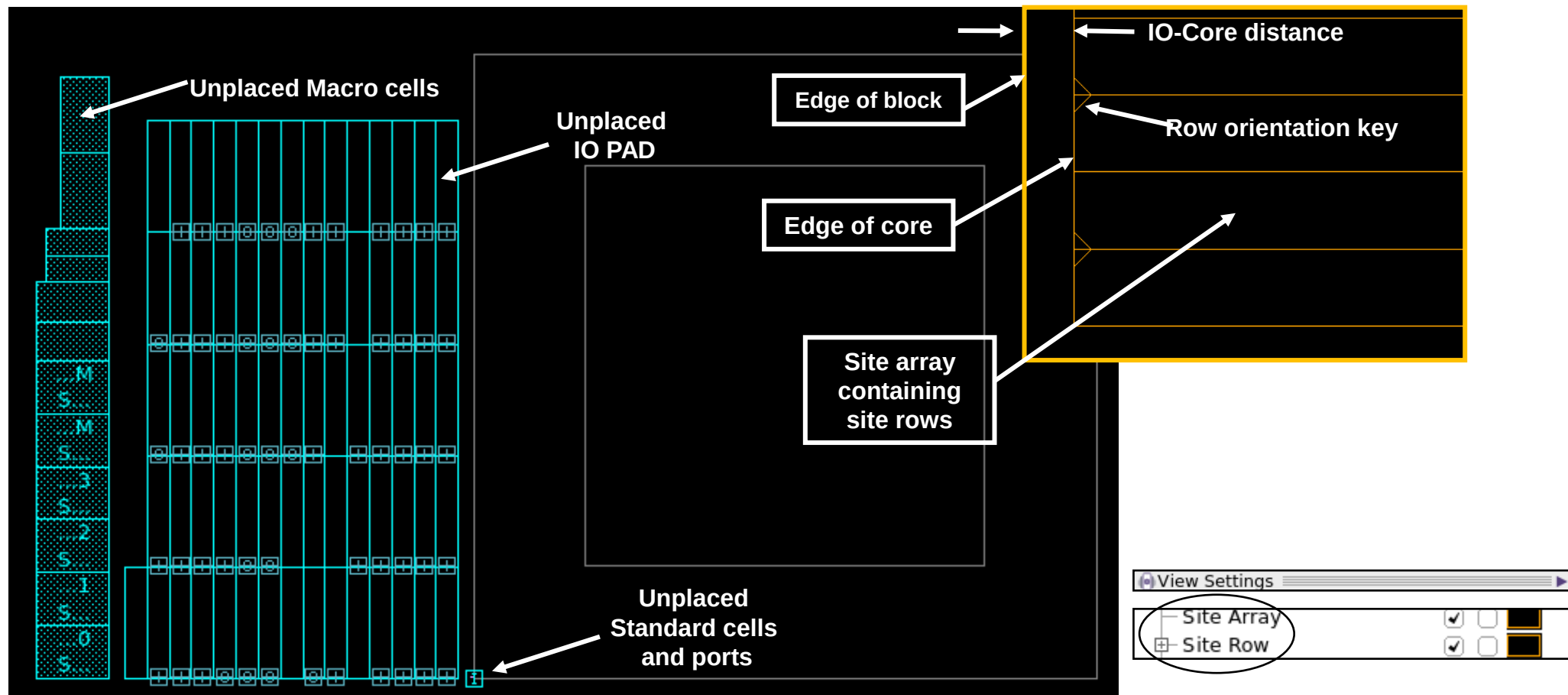


# Introduction to Floorplan : Keep-out-margin/Halo

- Halo is similar to Soft Blockage meaning it allows placement of buffers and inverters in its area
- Its basically a keep-out Macro margin Halo respects Macro while other Blockages respect location
- Halos of adjacent macros can overlap



# Floorplan After Initialization (read\_design)



# Create the initial Floorplan

- **Floorplan Initialization** (command search  – initialize\_floorplan)

- **-control\_type core | die**

Specifies whether the side\_length and side\_ratio options apply to the core or the die boundary (core is default)

**Boundary**  
☒ Core area ☐ Die area

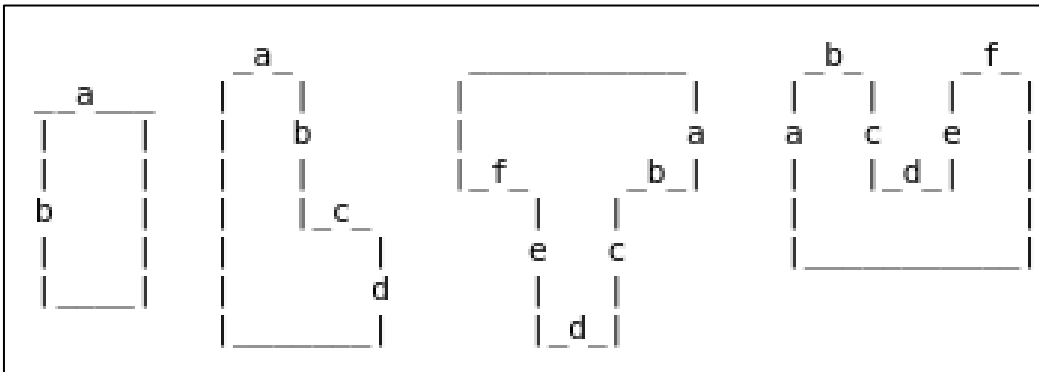
- **-shape Type (R | L | T | U)**

Specifies the shape to be used by the command. If the control\_type is die, this option applies to the die boundary shape. The argument to this option specifies a template shape used to determine the cell boundary and core shape of the rectilinear block. The following diagram shows the definition of the edges and the orientation of the R-, L-, T-, and U- rectilinear blocks. By default, the core shape is R (rectangular)

Shape: ☒ Template ☐ Custom

<no region>

Type: Rectangle ▼ Orientation: North ▼

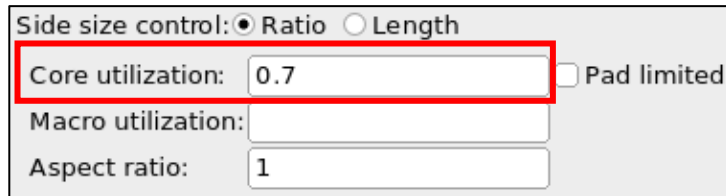


# Create the initial Floorplan

- **Floorplan Initialization**

- **-core\_utilization ratio**

Specifies the utilization of the core area. The utilization is the total area of the core occupied by all standard cells and macro cells divided by the total core area (The cell area includes all standard and macro cells)



Side size control: ☒ Ratio ☐ Length

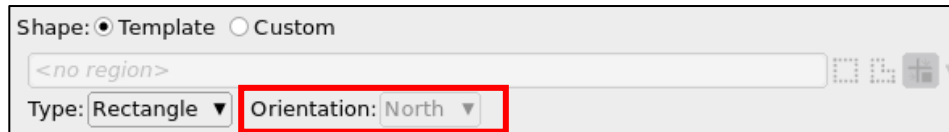
Core utilization:  ☐ Pad limited

Macro utilization:

Aspect ratio:

- **-orientation N | W | S | E**

Specifies one of four possible orientations for the specified rectilinear shape. The tool repositions the block to the specified orientation by rotating it in a clockwise direction. For -shape R, the orientation is always N



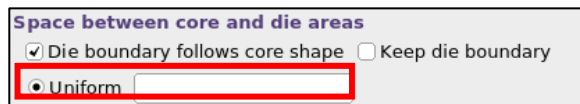
Shape: ☒ Template ☐ Custom

<no region>

Type:  Orientation:

- **-core\_offset { value | X\_Offset Y\_Offset | side\_1 ... side\_N }**

Specifies the distance between the side of the core and the side of the die boundary. If only one value is specified, the value is used for all sides



Space between core and die areas

☒ Die boundary follows core shape ☐ Keep die boundary

☒ Uniform

# Create the initial Floorplan

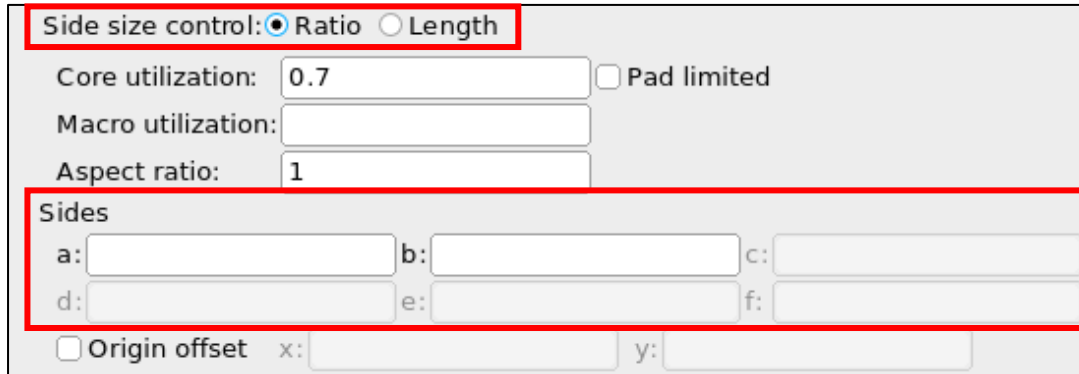
- **Floorplan Initialization**

- **-side\_ratio { side\_a side\_b side\_c side\_d side\_e side\_f }**

Specifies the relative proportion of the floorplan edges in relation to each other. If the control\_type is die, the side ratios apply to the boundary side settings

- **-side\_length { side\_a side\_b side\_c side\_d side\_e side\_f }**

Specifies the side lengths for the edges of the floorplan. If the control\_type is die, the side lengths apply to the die boundary



Side size control: ☒ Ratio ☐ Length

Core utilization: 0.7 ☐ Pad limited

Macro utilization:

Aspect ratio: 1

Sides

a:  b:  c:

d:  e:  f:

☐ Origin offset x:  y:

# Create the initial Floorplan

- Floorplan Initialization

- Defines standard cell Placement site array within The core area

- Supports various Core shapes:

- Rectangular
- L-shape
- T-shape
- U-shape
- Custom

Task Assistant - Design Planning

Floorplan Preparation->Floorplan Initialization

4 of 70

**Boundary**

☒ Core area ☐ Die area

Shape: ☒ Template ☐ Custom

<no region>

**Shape & Orientation**

Type: **U shape** Orientation: **West**

Side size control: ☒ Ratio ☐ Length

**Core utilization**

Core utilization: 0.7

Macro utilization:

Aspect ratio: 1

**Side size (ration value)**

Sides

a: 3 b: 1 c: 2

d: 2 e: 1 f: 2

☐ Origin offset x: y:

**Space between core and die areas**

☒ Die boundary follows core shape ☐ Keep die boundary

**Core offset from boundary**

☒ Uniform 40

☐ Horizontal/Vertical Vertical: Horizontal:

☐ Side by side side\_1 side\_2 ... side\_N

1: 200.064 2: 200.52 3: 200.064

4: 200.52 5: 6:

7: 8:

**Keep**

☐ All routes, placement except boundary

Placement: ☐ All ☐ Std cell ☐ Macro ☐ Block ☐ IO ☐ Physical only

Route: ☐ PG route ☐ Detail route

**Preview your design**

Preview

**Command**

initialize\_floorplan -shape U -orientation W -side\_ratio {3 1 2 2 1 2} -core\_offset {40}

Apply Script Default Help

Close

```
icc2 shell> initialize floorplan -shape U -orientation W -side_ratio {3 1 2 2 1 2} -core_offset {40}
```

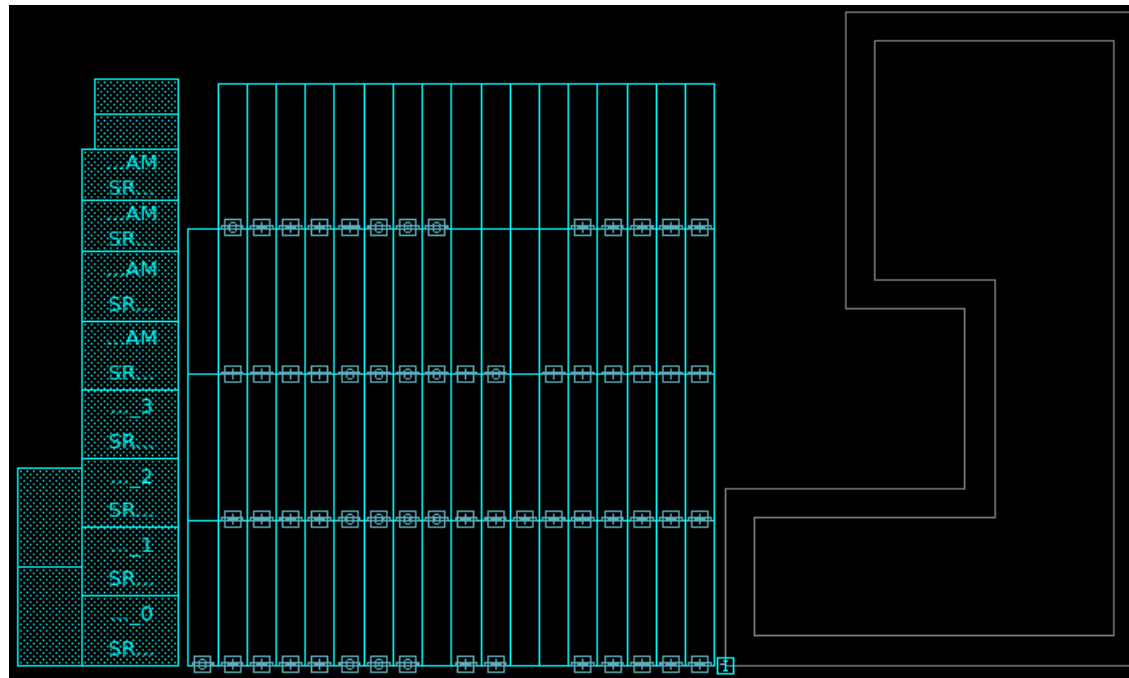
# Create the initial Floorplan

- Floorplan Initialization

Initialize\_floorplan also creates the routing tracks based on the definitions from your technology file

To define a floorplan other than the regular shapes listed above, select the “Custom” option for “Boundary/Shape” and provide the coordinates of the boundary. Here’s an example:

```
lcc2_shell > initialize_floorplan -shape U -orientation W -side_ratio {3 1 2 2 1 2} -core_offset {40}
```



# Place Macros and Standard Cells

- Example GUI for create\_placement of Macros and standard cells

The screenshot shows the 'Task Assistant - Design Planning' window, specifically the 'Cell Placement->Cell Placement' step (21 of 70). The 'Place hard macros' checkbox is checked. The 'Create blockages' dropdown is set to 'auto'. The 'Use floorplanning placement' checkbox is checked and highlighted with a red rectangle. To the right of this checkbox, a text annotation states: 'Specifies that floorplanning placement should be used (This includes macro placement and special effort levels which are tuned for speed)'. Other options include 'Use current placement as a seed for placement', 'Enable timing driven placement', 'Enable buffering aware timing driven initial placement', 'Enable congestion driven placement', and 'Perform incremental placement', all of which are unchecked. The 'Placement effort' and 'Congestion effort' dropdowns are both set to 'medium'. The 'From' dropdown is empty. The 'Host options' field is empty. At the bottom, there is a text area containing the following commands: 

```
set_app_options -name place.coarse.fix_hard_macros -value false
set_app_options -name plan.place.auto_create_blockages -value auto
create_placement -floorplan -effort medium
```

 The 'Script' button is highlighted with a blue border. Other buttons include 'Apply', 'Default', 'Help', and 'Close'.

```
icc2_shell> create_placement -floorplan -congestion -congestion_effort medium
```

# Place Macros and Standard Cells

- Example options for Create\_placement

The `-floorplan` option includes macro placement and special effort levels which are turned for speed

```
Usage: create_placement      # Create a coarse placement
      [-floorplan]          (Run design planning styled placement)
      [-host_options host_options_name]
                           (Host options to use)
      [-use_seed_locs]      (Use current cell locations as a seed for placement)
      [-timing_driven]      (Run timing-driven placement)
      [-congestion]         (Run congestion driven placement)
      [-congestion_effort string]
                           (Congestion driven placement effort)
      [-effort string]      (Placement effort)
      [-incremental]        (Run incremental placement)
```

# Configuring Macro and Standard Cell Placement

- After Auto Floorplan use `create_placement –floorplan` command
- Always start with the defaults
- Macros at the same hierarchy level and of the same size are automatically grouped into arrays
- Channels between macros are automatically sized based on the number of pins
- You can use **report\_app\_option** to report application options
- Soft placement blockages are automatically applied in thin channels (`plan.place.auto_create_blockages`)

```
icc2 shell> report app options plan.place.auto create blockages
```

```
*****
Report : app_option
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Aug  2 00:00:35 2023
*****
```

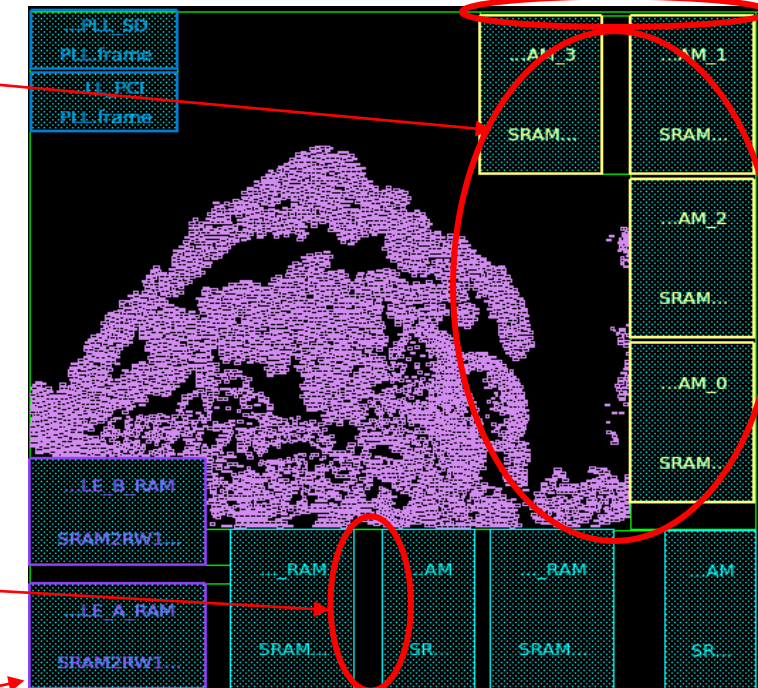
| Name                             | Type | Value | User-value | User-default | System-default | Scope | Status | Source |
|----------------------------------|------|-------|------------|--------------|----------------|-------|--------|--------|
| plan.place.auto_create_blockages | enum | auto  | --         | --           | auto           | block | normal | --     |

- The macro placement style is controlled using the application option `plan.macro.style` (`ON_EDGE` is default) - Macro placed on the edge of the core area (`plan.macro.style`)

```
icc2 shell> report app options plan.macro.style
```

```
*****
Report : app_option
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Aug  2 00:01:50 2023
*****
```

| Name             | Type | Value   | User-value | User-default | System-default | Scope | Status | Source |
|------------------|------|---------|------------|--------------|----------------|-------|--------|--------|
| plan.macro.style | enum | on_edge | --         | --           | on_edge        | block | normal | --     |



# Configuring Macro and Standard Cell Placement

- Application Options (Placement can be configured using `plan.place.*` and `plan.macro.*`)
  - **To disable auto-spacing set the application option**  
`plan.place.default_keepout` to false
  - **To manually specify macro spacing use:**  
`plan.macro.spacing_rule_heights`  
`plan.macro.spacing_rule_widths`
  - **In most cases, pins are only located on one of the macro edges (RAMs). By default, macros are flipped such that the sides with the pins are opposite each other. This reduces the number of channels necessary for routing to the pins. Fewer channels translates into less effort during implementation (buffering, routing)**  
**This is controlled using** (`plan.macro.auto_macro_array_minimize_channels`)
  - **Automatic blockage generation is controlled through the application option**  
`plan.place.auto_generate_blockages`
  - **Enable hard blockage creation, and control the max width of hard/soft blockages with:** `plan.place.auto_generate_soft_blockage_channel_width`      `plan.place.auto_generate_hard_blockage_channel_width`

# Macro Placement Styles – On-Edge vs. FreeForm

- By default, macros are placed around the core (on-edge)

- ❖ Application options are set using the "set\_app\_options" command

- **On-Edge**

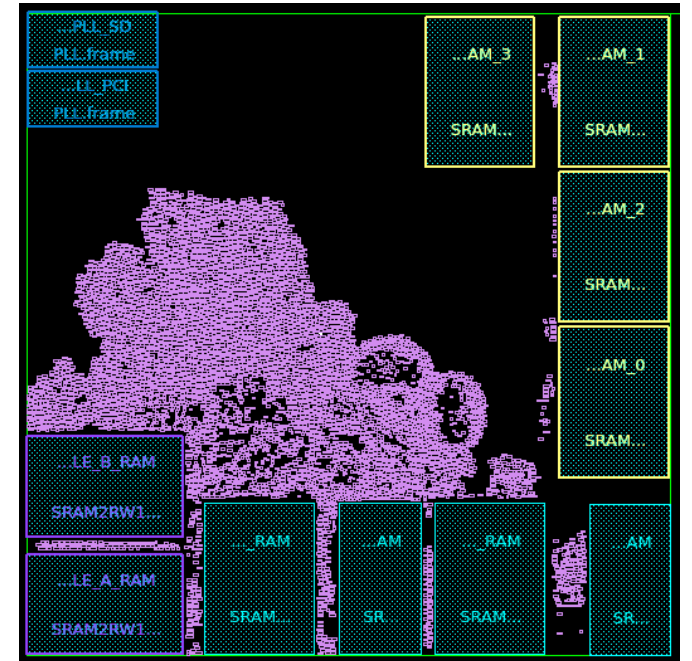
- On edge style is the default style
- It places macros along edges of the partitions  
(Partitions include blocks, voltage area, and move bounds)

```
set_app_options -name plan.macro.style \  
               -value on_edge  
  
create_placement -floorplan
```

- **Freeform**

- In freeform style, the placer places macros and standard cells simultaneously
- Macros will be treated as big STD cells and scattered in the partition

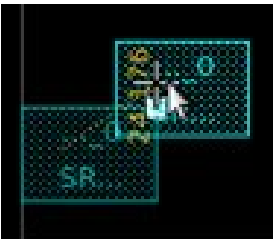
```
set_app_options -name plan.macro.style \  
               -value freeform  
  
create_placement -floorplan
```



# Manual Macro placement

- You can manually place the Macro using the toolbar on the left side of the ICC2

You can use the shortcut M key or click the corresponding icon, select the object you want to move, and then move it



After selecting the macros, you can specify the macro array by clicking the corresponding icon, and then set the spacing and rotate between the macros through the right click of the mouse.

**Select Macros to group**      **Macro array**      **Right click on the objects and select ungroup to ungroup objects**

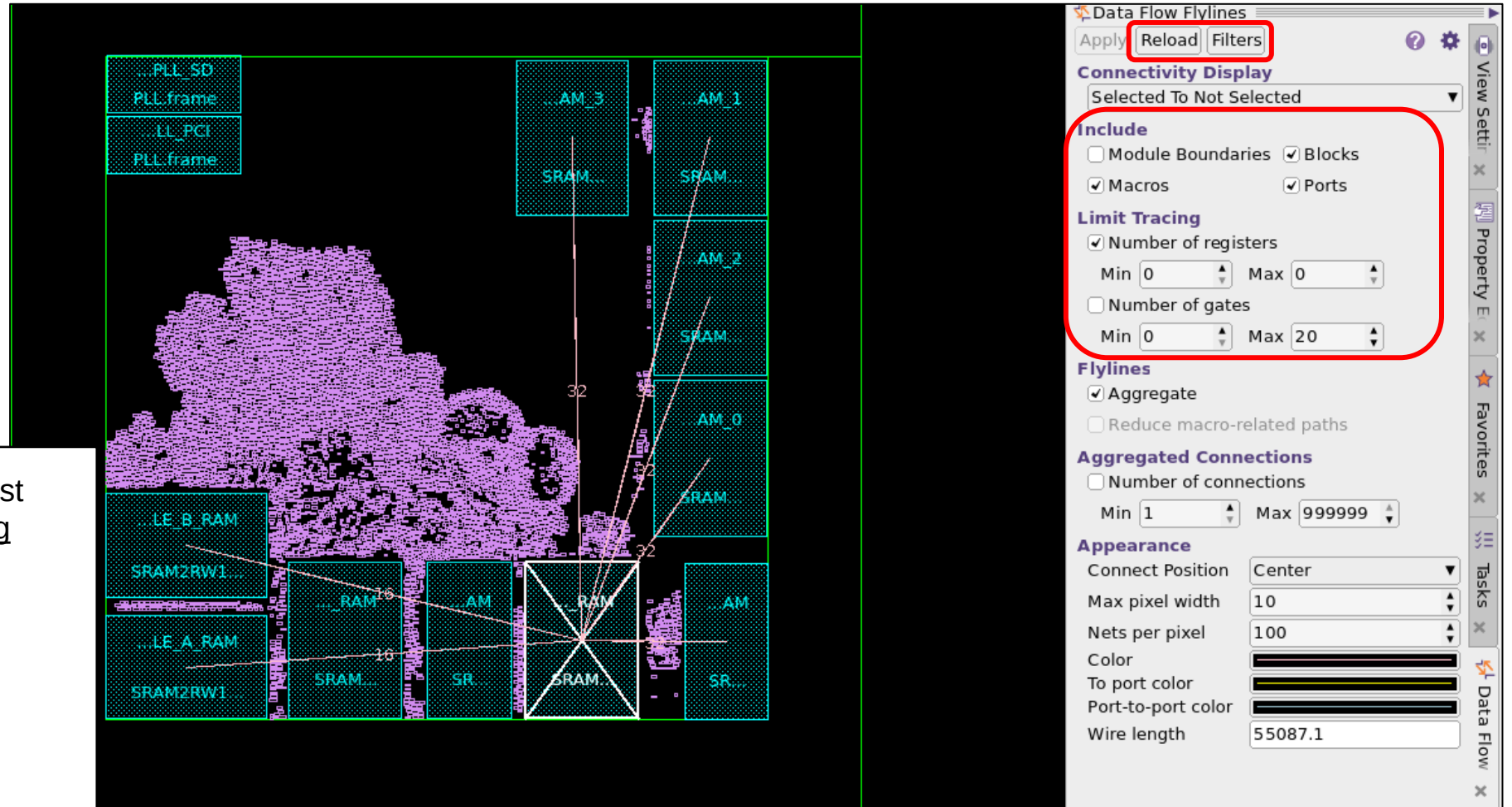
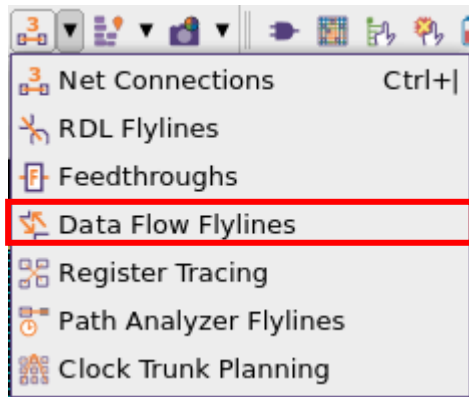
Object fixed & unrestricted icon

Align/Expand offset

You can align and distribute object use this icon

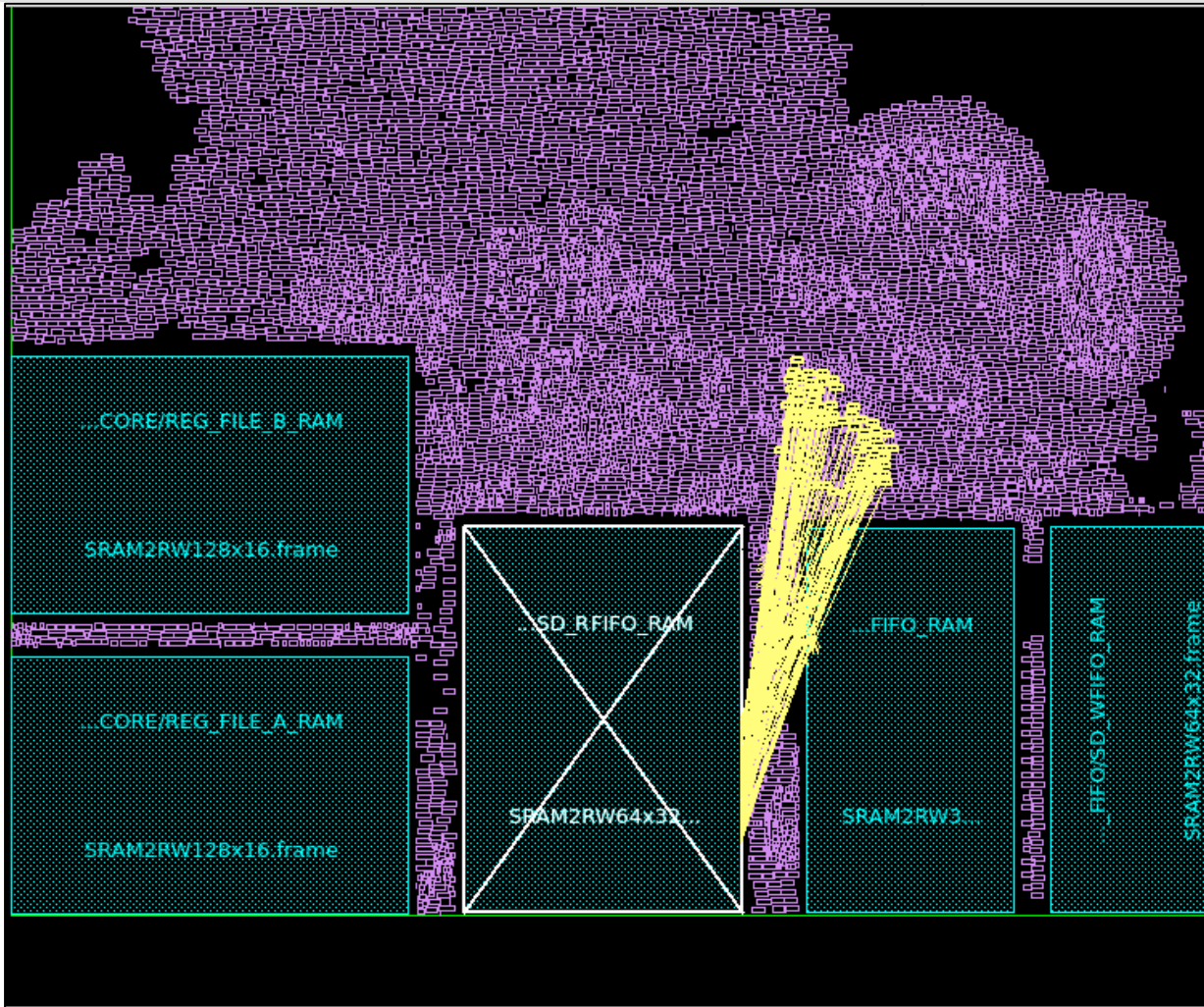
You can Rotate object use this icon

# Analyze Placement of Macros using Data Flow Flylines (DFF)



- DFF allows exploration of the netlist connectivity by using netlist tracing
- A “flyline” can be configured to pass through several levels of combinational gates as well as Registers
- This allows connectivity analysis beyond the simple “Net Connections”.
- Connections that terminate in registers are not shown

# Register Tracing



The main window displays a circuit diagram with a yellow flyline tracing connections from a macro to registers. The macro is labeled "SD\_RFIFO\_RAM" and is connected to "SRAM2RW64x32...". The flyline shows connections to "SRAM2RW128x16.frame", "SRAM2RW128x16.frame", "SRAM2RW3...", and "FIFO/SD\_WFIFO\_RAM".

**Register Tracing**

Apply

**Trace**

Selected To All

**Limit Tracing**

Max fanout 4

Max levels 1

**Show Levels**

☒ Level 1

**Highlight**

☒ Registers

☐ End points

☐ Direct end points

End point type:

☒ Pins ☐ Ports ☐ Both

☒ Show flylines

Show Table

**Net Connections** Ctrl+|

**RDL Flylines**

**Feedthroughs**

**Data Flow Flylines**

**Register Tracing**

**Path Analyzer Flylines**

**Clock Trunk Planning**

- Select one or more start-point objects (macros or non-clock input ports) to see their connectivity to one or more levels of registers, as well as direct and indirect connections to other macros or output ports
- Flylines only show connections to registers – no combinational logic connections are shown.

- Selecting a macro selects all of its data output pins (no PG, clock or scan pins)
- Standard cells/pins may also be selected, but due to runtime, only the first 100 of these start-points will be traced

# Place Pins (SUB Block Level – for reference only)

- **Syntax of place\_pins**

```
set_block_pin_constraints -self -allowed_layers "M3 M4" \  
                        -sides "1 2 3" | -exclude_sides "4 5 6"  
  
create_pin_constraint -type individual -ports clk ...  
  
place_pins -self
```

- **set\_block\_pin\_constraints (option)**

- **-self**

Applies constraints only to the top-level design. By default, the constraints are applied to block cells inside current top design.

- **-allowed\_layers** (Specifies the set of metal layers allowed for pin placement (list of Metal Layers))
  - **-sides** (Specifies the block sides on which the pin must be placed)
  - **-exclude\_sides** (Specifies the block edges on which pins cannot be placed)

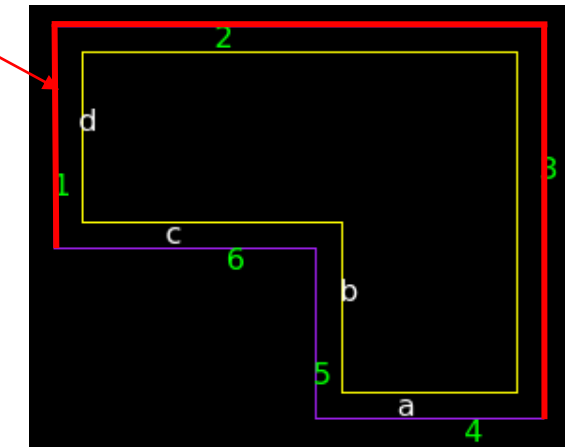
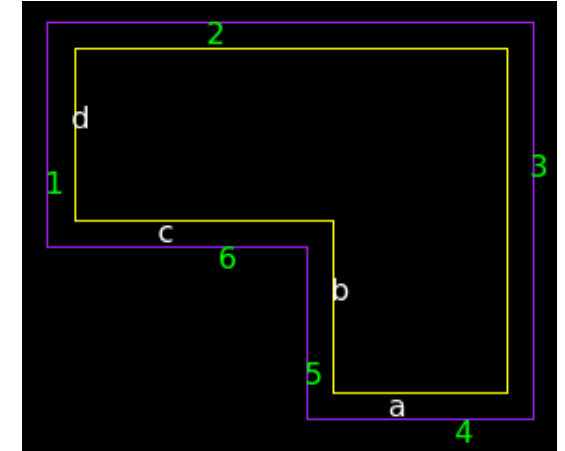
- **create\_pin\_constraint (option)**

- **-type** (Specifies the type of pin constraint to create)
  - **-ports** (Specifies the ports that receive the constraints)

- **place\_pins (option)**

- **-self**

Performs pin placement on the top-level design (By default, pin placement is performed on blocks within the current top-level design)



# Place I/O PADS (TOP design)

- IO information file
  - Pin/PAD locations
  - Edge and order for IO Placement
  - .tdf .io are common formats
- Example of IO information file

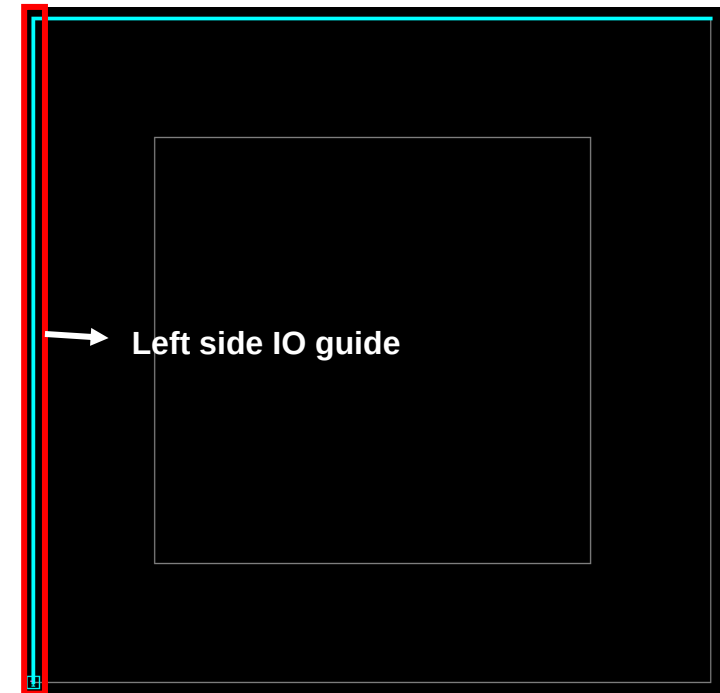
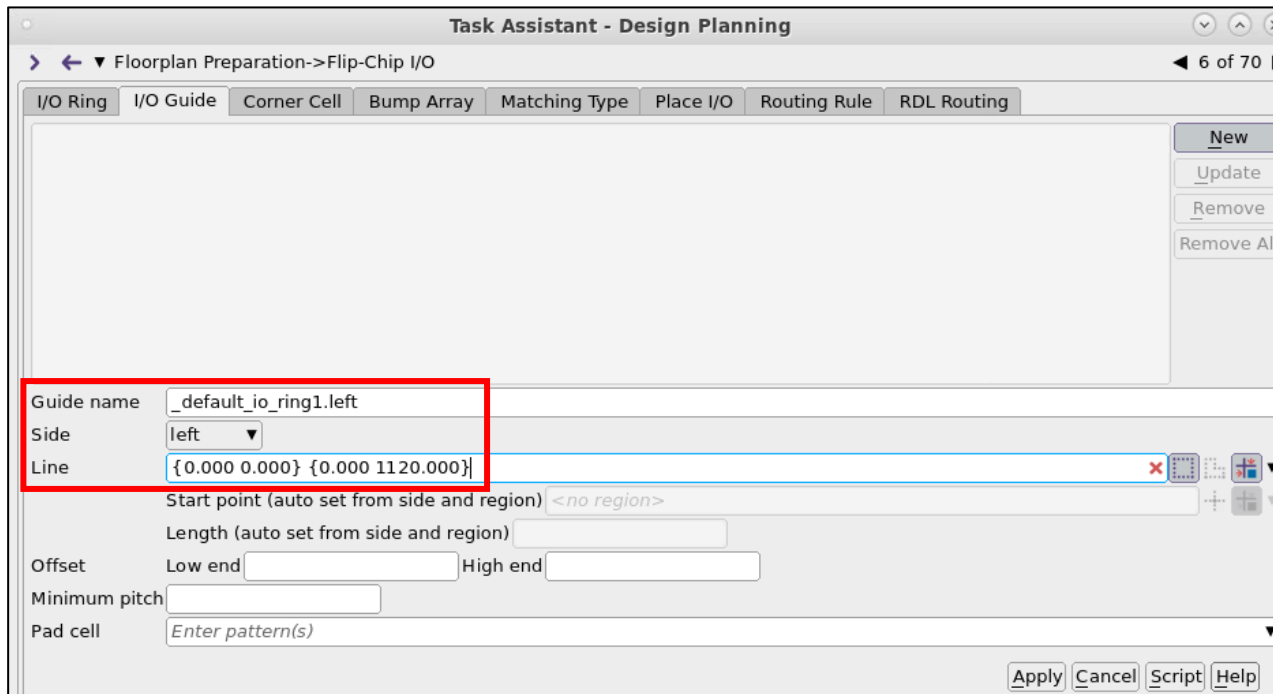
```
default_io_ring1.left
{
{order_only}
powersave_iopad
U_IOPOWER/VDD_L2
U_IOPOWER/VSS_L2
pad_iopad_0
pad_iopad_1
pad_iopad_2
pad_iopad_3
pad_iopad_4
pad_iopad_5
pad_iopad_6
pad_iopad_7
pad_iopad_8
pad_iopad_9
pad_iopad_10
U_IOPOWER/IOVDD_L1
U_IOPOWER/IOVSS_L1
pad_iopad_11
pad_iopad_12
pad_iopad_13
pad_iopad_14
pad_iopad_15
pidsel_iopad
U_IOPOWER/VSS_L1
U_IOPOWER/VDD_L1
pgnt_n_iopad
}
;
```

# Place I/O PADS (TOP design)

- Apply and known:
  - IO guide
  - IO Corner Cell
  - IO Filler Cell
  - IO information file

# Place I/O PADS (TOP design)

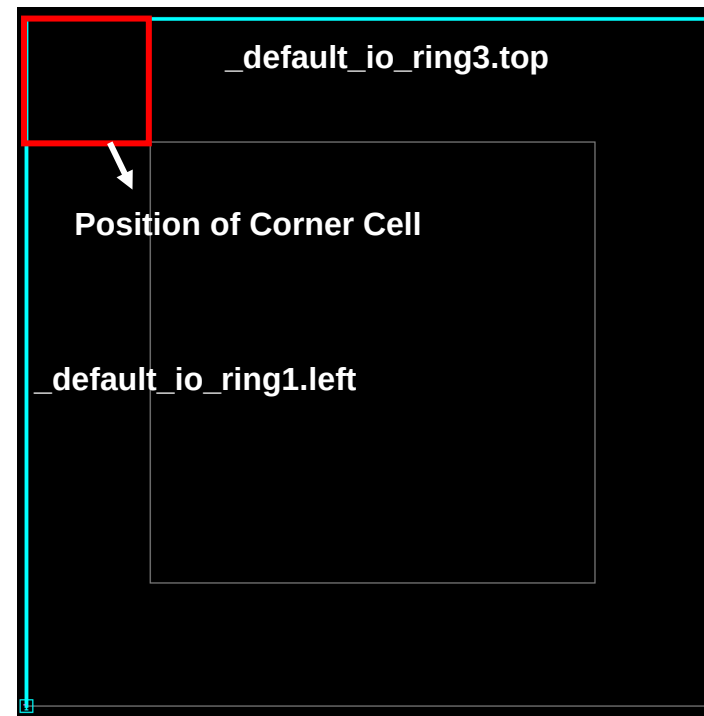
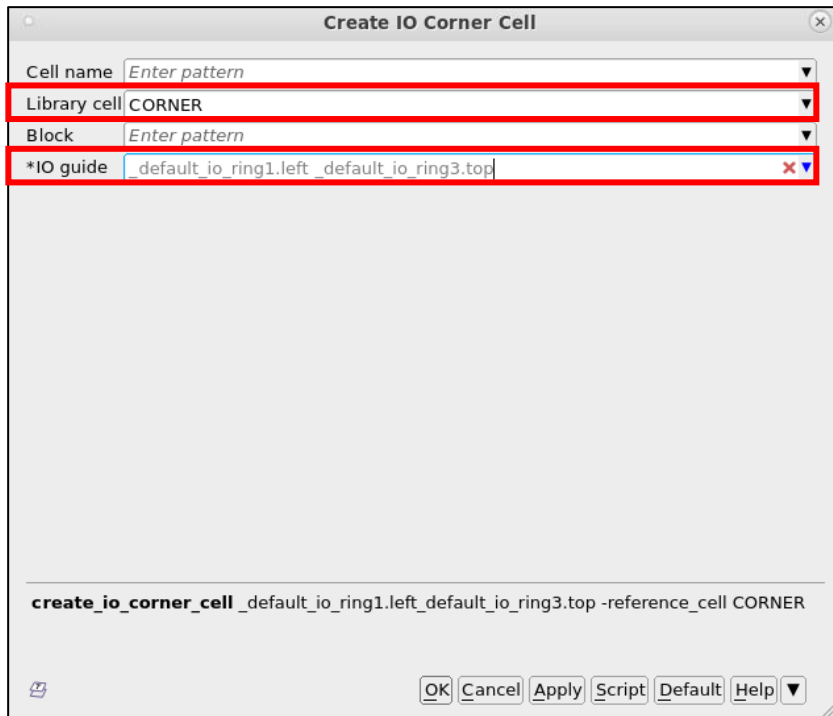
- create\_io\_guide
  - The create\_io\_guide command creates a line-based IO pad constraints for I/O placement
  - The I/O placer uses the I/O guide to properly place the pads within the region



```
icc2 shell> create_io_guide -name {_default_io_ring1.left} -side left -line {{0.000 0.000} 1120}  
{_default_io_ring1.left}
```

# Place I/O PADs (TOP design)

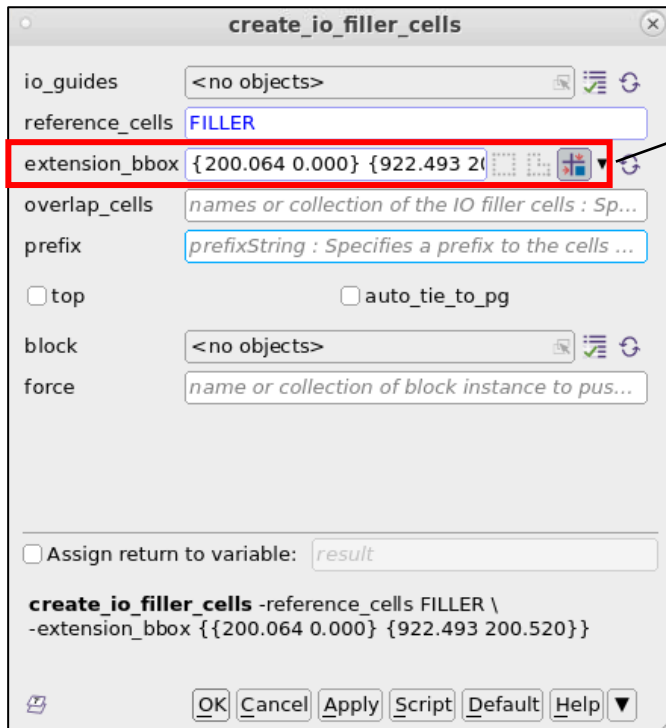
- `create_io_corner_cell`
  - This command places a corner cell at the intersection of two I/O guides
  - The two I/O guides must be perpendicular
  - The location is the crossing point of the two I/O guides



```
icc2_shell> create_io_corner_cell {_default_io_ring1.left _default_io_ring3.top} -reference_cell CORNER
```

# Place I/O PADS (TOP design)

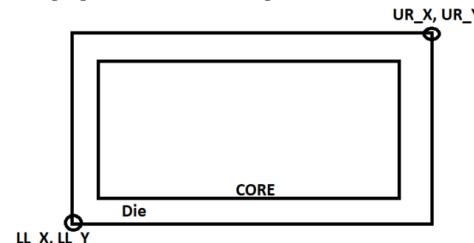
- `create_io_filler_cell`
  - This command inserts filler cells into I/O guides.
  - New filler cells are named `_added_filler_instance_index`, where index is a positive integer.



You can also select and insert the desired area with `extension_bbox` option instead of `IO_guide` option

## ❖ What is bbox (boundary box)?

- Bbox stands for boundary box
- where LL\_X is lower left of X coordinate, UR\_X is upper right of X coordinate, LL\_Y is lower left of Y coordinates, UR\_Y is upper right Y coordinate.
- Example of boundary box:  
`{LL_X LL_Y} {UR_X UR_Y}`



- Syntax of `create_io_filler_cell`

```
create_io_filler_cells -reference_cells "FILLER" \
-io_guides {default_io_ring2.right default_io_ring4.bottom}

create_io_filler_cells -reference_cells FILLER \
-extension_bbox {{200.064 0.000} {922.493 200.520}}
```

# WEEK 4 : LAB for Floorplan

# Floorplan : introduction to floorplan.tcl

- Review and check the variables in the floorplan.tcl script

```
#set current step & before step
set step "floorplan"
set before_step "read_design"
#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl
#open lib
open_lib ../01_read_design/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../01_read_design/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design

# Initialize floorplan
initialize_floorplan -side_length {939 939 939 939} -core_offset {250.548 251.004 250.548 251.004}

#create soft_blockage - an unnecessary area (only place buffer & inverter)
set core_bdry "[get_attr [get_core_area] bbox]"
set new_bdry "{425.5430 425.8460} {1014.4590 1014.1520}"
set core [compute_polygons -operation NOT -objects1 ${core_bdry} -objects2 ${new_bdry}]
create_placement_blockage -type soft -name core -boundary $core

#place PAD
source -e ./user_script/pad_place.tcl

#source FP.tcl
source ./FP_data/ORCA_floorplan_data/fp.tcl

#fix SRAM
set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]
change_selection $all_macros
set_fixed_object [get_selection]
change_selection

#check floorplan
check_legality
check_io_placement
check_mv_design |
#save floorplan
#write_floorplan -object $all_macros -output ./FP_data/ORCA_floorplan_data

#soft blockages for placement (between Macro)
read_def ./FP_data/all_soft_blockage.def
#write soft blockage def
#set sft_blk [get_placement_blockages -filter "blockage_type == soft"]
#change_selection $sft_blk
#write_def -objects [get_selection] ./FP_data/all_soft_blockage.def
#change_selection

#save & exit
save_lib -all
```

# Floorplan : Introduction to floorplan.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
< user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/02_floorplan/script
< user@linux script >> ls
01_design_setup_and_copy.tcl  03_create_soft_blockage.tcl  05_place_macro_and_check.tcl  full_script
02_initialize_floorplan.tcl  04_place_PAD_and_terminal.tcl  06_blockage_and_save.tcl
```

- **Review 01\_design\_setup\_and\_copy.tcl**

```
# module load
#module load icc2/2021.06-SP5
# open icc2_shell
#icc2_shell -gui | tee -i ./02_floorplan.log

#set current step & before step
set step "floorplan"
set before_step "read design"
#source design_setup.tcl
source ./icc2_common_env/design_setup.tcl
#open lib
open_lib ./nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ./nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
```

- **Review 02\_initialize\_floorplan.tcl**

```
# Initialize floorplan
initialize_floorplan -side_length {939 939 939 939} -core_offset {250.548 251.004 250.548 251.004}
```

# Floorplan : Introduction to floorplan.tcl

- Review and check the variables in each scripts

- **Review 03\_create\_soft\_blockage.tcl**

```
#create soft_blockage - an unnecessary area (only place buffer & inverter)
set core_bdry "[get_attr [get_core_area] bbox]"
set new_bdry "{425.5430 425.8460} {1014.4590 1014.1520}"
set core [compute_polygons -operation NOT -objects1 ${core_bdry} -objects2 ${new_bdry}]
create_placement_blockage -type soft -name core -boundary $core
```

- ❖ Polygon is a geometry region created by the user  
(compute\_polygons command performs a boolean geometric operation and returns the result as a geo\_mask)

- **Review 04\_place\_PAD\_and\_terminal.tcl**

```
#place PAD
source -e ./user_script/pad_place.tcl

#PAD place
#${WORK_DIR}/2_BACKEND/1_pnr/rundir/02_floorplan/user_script/pad_place.tcl

#create terminal
#${WORK_DIR}/2_BACKEND/1_pnr/rundir/02_floorplan/user_script/pad_create_terminal.tcl

#IO constraint
#${WORK_DIR}/2_BACKEND/1_pnr/rundir/02_floorplan/user_script/ORCA.io.ios
```

Script path for PAD place & create terminal

# Floorplan : Introduction to floorplan.tcl

- Review and check the variables in each scripts

- **Review 05\_place\_macro\_and\_check.tcl**

```
#source FP.tcl
source ./FP_data/ORCA_floorplan_data/fp.tcl ;# fp.tcl contains floorplan information

#fix SRAM
set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]
change_selection $all_macros
set_fixed_object [get_selection] ;# fixed all Macros after floorplan
change_selection

#check floorplan
check_legality

check_io_placement

check_mv_design
```

- **Review 06\_blockage\_and\_save.tcl**

```
#soft blockages for placement (between Macro)

read_def ./FP_data/soft_blockage.def

#write soft blockage def
set sft_blk [get_placement_blockages -filter "blockage_type == soft"]
change_selection $sft_blk
write_def -objects [get_selection] ./FP_data/all_soft_blockage.def ;# write def file (include soft blockage information)
change_selection

#save & exit
save_lib -all

exit
```

# Floorplan : Use ICC2 (IC Compiler II tool)

## 1. Invoke tool icc2\_shell

- **Recommend the icc2/2021.06-SP5 version**

Enter the command line by line in 02\_floorplan.tcl and check the result

>> icc2\_shell -gui | tee -i ./02\_floorplan.log ;# (| tee -i ./01\_read\_design.log) is the command to output log

```
< user@linux 02_floorplan >> icc2_shell -gui | tee -i ./02_floorplan.log

                                IC Compiler II (TM)

Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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https://solvnetplus.synopsys.com)

Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
icc2_shell>
```

## 2. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/02\_floorplanscript/01\_design\_setup\_and\_copy.tcl

- **set step & source design\_setup.tcl**

```
icc2_shell> set step "floorplan"
floorplan
icc2_shell> set before_step "read_design"
read_design
icc2_shell> #source design_setup.tcl
```

# Floorplan : Use ICC2 (IC Compiler II tool)

2. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/01_design_setup_and_copy.tcl`

- open lib

```
icc2_shell> open_lib ./nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/02_floorplan/nlib/ORCA_read_design.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_read_design.nlib}
```

Check that ndm is properly applied each time you open the nlib

- Copy `ORCA_read_design.nlib` to `ORCA_floorplan.nlib` & open `ORCA_floorplan.nlib`

```
icc2_shell> copy_lib -from ./nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
Information: Copying from library "ORCA_read_design.nlib" to "./nlib/ORCA_floorplan.nlib"
{ORCA_floorplan.nlib}
icc2_shell> close_lib -all
Closing all libraries...
1
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
```

# Floorplan : Use ICC2 (IC Compiler II tool)

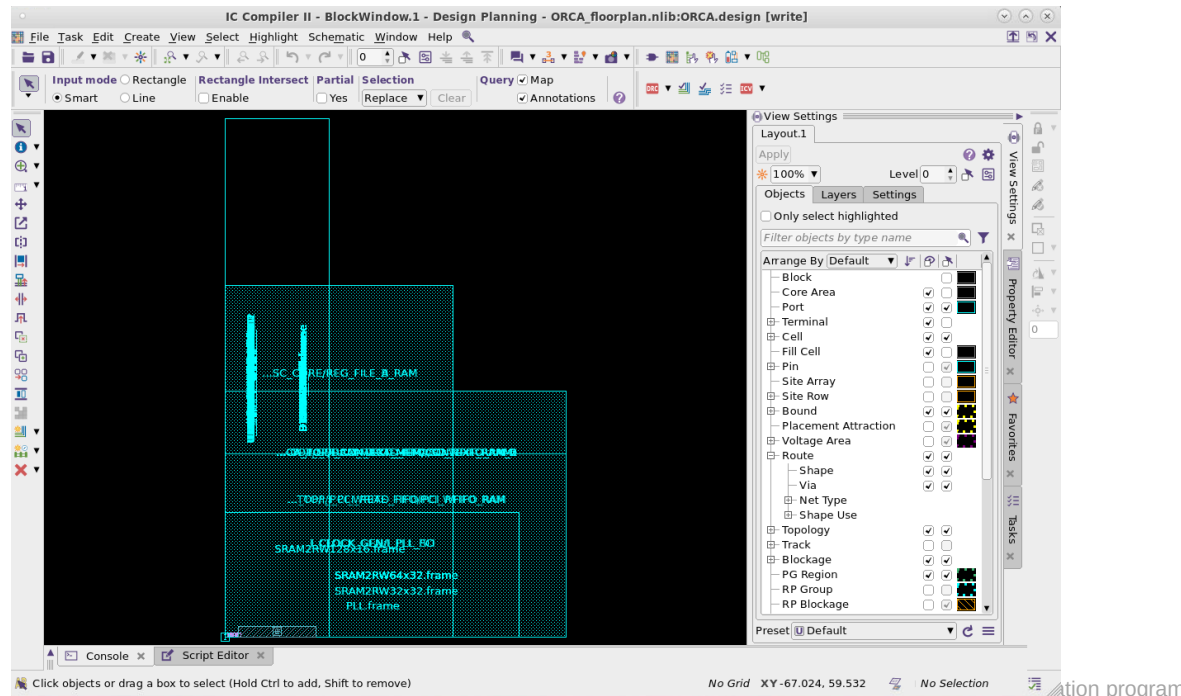
2. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/01_design_setup_and_copy.tcl`

- open block

```
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_floorplan.nlib:ORCA.design' in edit mode
{ORCA_floorplan.nlib:ORCA.design}
icc2_shell> link_block
Using libraries: ORCA_floorplan.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32rvt saed32sram
Visiting block ORCA_floorplan.nlib:ORCA.design
Design 'ORCA' was successfully linked.
1
```

Check using libraries

- Check the copied block in the gui window



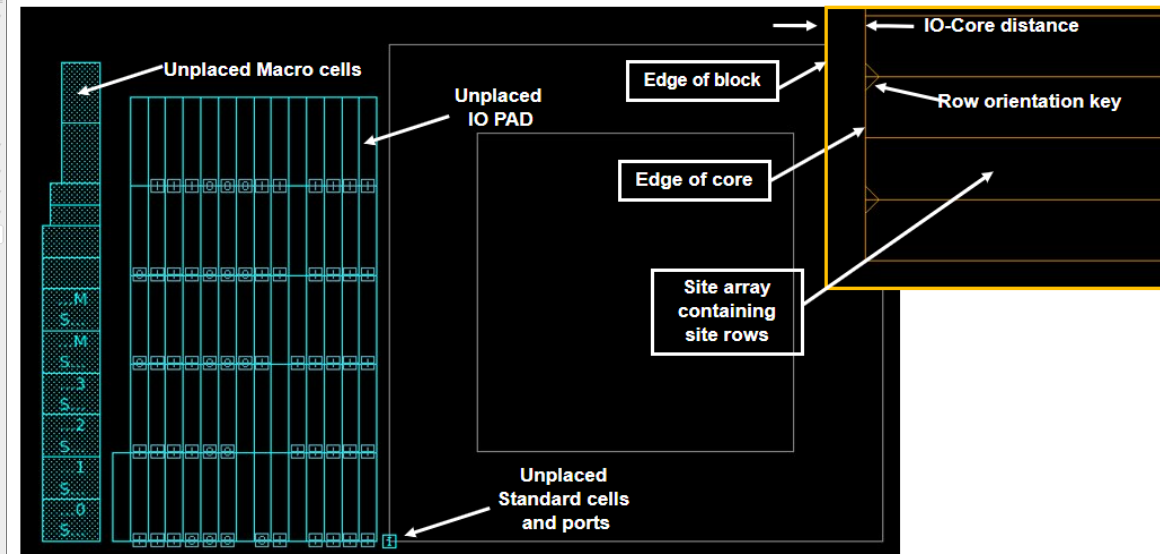
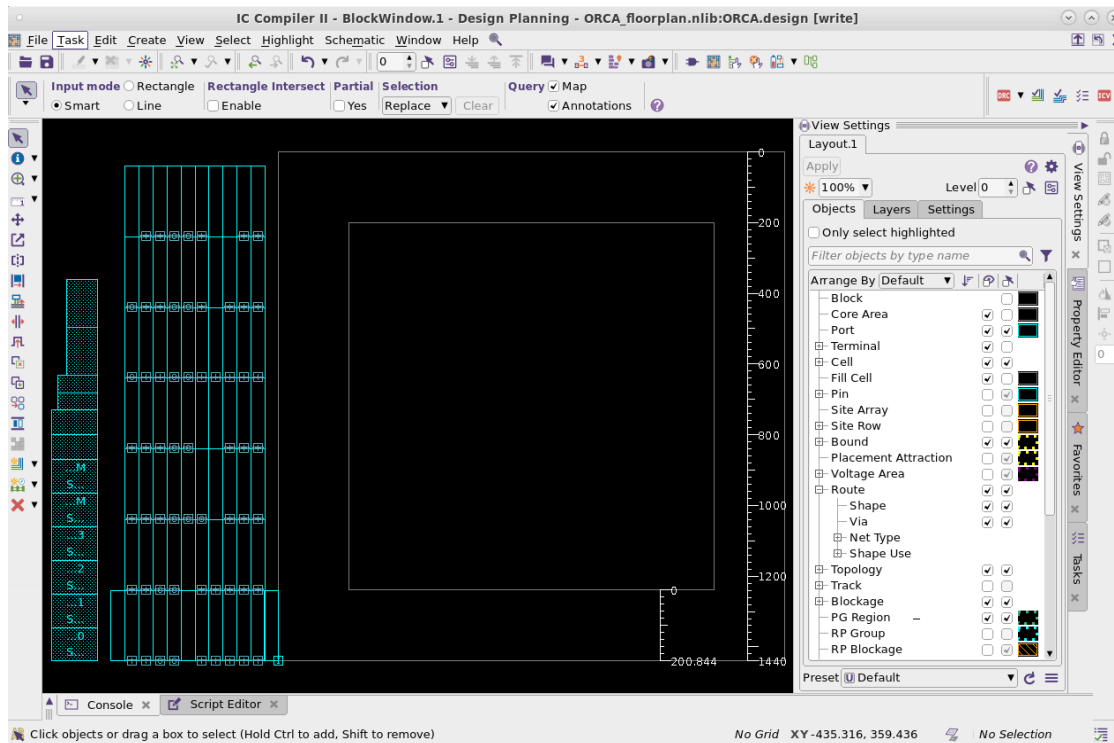
# Floorplan : Use ICC2 (IC Compiler II tool)

3. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/02_initialize_floorplan.tcl`

- open block

```
icc2_shell> initialize_floorplan -side_length {1039 1039 1039 1039} -core_offset {200.54 200.844}
844 200.54 200.844}
Removing existing floorplan objects
Creating core...
Warning: only 2 core side dimension is needed for rectangle shape core area. (DPI-504)
Core utilization ratio = 15.71%
Unplacing all cells...
Creating site array...
Creating routing tracks...
Initializing floorplan completed.
```

- Check floorplan initialization



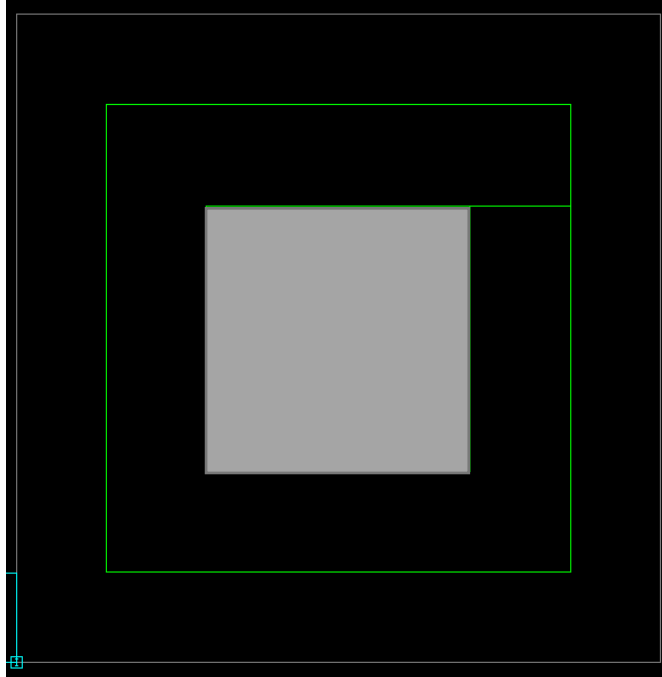
# Floorplan : Use ICC2 (IC Compiler II tool)

4. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/03_create_soft_blockage.tcl`

- Create soft blockages in unnecessary areas (only buffer & inverter can placed)

```
icc2_shell> set core_bdry "[get_attr [get_core_area] bbox]"  
{200.5400 200.8440} {1239.4600 1239.1560}  
icc2_shell> set new_bdry "{425.5430 425.8460} {1014.4590 1014.1520}"  
{425.5430 425.8460} {1014.4590 1014.1520}  
icc2_shell> set core [compute_polygons -operation NOT -objects1 ${core_bdry} -objects2 ${new_bdry}]  
sell386  
icc2_shell> create_placement_blockage -type soft -name core -boundary $core  
{core}
```

- Check soft blockage in core area



## Why create soft blockage in core area?

- ❖ This Design has a large chip size due to the number of PADs, so utilization is very low
- ❖ To prevent the cells being deployed from spreading too far, soft blockage was created to make the core area small
- ❖ In the area where the soft blockage is created, only the buffer of inverter for connection to the PAD is placed

# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- It is also good to Floorplan by source `04_pad_place_and_terminal.tcl`, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

(open `$WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/user_script/pad_place.tcl`)

```
#place PAD
source -e ./user_script/pad_place.tcl

#PAD place
#{$WORK_DIR}/2_BACKEND/1_pnr/rundir/02_floorplan/user_script/pad_place.tcl

#create terminal
#{$WORK_DIR}/2_BACKEND/1_pnr/rundir/02_floorplan/user_script/pad_create_terminal.tcl

#IO constraint
#{$WORK_DIR}/2_BACKEND/1_pnr/rundir/02_floorplan/user_script/ORCA.io.ios
```

- **Create IO guide & create IO corner cell**

```
create_io_guide -name {default_io_ring1.left} -side left -line {{0.000 0.000} 1440}
{ default_io_ring1.left}
icc2_shell> create_io_guide -name {default_io_ring2.right} -side right -line {{1440 1440} 1440}
{ default_io_ring2.right}
icc2_shell> create_io_guide -name {default_io_ring3.top} -side top -line {{0.000 1440} 1440}
{ default_io_ring3.top}
icc2_shell> create_io_guide -name {default_io_ring4.bottom} -side bottom -line {{1440 0.000} 1440}
{ default_io_ring4.bottom}
icc2_shell> create_io_corner_cell -reference_cell CORNER {default_io_ring1.left _default_io_ring3.top}
}
Information: The command 'create_io_corner_cell' cleared the undo history. (UNDO-016)
Corner cell placed successfully
1
icc2_shell> create_io_corner_cell -reference_cell CORNER {default_io_ring2.right _default_io_ring3.to
p}
Corner cell placed successfully
1
icc2_shell> create_io_corner_cell -reference_cell CORNER {default_io_ring2.right _default_io_ring4.bo
ttom}
Corner cell placed successfully
1
icc2_shell> create_io_corner_cell -reference_cell CORNER {default_io_ring1.left _default_io_ring4.bot
tom}
Corner cell placed successfully
1
```

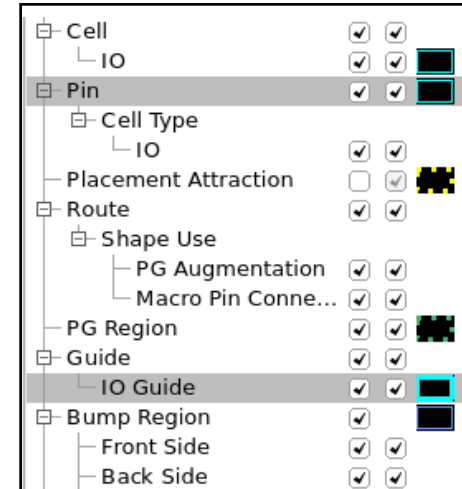
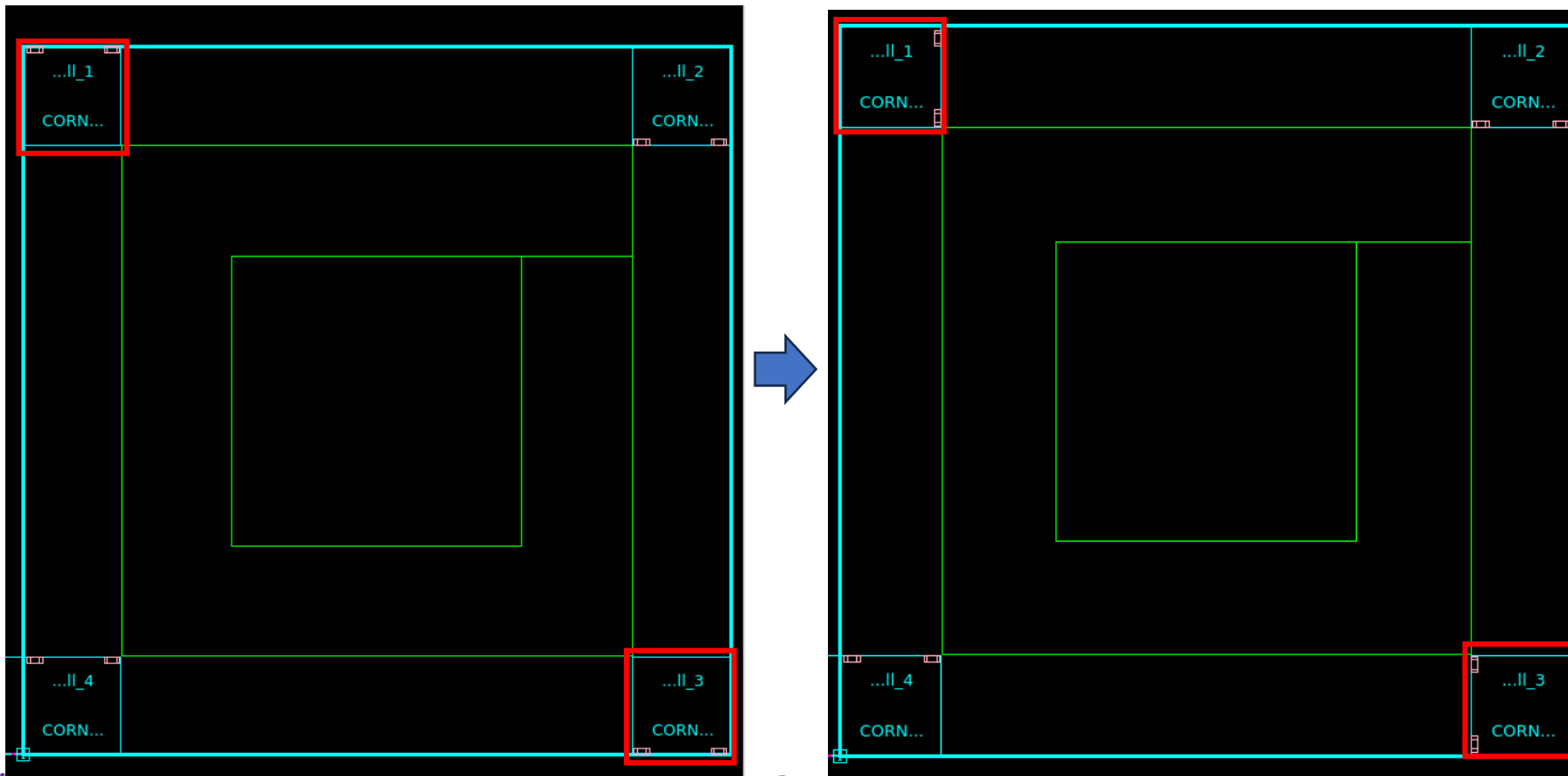
# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- Rotate Corner cell, check io guides (To match the pin direction connected to the IO PAD)

```
icc2_shell> move_objects [get_cells -filter "full_name == __added_corner_cell_1"] -rotate_by CW90
1
icc2_shell> move_objects [get_cells -filter "full_name == __added_corner_cell_3"] -rotate_by CW90
1
```

- Check io guides & Corner cells are properly create



Check pin & IO guide  
in View settings

# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- Remove signal\_io\_constraints & set io constraint file

```
icc2_shell> remove_signal_io_constraints
Information: The command 'remove_signal_io_constraints' cleared the undo history. (UNDO-016)
1
icc2_shell> set_signal_io_constraints -file ./user_script/ORCA.io.ios
Signal IO constraints set successfully
1
```

- Review ORCA\_io.ios file

(open `$WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/user_script/ORCA_io.ios`)

```
default_io_ring1.left default_io_ring4.bottom default_io_ring2.right default_io_ring3.top
{
{order_only}
{order_only}
{order_only}
{order_only}
powersave_iopad pdevsel_n_iopad prst_n_iopad pframe_n_iopad
U_IOPOWER/VDD_L2 U_IOPOWER/VDD_B2 U_IOPOWER/VDD_R1 U_IOPOWER/VDD_T2
U_IOPOWER/VSS_L2 U_IOPOWER/VSS_B2 U_IOPOWER/VSS_R1 U_IOPOWER/VSS_T2
pad_iopad_0 sdram_DQ_iopad_0 pc_be_iopad_0 pm66en_iopad
pad_iopad_1 sdram_DQ_iopad_1 pc_be_iopad_1 preq_n_iopad
pad_iopad_2 sdram_DQ_iopad_2 pc_be_iopad_2 ptrdy_n_iopad
pad_iopad_3 sdram_DQ_iopad_3 pc_be_iopad_3 sdram_CK_iopad
pad_iopad_4 sdram_DQ_iopad_4 pstop_n_iopad sdram_BWS_iopad_0
pad_iopad_5 sdram_DQ_iopad_5 pirdy_n_iopad sdram_BWS_iopad_1
pad_iopad_6 sdram_DQ_iopad_6 scan_en_iopad sdram_RW_iopad
pad_iopad_7 sdram_DQ_iopad_7 U_IOPOWER/IOVDD_R1 sdram_A_iopad_0
pad_iopad_8 sdram_DQ_iopad_8 U_IOPOWER/IOVSS_R1 sdram_A_iopad_1
pad_iopad_9 sdram_DQ_iopad_9 ppar_iopad U_IOPOWER/IOVDD_T1
pad_iopad_10 sdram_DQ_iopad_10 pserr_n_iopad U_IOPOWER/IOVSS_T1
U_IOPOWER/IOVDD_L1 U_IOPOWER/IOVDD_B1 sys_clk_iopad sdram_A_iopad_2
U_IOPOWER/IOVSS_L1 U_IOPOWER/IOVSS_B1 pclk_iopad sdram_A_iopad_3
pad_iopad_11 sdram_DQ_iopad_10 sdr_clk_iopad sdram_A_iopad_4
pad_iopad_12 sdram_DQ_iopad_11 U_IOPOWER/DVDD_PLL1 sdram_A_iopad_5
pad_iopad_13 sdram_DQ_iopad_12 U_IOPOWER/DVDD_PLL2 sdram_A_iopad_6
pad_iopad_14 sdram_DQ_iopad_13 U_IOPOWER/AVDD_PLL1 sdram_A_iopad_7
pad_iopad_15 sdram_DQ_iopad_14 U_IOPOWER/AVSS_PLL1 sdram_A_iopad_8
pidsel_iopad sdram_DQ_iopad_15 U_IOPOWER/AVDD_PLL2 sdram_A_iopad_9
U_IOPOWER/VSS_L1 sdram_LD_iopad U_IOPOWER/AVSS_PLL2 sdram_CKn_iopad
U_IOPOWER/VDD_L1 U_IOPOWER/VDD_B1 U_IOPOWER/VDD_R2 U_IOPOWER/VDD_T1
pgnt_n_iopad U_IOPOWER/VSS_B1 U_IOPOWER/VSS_R2 U_IOPOWER/VSS_T1
}
}
}
}
;
```

- ❖ Io.ios file contains IO PAD lists to be placed on the left, right, top, and bottom io\_guide
- ❖ This file also includes the order of IO PAD

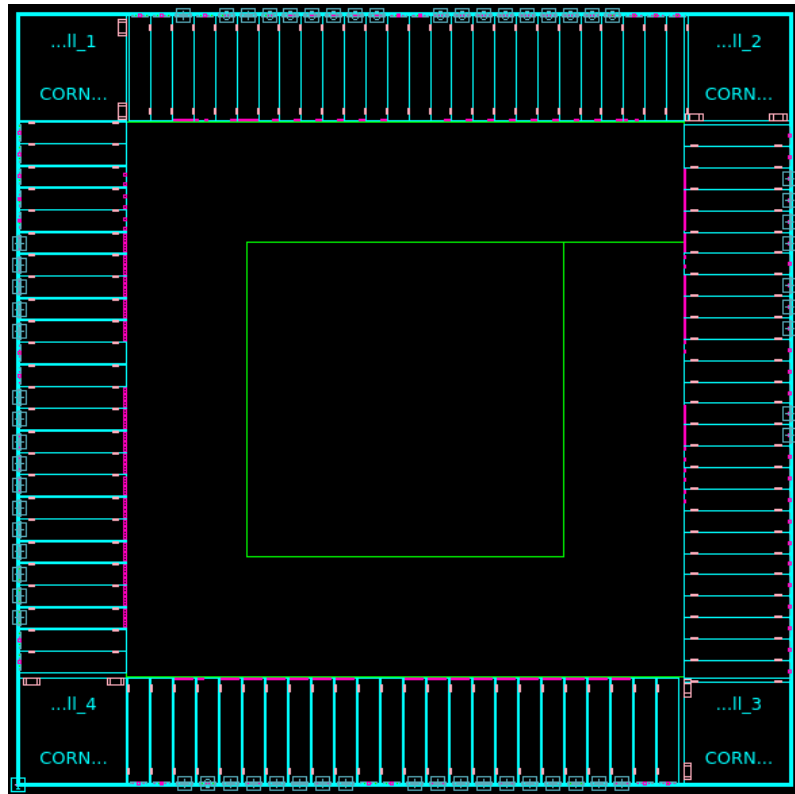
# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- Place IO pad

```
icc2_shell> place_io
Information: Starting 'place_io' (FLW-8000)
Information: Time: 2023-07-23 22:35:00 / Session: 3.96 hr / Command: 0.00 hr / Memory: 499 MB (FLW-8100)
Overall runtime for IO placement: 0.013u 0.000s 0:00.01e 98.1%
Information: Ending 'place_io' (FLW-8001)
Information: Time: 2023-07-23 22:35:00 / Session: 3.96 hr / Command: 0.00 hr / Memory: 499 MB (FLW-8100)
1
```

- Check IO PAD after placing



# Floorplan : Use ICC2 (IC Compiler II tool)

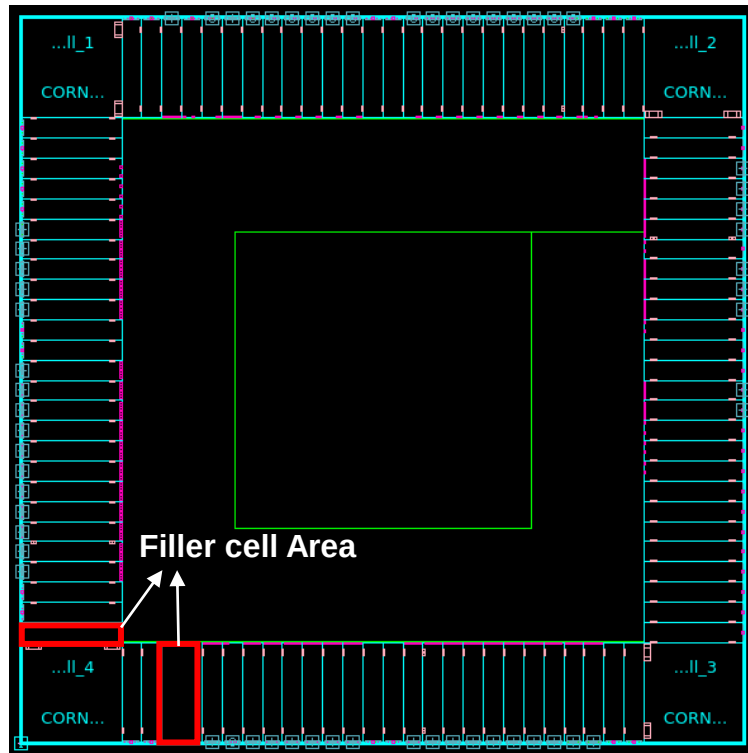
## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/02\_floorplan/script/04\_pad\_place\_and\_terminal.tcl

- IO PAD distribution for IO filler insertion

ex ) Left side

```
icc2_shell> set L_side_PAD "[get_cells -intersect "{0.0000 200.0000}" {201.2310 1440.0000}" \
?
-filter "design_type == pad && orientation == R270 || full_name == __adde
d_corner_cell_1"]"
{powersave iopad pidsel iopad pgnt n_iopad pad iopad 15 pad iopad 14 pad iopad 13 pad iopad 12 pad iop
ad 11 pad iopad 10 pad iopad 9 pad iopad 8 pad iopad 7 pad iopad 6 pad iopad 5 pad iopad 4 pad iopad 3
pad iopad 2 pad iopad 1 pad iopad 0 U_IOPPOWER/IOVDD_L1 U_IOPPOWER/IOVSS_L1 U_IOPPOWER/VDD_L1 U_IOPPOWER/
VDD_L2 U_IOPPOWER/VSS_L1 U_IOPPOWER/VSS_L2 __added_corner_cell_1}
icc2_shell> change_selection $L_side_PAD
icc2_shell> distribute_objects -side top
1
icc2_shell> change_selection
```

- Check IO PAD after distribute



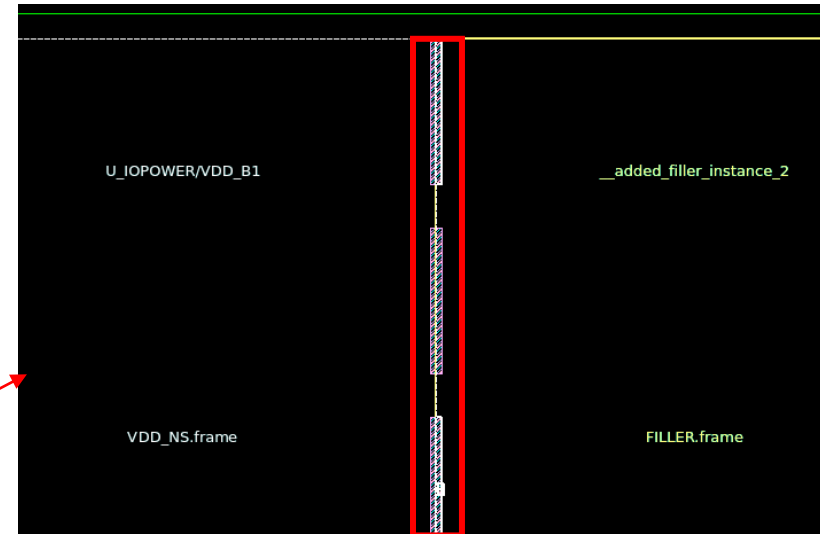
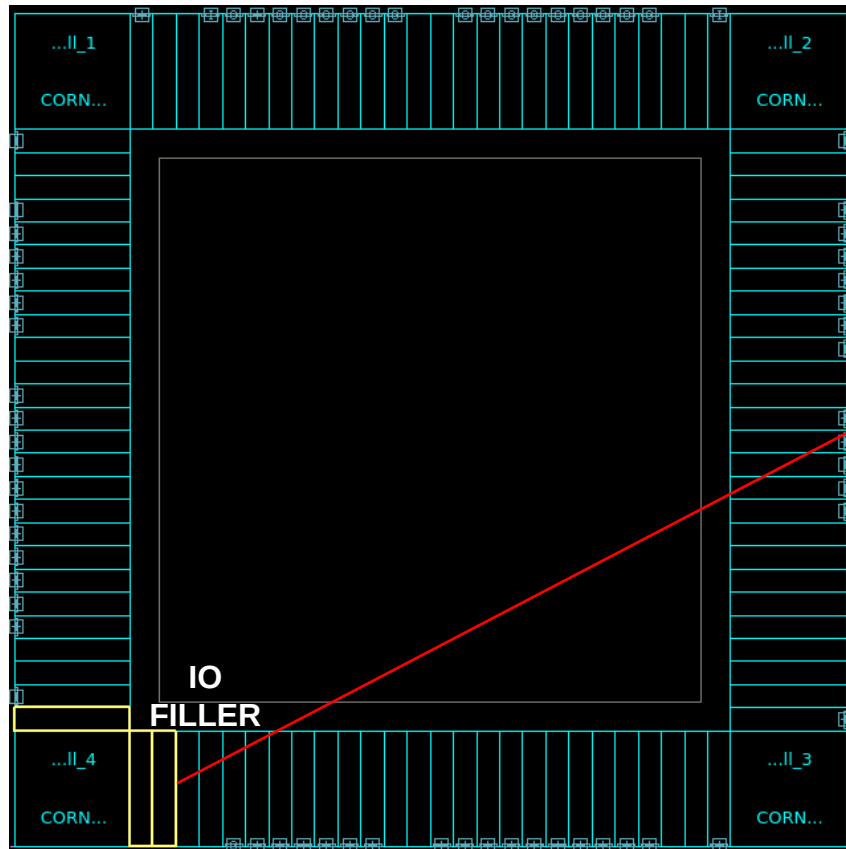
# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- Insert IO filler

```
icc2_shell> create_io_filler_cells -reference_cells "FILLER" -io_guides {_default_io_ring1.left _default_io_ring4.bottom}  
Information: The command 'create_io_filler_cells' cleared the undo history. (UNDO-016)  
Filler cells placed successfully  
1
```

- Check IO filler



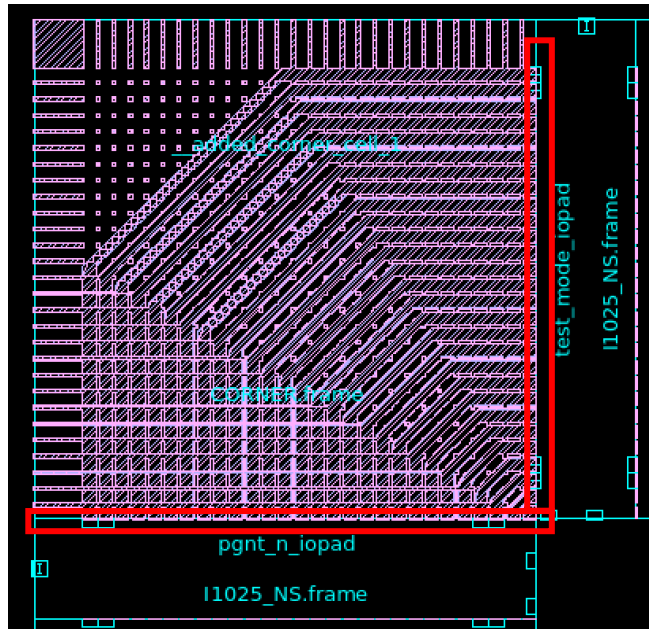
- ❖ The IO filler fills the empty space so that the connection between PADS is not broken

# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- **Connect\_pg\_net** for CORNER & FILLER cells (VDD, VDDIO, VSS, VSSIO)

```
icc2_shell> #CORNER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
icc2_shell> #FILLER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```



- ❖ **Connect Power & Ground net for Corner cells & Filler cells pg pins after create cells**

# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/04_pad_place_and_terminal.tcl`

- `Create_terminal > source -e ./user_script/pad_create_terminal.tcl`
- Creates a terminal on a port in the current design
- Terminals represent continuous, conservative ports in VHDL-AMSv (connection between Digital and Analog logic)

```
icc2_shell> source -e ./user_script/pad_create_terminal.tcl
```

```
##### Right & Left side #####
```

```
#####
```

```
set pad sys_clk iopad
```

```
#####
```

```
set pad_pins [get_pins -of $pad -filter "full_name =~ *PADIO*"]
```

```
set lay M5
```

```
set net [get_attr [get_nets -of $pad_pins] name]
```

```
set port [get_attr [get_nets -of $pad_pins] name]
```

```
set shapes [get_shapes -of $pad_pins -filter "layer.name =~ $lay"]
```

```
set term_shapes ""
```

```
foreach in_col x $shapes {
```

```
    set width [expr [lindex [get_attr $x bbox] 1 0] - [lindex [get_attr $x bbox] 0 0]]
```

```
    set height [expr [lindex [get_attr $x bbox] 1 1] - [lindex [get_attr $x bbox] 0 1]]
```

```
    if { $width < 4 || $height < 29 } {
```

```
    } else {
```

```
        append_to_col term_shapes $x
```

```
    }
```

```
}
```

```
change selection $term_shapes
```

```
set bbox [get_attribute [get_shapes $term_shapes] bbox]
```

```
set aa [create_shape -shape_type rect -shape_use user_route -layer $lay -boundary $bbox]
```

```
create_terminal -port $port -object $aa -name $net -direction "all"
```

Set PAD pins, net, port

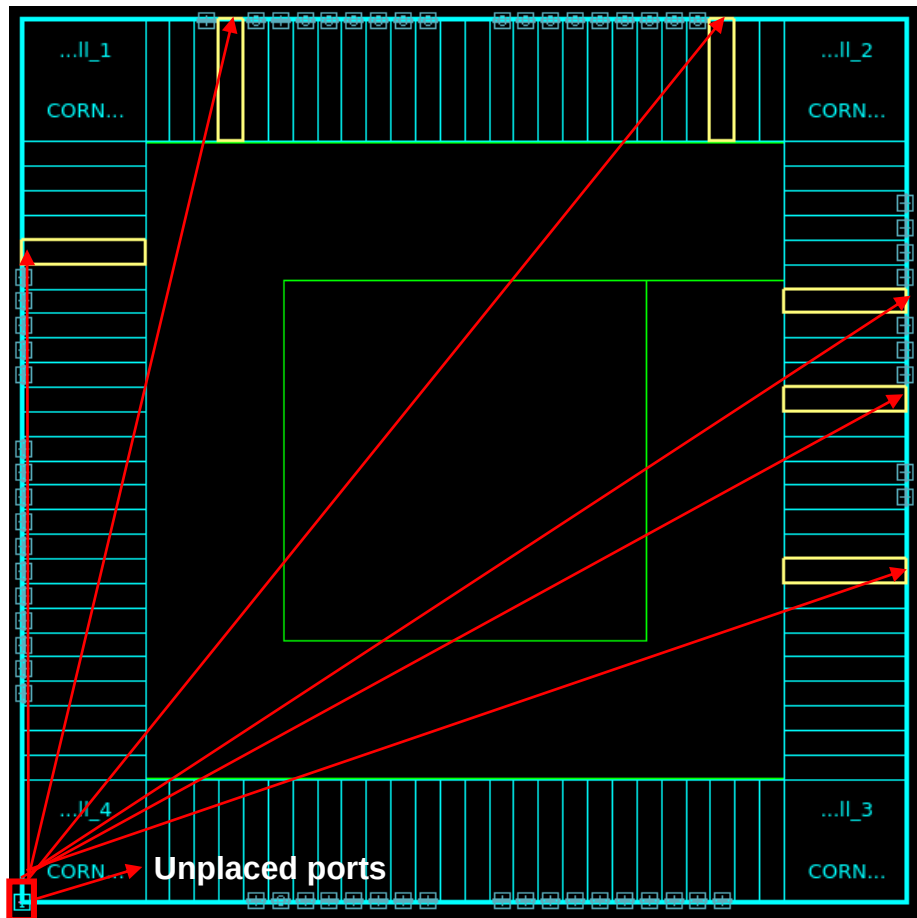
Set shapes for PAD pins

Set bbox for PAD pins shape, and create terminal (use PAD pins bbox)

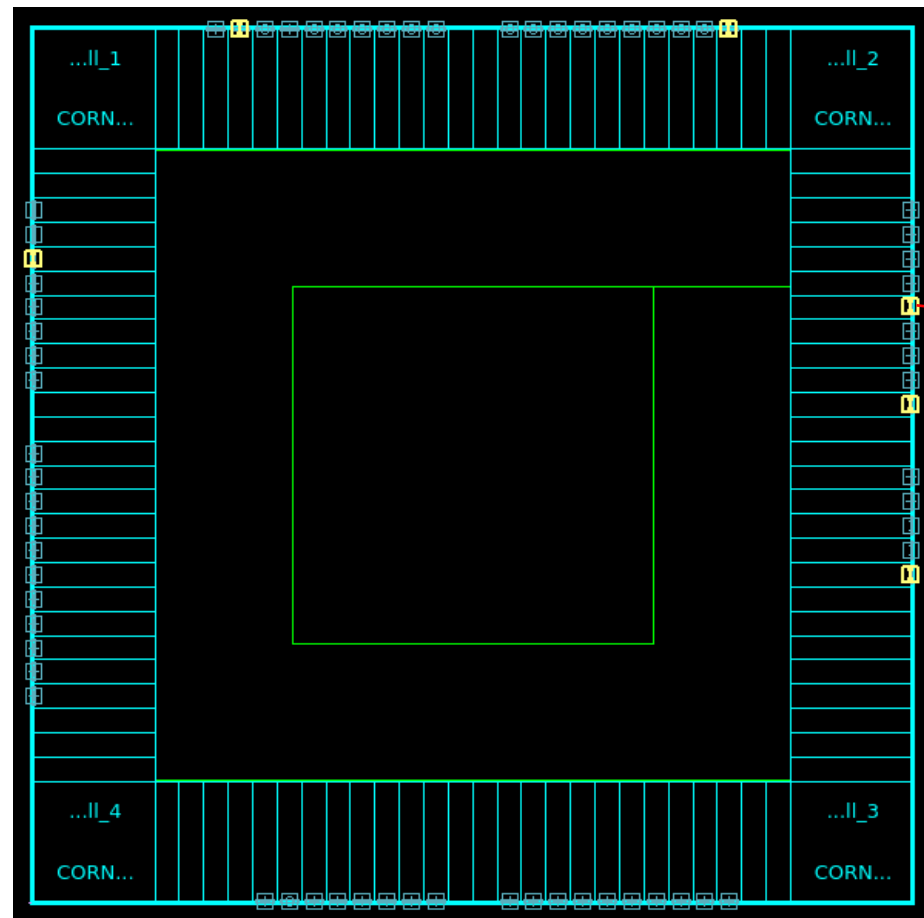
# Floorplan : Use ICC2 (IC Compiler II tool)

5. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/04_pad_place_and_terminal.tcl`

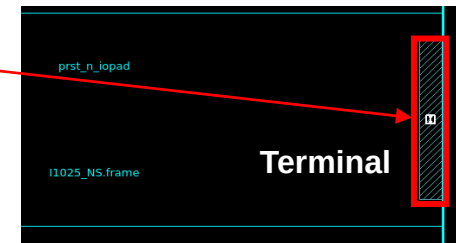
- Check after create terminals



<Before create terminal>

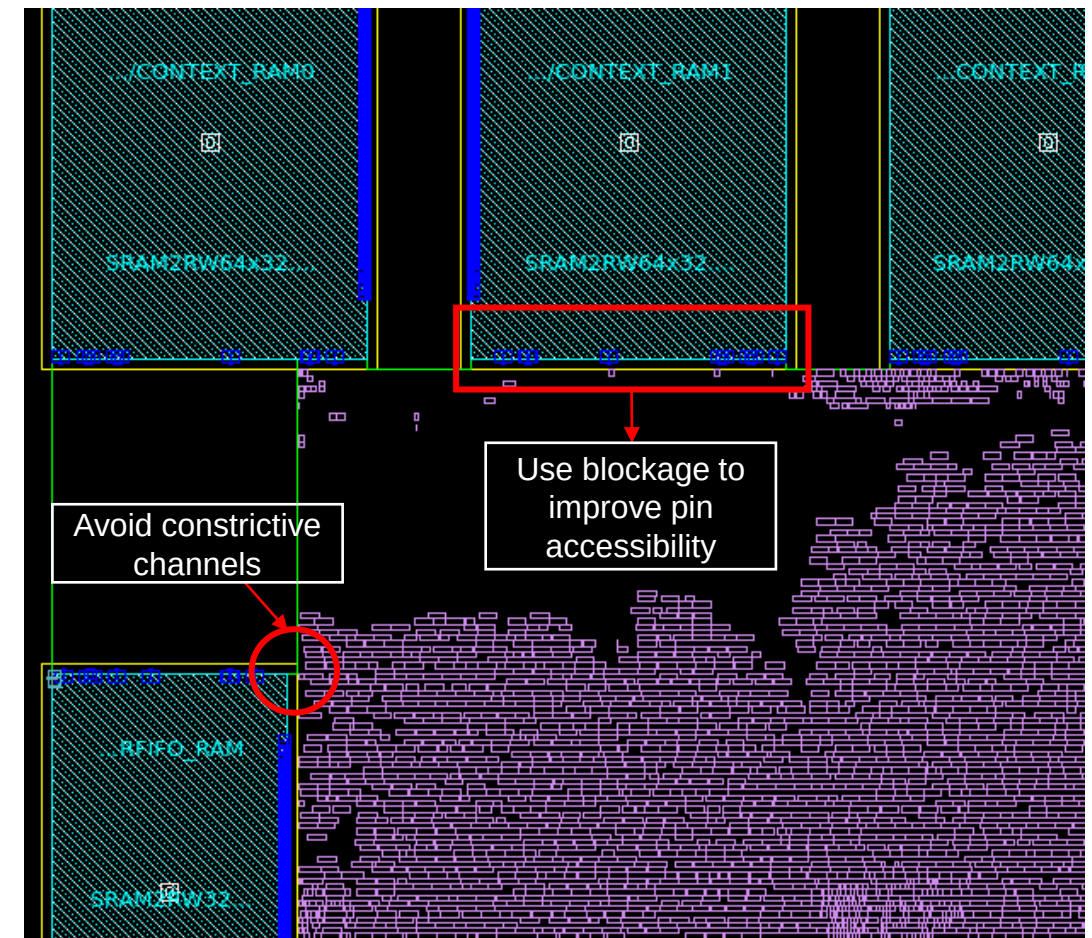
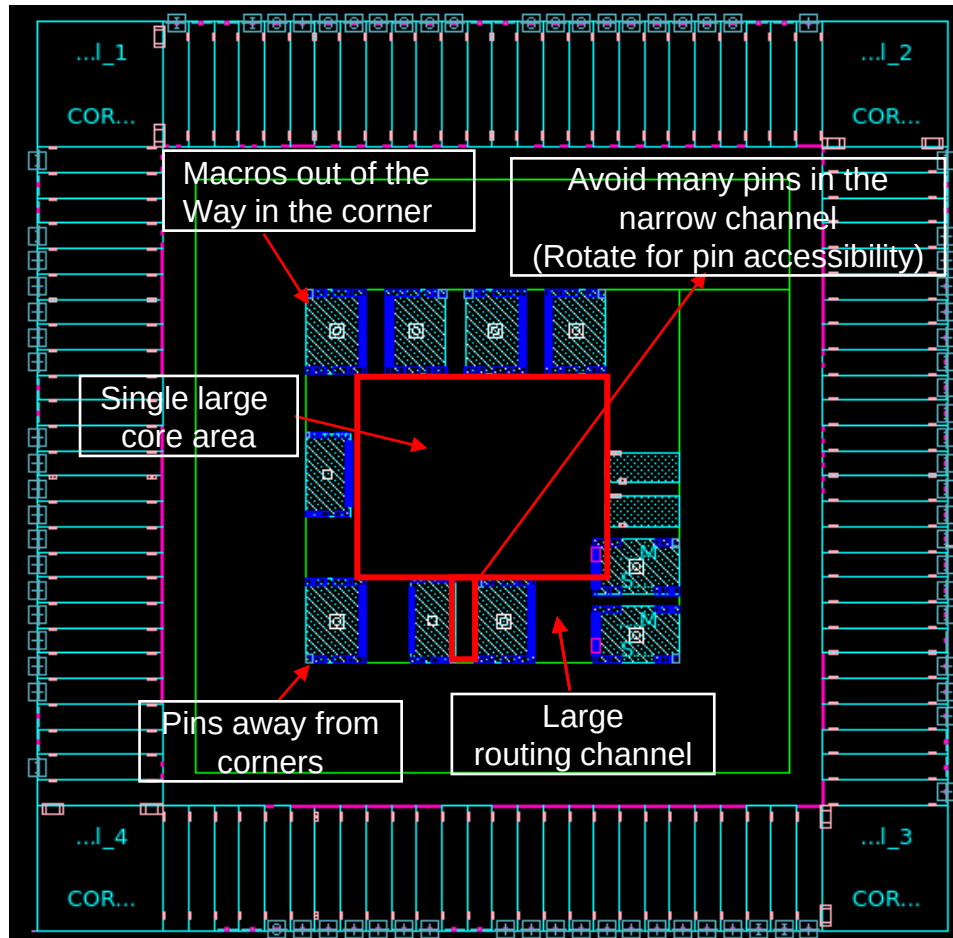


<After create terminal>



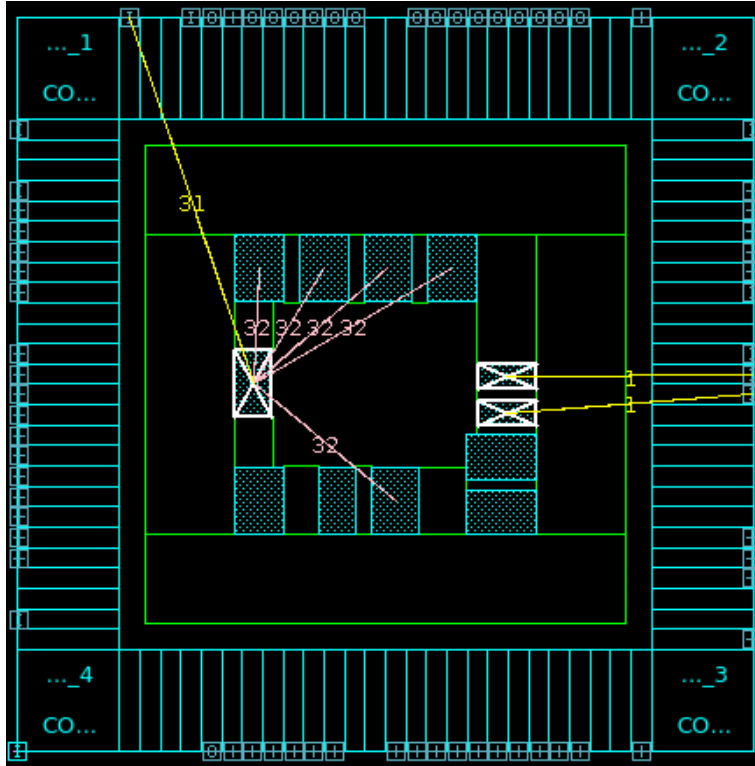
# Floorplan : Use ICC2 (IC Compiler II tool) Considerations

- Consideration of floorplan
  - Guidelines for a good floorplan

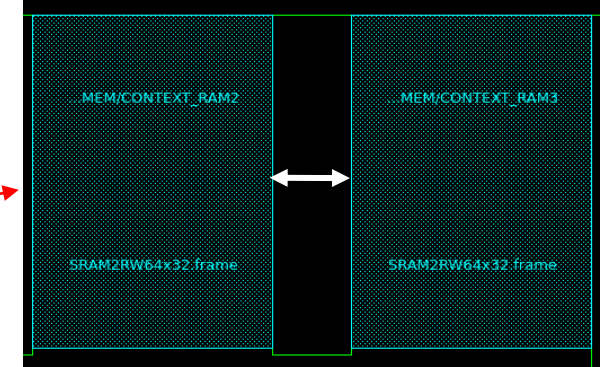
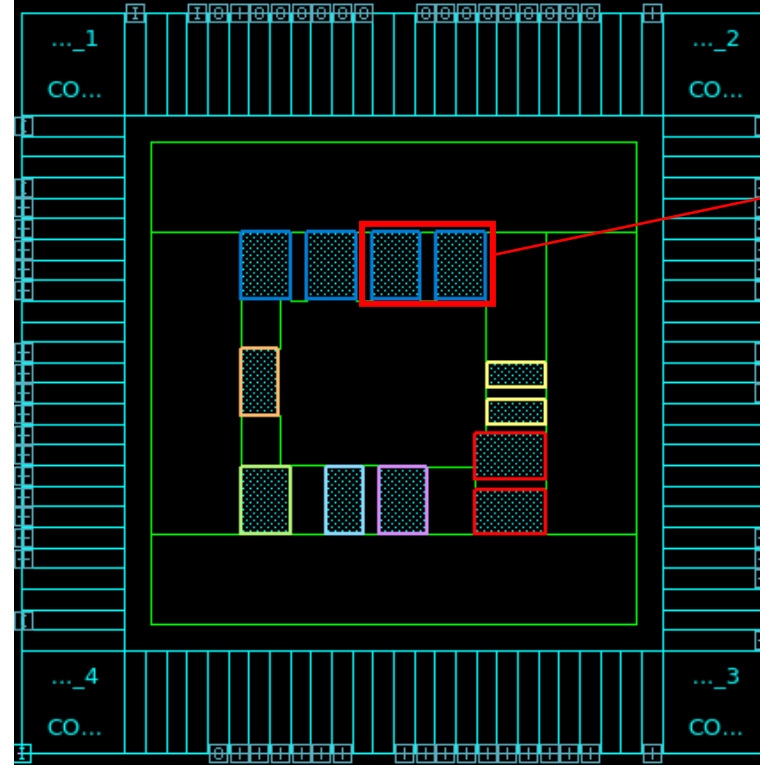


# Floorplan : Use ICC2 (IC Compiler II tool) Considerations

- Consideration of floorplan
  - Floorplan guide in ICC2



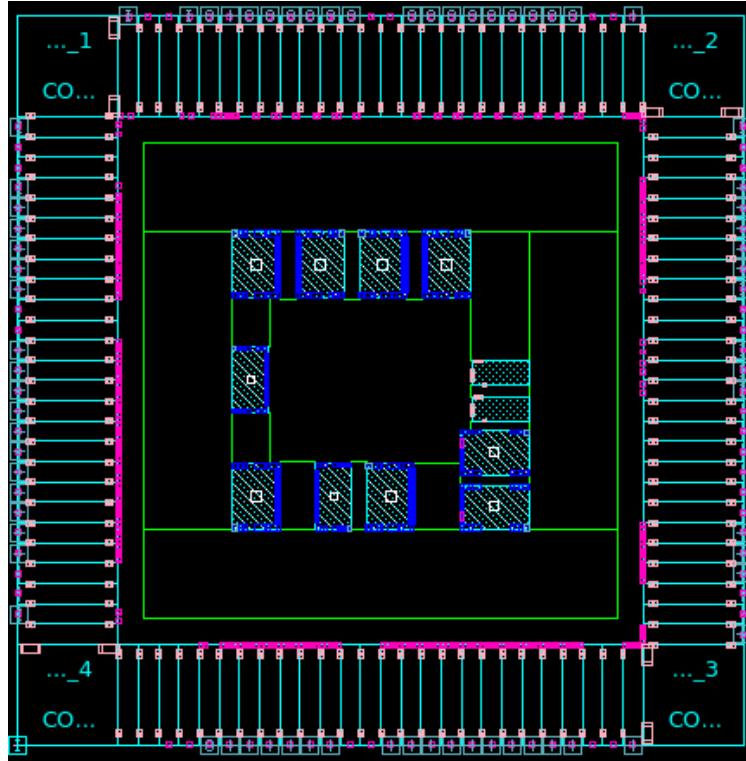
- Place macro in order of  
Macro/IP to IO connection  
Macro/IP to Macro connection
- Should not be cross-cross connection  
between Macro (check flylines)



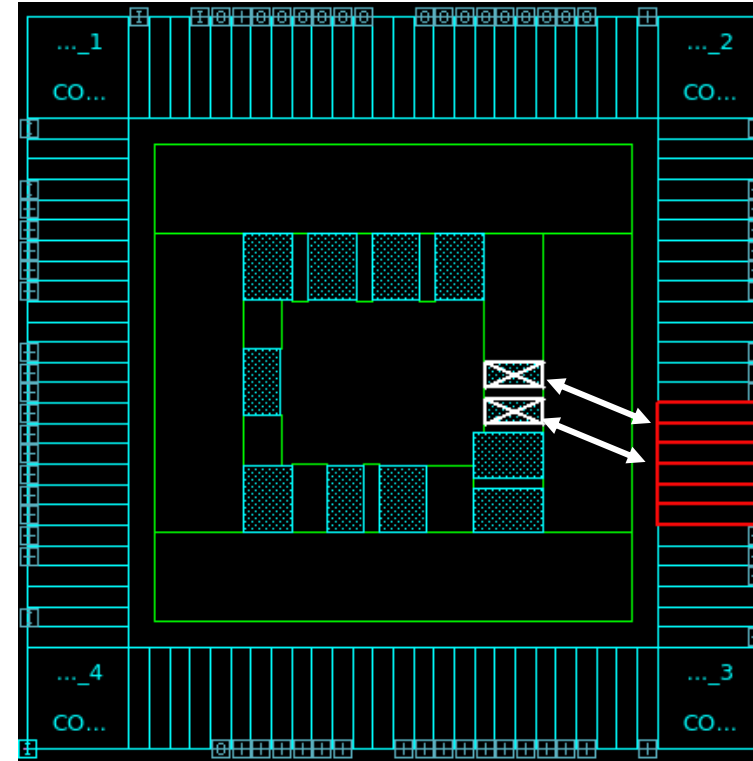
- Place Macro according to hierarchy
- Follow guidelines specific to foundry/technology related like spacing between Macro , alignment between Macro, Macro pin track alignment etc
- Should be sufficient spacing between Macro to avoid congestion
- Keep enough spacing around Macro for HALO(keepout margin) creation

# Floorplan : Use ICC2 (IC Compiler II tool) Considerations

- Consideration of floorplan
  - Floorplan guide in ICC2



- Direction of pins should be core facing for macros
- Check Macro orientation and Macro/IP should be place according to valid orientation



- Consider connections to fixed cells when placing Macros (Consider PAD and powerplan)

# Floorplan : Use ICC2 (IC Compiler II tool)

## 6. Practice manual Floorplan before sourcing 05\_place\_macro\_and\_check.tcl

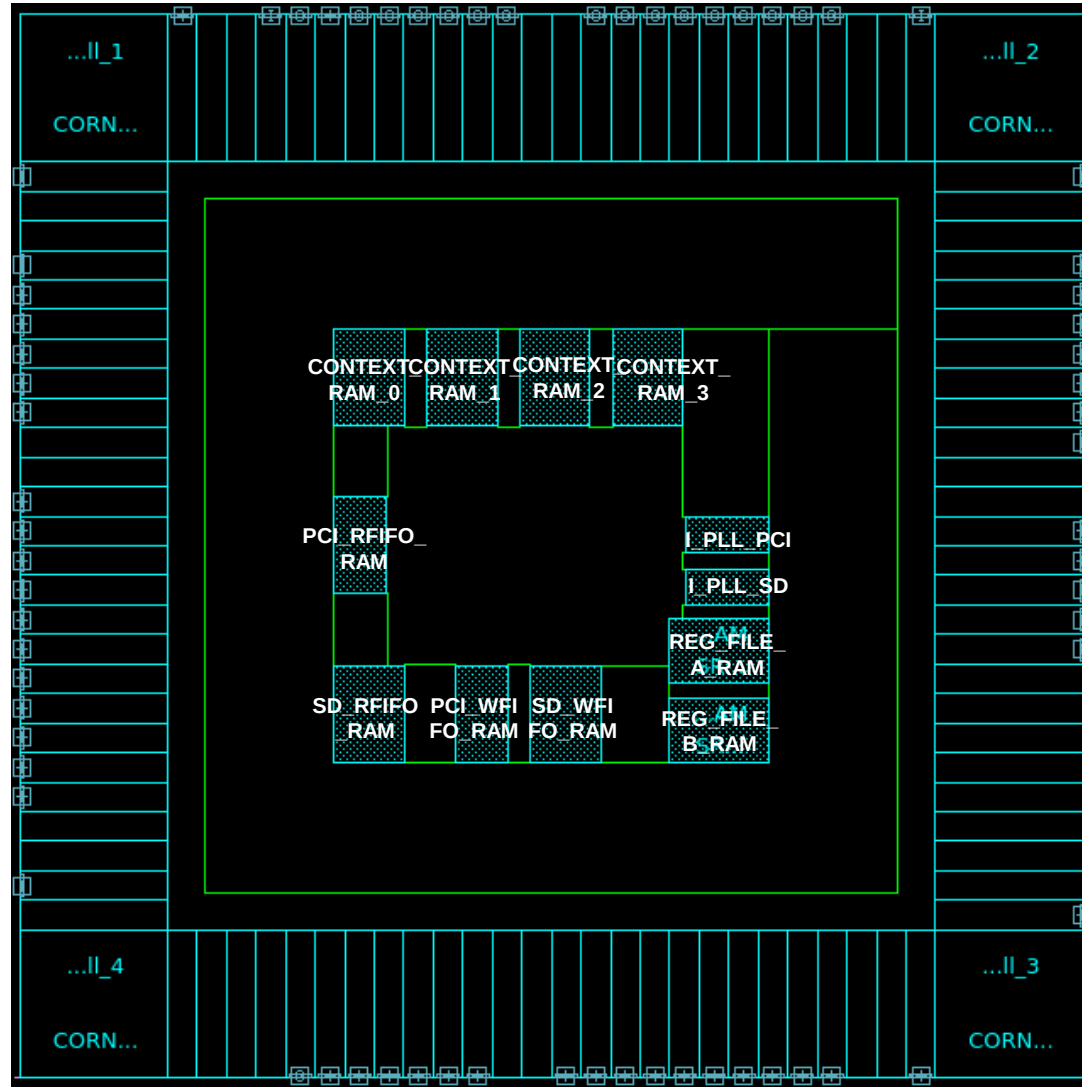
- Move Macros & PLL manually first before source script
- Floorplan Guide

| MACRO & PLL                                | Origin             | Orientation |
|--------------------------------------------|--------------------|-------------|
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0     | 520.4510 1014.1520 | MYR90       |
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_1     | 551.4510 1014.1520 | R270        |
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_2     | 772.2670 1014.1520 | MYR90       |
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3     | 803.2670 1014.1520 | R270        |
| I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM   | 496.3820 787.5740  | MYR90       |
| I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM  | 590.4510 425.8460  | MXR90       |
| I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM  | 520.4510 425.8460  | R90         |
| I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM | 786.1980 557.2520  | MYR90       |
| I_ORCA_TOP/I_RISC_CORE/REG_FILE_A_RAM      | 878.7860 620.8160  | R270        |
| I_ORCA_TOP/I_RISC_CORE/REG_FILE_B_RAM      | 878.7860 425.8460  | MXR90       |
| I_CLOCK_GEN/I_PLL_SD                       | 1014.4580 640.8160 | MY          |
| I_CLOCK_GEN/I_PLL_PCI                      | 1014.4580 709.9240 | MY          |

# Floorplan : Use ICC2 (IC Compiler II tool)

## 6. Practice manual Floorplan before sourcing 05\_place\_macro\_and\_check.tcl

- Floorplan guide in ICC2



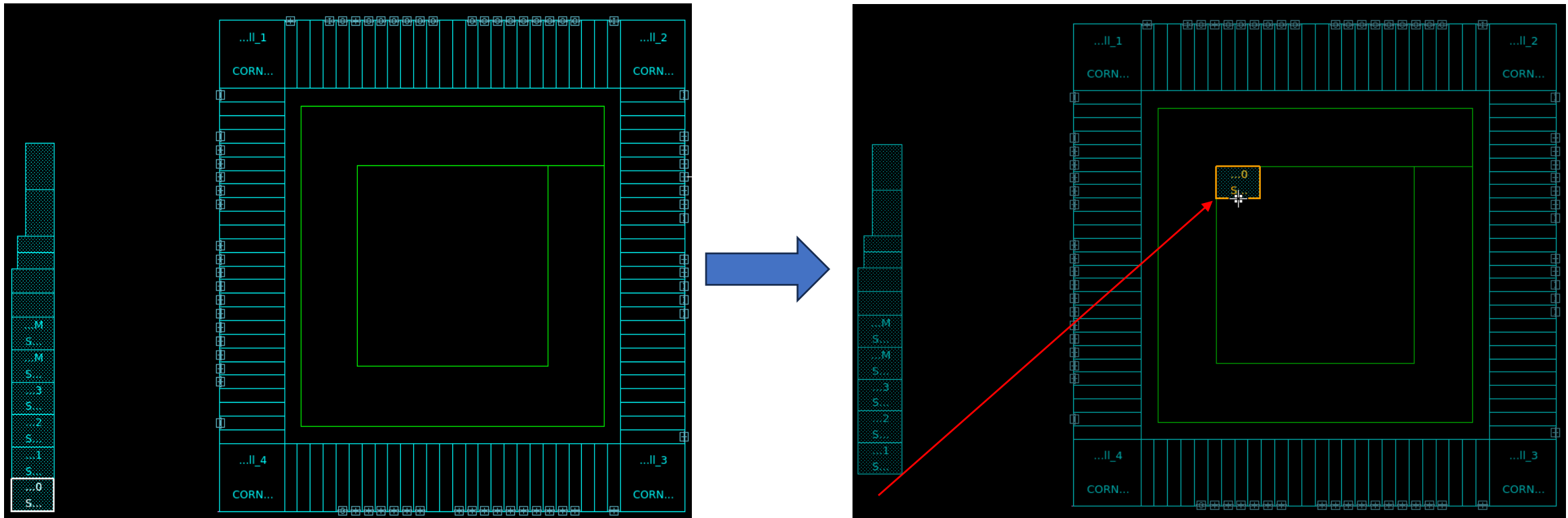
# Floorplan : Use ICC2 (IC Compiler II tool)

## 6. Practice manual Floorplan before sourcing 05\_place\_macro\_and\_check.tcl

- Move Macros

1. Find the Cells to be moved with get\_cells command and move it to the floorplan guide position

```
icc2_shell> change_selection [get_cells I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0]
```

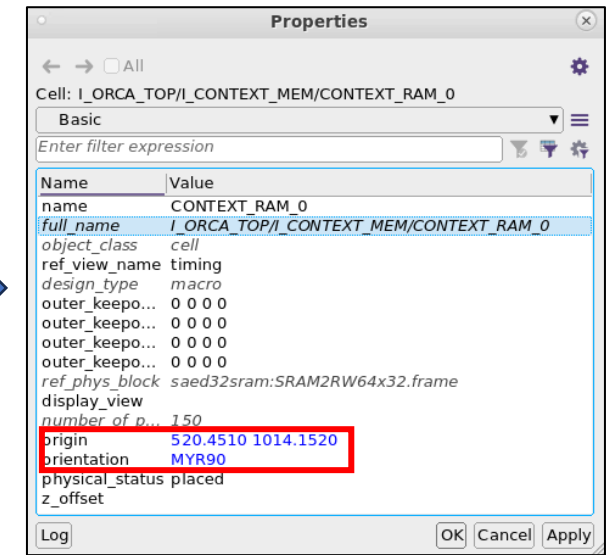
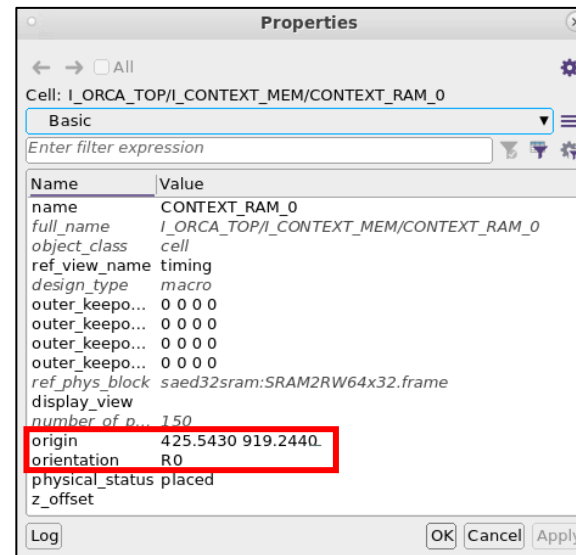
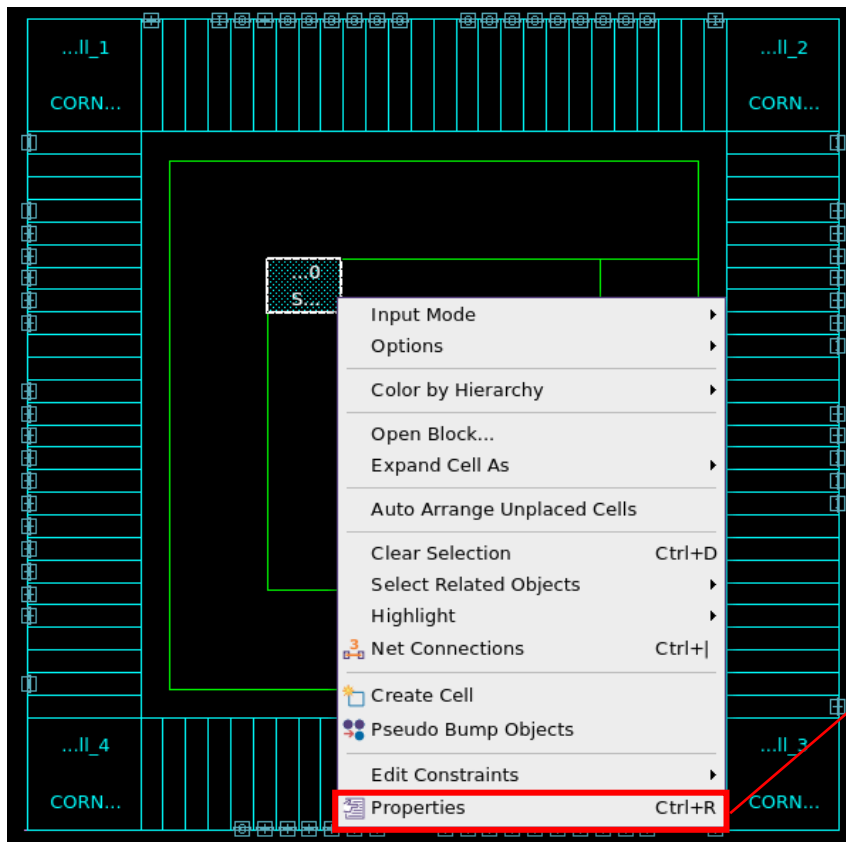


# Floorplan : Use ICC2 (IC Compiler II tool)

## 6. Practice manual Floorplan before sourcing 05\_place\_macro\_and\_check.tcl

- Check & change Macro's Origin & Orientation

2. Open the properties window of the moved cell, check the Origin & Orientation of the cell, and modify it according to the Floorplan guide

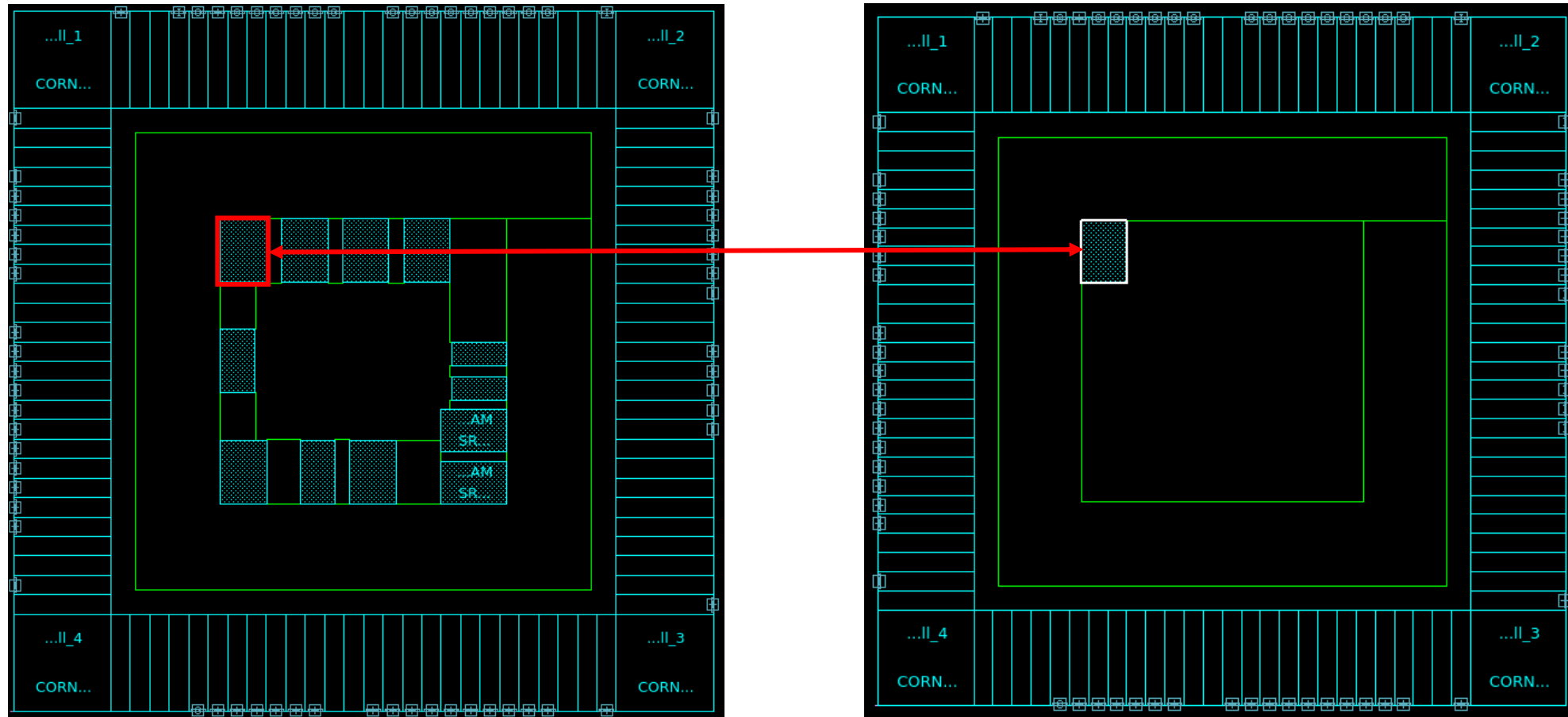


# Floorplan : Use ICC2 (IC Compiler II tool)

## 6. Practice manual Floorplan before sourcing 05\_place\_macro\_and\_check.tcl

- Check Macro & PLL placement

3. If the placement is finished, compare it to the Floorplan guide, and then place the other cells in the same way



# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/05_place_macro_and_check.tcl`

- Using `fp.tcl` to Macro place

```
icc2_shell> source ./FP_data/ORCA_floorplan_data/fp.tcl
{I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/KEEPOUT_hard_OUTER_9167}
icc2_shell>
icc2_shell> #fix SRAM
icc2_shell> set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]
{I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3 I_ORCA_TOP/I_RISC_CORE/REG_FILE_B_RAM I_ORCA_TOP/I_RISC_CORE/REG_FILE_A_RAM I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM}
icc2_shell> change_selection $all_macros
icc2_shell> set_fixed_object [get_selection]
1
```

## ❖ What is `fp.tcl`?

`fp.tcl` is file which floorplan information written to a files, using `write floorplan` command in ICC2

Open & Review > `$WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/FP_data/ORCA_floorplan_data`

The diagram illustrates the structure of the `fp.tcl` file. It shows two macro definitions. The first macro, `I_CLOCK_GEN/I_PLL_PCI`, is highlighted with a light blue background. Annotations point to specific parts of the code: one points to the macro name in the `set cellInst` command, labeled "Macro information (full name)", and another points to the `set_attribute` commands, labeled "Information about Macro (orientation, origin, status, keepout margin)". The second macro, `I_CLOCK_GEN/I_PLL_SD`, is shown below the first one.

```
#####
# Cells
#####

set cellInst [get_cells { I_CLOCK_GEN/I_PLL_PCI }]
set_attribute -quiet -objects $cellInst -name orientation -value MY
set_attribute -quiet -objects $cellInst -name origin -value { 1014.4580 \
709.9240 }
set_attribute -quiet -objects $cellInst -name status -value placed
create_keepout_margin -type hard -outer { 1.0000 1.0000 1.0000 1.0000 } { \
I_CLOCK_GEN/I_PLL_PCI }

set cellInst [get_cells { I_CLOCK_GEN/I_PLL_SD }]
set_attribute -quiet -objects $cellInst -name orientation -value MY
set_attribute -quiet -objects $cellInst -name origin -value { 1014.4580 \
640.8160 }
set_attribute -quiet -objects $cellInst -name status -value placed
create_keepout_margin -type hard -outer { 1.0000 1.0000 1.0000 1.0000 } { \
I_CLOCK_GEN/I_PLL_SD }
```

# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/05_place_macro_and_check.tcl`

- You can use this command to write floorplan

```
icc2_shell> write_floorplan -objects [get_selection] -output ./FP_data/FP_TEST
```

Select object  
(using write floorplan)

Output path and  
directory name

- Source `fp.tcl` & fixed all Macros

```
icc2_shell> source ./FP_data/ORCA_floorplan_data/fp.tcl
{I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/KEEPOUT_hard_OUTER_9167}
icc2_shell> set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]
{I_CLOCK_GEN/I_PLL_PCI_I_CLOCK_GEN/I_PLL_SD_I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0_I_ORCA_TOP/I_CONTEXT_MEM/CO
NTEXT_RAM_1_I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_2_I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3_I_ORCA_TOP/I_RISC_CO
RE/REG_FILE_B_RAM_I_ORCA_TOP/I_RISC_CORE/REG_FILE_A_RAM_I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM_I_ORCA_TOP/I
PCI_WRITE_FIFO/PCI_WFIFO_RAM_I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM_I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO
RAM}
icc2_shell> change_selection $all_macros
icc2_shell> set_fixed_object [get_selection]
1
icc2_shell> change_selection
```

- How to fix & unrestricted object manual



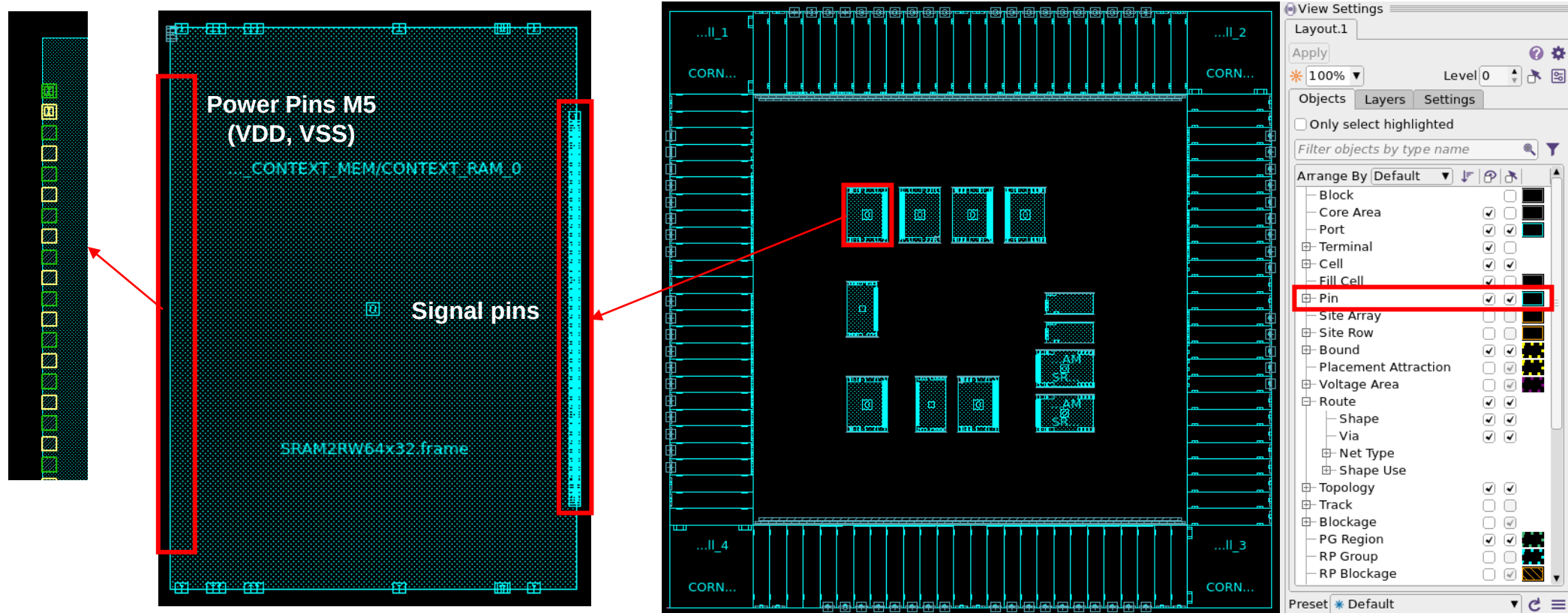
Select object and click this icon in toolbar  
(physical status = fixed)

Select object and click this icon in toolbar  
(physical status = unrestricted)

# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/01_read_design/script/05_place_macro_and_check.tcl`

- Check Macro & PLL pins direction & orientation
- Check Pins direction



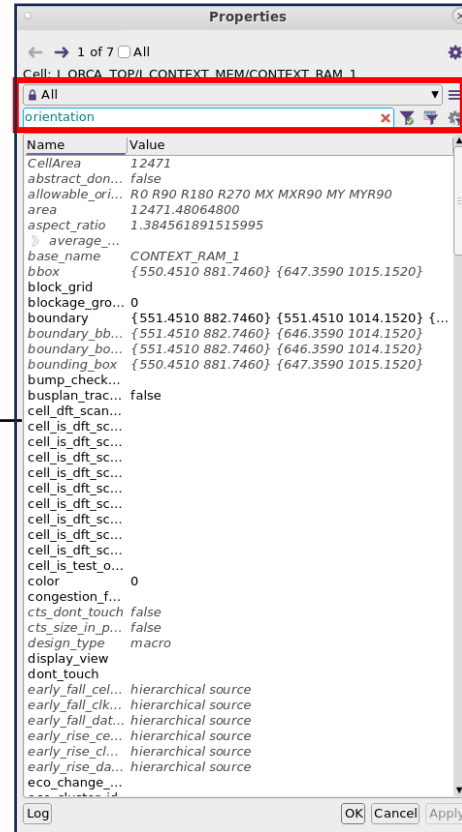
- ❖ The pins of Macro and PLL is M5 Layer
- ❖ M5 is a horizontal Layer, it should be placed in the horizontal direction considering the pin connection in the Powerplan step
- ❖ You can check Pin icon in View settings and check pins direction

# Floorplan : Use ICC2 (IC Compiler II tool)

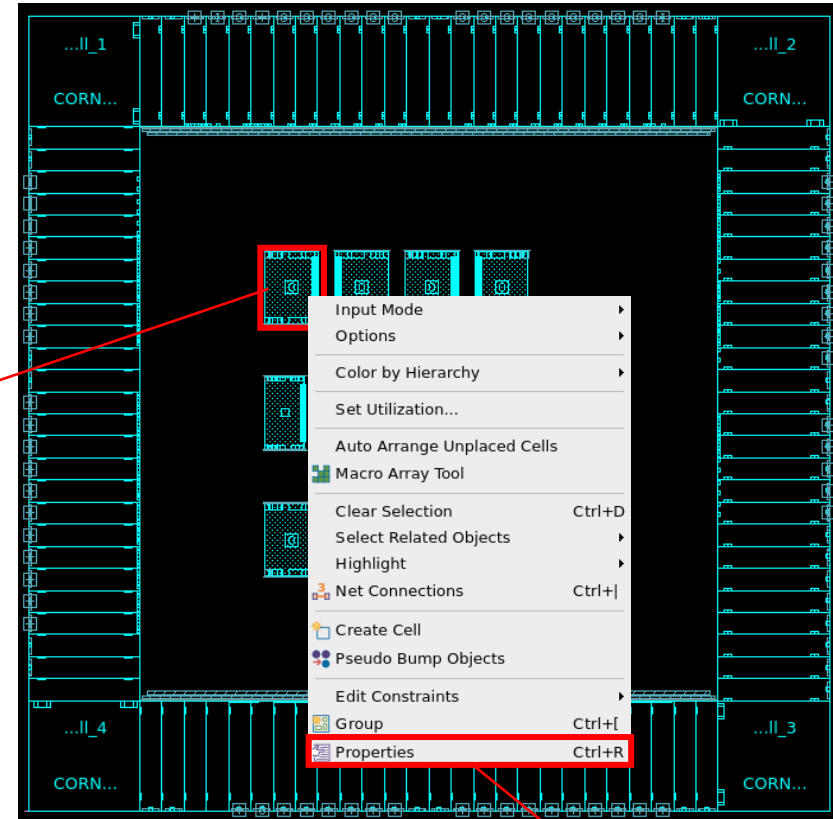
- Check orientation



Check orientation value



Check All & search orientation

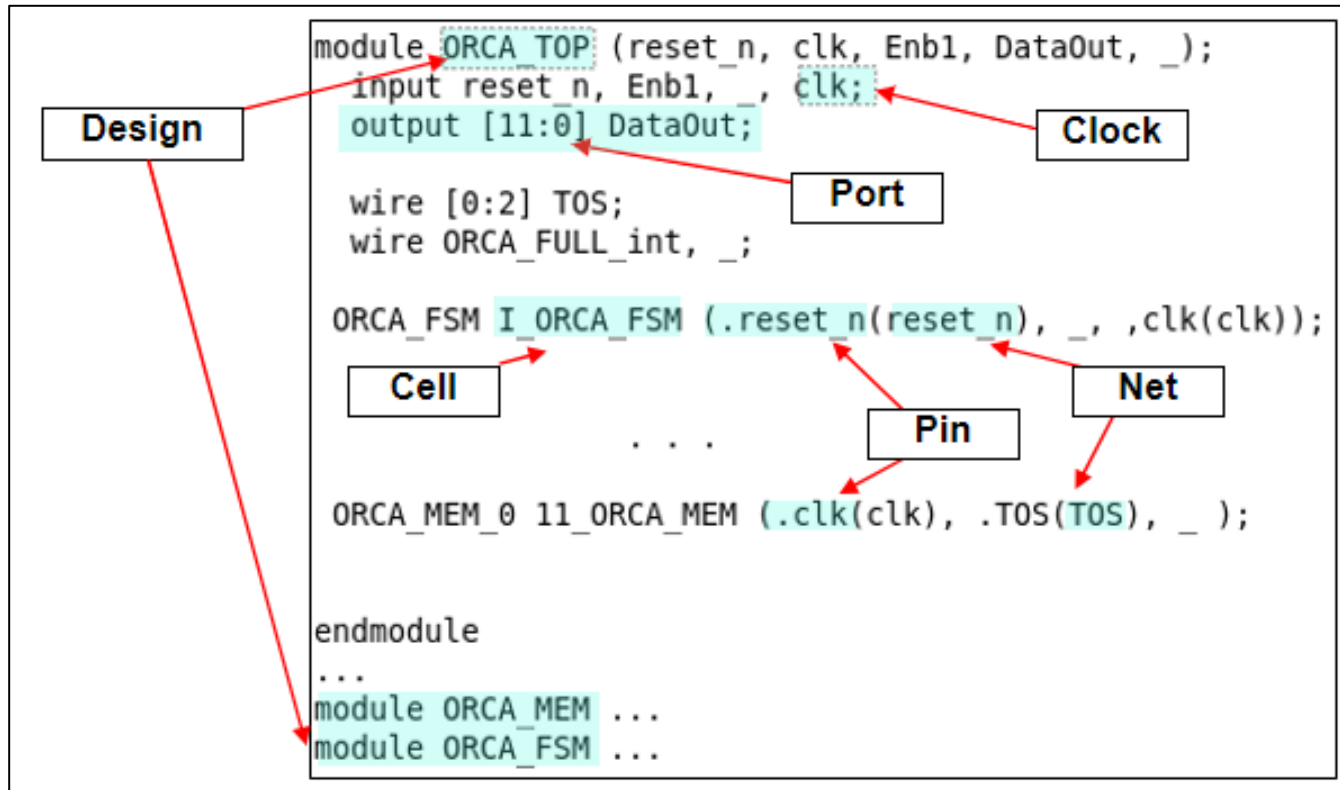


- ❖ Due to Macro and PLL's pin layer direction (M5), y-axis flip is possible when floorplan
- ❖ Rotating Macro and PLL is not recommended in this design
- ❖ Check orientation is applied to MXR90 or MYR90

- ❖ After selecting an object, you can check the property of the object by pressing the Ctrl + R key (short cut)
- ❖ You can also review it by clicking the mouse right key and then clicking properties

# Floorplan : Design Objects

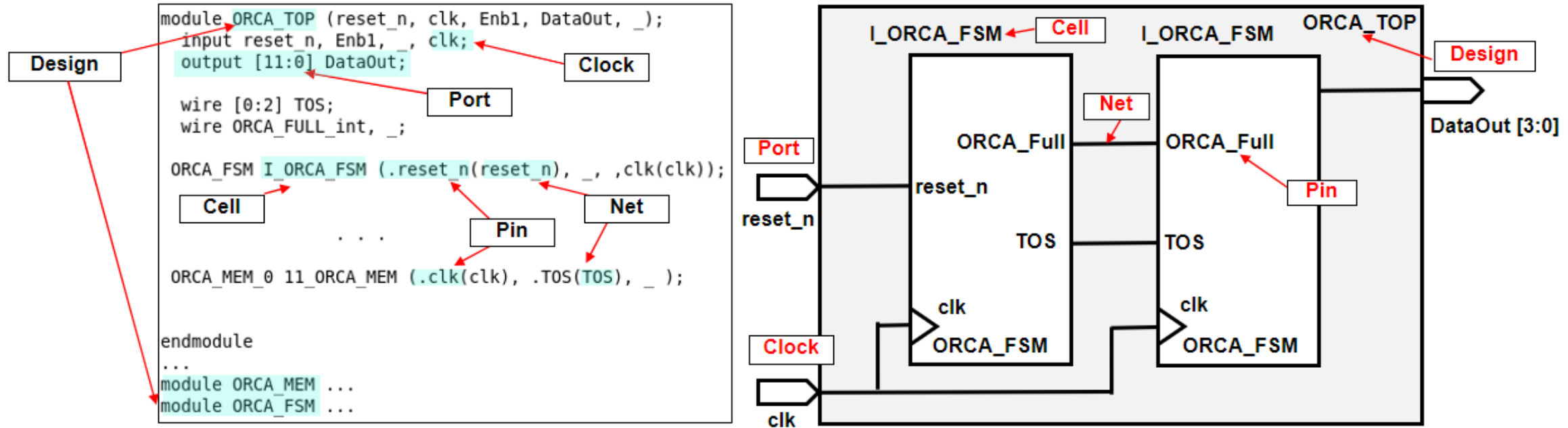
- Verilog Perspective



- Synthesis is the transformation of an idea into a manufacturable device to carry out an intended function

# Floorplan : Design Objects

- Verilog Perspective & Schematic Perspective



- Designs in DC memory: { ORCA\_TOP ORCA\_FSM ORCA\_MEM }
- Cells in ORCA\_TOP: { I\_ORCA\_FSM T1\_ORCA\_MEM }
- Pins in ORCA\_TOP: { I\_ORCA\_FSM/reset\_n I\_ORCA\_FSM/clk ... I1\_ORCA\_MEM/TOS }

\* Clock is a user-defined object within Design Compiler memory

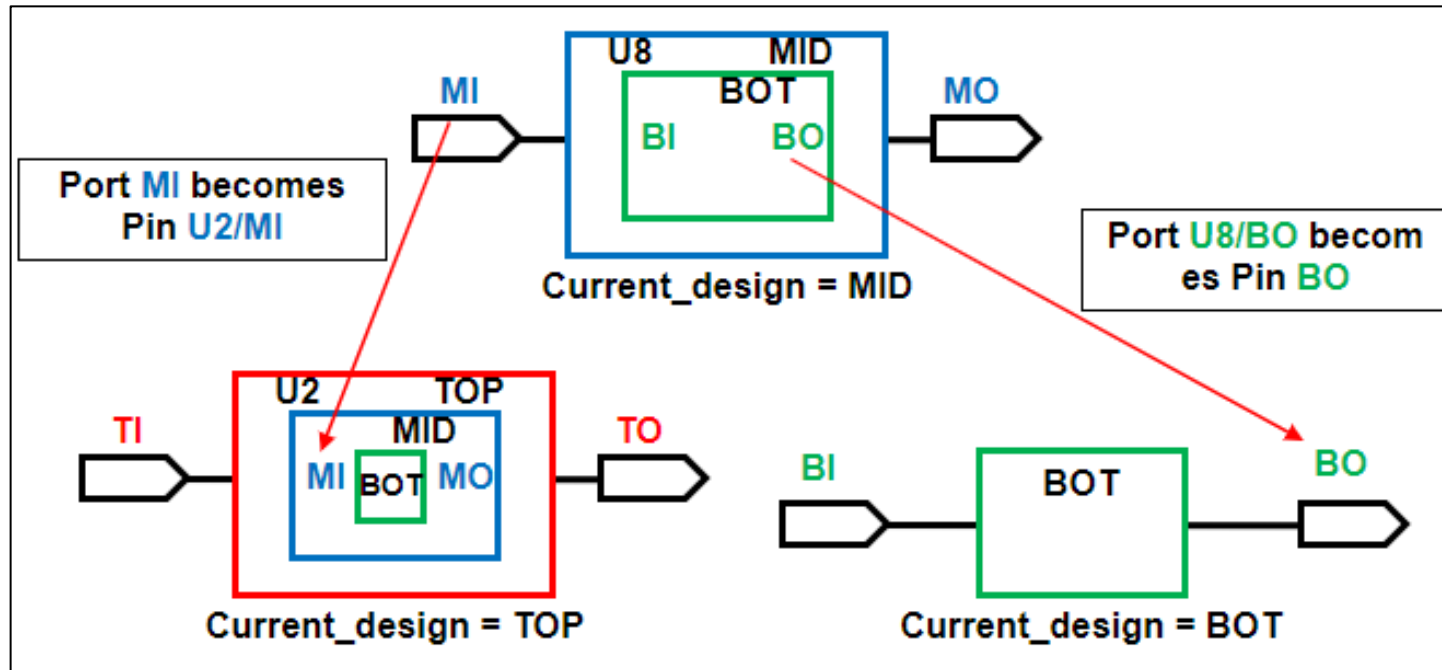
\* Pins are always associated with a specific instance or cell: cell\_name/pin\_name

# Floorplan : Design Objects

- Ports Versus Pins

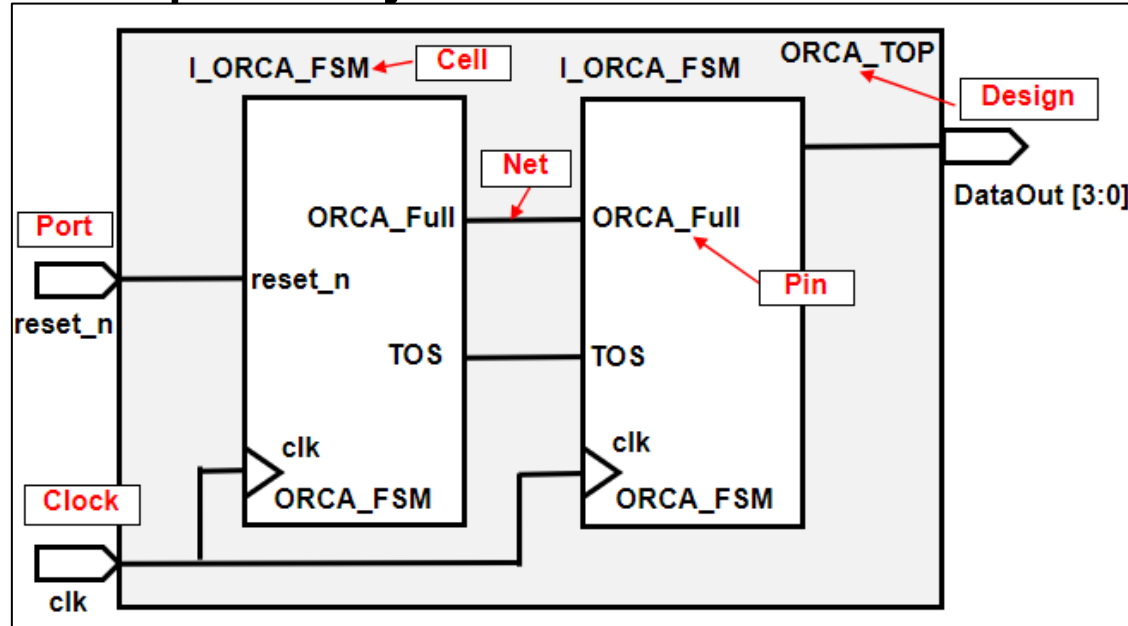
- 'Ports' are the inputs and outputs of the current design  
They become 'Pins' if the current design moves up to a parent design
- 'Pins' are the inputs and outputs of any cell that is instantiated in the current design  
They become ports if that instantiated design is made the current design

ex)



# Floorplan : Design Objects

- Multiple Objects with the Same Name



❖ set\_load 5 DataOut[1]

## Questions

Does "DataOut[1]" refer to the port or the net object?

Does it matter onto which object DC places the load?

- The DataOut[1] refers to BOTH the port and the net called DataOut[1] <- it is ambiguous
  - If the load is applied to the NET, the user-specified load over-rides any internally-calculated net capacitance
  - If the load is applied to the PORT, the load is ADDED to any internally calculated net capacitance, so the total load on the register will be higher, and the delay will be larger
- In the example above DC decides which of the two objects it will apply the load to – it turns out that for this command DC chooses a port object
- What if that is NOT what you had intended? -> To avoid ambiguity issues it is highly recommended to always specify the object type

# Floorplan : Use ICC2 (IC Compiler II tool)

- **ICC2 tcl command**
  - **ICC2 provides a full set of object classes and associated tcl commands to create, remove, query, and modify all primary design data**
    - Logical objects, such as ports, cells, and nets
    - Timing data, such as, modes and corners
    - Physical objects, such as shapes, vias, tracks, and blockages
    - Reference library objects, such as libraries, library cells, and lib pins
  - **Most design objects have basic commands for object creation, removal, and querying**
    - To create an object: `create_<object_class>`
    - To remove an object: `remove_<object_class>`
    - To create a collection of objects: `get_<object_class>`
    - To move a collection of objects: `move_<object_class>`
    - To mirror a collection of objects: `flip_<object_class>`
    - To rotate a collection of objects: `rotate_<object_class>`

# Floorplan : Use ICC2 (IC Compiler II tool)

- **ICC2 Query command**

- **get\_nets command**

Creates a collection of nets from the netlist. You can assign these nets to a variable or pass them to another command

```
icc2_shell> get_nets *_clk  
{buf_sdram_clk buf_sys_2x_clk buf_sys_clk net_sdram_clk net_sys_clk s_sdram_clk s_sys_2x_clk s_sys_clk sdr_clk sys_clk}
```

- **get\_cells (Querying Objects Associated with Objects)**

Creates a collection of cells from the current design relative to the current instance. You can assign these cells to a variable or pass them into another command

```
icc2_shell> get_cells -of_objects [get_nets net_sdram_clk]  
{sdr_clk iopad I CLOCK GEN}
```

❖ You can find objects associated with, or connected to, specific objects by using the `-of_objects` option

(for example, you can find the cells associated with other cells, or with pins, nets bounds, voltage areas, voltage area shapes, etc :)

- **get\_cells (Querying Objects/Hierarchy)**

Object names must match the logical hierarchy name attribute

```
icc2_shell> get_cells I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD*  
{I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM}
```

**Hierarchy**

## Floorplan : Use ICC2 (IC Compiler II tool)

- **ICC2 Query command**

- **get\_flat\_cells** (Querying Objects/Hierarchy)

Creates a collection of leaf cells that match certain criteria in the current design

(leaf cell is a cell that does not contain any instances of other cells)

```
icc2 shell> get flat_cells *I PCI_READ_FIFO*U*
{I_ORCA_TOP/I_PCI_READ_FIFO/U3 I_ORCA_TOP/I_PCI_READ_FIFO/U4 I_ORCA_TOP/I_PCI_READ_FIFO/U5 I_ORCA_TOP/I_PCI_READ_FIFO/U6 I_ORCA_TOP/I_PCI_READ_FIFO/U7 I_ORCA_TOP/I_PCI_READ_FIFO/U8 I_ORCA_TOP/I_PCI_READ_FIFO/U9 I_ORCA_TOP/I_PCI_READ_FIFO/U10 I_ORCA_TOP/I_PCI_READ_FIFO/U11 I_ORCA_TOP/I_PCI_READ_FIFO/U12 I_ORCA_TOP/I_PCI_READ_FIFO/U13 I_ORCA_TOP/I_PCI_READ_FIFO/U14 I_ORCA_TOP/I_PCI_READ_FIFO/U15 I_ORCA_TOP/I_PCI_READ_FIFO/U16 I
```

- ❖ The `*flat*` commands use the `full_name` attribute of the object for searching
- ❖ Also available: `get_flat_nets`, `get_flat_pins`
- ❖ You can match using regular expressions with the `-regex` option

- **get\_cells –hierarchical** (returns leaf and hierarchy cells with the name attribute matching the pattern, at all levels of hierarchy)

- To return only leaf cells

```
icc2_shell> get cells -hierarchical *mux*
{I_ORCA_TOP/I_PARSER/pci_w_mux_select_reg_0 I_ORCA_TOP/I_PARSER/pci_w_mux_select_reg_1 I_ORCA_TOP/I_PARSER/sd_w_mux_select_reg_0 I_ORCA_TOP/I_PARSER/sd_w_mux_select_reg_1 I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK I_ORCA_TOP/I_SDRAM_IF/sd_mux_CKn I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0 I_ORCA_T
```

- To return cells that are at a particular hierarchy

```
icc2_shell> get_cells -hierarchical *mux* -filter "!is_hierarchical"
```

- To return cells that are at or below a particular hierarchy

```

{I_ORCA_TOP/I_SDRAM_IF/sd_mux CK I_ORCA_TOP/I_SDRAM_IF/sd_mux CKn I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_0 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_1 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_10 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_11 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_12 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_13 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_14 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_15 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_2 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_3 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_4 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_5 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_6 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_7 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_8 I_ORCA_TOP/I_SDRAM_IF/sd_mux dq_out_9}

```

# Floorplan : Use ICC2 (IC Compiler II tool)

- ICC2 Query command

- `get_cells` (Querying objects by Location)

- `-at` option (Returns objects at the specified point)

```
icc2_shell> get_cells -at {803.2670 1014.1520}  
{I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3}
```

- `-within` option (Returns objects that are strictly inside the queried-region)

Does not include objects that are touching the queried-region from the inside

```
icc2_shell> get_cells -within {{0 0} {803.2670 1014.1520}}  
{I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I  
_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM}
```

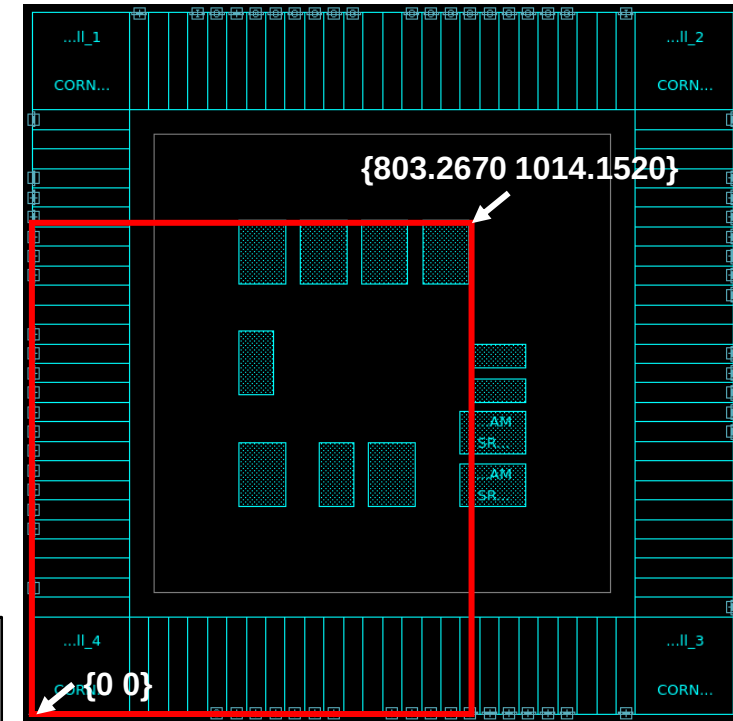
- `-touching` option (Returns objects that are strictly inside the queried-region,  
including objects that are touching the queried-region from the inside)

```
icc2_shell> get_cells -touching {{0 0} {803.2670 1014.1520}}  
{powersave_iopad pad_iopad_13 pad_iopad_12 pad_iopad_11 pad_iopad_10 pad_iopad_9 pad_iopad_8 pad_iopad_7 pad_iopad_6 pad_iopad_5 pad_iopad_4 pad_iopad_3 pad_iopad_2 pad_iopad_1 pad_iopad_0 sdram_LD_iopad sdram_DQ_iopad_15 sdram_DQ_iopad_14 sdram_DQ_iopad_13 sdram_DQ_iopad_12 sdram_DQ_iopad_11 sdram_DQ_iopad_10 sdram_DQ_iopad_9 sdram_DQ_iopad_8 U IOPOWER/IOVDD_L1 U IOPOWER/IOVDD_B1 U IOPOWER/IOVSS_L1 U IOPOWER/IOVSS_B1 U IOPOWER/VDD_L2 U IOPOWER/VDD_B1 U IOPOWER/VSS_L2 U IOPOWER/VSS_B1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3 I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM __added_corner_cell_4 __added_filler_instance_0 __added_filler_instance_1 __added_filler_instance_2}
```

- `-intersect` option (Returns objects that are overlapping with queried-region as

Well as objects that are touching the queried-region from the inside or the outside)

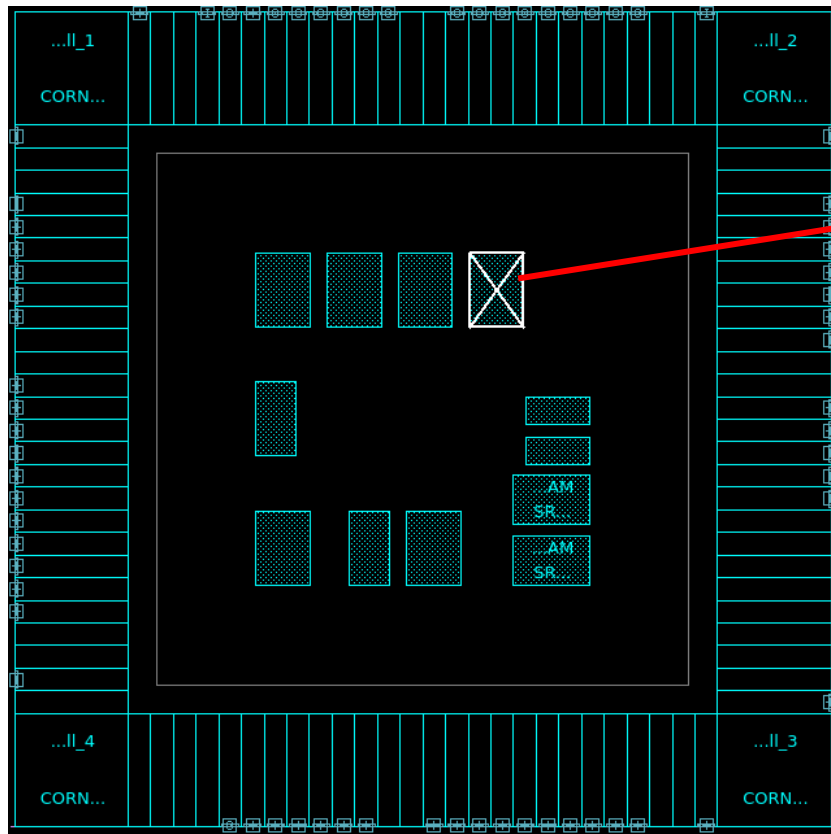
```
icc2_shell> get_cells -intersect {{0 0} {803.2670 1014.1520}}  
{powersave_iopad pad_iopad_14 pad_iopad_13 pad_iopad_12 pad_iopad_11 pad_iopad_10 pad_iopad_9 pad_iopad_8 pad_iopad_7 pad_iopad_6 pad_iopad_5 pad_iopad_4 pad_iopad_3 pad_iopad_2 pad_iopad_1 pad_iopad_0 sdram_LD_iopad sdram_DQ_iopad_15 sdram_DQ_iopad_14 sdram_DQ_iopad_13 sdram_DQ_iopad_12 sdram_DQ_iopad_11 sdram_DQ_iopad_10 sdram_DQ_iopad_9 sdram_DQ_iopad_8 sdram_DQ_iopad_7 U IOPOWER/IOVDD_L1 U IOPOWER/IOVDD_B1 U IOPOWER/IOVSS_L1 U IOPOWER/IOVSS_B1 U IOPOWER/VDD_L2 U IOPOWER/VDD_B1 U IOPOWER/VSS_L2 U IOPOWER/VSS_B1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3 __added_corner_cell_4 __added_filler_instance_0 __added_filler_instance_1 __added_filler_instance_2}
```



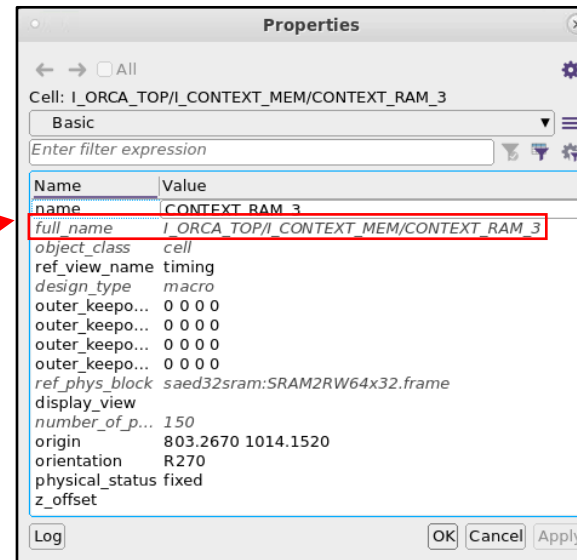
# Floorplan : Use ICC2 (IC Compiler II tool)

- **Change\_selection command**
  - Changes the selection in the GUI, taking a collection of objects and changing the selection according to the type of change specified

```
(icc2_shell> change_selection [get_cells I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3]inds)
```



Object



- ❖ If you used the change\_selection command to select an object, open the properties window to verify that the object you want to select is correct

# Floorplan : Use ICC2 (IC Compiler II tool)

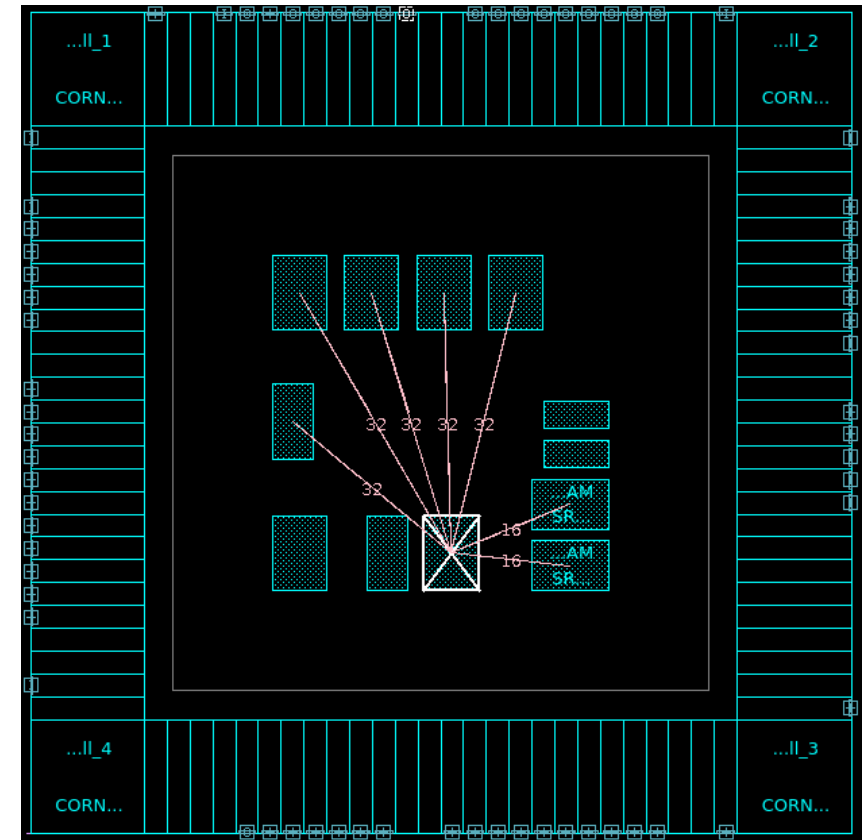
- **Data Flow Flylines (DFF)**

The "Data Flow Flylines Tool" (DFF) supports exploration of the netlist connectivity of a design by using netlist tracing. DFF recognizes the following netlist elements: Module Boundaries, Blocks, Macros and Ports and can display connectivity between them in various configurations

- **Display modes**

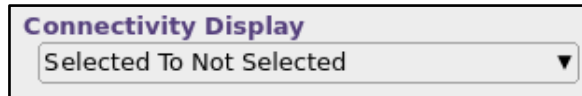
Aggregated Flylines (default)

- ❖ Aggregated flylines are displayed when the Aggregate by block option is enabled (the default). In this mode, the source and destination pin of the generated flylines are aggregated by their parent macro cell, block, cell, or module boundary. These aggregated flylines are displayed as a line
- ❖ The tool also displays a count of the number of connections between the connected object pair. The types of objects that are included as source or destination objects is determined by the options you select in the Include section



# Floorplan : Use ICC2 (IC Compiler II tool)

- **Data Flow Flylines (DFF)**
- **Connectivity display**



Use this option to control which objects have connections displayed:

- **All**  
Display all paths found by the tracer
- **Selected To All**  
Display paths from selected objects to all connected objects
- **Selected To Selected**  
Display paths from selected objects to connected objects if they are selected
- **Selected To Not Selected**  
Display paths from selected objects to connected objects if they are not selected

# Floorplan : Use ICC2 (IC Compiler II tool)

- **Data Flow Flylines (DFF)**
  - **Include**

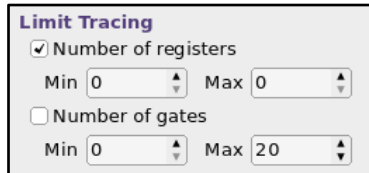
| Include                                    |                                            |
|--------------------------------------------|--------------------------------------------|
| <input type="checkbox"/> Module Boundaries | <input checked="" type="checkbox"/> Blocks |
| <input checked="" type="checkbox"/> Macros | <input checked="" type="checkbox"/> Ports  |

The options in the Include section determine what types of connections are shown from the selected objects

- **Module Boundaries**  
Show connections to Module Boundaries (if present)
- **Blocks**  
Shows connections to blocks (this option selected by default)
- **Macros**  
Show connections to macros
- **Ports**  
Show connections to ports.

# Floorplan : Use ICC2 (IC Compiler II tool)

- **Data Flow Flylines (DFF)**
- **Limit Tracing**

A screenshot of the 'Limit Tracing' dialog box in the ICC2 tool. The dialog has a title bar 'Limit Tracing'. It contains two sections. The first section is 'Number of registers', which is checked with a checkbox. Below it are two spinners: 'Min' set to 0 and 'Max' set to 0. The second section is 'Number of gates', which is unchecked. Below it are two spinners: 'Min' set to 0 and 'Max' set to 20.

This option specifies the minimum and maximum number of registers between two objects

- **Number of registers**

This option specifies the minimum and maximum number of registers between two objects

- **Number of gates**

This option specifies the minimum and maximum number of gates between two objects

- **Min**

Specifies the minimum number of registers for a flyline. If the number of aggregated connections is below this amount, the flyline is not displayed

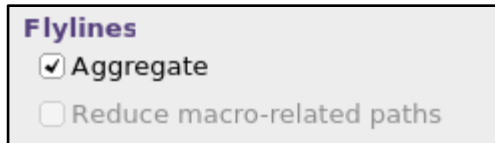
- **Max**

Specifies the maximum number of registers for a flyline. If the number of aggregated connections is above this amount, the flyline is not displayed

# Floorplan : Use ICC2 (IC Compiler II tool)

- **Data Flow Flylines (DFF)**

- **Flylines**

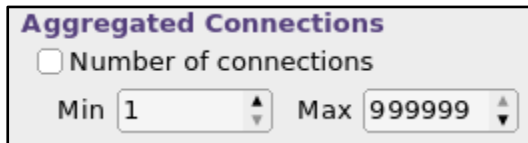


A configuration panel titled "Flylines" with a light gray background. It contains two options: "Aggregate" which is checked with a small square icon, and "Reduce macro-related paths" which is unchecked with a small square icon.

- **Aggregate by block**

Selecting this option groups individual flylines between two objects as a single wide flyline. When deselected, the tool displays individual flylines

- **Aggregated Connections**



A configuration panel titled "Aggregated Connections" with a light gray background. It contains one option: "Number of connections" which is unchecked with a small square icon. Below this option are two spinners: "Min" with the value "1" and "Max" with the value "999999".

- **Number of connections**

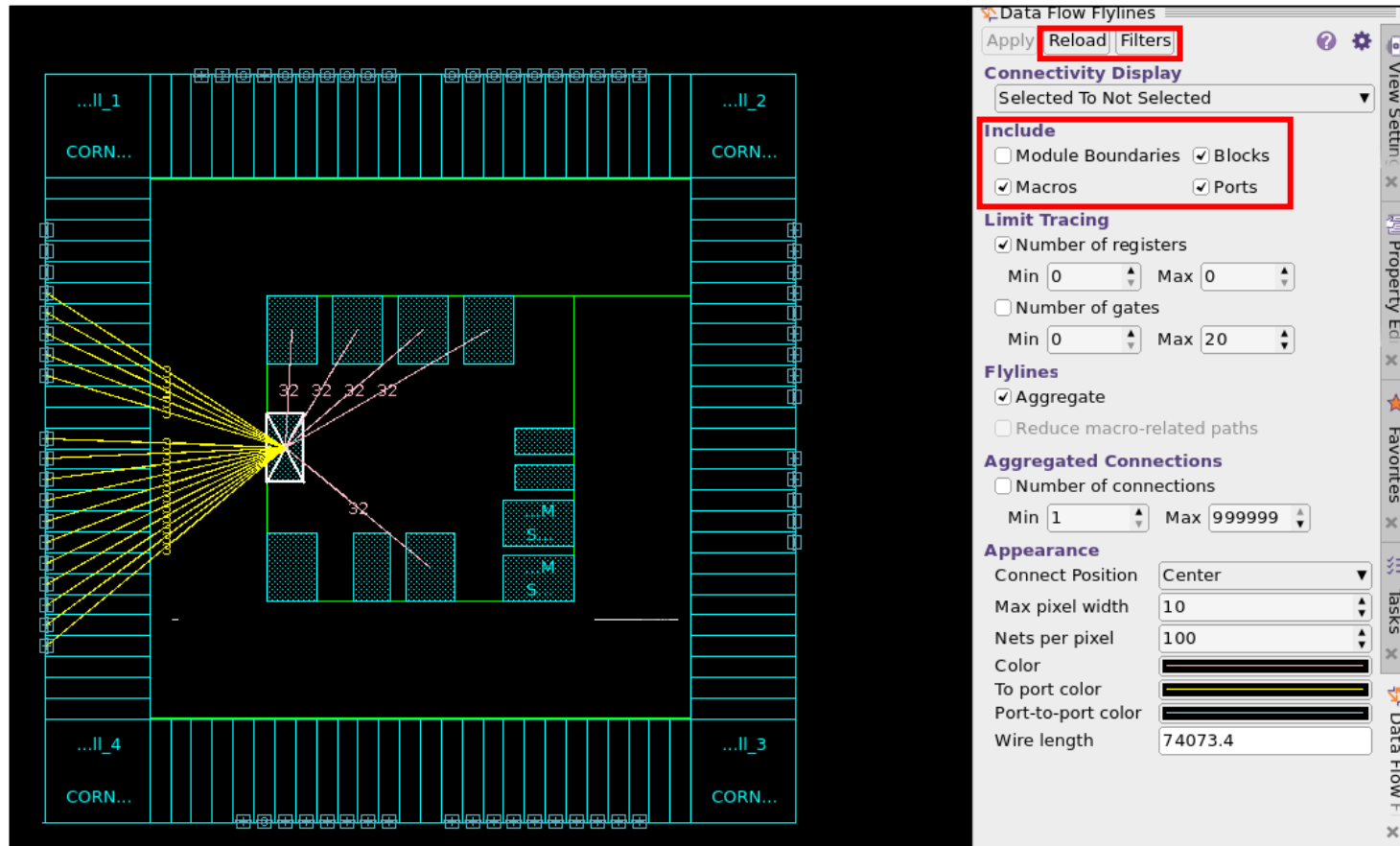
Selecting this option filters flylines based on the Min and Max settings. If this option is selected and the connection count is less than the minimum or greater than the maximum, the flyline is not displayed

# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/05_place_macro_and_check.tcl`

- **Check Data Flow Flylines (DFF)**

- ❖ The "Data Flow Flylines Tool" (DFF) supports exploration of the netlist connectivity of a design by using netlist tracing.
- ❖ DFF recognizes the following netlist elements: Module Boundaries, Blocks, Macros and Ports and can display connectivity between them in various configurations.
- ❖ The tool is modal and can be displayed or hidden by clicking the Data Flow Flylines button in the toolbar or selecting View > Flylines > Data Flow Flyline from the menu bar.

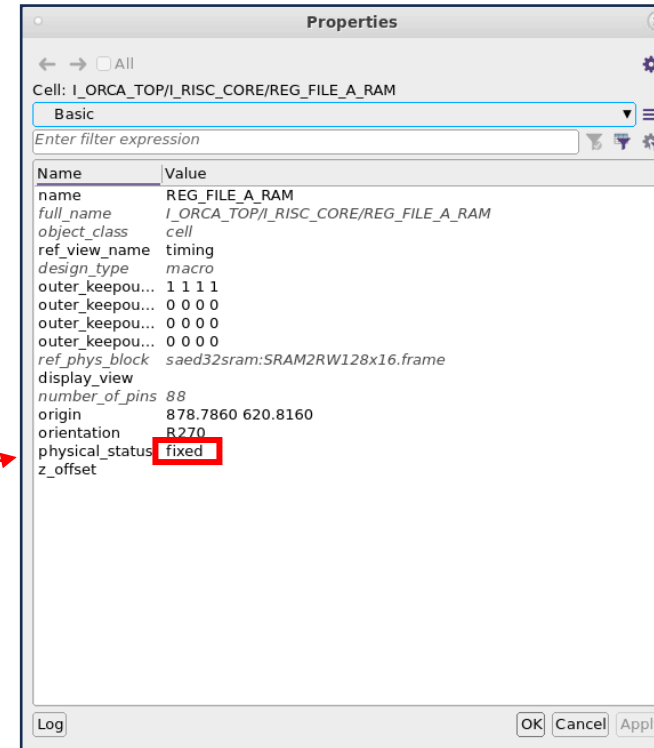
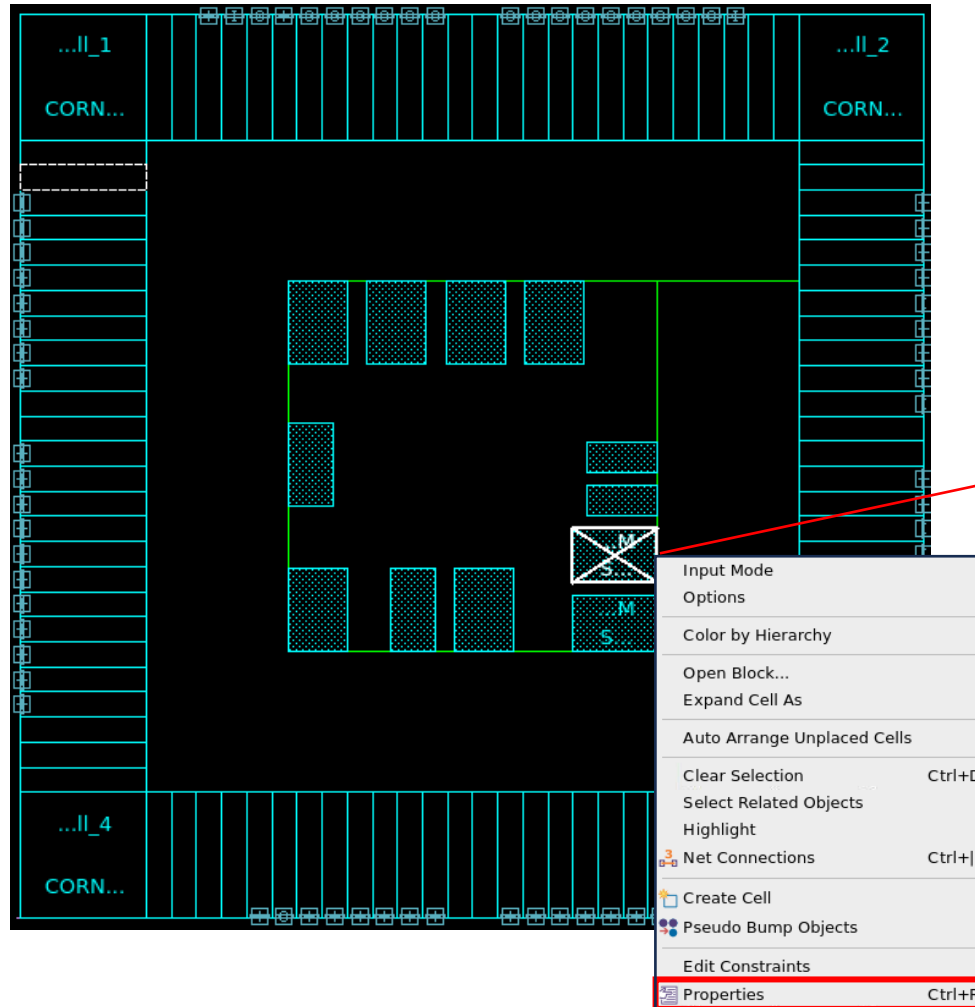


Check & Reload in this way

# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/05_place_macro_and_check.tcl`

- Check Macro placement & status (fixed)



- ❖ After selecting an object, you can check the property of the object by pressing the Ctrl + R key (short cut)
- ❖ You can also review it by clicking the mouse right key and then clicking properties

# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/05_place_macro_and_check.tcl`

- Check floorplan

- ❖ `check_legality`

```
icc2_shell> check_legality
```

```
Information: The command 'check_legality' cleared the undo history. (UNDO-016)
```

|                         |           |            |                                   |                            |           |            |                                                   |   |   |   |                                                 |  |  |  |  |
|-------------------------|-----------|------------|-----------------------------------|----------------------------|-----------|------------|---------------------------------------------------|---|---|---|-------------------------------------------------|--|--|--|--|
| *****                   |           |            |                                   | VIOLATIONS BY SUBCATEGORY: |           |            |                                                   |   |   |   |                                                 |  |  |  |  |
| Report : Legality       |           |            |                                   | MOVABLE                    | APP-FIXED | USER-FIXED | DESCRIPTION                                       | 0 | 0 | 0 | A cell is in the wrong region.                  |  |  |  |  |
| *****                   |           |            |                                   |                            |           |            |                                                   | 0 | 0 | 0 | A cell is outside its hard bound.               |  |  |  |  |
|                         |           |            |                                   |                            |           |            |                                                   | 0 | 0 | 0 | A cell is in the wrong voltage area.            |  |  |  |  |
|                         |           |            |                                   |                            |           |            |                                                   | 0 | 0 | 0 | A cell violates an exclusive movebound.         |  |  |  |  |
| VIOLATIONS BY CATEGORY: |           |            |                                   |                            |           |            |                                                   |   |   |   |                                                 |  |  |  |  |
| MOVABLE                 | APP-FIXED | USER-FIXED | DESCRIPTION                       |                            |           |            |                                                   |   |   |   |                                                 |  |  |  |  |
| 0                       | 0         | 0          | Two objects overlap.              | 0                          | 0         | 0          | Two cells overlap.                                | 0 | 0 | 0 | Two cells violate cts margins.                  |  |  |  |  |
| 0                       | 0         | 0          | A cell violates a pnet.           | 0                          | 0         | 0          | A cell overlaps a blockage.                       | 0 | 0 | 0 | Two cells violate coloring.                     |  |  |  |  |
| 0                       | 0         | 0          | A cell is illegal at a site.      | 0                          | 0         | 0          | A cell keepout margin overlaps a blockage.        |   |   |   |                                                 |  |  |  |  |
| 0                       | 0         | 0          | A cell is not aligned with a site | 0                          | 0         | 0          | A cell violates a pnet.                           | 0 | 0 | 0 |                                                 |  |  |  |  |
| 0                       | 0         | 0          | A cell has an illegal orientation | 0                          | 0         | 0          | A cell is illegal at a site.                      |   |   |   | check_legality for block design ORCA succeeded! |  |  |  |  |
| 0                       | 0         | 0          | A cell spacing rule is violated.  | 0                          | 0         | 0          | A cell violates pin-track alignment rules.        |   |   |   |                                                 |  |  |  |  |
| 0                       | 0         | 0          | A layer rule is violated.         | 0                          | 0         | 0          | A cell is illegal at a site.                      |   |   |   | check_legality succeeded.                       |  |  |  |  |
| 0                       | 0         | 0          | A cell is in the wrong region.    | 0                          | 0         | 0          | A cell violates legal index rule.                 |   |   |   | *****                                           |  |  |  |  |
| 0                       | 0         | 0          | Two cells violate cts margins.    | 0                          | 0         | 0          | A cell has the wrong variant for its location.    |   |   |   |                                                 |  |  |  |  |
| 0                       | 0         | 0          | Two cells violate coloring.       | 0                          | 0         | 0          | A cell is not aligned with a site.                |   |   |   | 1                                               |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A cell is not aligned with the base site.         |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A cell is not aligned with an overlaid site.      |   |   |   |                                                 |  |  |  |  |
| TOTAL 0 Violations.     |           |            |                                   | 0                          | 0         | 0          | A cell has an illegal orientation.                |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A cell spacing rule is violated.                  |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A spacing rule is violated in a row.              |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A spacing rule is violated between adjacent rows. |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A cell violates vertical abutment rule.           |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A cell violates metal spacing rule.               |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer rule is violated.                         |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer VTH rule is violated.                     |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer OD rule is violated.                      |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer OD max-width rule is violated.            |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer ALL_OD corner rule is violated.           |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer max-vertical-length rule is violated.     |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | A layer TPO rule is violated.                     |   |   |   |                                                 |  |  |  |  |
|                         |           |            |                                   | 0                          | 0         | 0          | Filler cell insertion cannot satisfy layer rules. |   |   |   |                                                 |  |  |  |  |

# Floorplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/05_place_macro_and_check.tcl`

- Check floorplan

## ❖ `check_io_placement`

```
icc2_shell> check_io_placement
----- Start of Overlap Check -----
----- End of Overlap Check -----

----- Start of Bump Assignment Check -----
----- End of Bump Assignment Check -----

----- Start of Pad to Guide Assignment Check -----
----- End of Pad to Guide Assignment Check -----

----- Start of Signal I/O constraints Check -----
----- End of Signal I/O constraints Check -----

----- Start of Power I/O Constraints Check -----
----- End of Power I/O Constraints Check -----

----- Start of Unplaced Pads Check -----
----- End of Unplaced Pads Check -----

----- Start of Pad Flip Check -----
----- End of Pad Flip Check -----

----- Start of Gap Check -----
----- End of Gap Check -----

----- Start of Min Pitch Check -----
----- End of Min Pitch Check -----

No violation has been found.
```

❖ Performs I/O placement checking

❖ You must have to fix all the violations and move on to the next step

# Floorplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell > source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/02\_floorplan/script/05\_place\_macro\_and\_check.tcl

### • Check floorplan

### ❖ check\_mv\_design (Checks for violations in a multi-voltage design)

```
icc2_shell> check_mv_design
*****
Report : check_mv_design
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Aug 27 16:51:45 2023
*****
```

```
----- Power domain rule -----
No errors or warnings.
```

```
----- Supply set rule -----
No errors or warnings.
```

```
----- Supply net rule -----
No errors or warnings.
```

```
----- Supply port rule -----
No errors or warnings.
```

```
----- Isolation strategy rule -----
No errors or warnings.
```

```
----- Level shifter strategy rule -----
No errors or warnings.
```

```
----- Retention strategy rule -----
No errors or warnings.
```

```
----- Power switch strategy rule -----
No errors or warnings.
```

```
----- Repeater rule -----
No errors or warnings.
```

```
----- Terminal boundary rule -----
No errors or warnings.
```

```
----- Isolation cell rule -----
No errors or warnings.
```

```
----- Level shifter cell rule -----
No errors or warnings.
```

```
----- Retention cell rule -----
No errors or warnings.
```

```
----- Switch cell rule -----
No errors or warnings.
```

```
----- PGMUX rule -----
No errors or warnings.
```

```
----- Diode cell rule -----
No errors or warnings.
```

```
----- Model rule -----
No errors or warnings.
```

```
----- Isolation rule -----
No errors or warnings.
```

```
----- Voltage shifting rule -----
```

```
Warning: Driver pin 'scan_en' (supply net 'VDD' [0.95V]) cannot drive load pin 'scan_en_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'test_mode' (supply net 'VDD' [0.95V]) cannot drive load pin 'test_mode_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'power_save' (supply net 'VDD' [0.95V]) cannot drive load pin 'powersave_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'sys_clk' (supply net 'VDD' [0.95V]) cannot drive load pin 'sys_clk_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pclk' (supply net 'VDD' [0.95V]) cannot drive load pin 'pclk_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'prst_n' (supply net 'VDD' [0.95V]) cannot drive load pin 'prst_n_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pidsel' (supply net 'VDD' [0.95V]) cannot drive load pin 'pidsel_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pgnt_n' (supply net 'VDD' [0.95V]) cannot drive load pin 'pgnt_n_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pm66en' (supply net 'VDD' [0.95V]) cannot drive load pin 'pm66en_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'sdr_clk' (supply net 'VDD' [0.95V]) cannot drive load pin 'sdr_clk_iopad/PADIO' (supply net 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Information: Total 10 MV-012 violations. (MV-080)
```

```
----- Tie-off connection rule -----
No errors or warnings.
```

```
----- Analog net rule -----
No errors or warnings.
```

```
----- Physical block pin rule -----
No errors or warnings.
```

```
----- PG pin rule -----
No errors or warnings.
```

```
----- Signal pin rule -----
No errors or warnings.
```

```
----- Summary -----
```

```
Information: Total 0 error(s) and 10 warning(s) from check_mv_design. (MV-082)
1
```

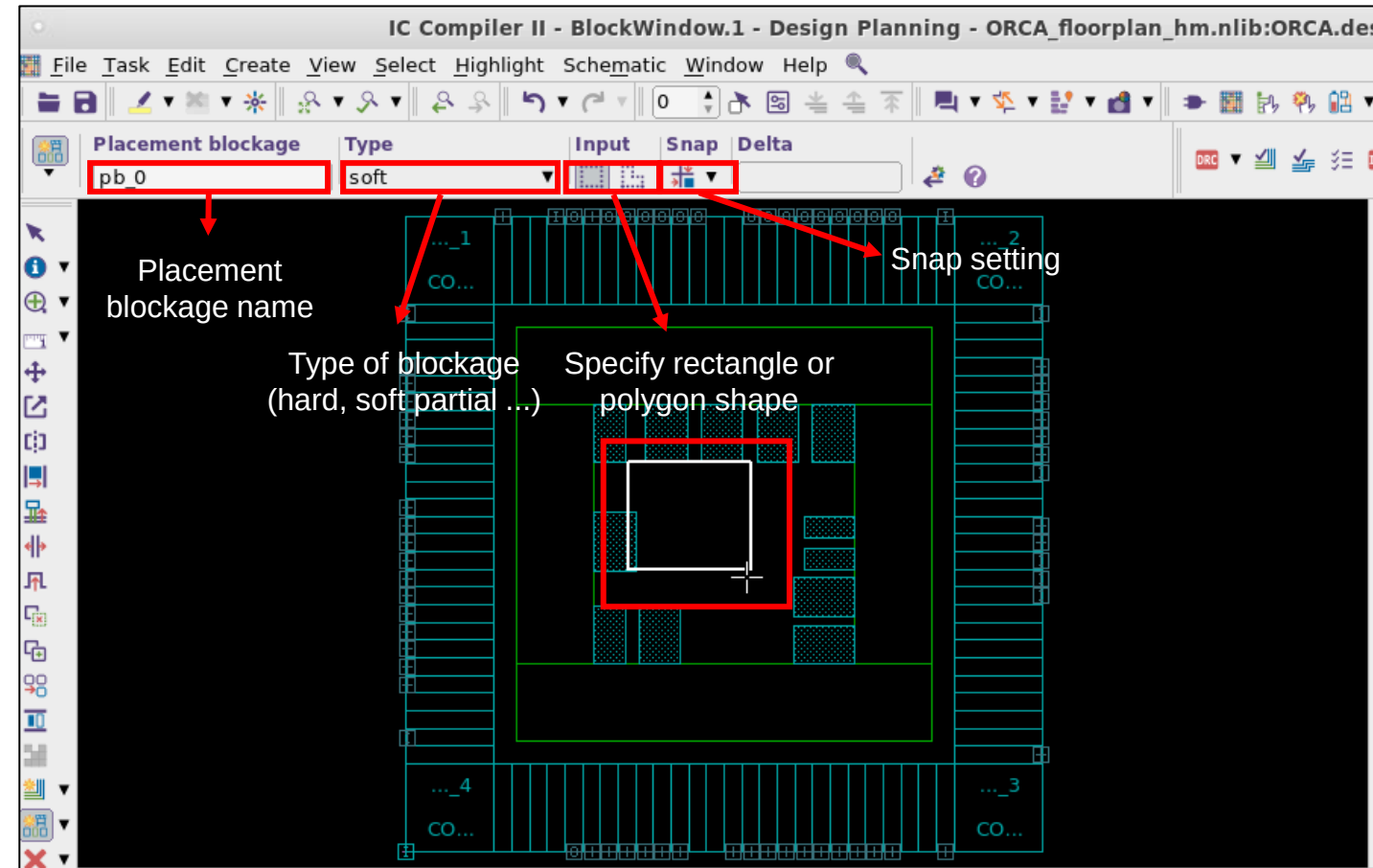
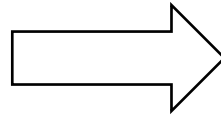
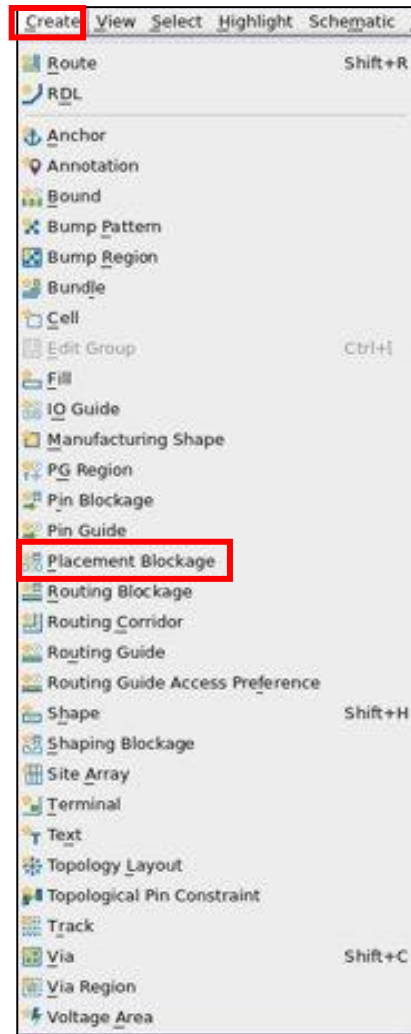
❖ MV-012 Warning due to voltage difference between supply net (VDD) and PAD's VDDIO pin (described in upf)

< UPF file >

```
# add port state inform 20230821 add
add_port_state VDD -state {VDD_ON 0.95}
add_port_state VDDIO -state {VDDIO_ON 2.25}
add_port_state AVDD1 -state {AVDD1_ON 2.25}
add_port_state AVDD2 -state {AVDD2_ON 2.25}
add_port_state DVDD1 -state {DVDD1_ON 0.95}
add_port_state DVDD2 -state {DVDD2_ON 0.95}
add_port_state VSS -state {VSS_OFF 0.00}
add_port_state VSSIO -state {VSSIO_OFF 0.00}
add_port_state AVSS -state {AVSS_OFF 0.00}
```

# Floorplan : Use ICC2 (IC Compiler II tool)

- Create the soft & hard blockage manually before read def



When the (name, type, input, snap setting is finished, you can drag to the desired area and create placement blockage)

Click create – Placement Blockage

# Floorplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/06_blockage_and_save.tcl`

- Read def file to recall soft blockages

```
icc2_shell> read_def ./FP_data/soft_blockage.def
Information: Loading DEF file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/02_floorplan/FP_data/soft_blockage.def' (FILE-007)
Information: Reading 'VERSION' statement. (DEFR-016)
Information: Reading 'DIVIDERCHAR' statement. (DEFR-016)
Information: Reading 'BUSBITCHARS' statement. (DEFR-016)
Information: Reading 'DESIGN' statement. (DEFR-016)
Information: Reading 'UNITS' statement. (DEFR-016)
Information: Reading 'BLOCKAGES' section. (DEFR-016)
-----
Number of Processed/Read DEF Constructs
-----
VERSION                : 1/1
DIVIDERCHAR             : 1/1
BUSBITCHARS            : 1/1
DESIGN                  : 1/1
UNITS                   : 1/1
BLOCKAGES               : 12/12
1
```

## ❖ What is .def file?

Def file in vlsi stands for Design Exchange File, which is an industry standard file used for representing logic and connectivity of an integrated Circuit (IC) in ASCII format

## ❖ Write def file command

```
#If you want write def, use this command
#write_def -objects [get_selection] ./FP_data/soft_blockage.def
```

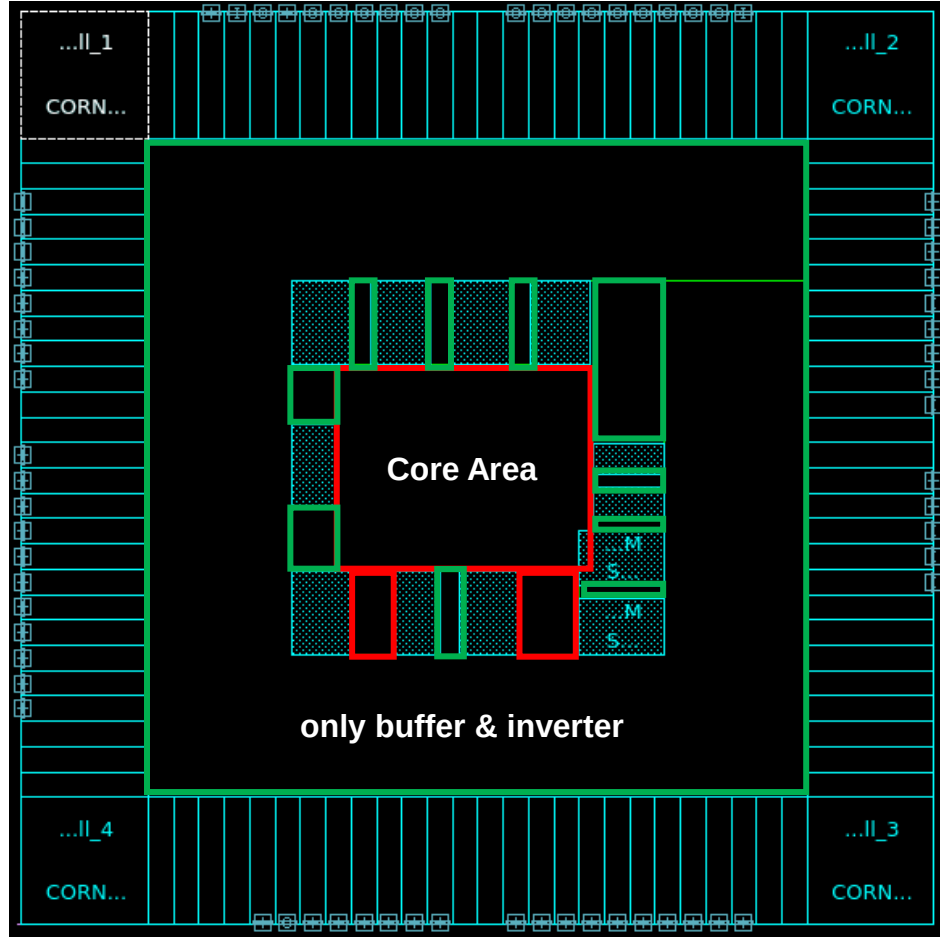
Select object  
(using write floorplan)

Output path and  
file name

# Floorplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/06_blockage_and_save.tcl`

- Check soft blockages between Macros



- Core std cell place area
- Soft blockage

❖ By creating a soft blockage between Macro, STD cells are prevented from being placed between Macro

# Floorplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplanscript/06_blockage_and_save.tcl`

- Write def file for soft blockages (optional)

```
icc2_shell> set sft_blk [get_placement_blockages -filter "blockage_type == soft"]
{core PB_1 PB_2 PB_3 PB_4 PB_5 PB_6 PB_7 PB_8 PB_9 PB_10 PB_11 PB_12}
icc2_shell> change_selection $sft_blk
icc2_shell> write_def -objects [get_selection] ./FP_data/all_soft_blockage.def
*****
Report : Data Mismatches
Version: S-2021.06-SP5
Date   : Mon Jul 24 23:49:36 2023
*****
No mismatches exist on the design.
Information: DEF BLOCKAGES section is filtered by the -objects option list; 13 blockages written to this section. (DEFW-009)
-----
Number of Written DEF Constructs
-----
VERSION                : 1
DIVIDERCHAR             : 1
BUSBITCHARS            : 1
DESIGN                  : 1
UNITS                   : 1
BLOCKAGES               : 13
1
icc2_shell> change_selection
```

- Check that the .def write correctly

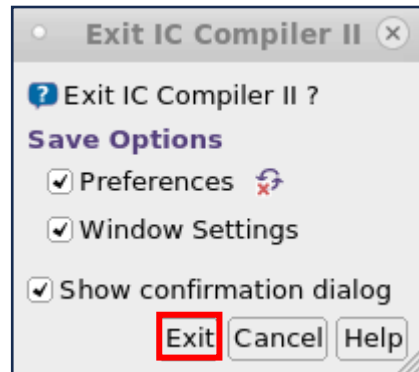
```
< user@linux FP_data >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/02_floorplan/FP_data
< user@linux FP_data >> ls
all_soft_blockage.def  FP_TEST  ORCA_floorplan_data  soft_blockage.def
```

# Floorplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell > source $WORK_DIR/2_BACKEND/1_pnr/rundir/02_floorplan/script/06_blockage_and_save.tcl`

- **Save & Exit ICC2**

```
icc2_shell> save_lib -all
Information: The command 'save_lib' cleared the undo history. (UNDO-016)
Saving all libraries...
1
icc2_shell> exit
```



- **Check that the .nlib is saved correctly**

```
< user@linux nlib >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/02_floorplan/nlib
< user@linux nlib >> ls
ORCA_floorplan.nlib  ORCA_read_design.nlib
```

# WEEK 4 : Exercise for Floorplan

# Floorplan : Use ICC2 (IC Compiler II tool) HomeWork

- Unlike the floorplan you practiced, we proceed with the P&R flow after placing the Macro & PLL
  - **Input files**
    - Fp.tcl** - `${WORK_DIR}/2_BACKEND/1_pnr/02_floorplan/Fp_data/ORCA_homework_floorplan_data/fp.tcl`
    - Soft\_blockage.def** - `${WORK_DIR}/2_BACKEND/1_pnr/02_floorplan/Fp_data/ORCA_homework_soft_blockage.def`
  - **Guidelines**
    - ❖ It is the same as the previous exercise, but fp.tcl and soft\_blockage.def are applied differently.
    - ❖ It is recommended not to source fp.tcl (look at the guide and Place Macro & PLL as a manually)
    - ❖ It is also recommended to draw the soft blockage manually
    - ❖ After floorplan, All Macro & PLL must be fixed

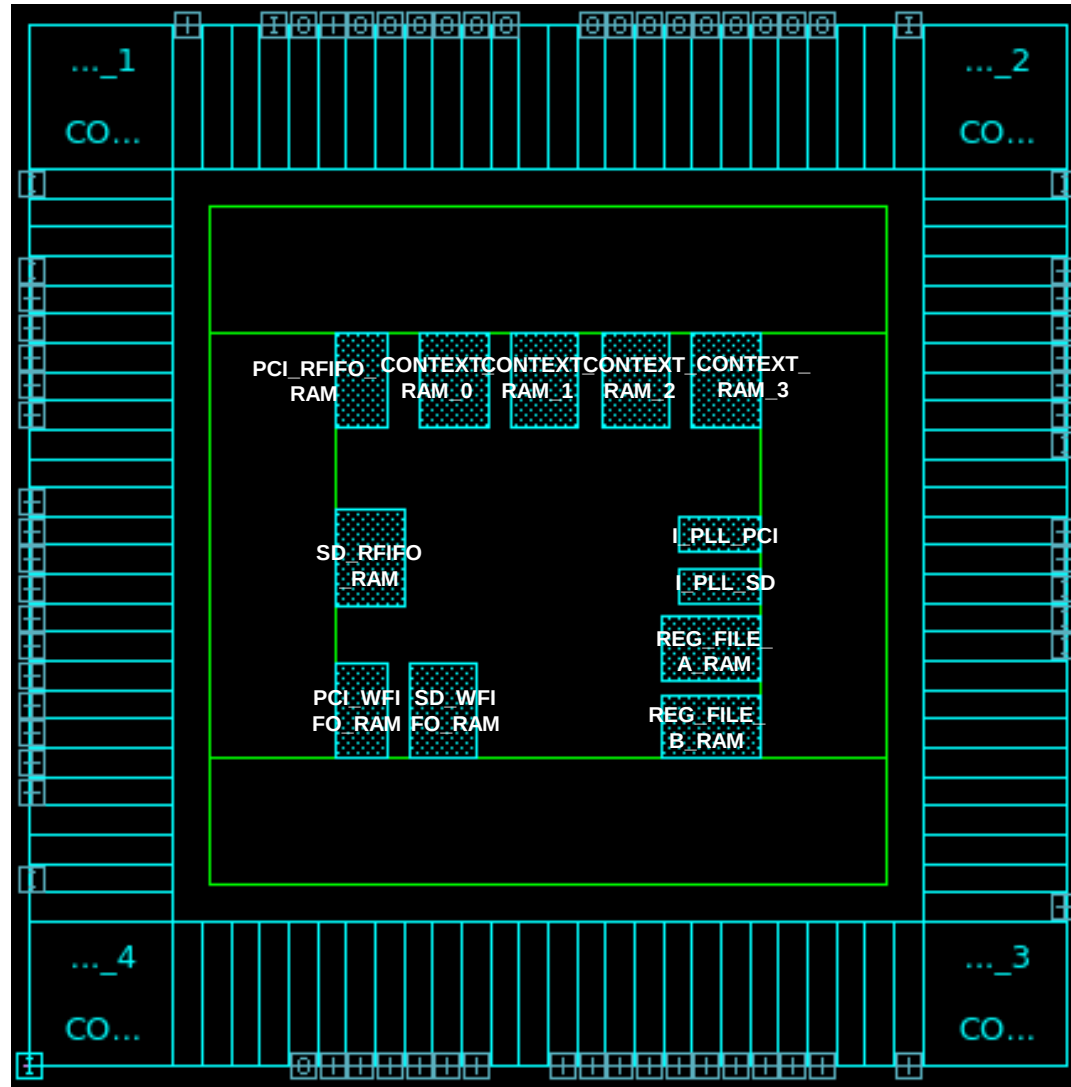
# Floorplan : Use ICC2 (IC Compiler II tool) HomeWork

- Practice manual Floorplan before sourcing `${WORK_DIR}/2_BACKEND/1_pnr/02_floorplan/Fp_data/ORCA_homework_floorplan_data/fp.tcl`
  - Move Macros & PLL manually first before source `fp.tcl`
  - Floorplan Guide

| MACRO & PLL                                      | Origin             | Orientation |
|--------------------------------------------------|--------------------|-------------|
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_0           | 636.7350 1014.1520 | MYR90       |
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_1           | 667.7350 1014.1520 | R270        |
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_2           | 888.5510 1014.1520 | MYR90       |
| I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM_3           | 919.5510 1014.1520 | R270        |
| I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM         | 496.3820 1014.1520 | MYR90       |
| I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM        | 496.3810 425.8460  | R90         |
| I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM        | 520.4510 637.1090  | R90         |
| I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM       | 621.2900 557.2520  | MYR90       |
| I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/REG_FILE_A_RAM | 878.7860 620.8160  | R270        |
| I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/REG_FILE_B_RAM | 878.7860 425.8460  | MXR90       |
| I_CLOCK_GEN/I_PLL_SD                             | 1014.4580 640.8160 | MY          |
| I_CLOCK_GEN/I_PLL_PCI                            | 1014.4580 709.9240 | MY          |

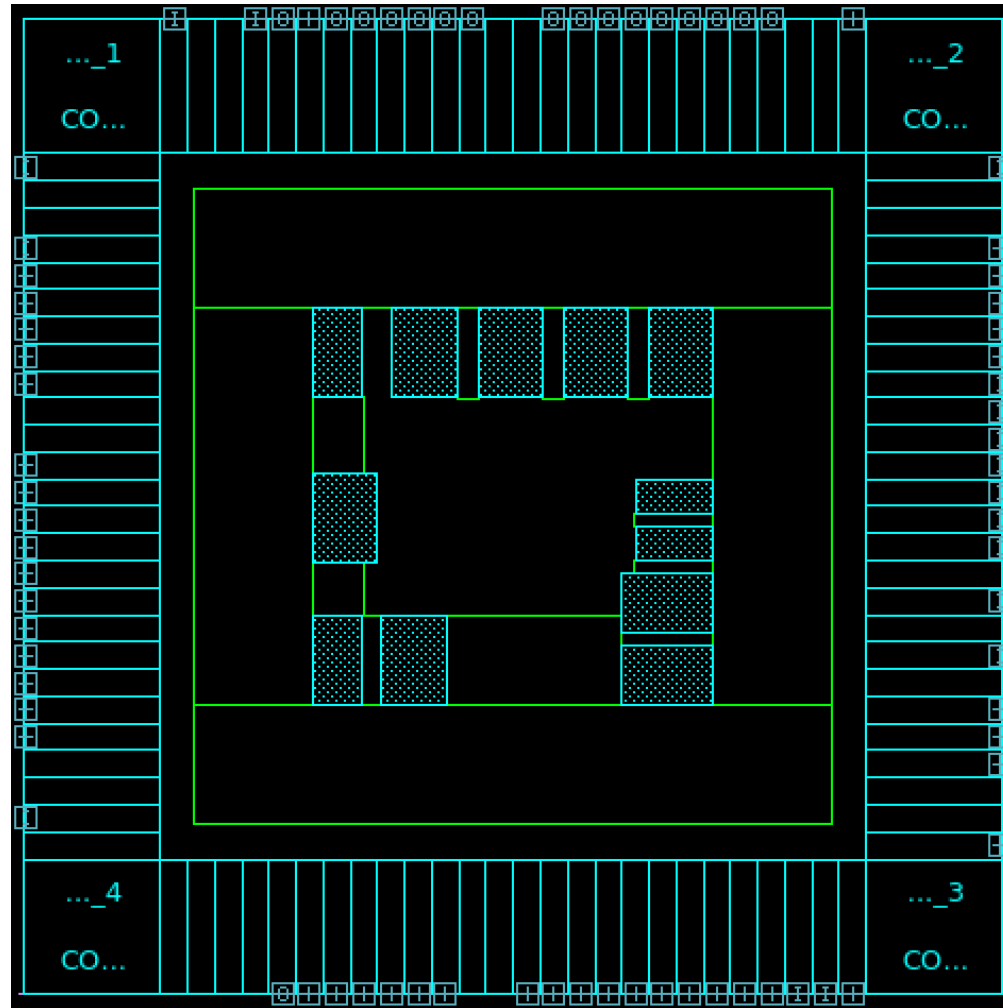
# Floorplan : Use ICC2 (IC Compiler II tool) HomeWork

- Practice manual Floorplan before sourcing fp.tcl
  - Floorplan guide in ICC2



# Floorplan : Use ICC2 (IC Compiler II tool) HomeWork

- Practice manual Floorplan before read `soft_blockage.def` (between Macro & PLL)  
`Read_def {WORK_DIR}/2_BACKEND/1_pnr/02_floorplan/Fp_data/ORCA_homework_soft_blockage.def`
- Floorplan guide in ICC2



# LAB WEEK 5 : Introduction to Powerplan

# Introduction to Powerplan : Overview to Powerplan

- What is Powerplan
  - Power planning means to provide power to the every Macros, STD cells, and all other cells are present in the design
  - Powerplan includes  
(Creating Power Ring, Stripes/Mesh/Grid, and STD cell Power rails)
  - Power and Ground nets are usually laid out on the Metal Layers  
(In this create Power and Ground structure for both IO pads and Core logic)
  - Powerplan also includes Power Via insertion
- Goals of Powerplan
  - Provide Power to every Macros, STD cells, and all other cells
  - Improve IR drop issue

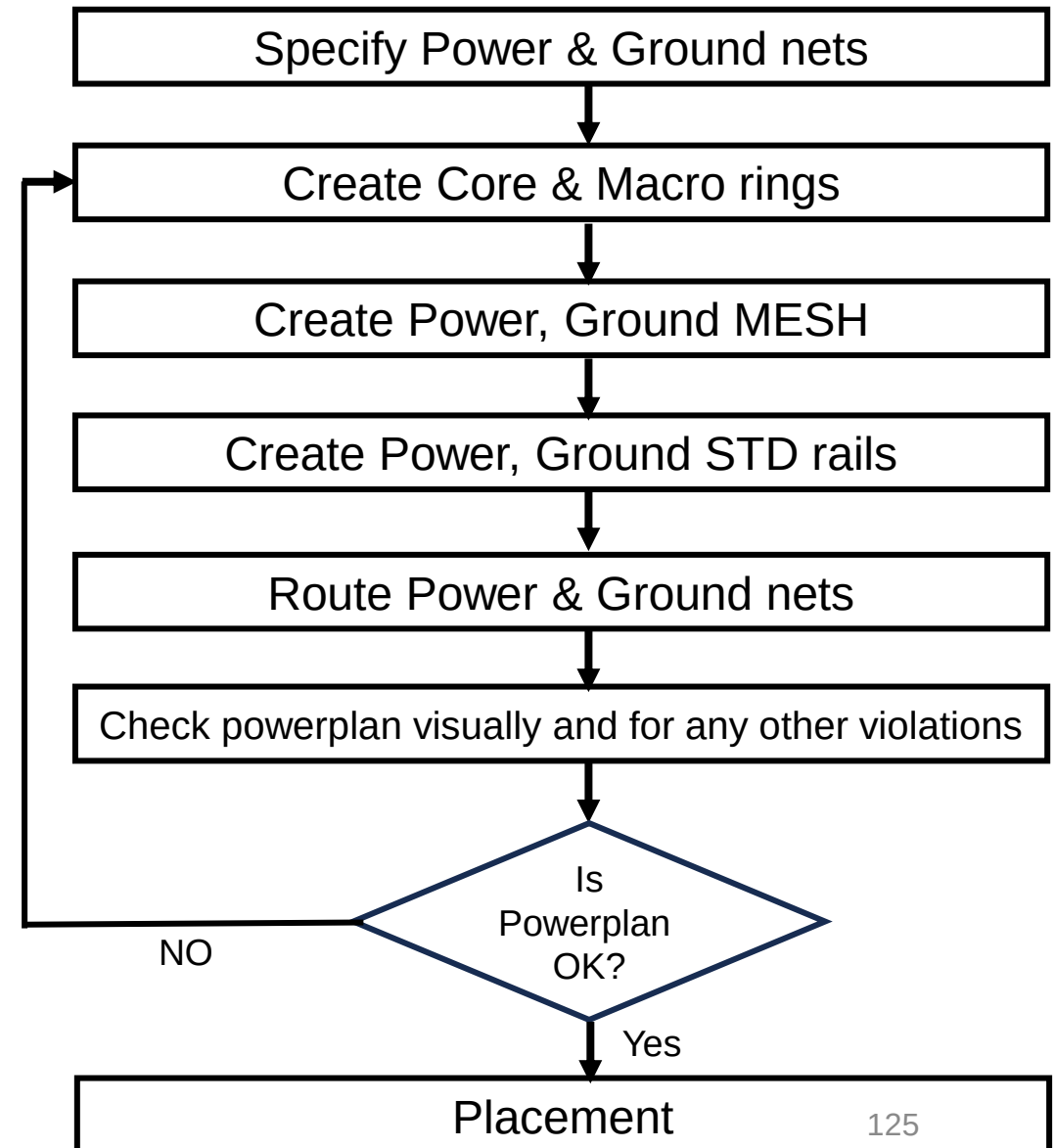
# Introduction to Powerplan : Overview to Powerplan

- Power Planning involves
  - Calculating number of power pins required
  - Number of Rings and Stripes
  - Width of Ring and Stripes
  - IR DROP
- Powerplan outputs
  - Power Mesh created
  - STD cell power rails are created
  - Macro & Core Ring created
  - Power PAD power connection created
  - Power via created
  - Power Stripes are created

# Introduction to Powerplan

- **Powerplan Overview**

- This module provides an overview of some of the key steps in Powerplan :
- **Defining**
  - Tech file (.tf)
  - Netlist
  - Library files (.lib & .lef)
  - UPF
- **Overview to Powerplan (directory, input files, script)**
- **Powerplan**

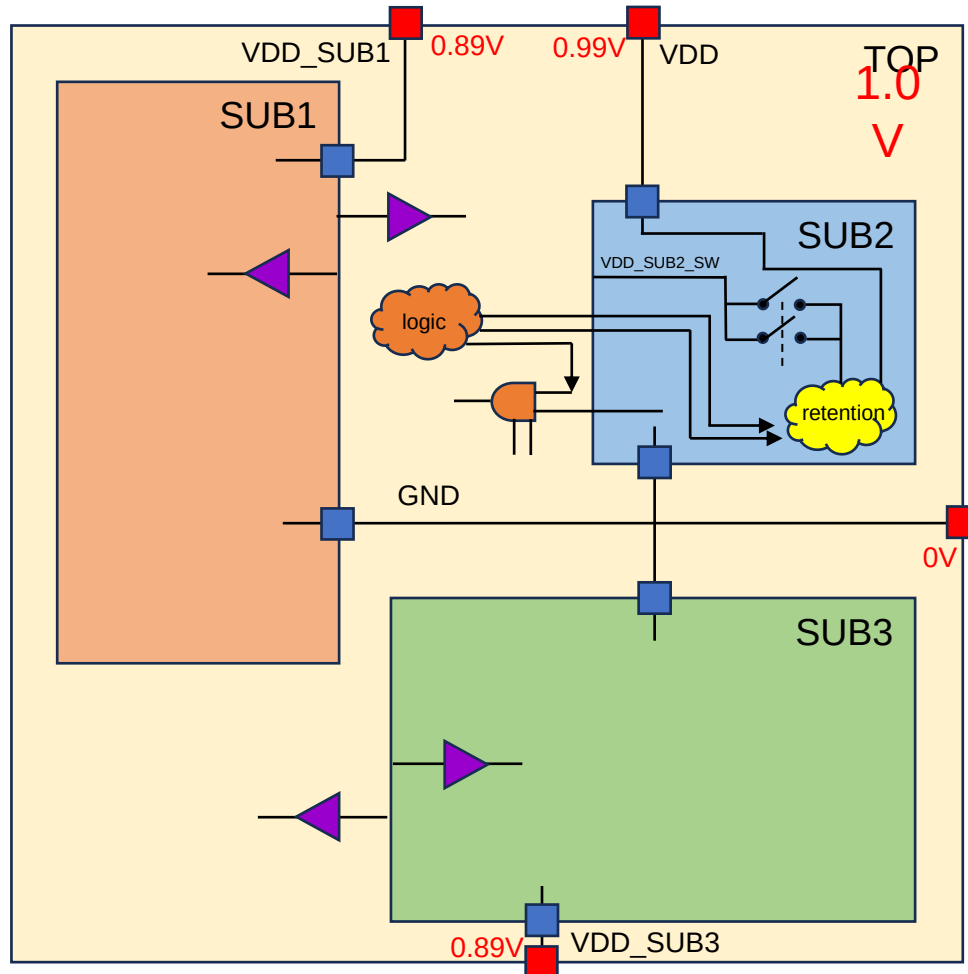


# Introduction to Powerplan : Terminologies & Definitions

- UPF (Unified Power Format)
  - UPF is a power specification file used in the design cycle to implement low-power design approaches
  - Power Distribution Architecture
    - Power domains
    - Supply rails
    - Shutdown control (Low Power Design)
  - Power Strategy
    - Power state tables (Legal combination of states of each Power Domain)
    - Operating voltages
  - Usage of Special cells (Low Power Design)
    - Isolation cells
    - Level shifters
    - Power switches
    - Retention registers

# Introduction to Powerplan : Terminologies & Definitions

- UPF – Example



# Design Power Spec

4 Power Domains, 2 Hierarchies

Top Level: 1.0V constant

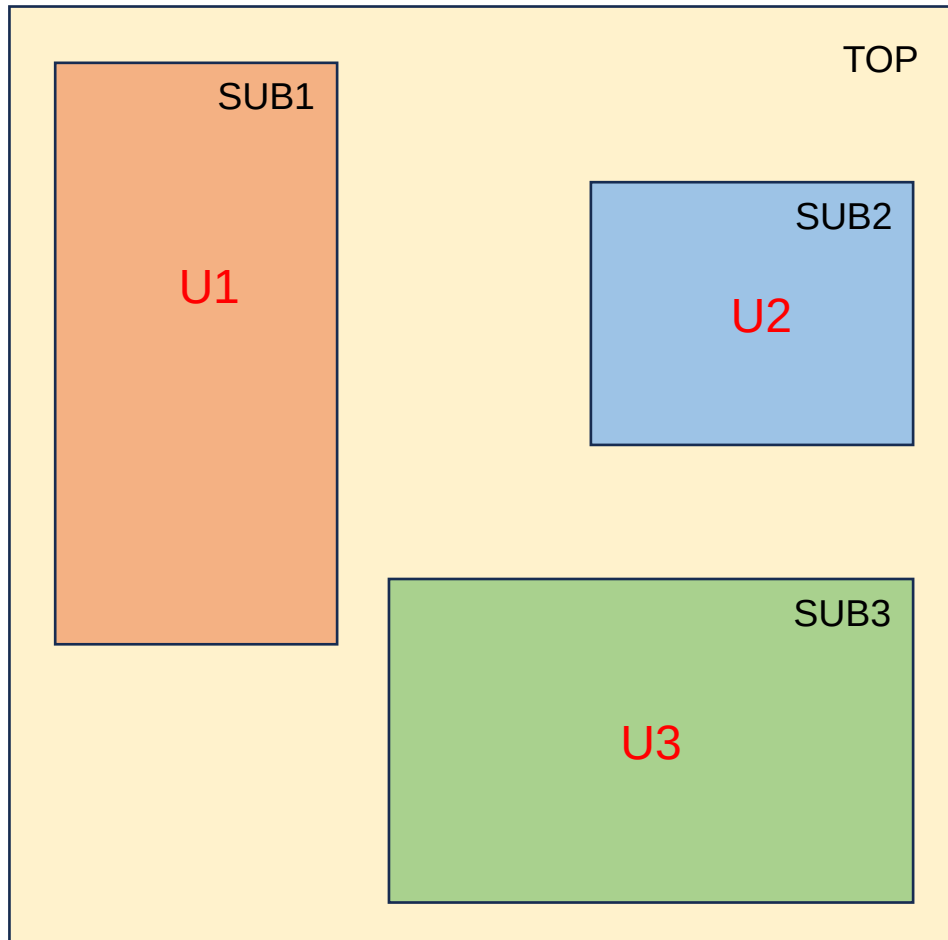
Secondary Level SUB1 : 0.89V constant

Secondary Level SUB2 : 0.99V switched constant

Secondary Level SUB3 : 0.89V constant

# Introduction to Powerplan : Terminologies & Definitions

- UPF – Domain Creation

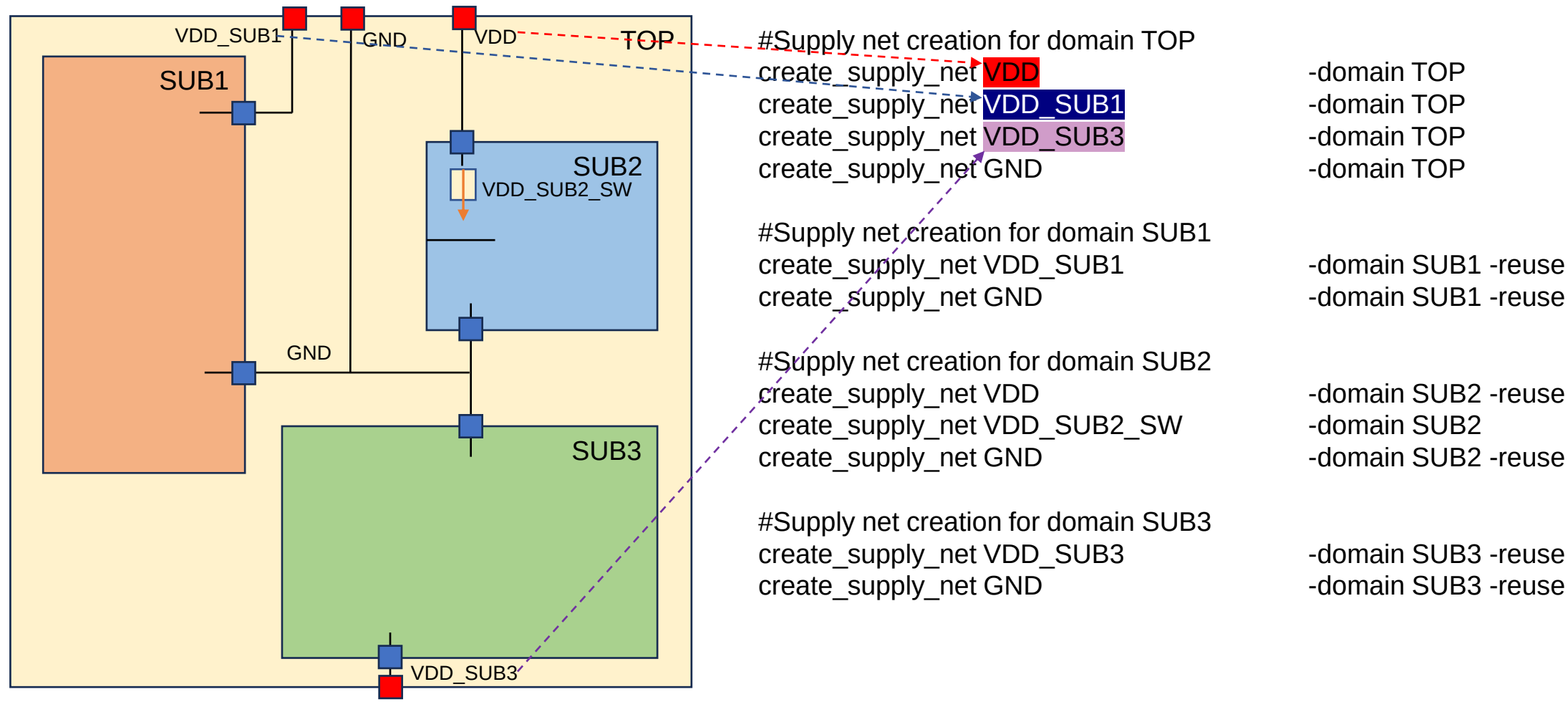


# Define Power domain

```
create_power_domain TOP    -include_scope  
create_power_domain SUB1   -elements {U1}  
create_power_domain SUB2   -elements {U2}  
create_power_domain SUB3   -elements {U3}
```

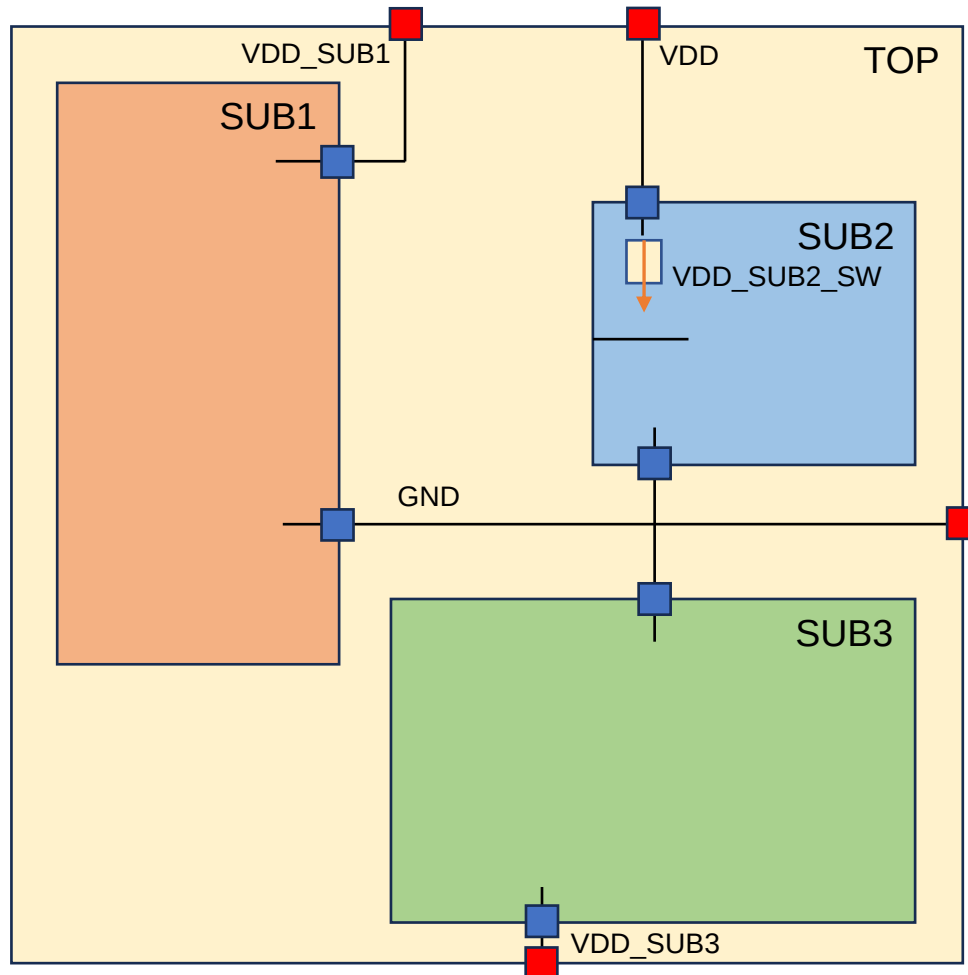
# Introduction to Powerplan : Terminologies & Definitions

- UPF – Supply Network Creation : create\_net



# Introduction to Powerplan : Terminologies & Definitions

- UPF – Supply Network Creation : create\_port / connect\_supply\_net



```
create_supply_port VDD  
create_supply_port VDD_SUB1  
create_supply_port VDD_SUB3  
create_supply_port GND
```

-domain TOP  
-domain TOP  
-domain TOP  
-domain TOP

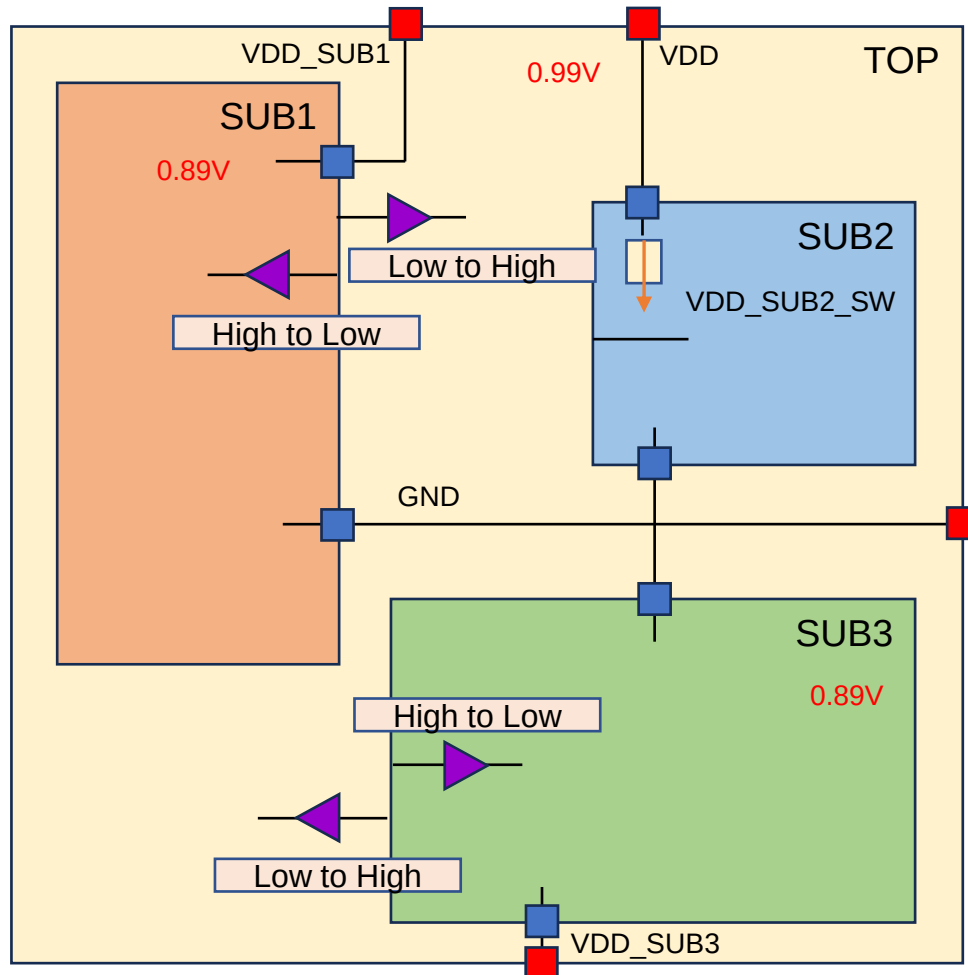
```
connect_supply_net VDD  
connect_supply_net VDD_SUB1  
connect_supply_net VDD_SUB3  
connect_supply_net GND
```

-ports VDD  
-ports VDD\_SUB1  
-ports VDD\_SUB3  
-ports GND

```
set_domain_supply_net TOP \  
-primary_power_net VDD \  
-primary_ground_net GND  
set_domain_supply_net SUB1 \  
-primary_power_net VDD_SUB1 \  
-primary_ground_net GND  
set_domain_supply_net SUB2 \  
-primary_power_net VDD_SUB2_SW \  
-primary_ground_net GND  
set_domain_supply_net SUB3 \  
-primary_power_net VDD_SUB3 \  
-primary_ground_net GND
```

# Introduction to Powerplan : Terminologies & Definitions

## • UPF – Level Shifter Strategy



```
set_level_shifter SUB1_to_TOP \  
-domain SUB1 \  
-applies_to outputs \  
-rule low_to_high \  
-location parent
```

SUB1 to TOP path  
Level shifter : low to high

```
set_level_shifter TOP_to_SUB1 \  
-domain SUB1 \  
-applies_to inputs \  
-rule high_to_low \  
-location self
```

TOP to SUB1 path  
Level shifter : high to low

```
set_level_shifter SUB3_to_TOP \  
-domain SUB3 \  
-applies_to outputs \  
-rule low_to_high \  
-location parent
```

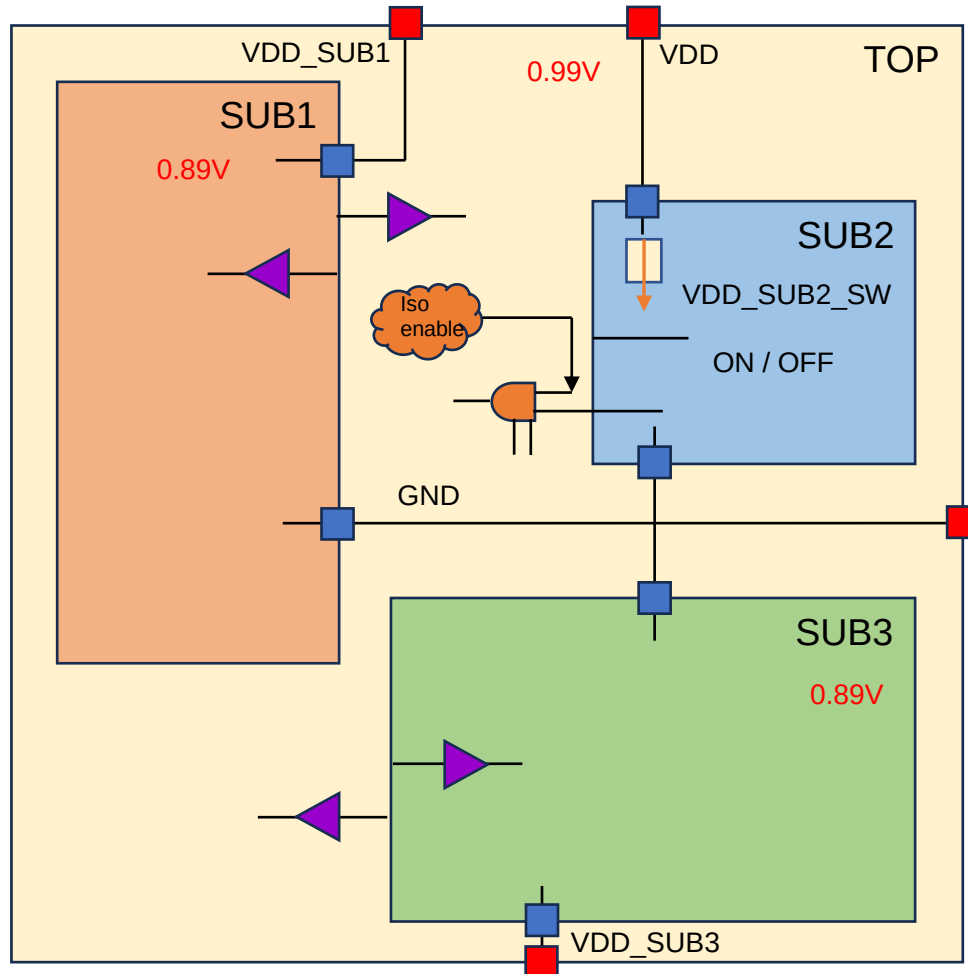
SUB3 to TOP path  
Level shifter : low to high

```
set_level_shifter TOP_to_SUB3 \  
-domain SUB3 \  
-applies_to inputs \  
-rule high_to_low \  
-location self
```

TOP to SUB3 path  
Level shifter : high to low

# Introduction to Powerplan : Terminologies & Definitions

- UPF – Isolation Cell Strategy

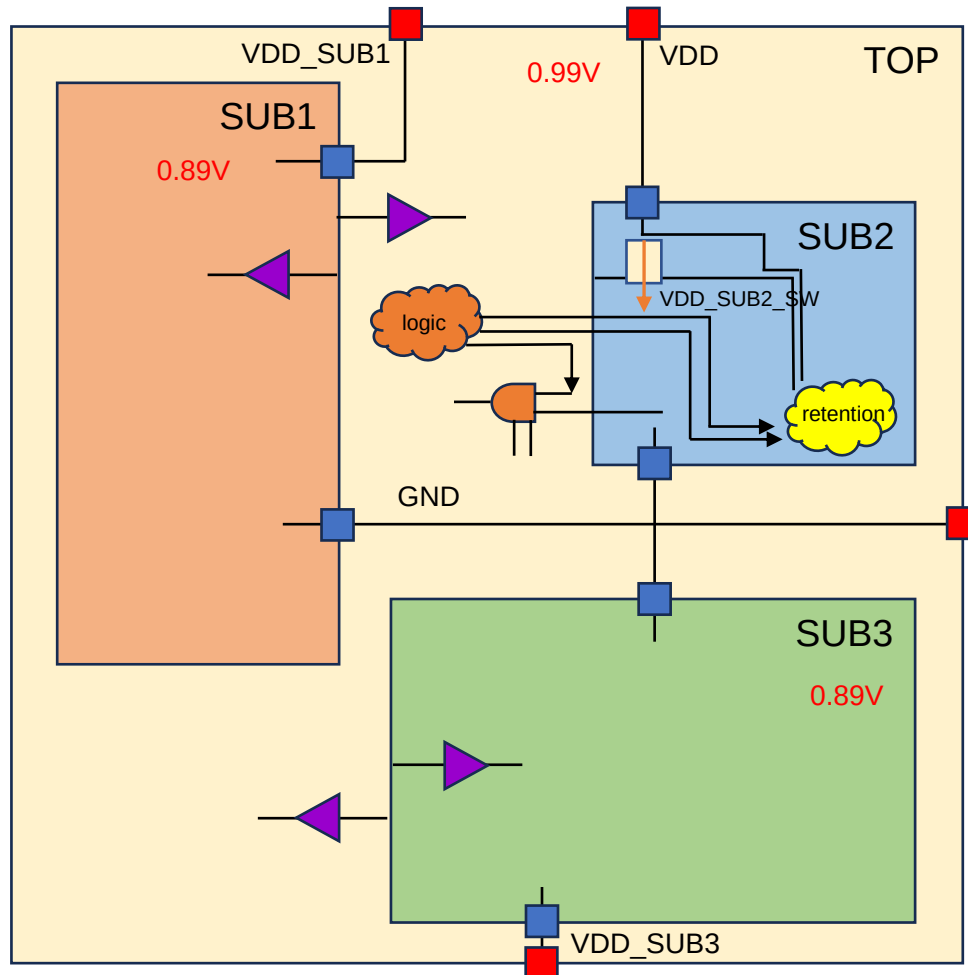


```
set_isolation ISO \  
-domain SUB2 \  
-isolation_power_net VDD \  
-isolation_ground_net GND \  
-clamp_value 0
```

```
set_isolation_control ISO \  
-domain SUB2 \  
-isolation_signal iso_enable \  
-isolation_sense high \  
-location parent
```

# Introduction to Powerplan : Terminologies & Definitions

- UPF – Retention Cell Strategy

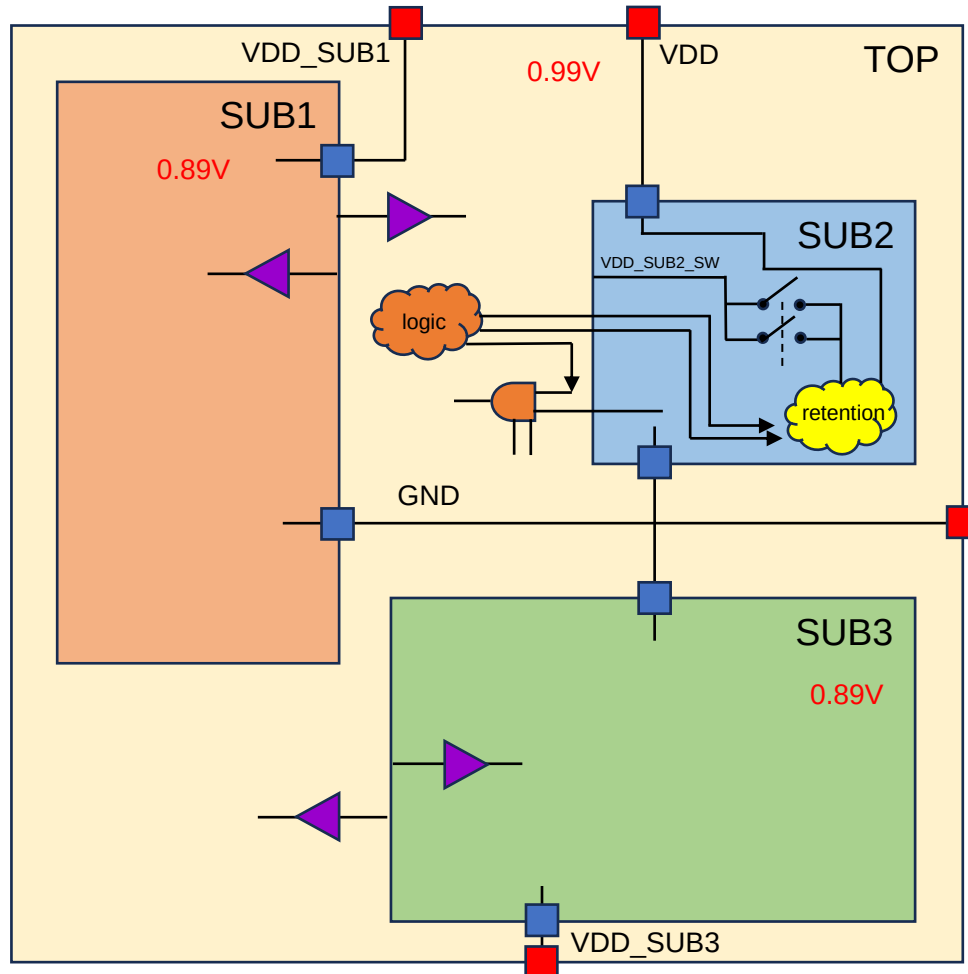


```
set_retention retention_key \  
-domain SUB2 \  
-retention_power_net VDD \  
-elements {inst_name}
```

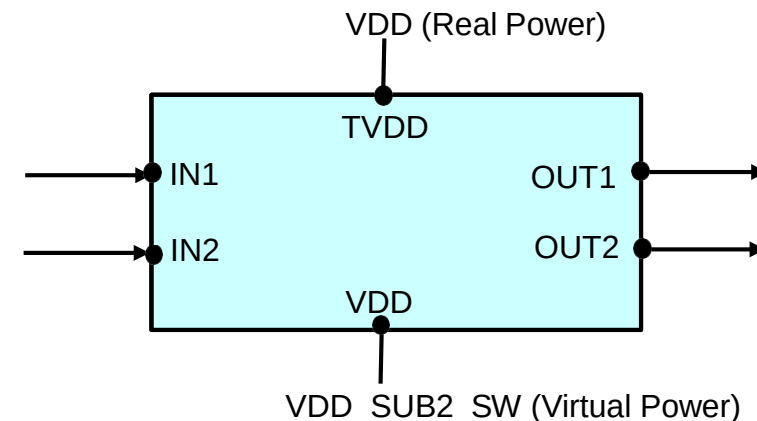
```
set_retention_control retention_key \  
-domain SUB2 \  
-save_signal {control_reg1/pin:Q high} \  
-restore_signal {control_reg2/pin:Q low}
```

# Introduction to Powerplan : Terminologies & Definitions

- UPF – Switch Cell Creation

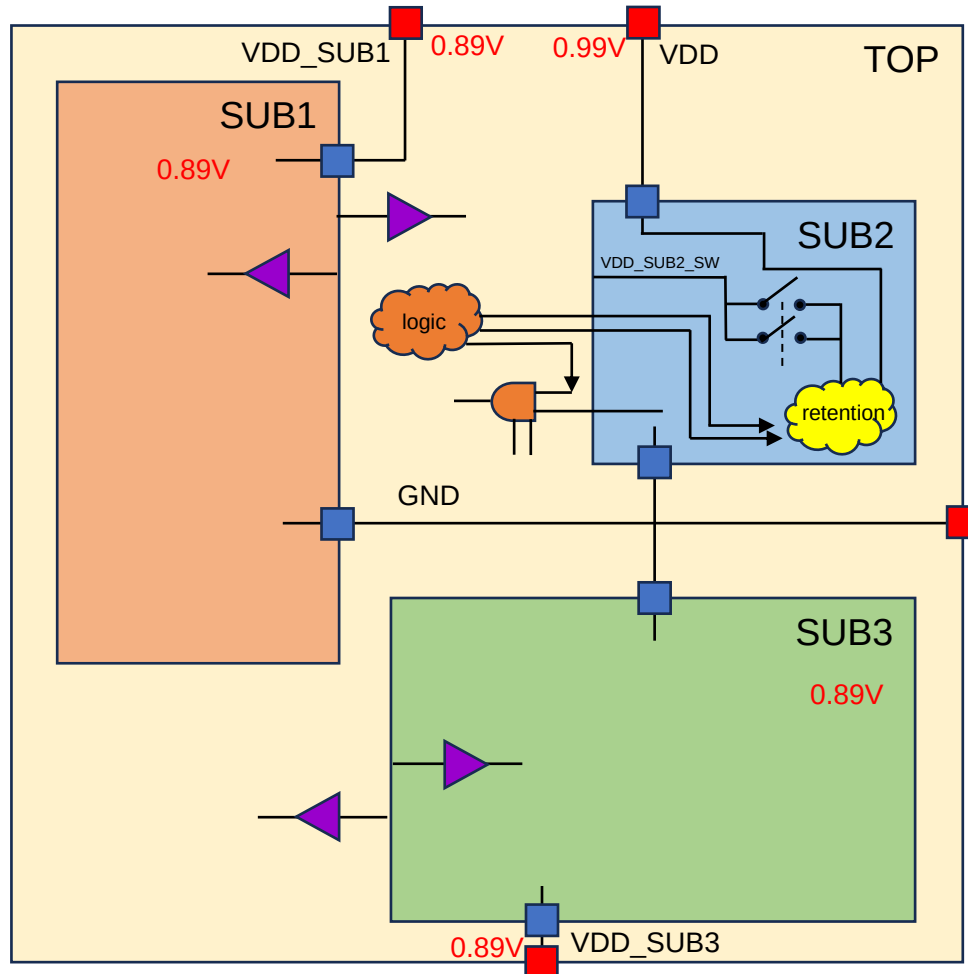


```
create_power_switch SUB2_SW \  
-domain SUB2 \  
-input_supply_port {TVDD VDD} \  
-output_supply_port {VDD VDD_SUB2_SW} \  
-control_port {IN1 SE1} \  
-control_port {IN2 SE2} \  
-ack_port {OUT1 SE_ack1} \  
-ack_port {OUT2 SE_ack2} \  
-on_state {SW_on TVDD {IN1 & IN2}} \  
-off_state {SW_off {!IN1 & !IN2}}
```



# Introduction to Powerplan : Terminologies & Definitions

- UPF – Port State Information



```
add_port_state VDD \  
-state {VDD_N 0.99}
```

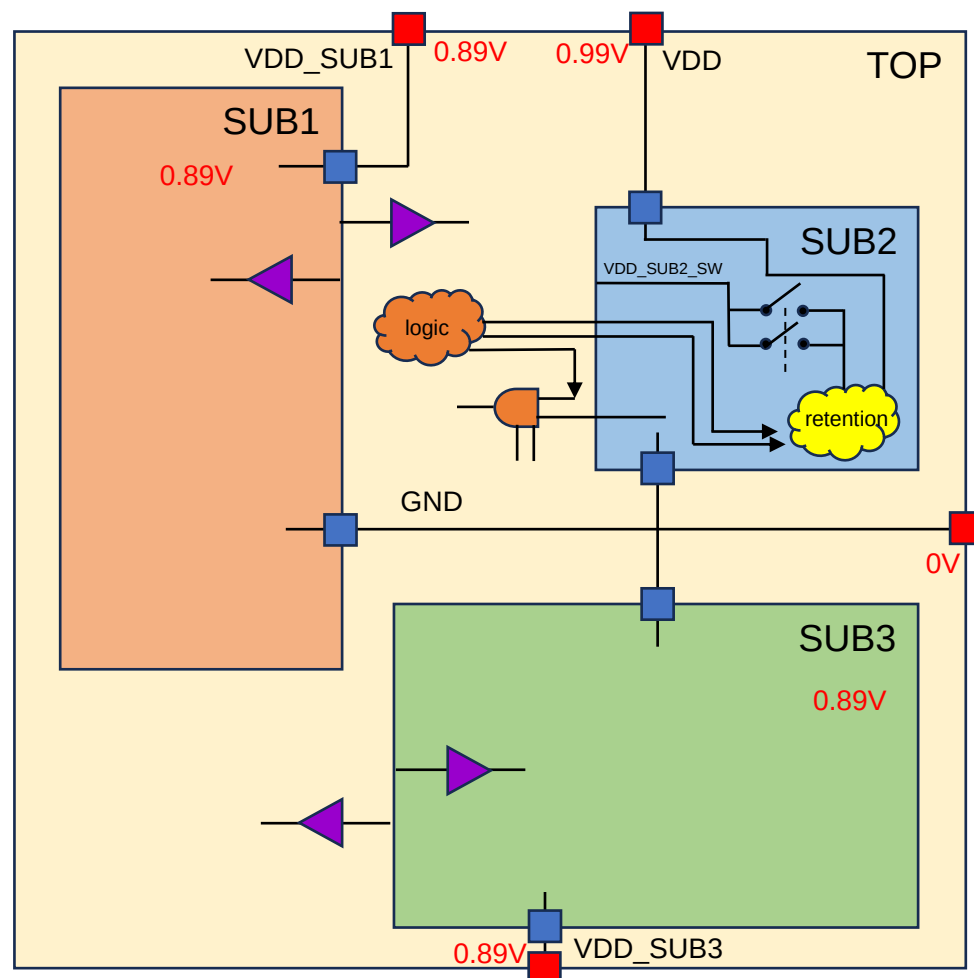
```
add_port_state VDD_SUB1 \  
-state {SUB1_H 0.89}  
-state {SUB1_L 0.69}
```

```
add_port_state VDD_SUB3 \  
-state {SUB3_H 0.89}  
-state {SUB3_L 0.69}
```

```
add_port_state GND \  
-state {default 0}
```

# Introduction to Powerplan : Terminologies & Definitions

- UPF – Port State Information



```
create_pst PST -supplies \
{ VDD u2/VDD_SUB2_SW VDD_SUB1 VDD_SUB3 }
```

```
add_pst_state PST0 -pst PST -state \
{ VDD_N SW_on SUB1_H SUB3_H }
```

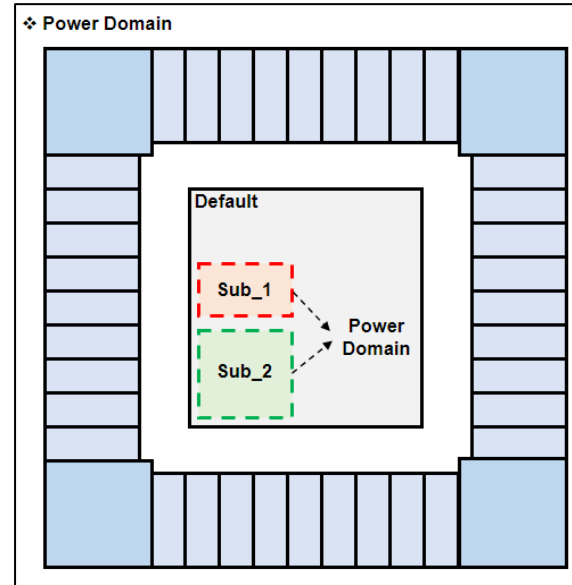
```
add_pst_state PST1 -pst PST -state \
{ VDD_N SW_off SUB1_L SUB3_L }
```

|      | VDD   | VDD_SUB0_SW | VDD_SUB1 | VDD_SUB3 |
|------|-------|-------------|----------|----------|
| PST0 | VDD_N | SW_on       | SUB1_H   | SUB3_H   |
| PST1 | VDD_N | SW_off      | SUB1_L   | SUB3_L   |

# Introduction to Powerplan : Terminologies & Definitions

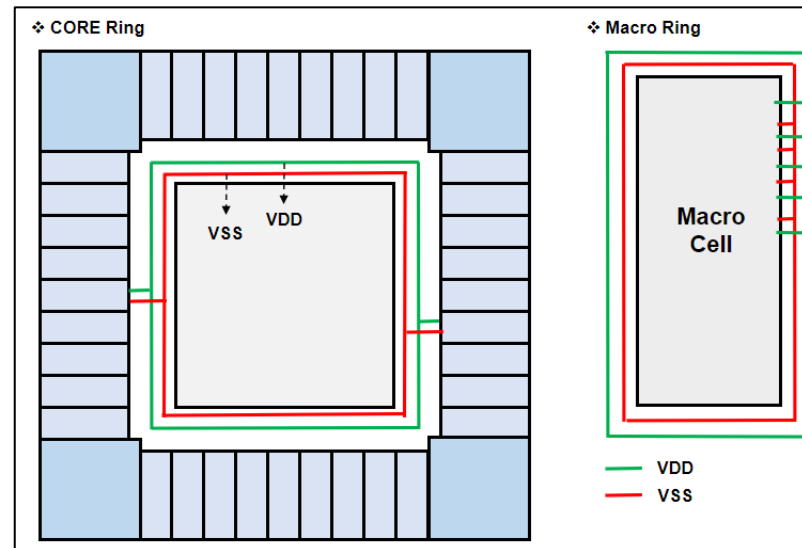
- **Power Domain**

- Group of elements which share a common set of power supply requirements
- Voltage domains do not have the same voltage values, so level shifters are required
- Power domains require isolation cells and power on-off control



- **Rings**

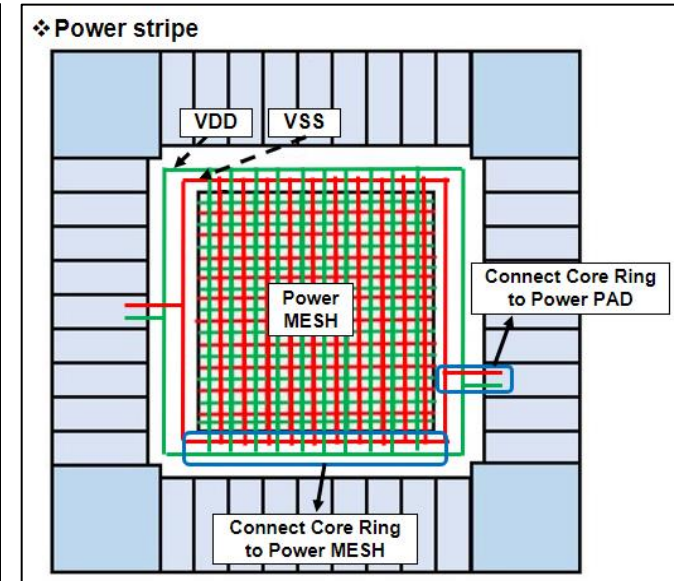
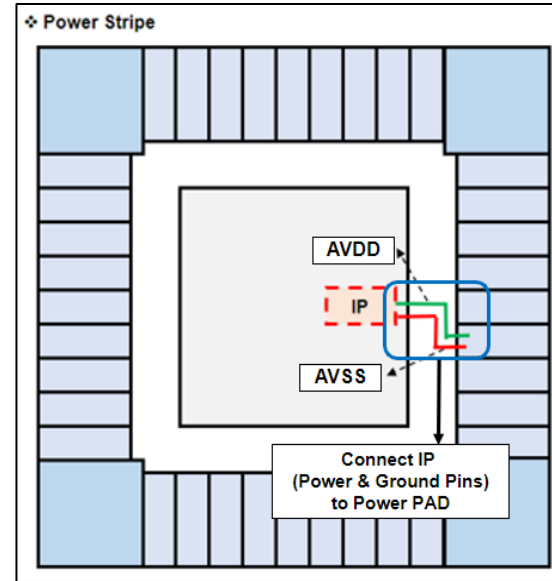
- VDD and VSS Rings are formed around the Macro and Core area



# Introduction to Powerplan : Terminologies & Definitions

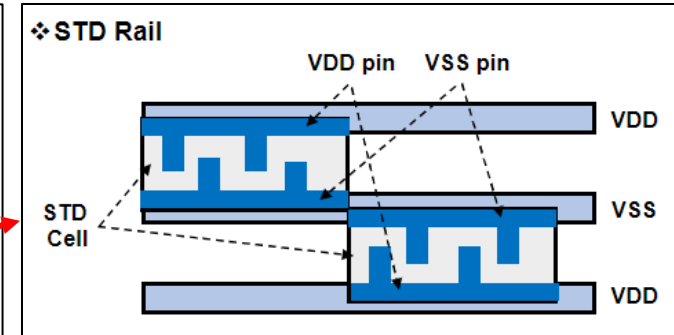
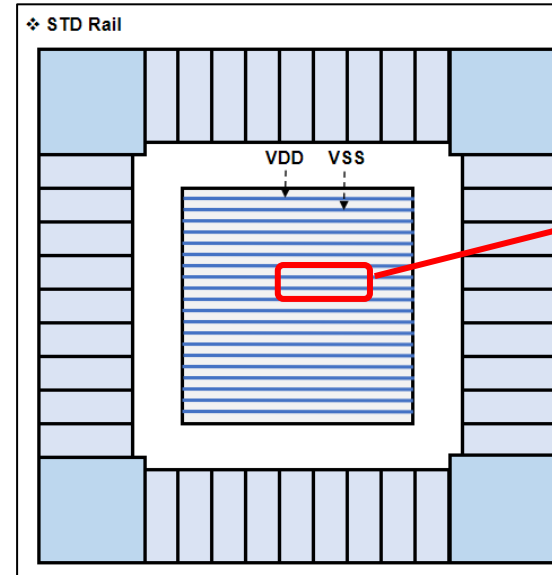
- **Stripes**

- Carries VDD and VSS around the chip
- Carries VDD and VSS from Rings across the chip
- Power Stripes are created in the Core Area to tap power from Core Rings to the core area



- **STD Rails (Special Route)**

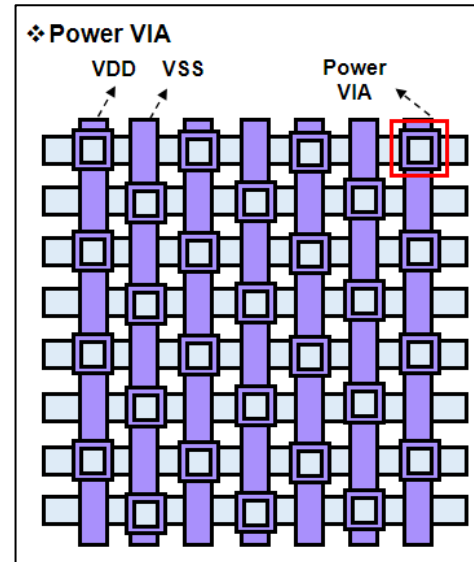
- Connect VDD and VSS to the STD cell
- STD cell Rails are created to tap power from Power stripes to STD cell Power/Ground Pins



# Introduction to Powerplan : Terminologies & Definitions

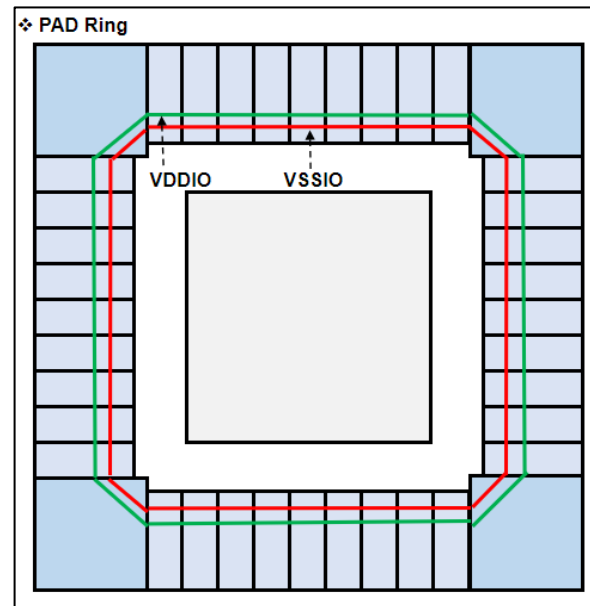
- **Power Vias**

- Insert all Power Vias between Ring & Grid, Grid & Rail and Vertical Grid & Horizontal Grid
- Power vias are used to carry VDD and VSS around the chip
- Power vias are used to connect VDD and VSS to the standard cells and net
- Power vias are inserted in defined power domains (Multi voltage design) and they are connected to lower metal layers using vias



- **PAD Rings**

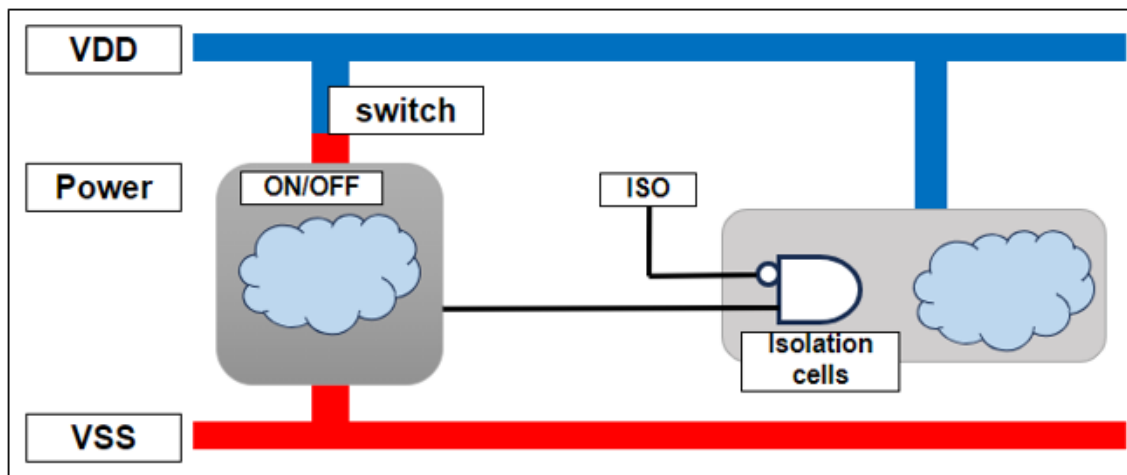
- In VLSI, the core of the chip is surrounded by a ring of PADs
- Connects IO Power and Ground Rings to PADs
- The minimum PAD to PAD distance and gaps (recommend abut) around the core are important factors to consider in the design of the PAD rings



# Introduction to Powerplan : Terminologies & Definitions

- **Isolation Cells (Low Power Design)**

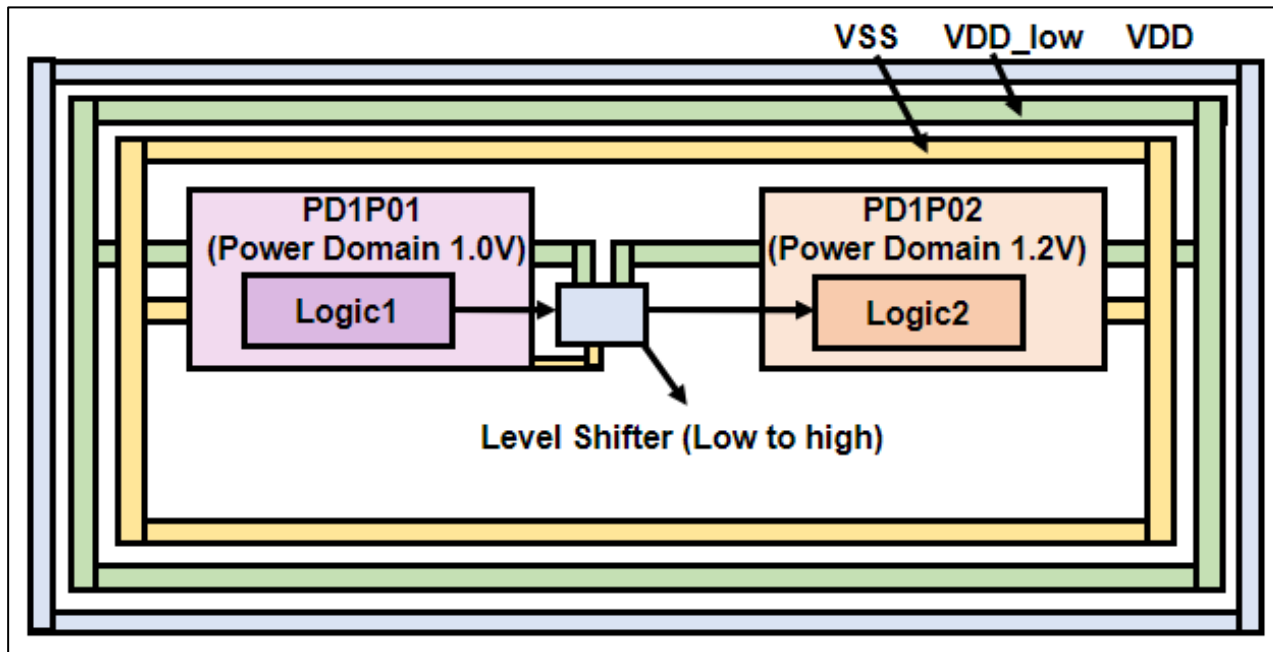
- ❖ Powered off domains do not drive their outputs anymore and these outputs become floating nodes. This could be a problem when other active domains gets these floating nodes as input
- Isolation cells (also called “clamps”) keep the turned off sub-block outputs at a predefined value. This is how the shut-down sub-block does not corrupt other active sub-block functionality
- Isolation cells are powered by a constant supply and drive 0, 1 or latch the old value of the turned off domain
- Isolation cells pass logic values during the normal mode of operation, but clamp it to some specified value when a control signal is asserted
- Isolation cell clamps the output of powered down block to a specified value ('0', '1', last)
  - Gate type clamp cells (AND, OR)
  - Transistor type clamp (pull-up, pull-down)



# Introduction to Powerplan : Terminologies & Definitions

- **Level Shifters (Low Power Design)**

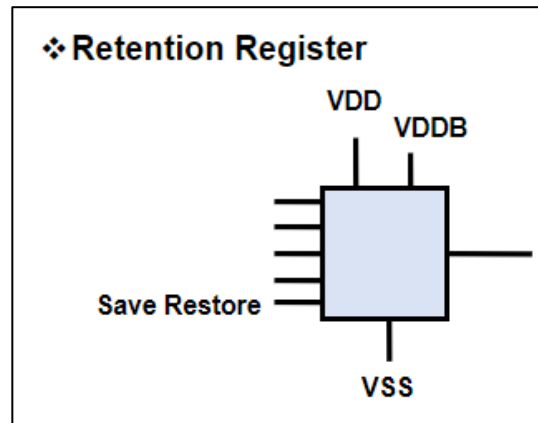
- Level shifters have the minimal functionality of a buffer.
- Necessary as most low-power designs have multi-voltage domains and/or employ dynamic voltage scaling.
- Level shifter swings a logic value in one voltage domain to the same logic value in another voltage domain
- 'Up' level shifter swings a logic value from a lower voltage domain to the same logic value in a higher voltage domain
- 'Down' level shifter swings a logic value from a higher voltage domain to the same logic value in a lower



# Introduction to Powerplan : Terminologies & Definitions

- **Retention Registers (Low Power Design)**

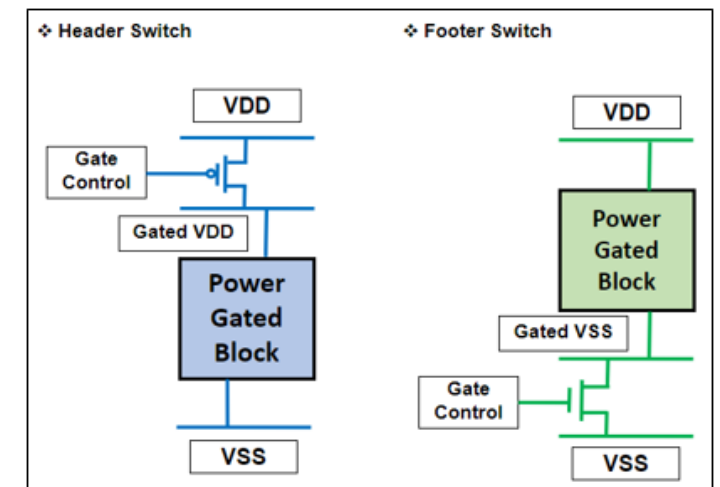
- In order to reduce power consumption, memories are shut down where their power domain is switched off or when they are not in use. Registers are corrupted when power is switched off
- Some memories need to retain their values for fast wake-up (For these memories, only the memory array stays powered on during the shut-down while the peripheral interfaces are powered off)
- Retention registers keep their previous active value after being shut down
- Retention registers save state information before a power domain is switched off and restore it when the power is turned back on
- Retention registers comprise of two circuits.
  - Standard register logic, supplied by primary power VDD
  - Shadow latch retention circuitry, with alternate supply VDDB



# Introduction to Powerplan : Terminologies & Definitions

- **Power Switches (Low Power Design)**

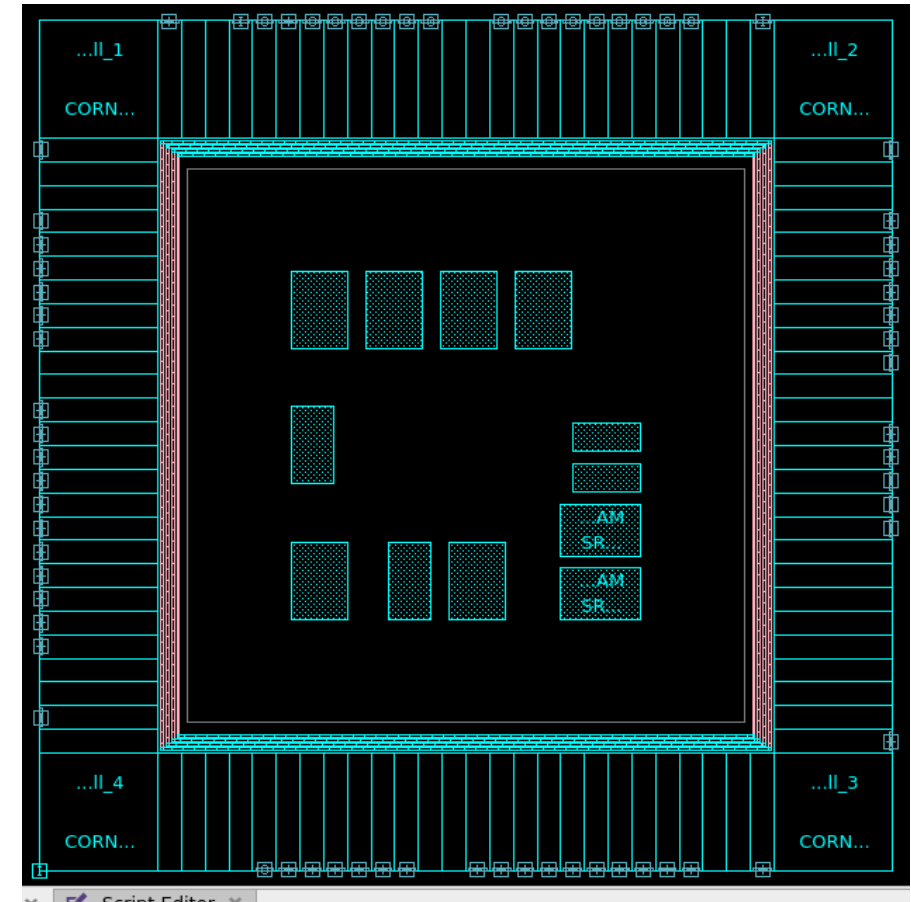
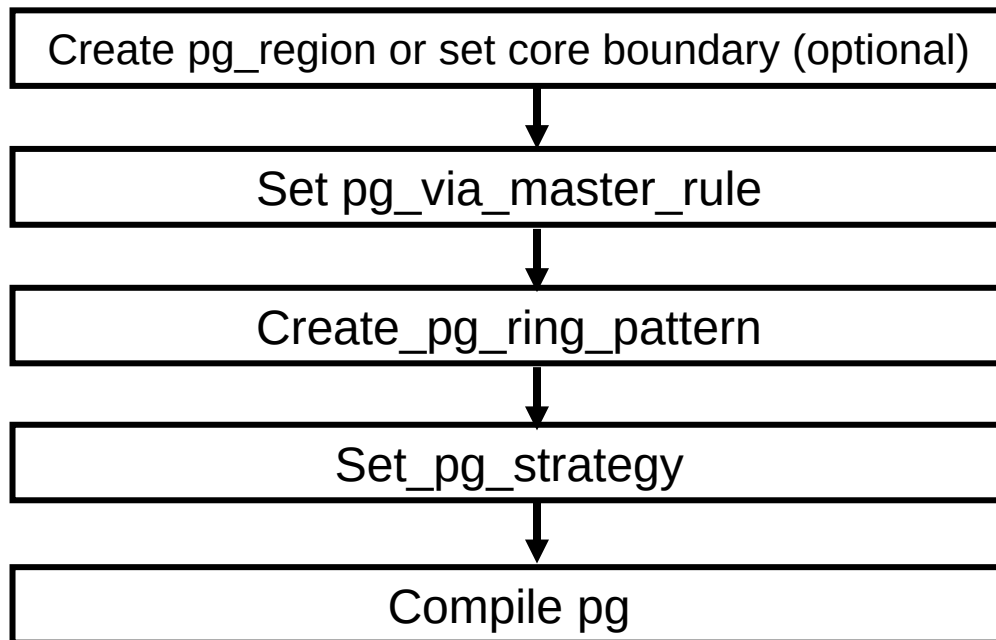
- Power switches are required to Gate the power supply of gated domain when not required
- Power switches are inserted in power mesh & supply to all gated domain cells will be through power switches (Single/few switches are not enough)
- There are 2 types of Power Switches
  - Header
    - The header switch is implemented by PMOS transistors to control VDD supply
    - PMOS transistor is less leaky than NMOS transistor of a same size (Header Switches turn off VDD and keep VSS on)
    - The disadvantage of the header switch is that PMOS has lower drive current than NMOS of a same size
  - Footer
    - The footer switch is implemented by NMOS transistor to control VSS supply
    - The advantage of footer switch is the high drive and hence smaller area
    - NMOS is leakier than PMOS and a Designs become more sensitive to ground noise



# Introduction to Powerplan : Create Core Ring

- **Create Core Ring**

- Core Ring is that VDD and VSS Rings are formed around the Core area
- Step of create Core Ring



# Introduction to Powerplan : Create Core Ring

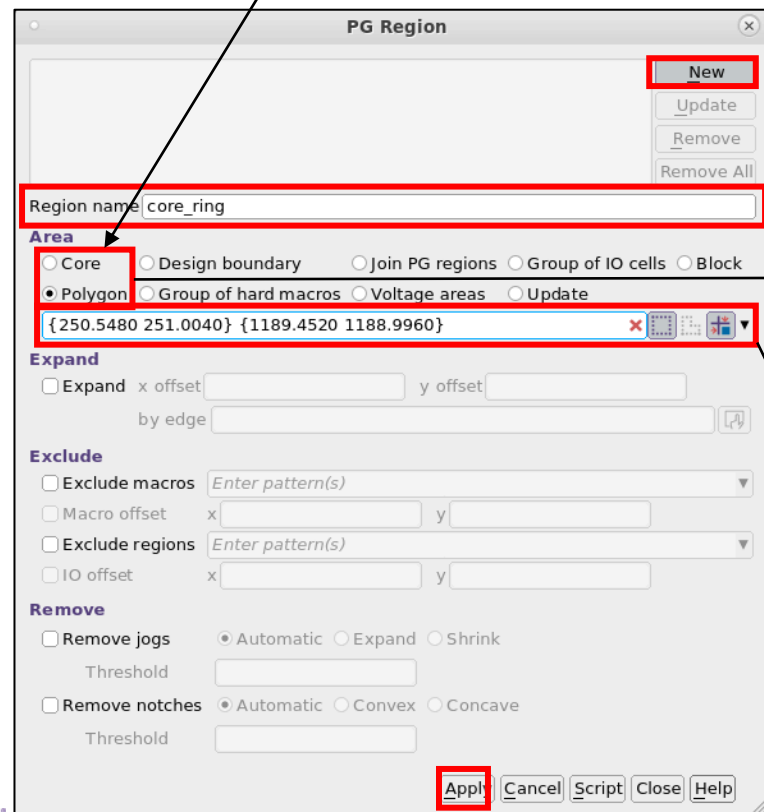
- **Create pg\_region or set core boundary (Use when you want to specify a Core Ring boundary)**

- set core\_bdry "[get\_attr [get\_core\_area] bbox]"

(You can use this command to get the bbox value of the core boundary)

```
icc2_shell> set core_bdry "[get_attr [get_core_area] bbox]"  
{250.5480 251.0040} {1189.4520 1188.9960}
```

- **Create\_polygon**



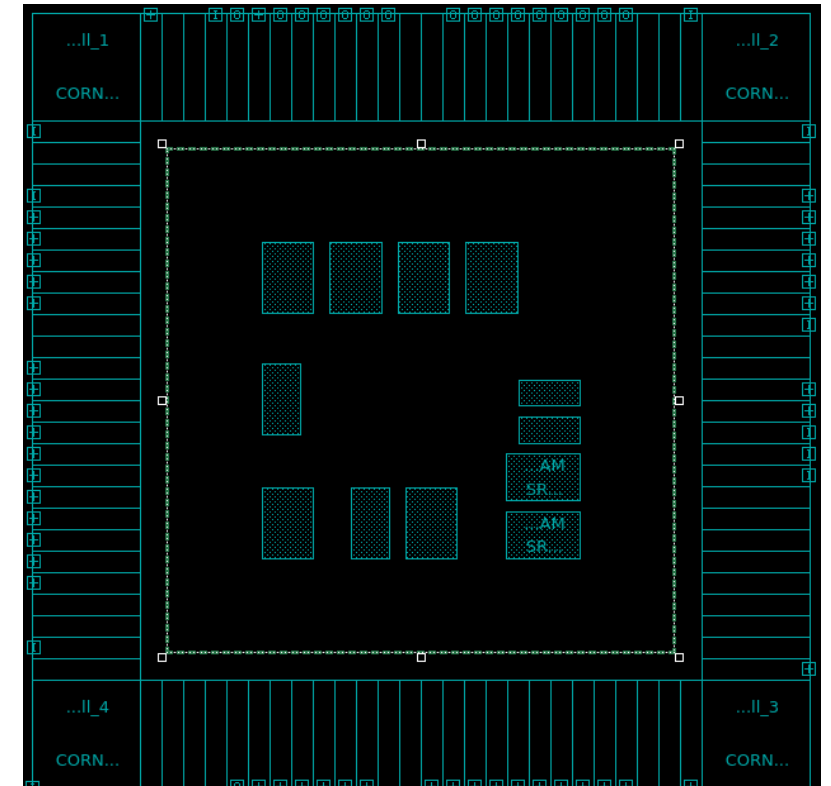
PG\_region\_name

Area  
Core - Core Area boundary  
Polygon – customization

Boundary  
(if you select polygon)

- ICC2 Command

```
create pg region CORE -polygon "{250.5480 251.0040} {1189.4520 1188.9960}"
```



# Introduction to Powerplan : Create Core Ring

- **set\_pg\_via\_master\_rule**

- Creates a new via master rule for the power and ground network
- The via rule is defined by the contact code, location of contact code inside the intersection box, cut spacing, multiplication, and so on

- **set\_pg\_via\_master\_rule**

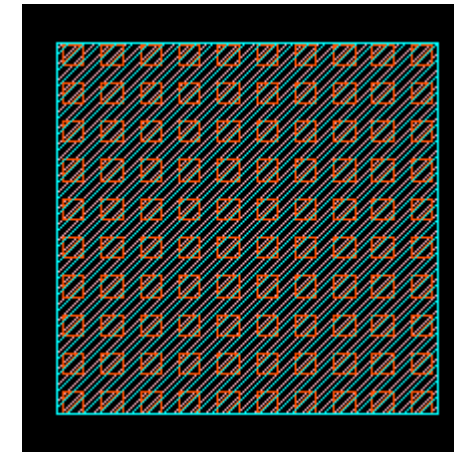


via\_master\_rule name

via\_array\_dimension  
(column x row = 10 x 10)

- ICC2 Command

```
set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10}
```



<10 x 10 via>

# Introduction to Powerplan : Create Core Ring

- **create\_pg\_ring\_pattern**

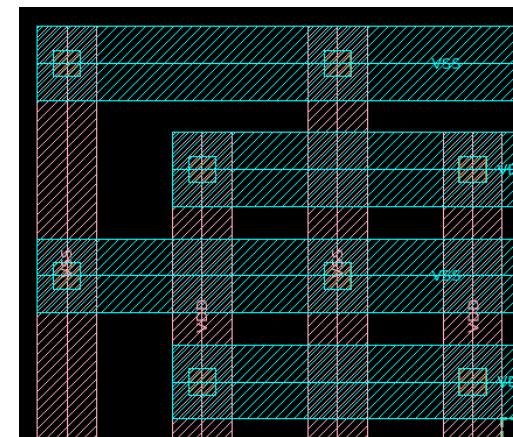
- The pattern created by this command is associated with a region or area of the design by using the set\_pg\_strategy command
- The actual ring is inserted into the design by the compile\_pg command
- **Create\_pg\_ring\_pattern**

- **ICC2 Command**

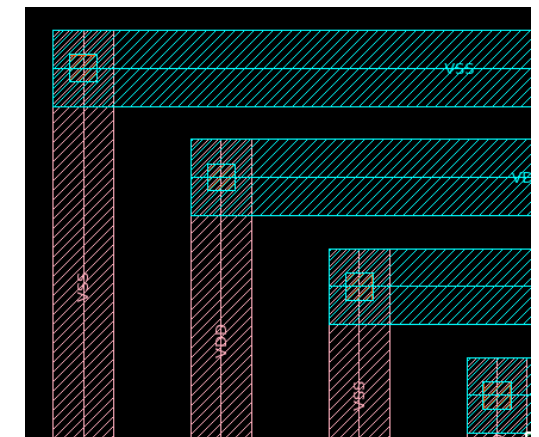
```
create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 \  
-horizontal_width {5} -horizontal_spacing {2.188} \  
-vertical_layer M6 -vertical_width {4} -vertical_spacing {5.12} \  
-corner_bridge false \  
-via_rule { {intersection: adjacent} {via_master : pgvia_10x10} }
```

The screenshot shows the 'create\_pg\_ring\_pattern' dialog box with various fields and annotations:

- Pattern name:** \*pattern\_name core\_ring\_pattern
- Horizontal & Vertical Layer width:** horizontal\_width 5, vertical\_width 4
- Horizontal & Vertical Layer:** horizontal\_layer M7, vertical\_layer M6
- Horizontal & Vertical Layer Spacing:** horizontal\_spacing 2.188, vertical\_spacing 5.12
- Corner bridge:** corner\_bridge false
- Via\_rule:** via\_rule ent} {via\_master : pgvia\_10x10}
- Annotations:** Red boxes highlight the fields for pattern name, widths, layers, spacings, corner bridge, and via rule. Arrows point from these fields to labels on the right.
- Buttons:** OK, Cancel, Apply, Script, Default, Help
- Footer:** (Fields with \* are required)



<corner bridge true>



<corner bridge false>

## {intersection : adjacent} option

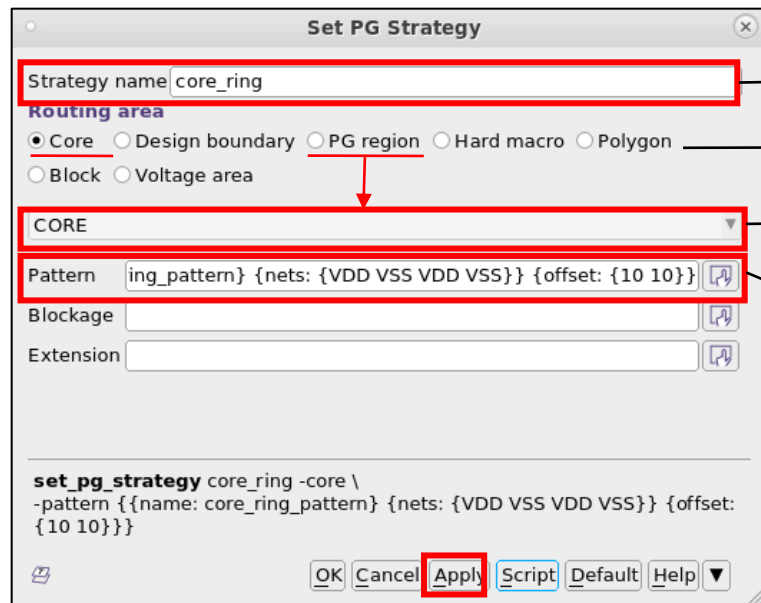
Intersection is the keyword, and adjacent refers to all intersections between any two ring segments in different directions only in adjacent layers

# Introduction to Powerplan : Create Core Ring

- **set\_pg\_strategy**

- Specifies the power ground network creation strategy

- **set\_pg\_strategy**



Strategy name

PG\_region name

- ICC2 Command

```
set_pg_strategy core_ring \  
-core \  
-pattern {{name: core_ring_pattern}} \  
{nets: {VDD VSS VDD VSS}} {offset: {10 10}}
```

**-core option**

Uses the core area as the power ground region

**-pg region option**

Uses the power ground regions created by the create\_pg\_region command as the power ground regions

**name option**

pg\_pattern name

**Nets option**

Power & ground net name

**Offset option**

The offset keyword followed by {x\_offset y\_offset} specifies the horizontal and vertical offsets when creating the first pattern

# Introduction to Powerplan : Create Core Ring

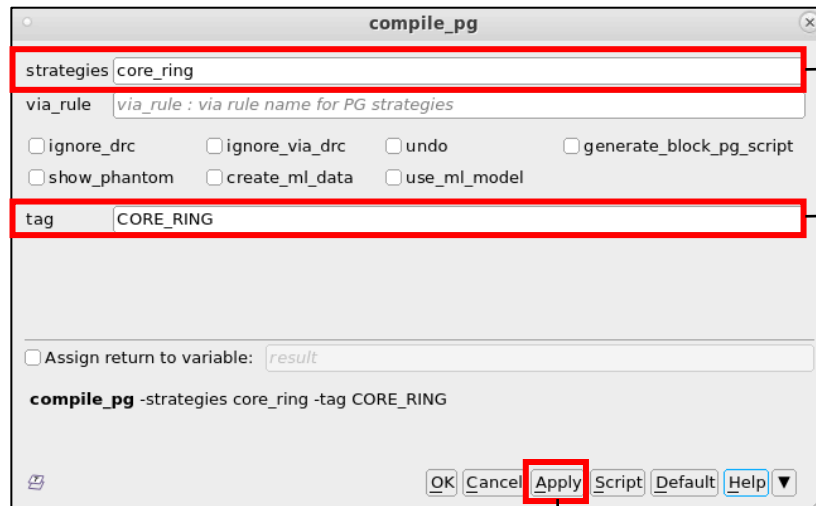
- **compile\_pg**

- Creates a power and ground network including straps, rings, macro connections, standard cell connections and via connections based on the specified power ground strategies

- **compile\_pg**

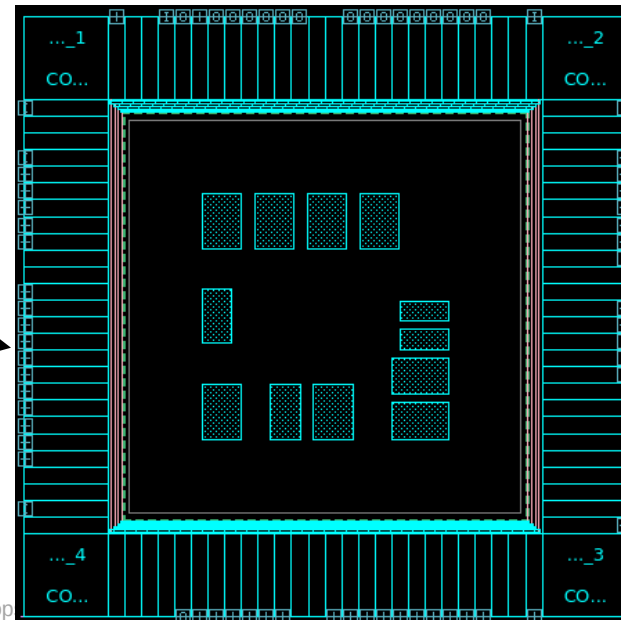
- ICC2 Command

```
compile_pg -strategies {core_ring} -tag {CORE_RING}
```



Strategy name

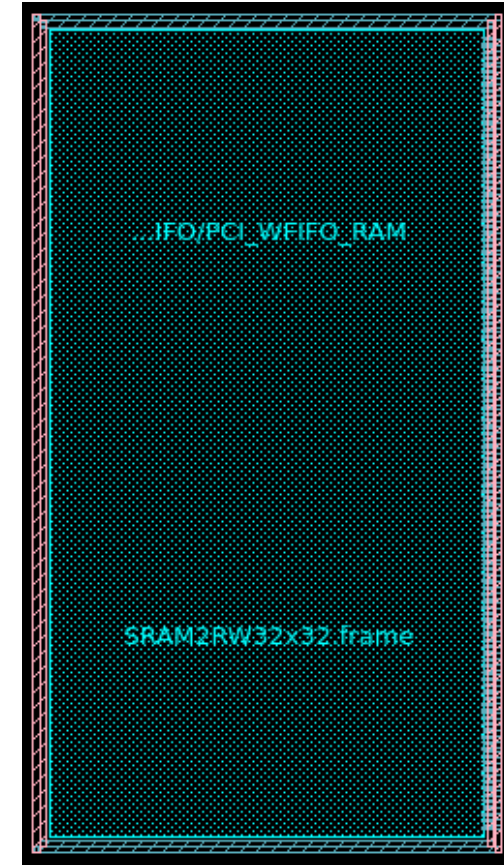
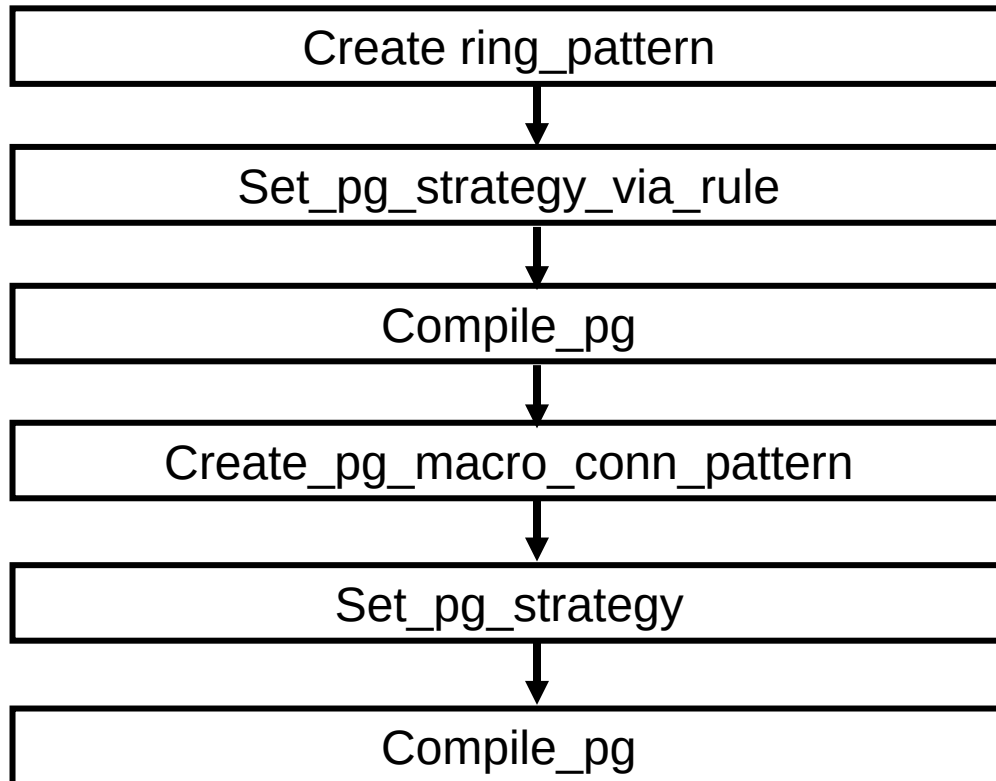
-tag option  
Specifies a tag name to assign to all new vias and shapess created by this command



# Introduction to Powerplan : Create Macro Ring

- **Create Macro Ring**

- Macro Ring is VDD and VSS Rings are formed around the Macro area
- Step of create Macro Ring



# Introduction to Powerplan : Create Macro Ring

- **create\_pg\_ring\_pattern**

- The pattern created by this command is associated with a region or are a of the design by using the set\_pg\_strategy command
- The actual ring is inserted into the design by the compile\_pg command

- ICC2 Command

```
create_pg_ring_pattern Macro_ring -horizontal_layer M5 \  
-horizontal_width {1} -vertical_layer M6 -vertical_width {1} -corner_bridge false
```

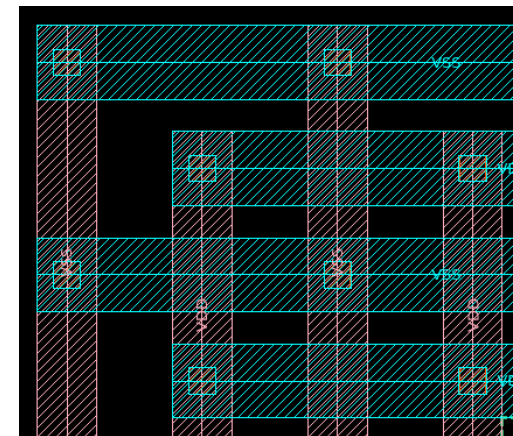
- **Create\_pg\_ring\_pattern**

The screenshot shows the 'create\_pg\_ring\_pattern' dialog box with several fields highlighted by red boxes and arrows pointing to labels:

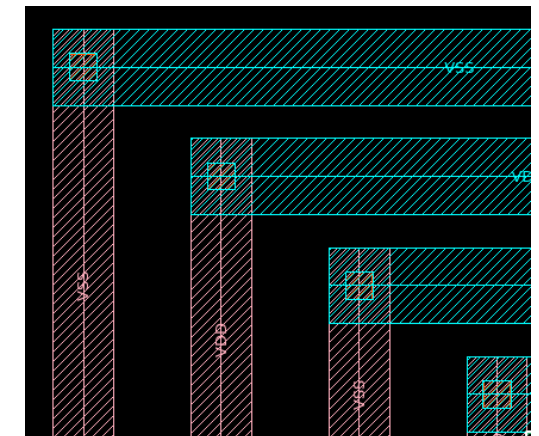
- \*pattern\_name**: Macro\_ring (labeled "Pattern name")
- horizontal\_width**: 1 (labeled "Horizontal & Vertical Layer width")
- vertical\_width**: 1 (labeled "Horizontal & Vertical Layer width")
- horizontal\_layer**: M5 (labeled "Horizontal & Vertical Layer")
- vertical\_layer**: M6 (labeled "Horizontal & Vertical Layer")
- corner\_bridge**: false (labeled "Corner bridge")

Other fields include: horizontal\_spacing, vertical\_spacing, side\_width, side\_layer, side\_spacing, via\_rule, track\_alignment, and parameters. A note at the bottom states: "(Fields with \* are required)".

At the bottom of the dialog, there is a checkbox for "Assign return to variable:" with the value "result". Below this, the command is displayed: `create_pg_ring_pattern -horizontal_width 1 \  
-vertical_width 1 -horizontal_layer M5 \  
-vertical_layer M6 \  
-corner_bridge false Macro_ring`. The "Apply" button is highlighted with a red box.



<corner bridge true>



<corner bridge false>

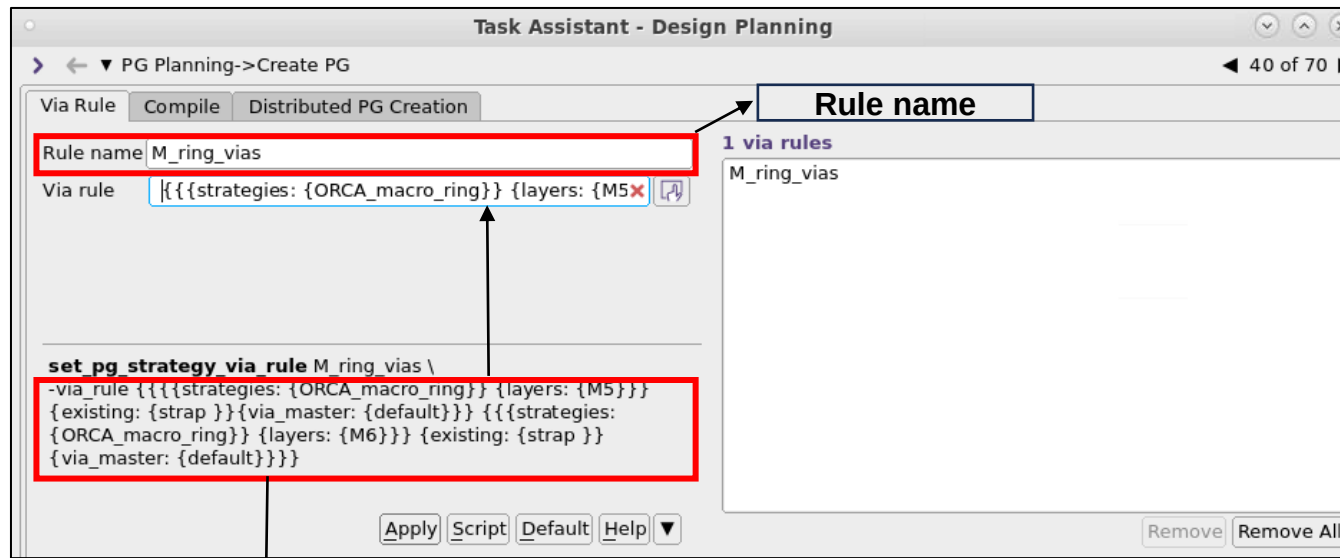
# Introduction to Powerplan : Create Macro Ring

- **set\_pg\_strategy\_via\_rule**

- Specifies the via rule between strategies and collections of existing shapes
- **set\_pg\_strategy\_via\_rule**

- **ICC2 Command**

```
set_pg_strategy_via_rule M ring vias -via_rule { \
    {{strategies: {ORCA_macro_ring}} {layers: {M5}}} \
    {existing: {strap }} {via_master: {default}}} \
    {{strategies: {ORCA_macro_ring}} {layers: {M6}}} \
    {existing: {strap }} {via_master: {default}}} \
}
```



- **-via\_rule option**

### Strategies –name of strategy

## Layers – Horizontal & Vertical Layers

**Existing** – specified by shape type (strap, macro\_conn, follow\_pin, std\_conn)

**Via\_master** – via master rules are defined by set\_pg\_via\_master\_rule command (NIL can be used as a keyword to indicate that no via is created)

# Introduction to Powerplan : Create Macro Ring

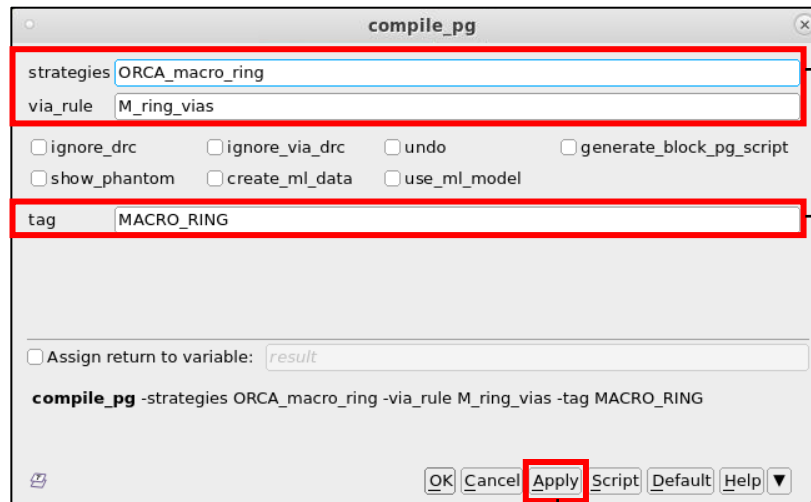
- **compile\_pg**

- Creates a power and ground network including straps, rings, macro connections, standard cell connections and via connections based on the specified power ground strategies

- ICC2 Command

```
compile_pg -strategies {ORCA_macro_ring} -via_rule {M_ring_vias} \
          -tag {MACRO_RING}
```

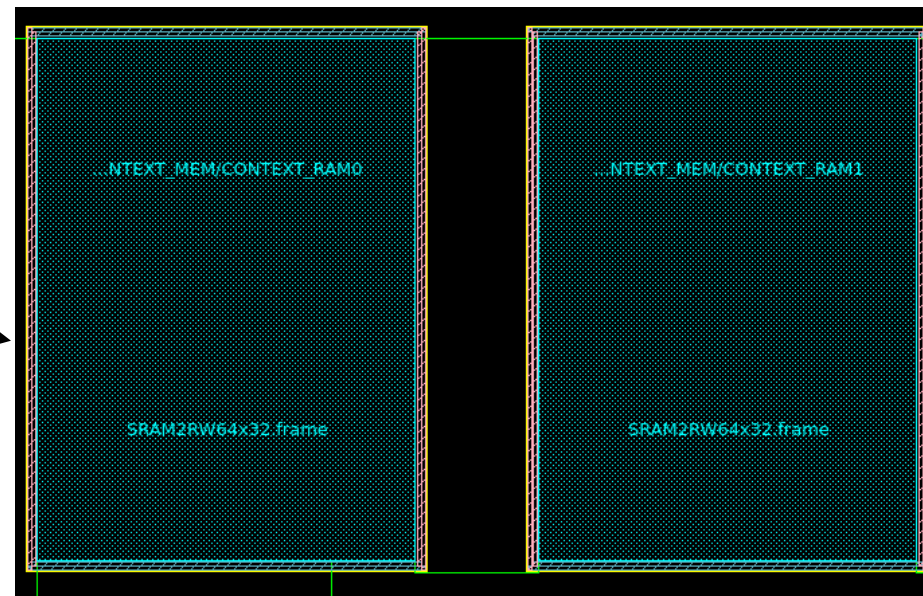
- **compile\_pg**



Strategy & Via\_rule name

-tag option

Specifies a tag name to assign to all new vias and shapess created by this command



# Introduction to Powerplan : Create Macro Ring

- **create\_pg\_macro\_conn\_pattern**

- The pattern created by this command is associated with a region or area of the design by using the set\_pg\_strategy command
- The actual ring is inserted into the design by the compile\_pg command

- **ICC Command**

```
create_pg_macro_conn_pattern Macro_ring_pin -pin_conn_type scattered_pin \
                                         -layers {M5 M6}
```

- **Create\_pg\_macro\_conn\_pattern**

The screenshot shows the 'create\_pg\_macro\_conn\_pattern' dialog box. Three fields are highlighted with red rectangles and labeled with arrows: '\*pattern\_name' is labeled 'Pattern name', '\*pin\_conn\_type' is labeled 'Pin\_conn\_type', and 'layers' is labeled 'Pin Layers'. The 'layers' field contains the text 'M5 M6'. At the bottom, the 'Assign return to variable' checkbox is checked with 'result' in the adjacent text field. The 'Apply' button is highlighted with a red rectangle. Below the dialog box, the generated command is shown: `create_pg_macro_conn_pattern \`  
`-pin_conn_type scattered_pin \`  
`-layers {M5 M6} Macro_ring_pin`

## **scattered\_pin**

The scattered\_pin keyword refers to the irregular pin shapes scattered inside the macro; extension straps will be created to connect scattered pins to the external power network

## **long\_pin**

The long\_pin keyword refers to long pins in the macro; orthogonal straps will be created for connection

## **ring\_pin**

The ring\_pin keyword refers to internal ring pins inside the macro; finger connections will be created to connect the ring pin to a strap or external macro ring in the external power network

# Introduction to Powerplan : Create Macro Ring

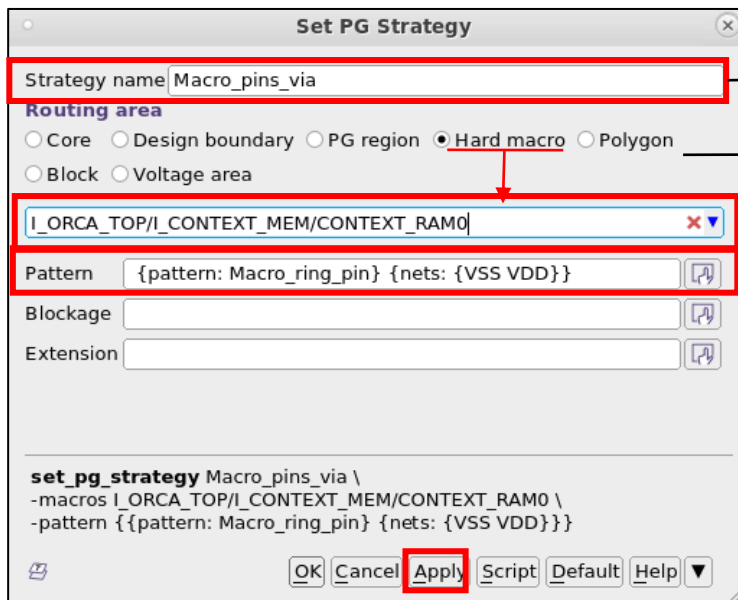
- **set\_pg\_strategy**

- Specifies the power ground network creation strategy

- **set\_pg\_strategy**

- ICC2 Command

```
set_pg_strategy Macro_pins_via -macros "I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0" \
-pattern { {pattern: Macro_ring_pin} {nets: {VSS VDD}} }
```



Strategy name

**-macro option**

Uses the specified Macros as the power ground region

Hard Macro name

**name option**

pg\_pattern name

**Nets option**

Power & ground net name

# Introduction to Powerplan : Create Macro Ring

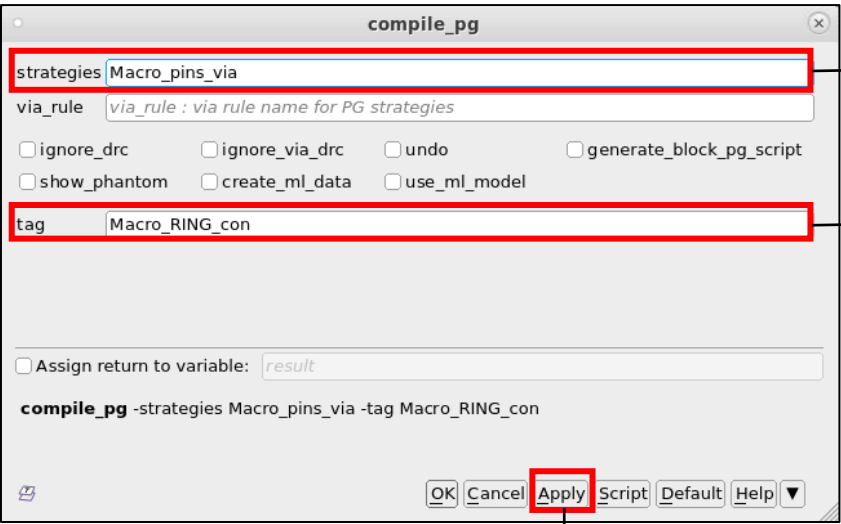
- **compile\_pg**

- Creates a power and ground network including straps, rings, macro connections, standard cell connections and via connections based on the specified power ground strategies

- **compile\_pg**

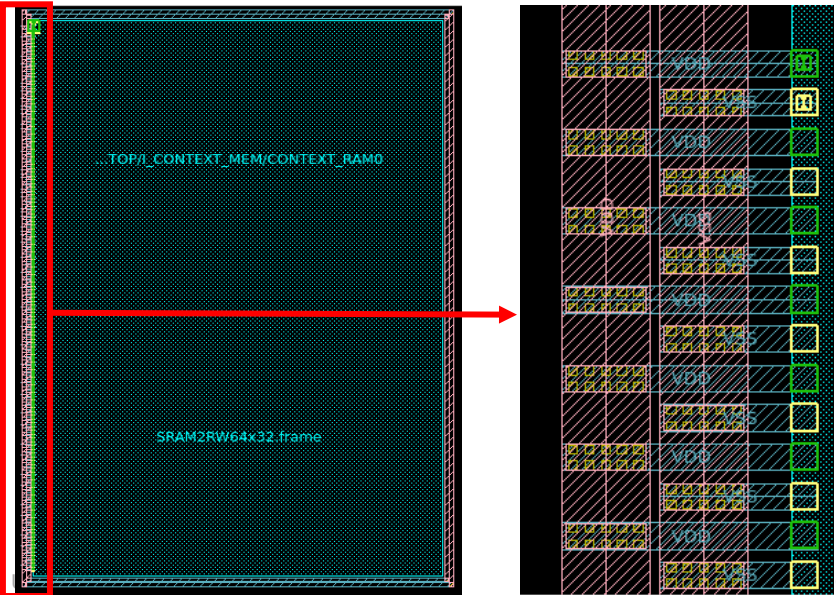
- ICC2 Command

```
compile_pg -strategies {Macro_pins_via} -tag {Macro_RING_con}
```



Strategy name

-tag option  
Specifies a tag name to assign to all new vias and shapess created by this command

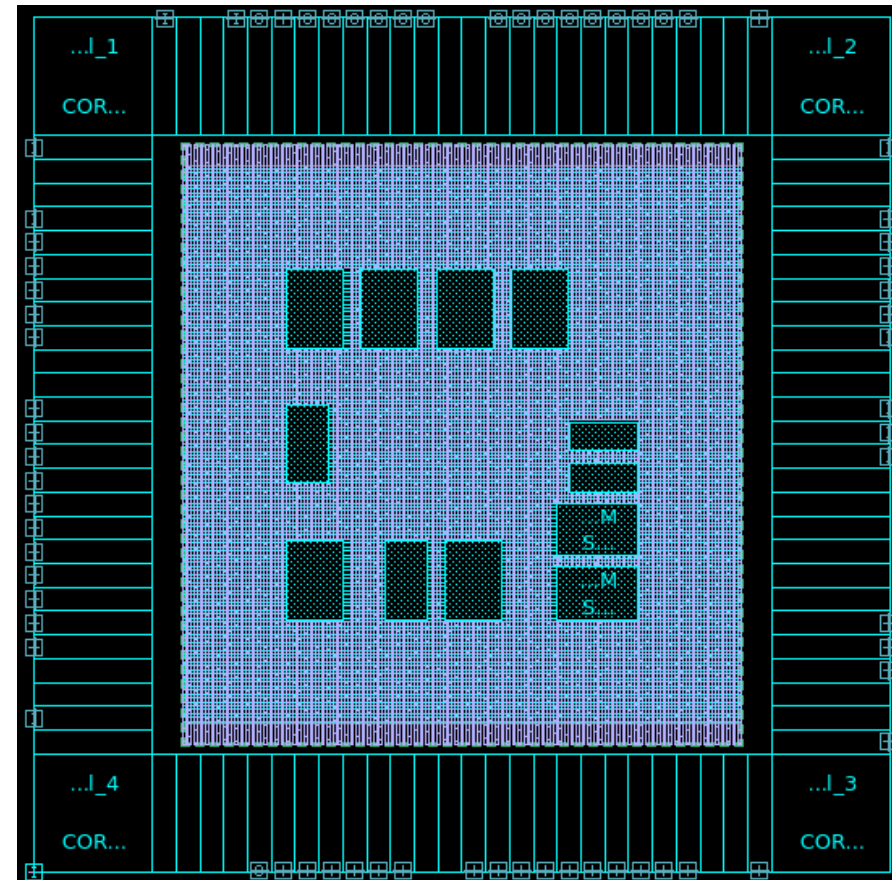
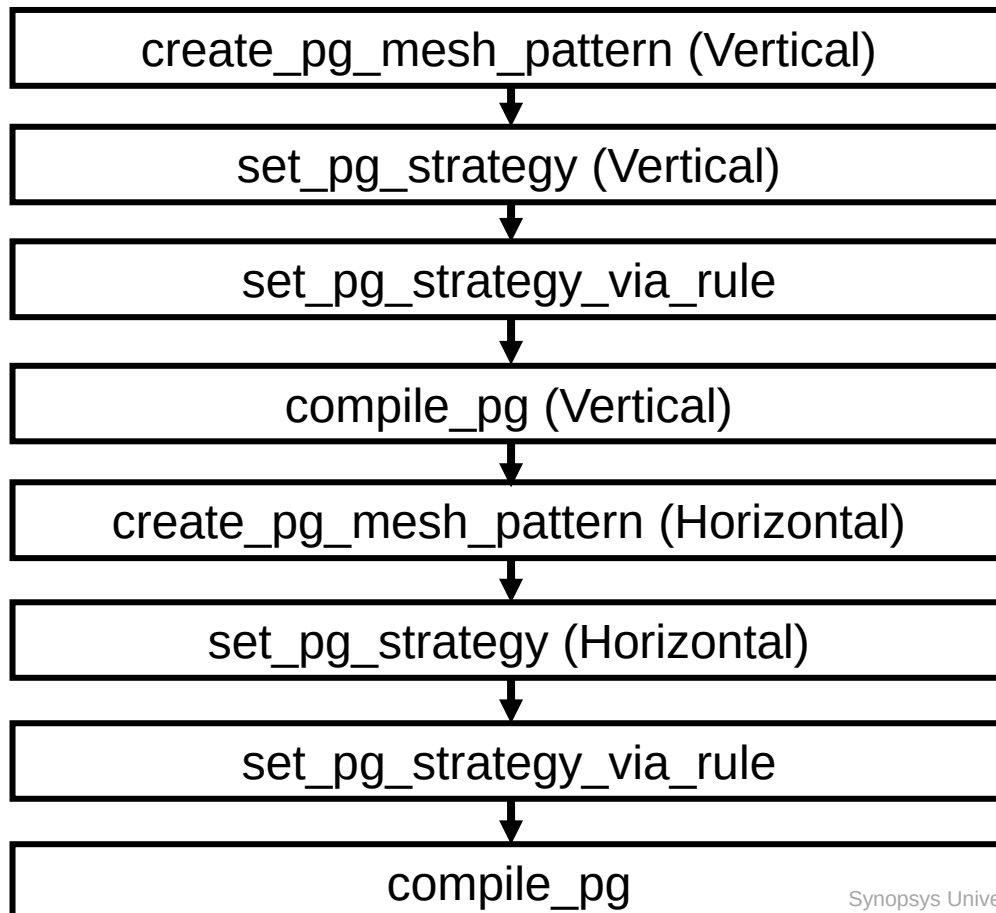


Connecting Macro's power & ground (staggered)pin to Macro Ring

# Introduction to Powerplan : Create Power MESH

- **Create Power MESH (Horizontal & Vertical)**

- Power MESH created in the core area to tap power from core rings to the core area (Carries power & ground from rings across the chip)
- Step of create Power MESH



# Introduction to Powerplan : Create Power MESH (Vertical)

- **create\_pg\_mesh\_pattern**

- Creates a power ground mesh pattern
- The mesh pattern can be instantiated in a design set\_pg\_strategy command to create power ground mesh

- **Create\_pg\_mesh\_pattern**

- ICC Command :

```
create_pg_mesh_pattern METAL8_PG_PATTERN -layers {{vertical_layer: M8} \
    {width: 4.64} \
    {spacing: 5.088} \
    {pitch: 19.456} \
    {trim: false}} \
    -parameters {4.64 19.456 5.088}
```

create\_pg\_mesh\_pattern

\*(positional:1) METAL8\_PG\_PATTERN

\*layers {vertical\_layer: M8} {width: 4.64} {spacing: 5.088}

parameters 4.64 19.456 5.088

via\_rule via\_rule : via rule between straps in this PG me...  
(Fields with \* are required)

☐ Assign return to variable: result

create\_pg\_mesn\_pattern \  
-layers {{vertical\_layer: M8} {width: 4.64} {spacing: 5.088}  
{pitch: 19.456} {trim: false}} \  
-parameters {4.64 19.456 5.088} METAL8\_PG\_PATTERN

OK Cancel Apply Script Default Help

Pattern name

Vertical Layer (M8)  
(width spacing pitch trim)

Specifies the list of parameters  
(width pitch spacing)

Specifies the layer configuration for this mesh pattern. Each layer is specified by a *layer\_expr* expression in the following syntax:

```
{horizontal_layer | vertical_layer: layer_name | @var}  
{width: minimum | value_list | @var}  
{spacing: interleaving | minimum | value_list | @var}  
{offset: offset_value | @var}  
{pitch: pitch_value | @var}  
{trim : true | false | @var}  
{track_alignment : track | half_track | auto | @var}  
{mask : mask_one | mask_two | mask_three | mask_one_mask_two_alternate |  
    mask_two_mask_one_alternate | @var}  
{mask_constraint : constraint_name}
```

# Introduction to Powerplan : Create Power MESH (Vertical)

- **set\_pg\_strategy**

- Specifies the power ground network creation strategy
- **set\_pg\_strategy**

- **ICC2 Command**

```
set_pg_strategy METAL8_MAPPING_STRATEGY \  
-pg_regions MESH \  
-pattern \  
  {name: METAL8_PG_PATTERN} \  
  {nets: {VDD VSS}} \  
  {parameters: {4.64 19.456 5.088}} \  
  {offset_start: {0 0}} \  
-blockage {placement_blockages: all} \  
-extension {{stop: core_boundary} {direction: L B R T }}
```

Set PG Strategy

Strategy name: METAL8\_MAPPING\_STRATEGY

Routing area

☒ Core ☐ Design boundary ☐ PG region ☐ Hard macro ☐ Polygon

☐ Block ☐ Voltage area

Enter pattern(s)

Pattern: rameters: {4.64 19.456 5.088} {offset\_start: {0 0}}

Blockage: placement\_blockages: all

Extension: {stop: core\_boundary} {direction: L B R T}

set\_pg\_strategy METAL8\_MAPPING\_STRATEGY -core \  
-pattern {{name: METAL8\_PG\_PATTERN} {nets: {VDD VSS}} \  
{parameters: {4.64 19.456 5.088}} {offset\_start: {0 0}} \  
-blockage {placement\_blockages: all} \  
-extension {{stop: core\_boundary} {direction: L B R T }} \  
OK Cancel Apply Script Default Help

## Strategy name

### -core option

Uses the core area as the power ground region

### -pattern option

- Specifies the power ground pattern to be instantiated in this strategy
- Pattern can be wire, composite, mesh, ring, macro connection, std cell rail connection or special pattern
- Only one pattern can be specified and this is a required option

```
{name: pattern_name}  
{nets: real_net_list}  
{parameters: var_list}  
{offset: {x_offset y_offset}}  
{offset_start: boundary | {x y}}
```

### -blockage option

- Specifies the routing blockage in the power ground network
- Blockages are separated by brace and contain the keywords nets;, layers;, and a target specification as follows;

```
{{nets: nets}  
{side : id_list}  
{layers : layer_list}  
{direction: directions}  
{stop: stop_target}}
```

### -Extension option

- Specifies how a group of power and ground nets should extend beyond the power grid
- Each specification contains the keywords nets, side, layers, direction and stop as follows:

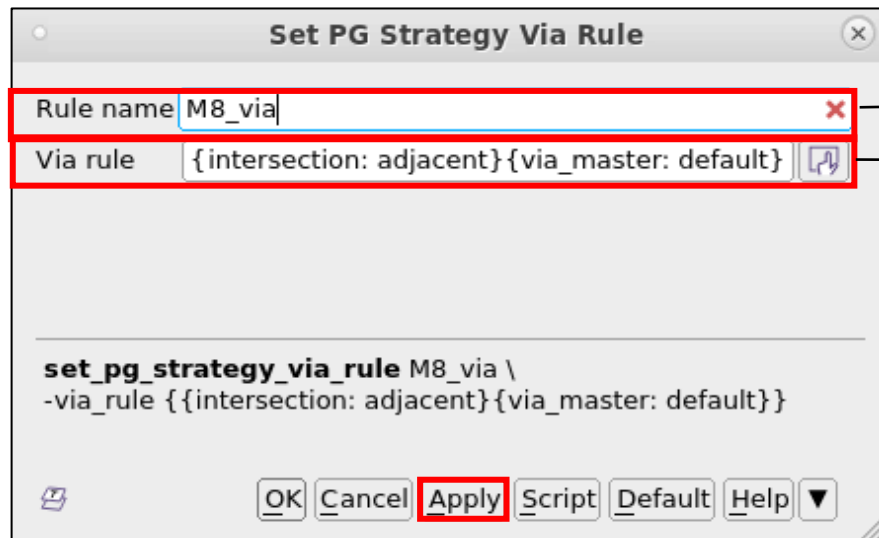
# Introduction to Powerplan : Create Power MESH (Vertical)

- **set\_pg\_strategy\_via\_rule**

- Specifies the via rule between strategies and collections of existing shapes
- **set\_pg\_strategy\_via\_rule**

- **ICC2 Command**

```
set_pg_strategy_via_rule M8_via \  
-via_rule {{intersection: adjacent} {via_master:default}}
```



**Via rule name**

**-via rule option**

Specifies the via rule via\_rule\_spec between strategies, and also between the strategies and existing shapes

```
{intersection : all}      {via_def} |  
{intersection : adjacent} {via_def} |  
list_of_filter_rules
```

- "{intersection : all}", where intersection is the keyword and all refers to all intersections between any two orthogonal shapes including straps, rings, standard cell rails created by strategies or any existing shapes in any different layers

- The second method is "{intersection : adjacent}", where intersection is the keyword and adjacent refers to all intersections between any two orthogonal shapes only in adjacent layers

```
{via_master : via_master_list}
```

- where via\_master is the keyword for both via masters and via master rules(Via masters include the via contact code defined in the technology file or user-defined via cells)

# Introduction to Powerplan : Create Power MESH (Horizontal)

- **create\_pg\_mesh\_pattern**

- Creates a power ground mesh pattern
- The mesh pattern can be instantiated in a design set\_pg\_strategy command to create power ground mesh

- **Create\_pg\_mesh\_pattern**

create\_pg\_mesh\_pattern

\*(positional:1) METAL7\_PG\_PATTERN

\*layers {width: 1.104} {spacing: 5.584} {pitch: 13.376} {trim: false}

parameters 1.104 13.376 5.584

via\_rule via\_rule : via rule between straps in this PG mes... (Fields with \* are required)

☐ Assign return to variable: result

create\_pg\_mesn\_pattern \

-layers {{horizontal\_layer: M7} {width: 1.104}{spacing: 5.584} {pitch: 13.376} {trim: false}} \

-parameters {1.104 13.376 5.584} METAL7\_PG\_PATTERN

OK Cancel **Apply** Script Default Help

Pattern name

Vertical Layer (M7)  
(width spacing pitch trim)

Specifies the list of parameters  
(width pitch spacing)

Specifies the layer configuration for this mesh pattern. Each layer is specified by a *layer\_expr* expression in the following syntax:

```
{horizontal_layer | vertical_layer: layer_name | @var}
{width: minimum | value_list | @var}
{spacing: interleaving | minimum | value_list | @var}
{offset: offset_value | @var}
{pitch: pitch_value | @var}
{trim : true | false | @var}
{track_alignment : track | half_track | auto | @var}
{mask : mask_one | mask_two | mask_three | mask_one_mask_two_alterate |
mask_two_mask one_alterate | @var}
{mask_constraint : constraint_name}
```

- **Command**

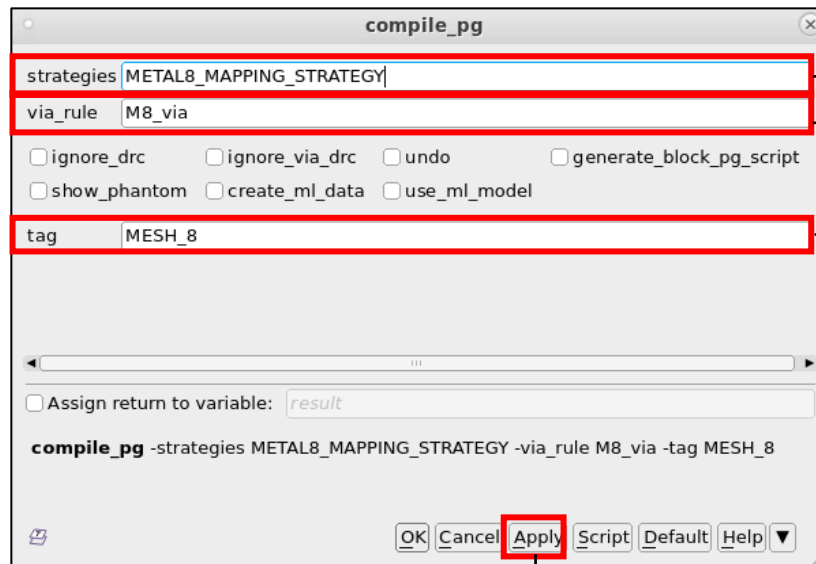
```
create_pg_mesh_pattern METAL7_PG_PATTERN -layers {{horizontal_layer: M7} \
{width: 1.104} \
{spacing: 5.584} \
{pitch: 13.376} \
{trim: false}} \
-parameters {1.104 13.376 5.584}
```

# Introduction to Powerplan : Create Power MESH (Vertical)

- **compile\_pg**

- Creates a power and ground network including straps, rings, macro connections, standard cell connections and via connections based on the specified power ground strategies

- **compile\_pg**

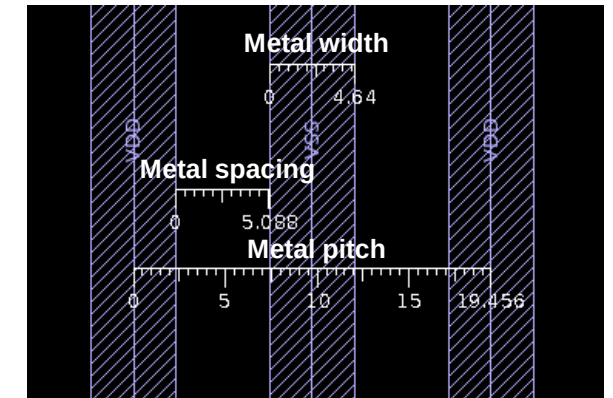
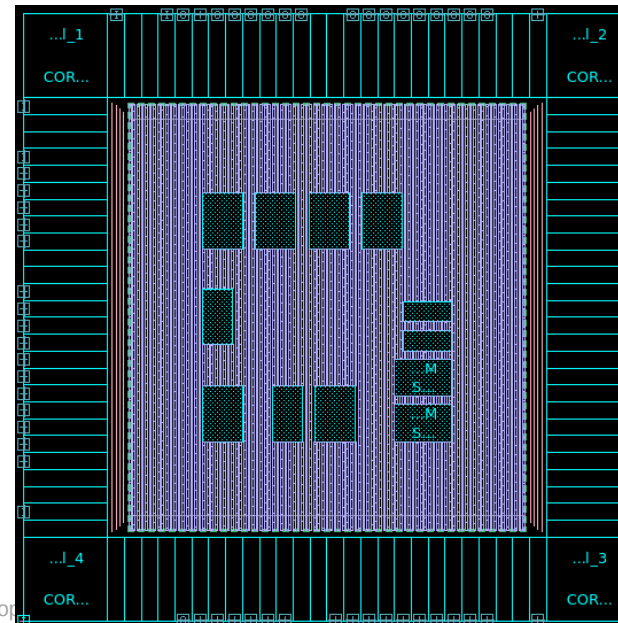


Strategy name

Via rule name

-tag option

Specifies a tag name to assign to all new vias and shapess created by this command



- ICC2 Command

```
compile_pg -strategies {METAL8_MAPPING_STRATEGY} -via_rule {M8_via} -tag MESH_8
```

# Introduction to Powerplan : Create Power MESH (Horizontal)

- **set\_pg\_strategy**

- Specifies the power ground network creation strategy
- **set\_pg\_strategy**

- **ICC2 Command**

```
set_pg_strategy METAL7_MAPPING_STRATEGY \  
-core \  
-pattern {{name: METAL7_PG_PATTERN} \  
  {nets: {VDD VSS}} \  
  {parameters: {1.104 13.376 5.584}} \  
  {offset_start: {0 0}}} \  
-blockage {placement_blockages: all} \  
-extension {{stop: core boundary} {direction: L B R T }}
```

Set PG Strategy

Strategy name: METAL7\_MAPPING\_STRATEGY

Routing area

☒ Core ☐ Design boundary ☐ PG region ☐ Hard macro ☐ Polygon

☐ Block ☐ Voltage area

Enter pattern(s)

Pattern: {parameters: {1.104 13.376 5.584}} {offset\_start: {0 0}}

Blockage: placement\_blockages: all

Extension: {stop: core boundary} {direction: L B R T}

set\_pg\_strategy METAL7\_MAPPING\_STRATEGY -core \  
-pattern {{name: METAL7\_PG\_PATTERN} {nets: {VDD VSS}} \  
{parameters: {1.104 13.376 5.584}} {offset\_start: {0 0}}} \  
-blockage {placement\_blockages: all} \  
-extension {{stop: core boundary} {direction: L B R T }}

OK Cancel Apply Script Default Help

## Strategy name

### -core option

Uses the core area as the power ground region

### -pattern option

- Specifies the power ground pattern to be instantiated in this strategy
- Pattern can be wire, composite, mesh, ring, macro connection, std cell rail connection or special pattern
- Only one pattern can be specified and this is a required option

```
{name: pattern_name}  
{nets: real_net_list}  
{parameters: var_list}  
{offset: {x_offset y_offset}}  
{offset_start: boundary | {x y}}
```

### -blockage option

- Specifies the routing blockage in the power ground network
- Blockages are separated by brace and contain the keywords nets;, layers;, and a target specification as follows;

```
{{nets: nets}  
 {side : id_list}  
 {layers : layer_list}  
 {direction: directions}  
 {stop: stop_target}}
```

### -Extension option

- Specifies how a group of power and ground nets should extend beyond the power grid
- Each specification contains the keywords nets, side, layers, direction and stop as follows:

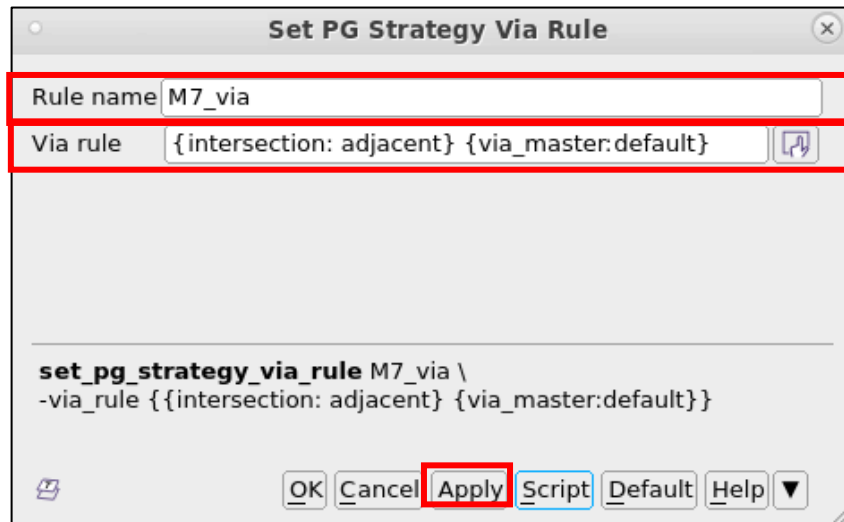
# Introduction to Powerplan : Create Power MESH (Horizontal)

- **set\_pg\_strategy\_via\_rule**

- Specifies the via rule between strategies and collections of existing shapes
- **set\_pg\_strategy\_via\_rule**

- ICC2 Command

```
set_pg_strategy_via_rule M7_via \  
-via_rule {{intersection: adjacent} {via_master:default}}
```



## Via rule name

### -via rule option

Specifies the via rule via\_rule\_spec between strategies, and also between the strategies and existing shapes

```
{intersection : all}      {via_def} |  
{intersection : adjacent} {via_def} |  
list_of_filter_rules
```

- "{intersection : all}", where intersection is the keyword and all refers to all intersections between any two orthogonal shapes including straps, rings, standard cell rails created by strategies or any existing shapes in any different layers

- The second method is "{intersection : adjacent}", where intersection is the keyword and adjacent refers to all intersections between any two orthogonal shapes only in adjacent layers

```
{via_master : via_master_list}
```

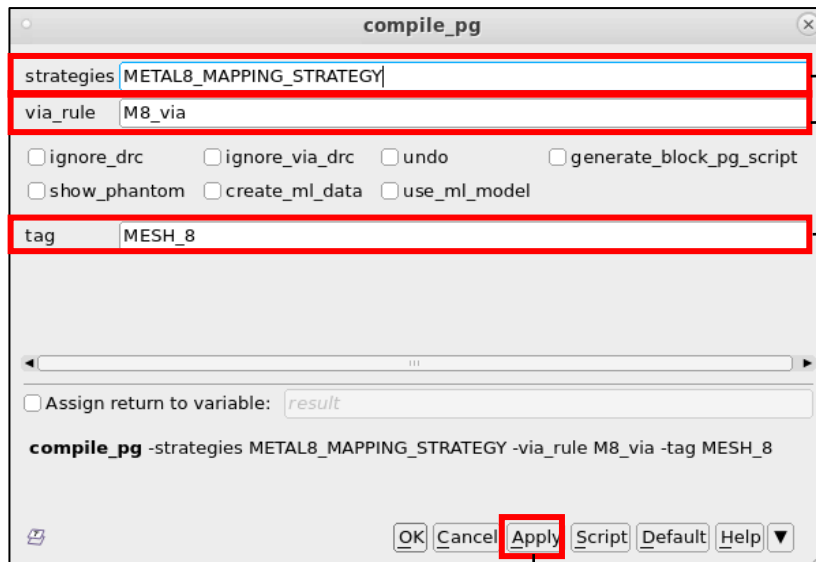
- where via\_master is the keyword for both via masters and via master rules(Via masters include the via contact code defined in the technology file or user-defined via cells)

# Introduction to Powerplan : Create Power MESH (Horizontal)

- **compile\_pg**

- Creates a power and ground network including straps, rings, macro connections, standard cell connections and via connections based on the specified power ground strategies

- **compile\_pg**

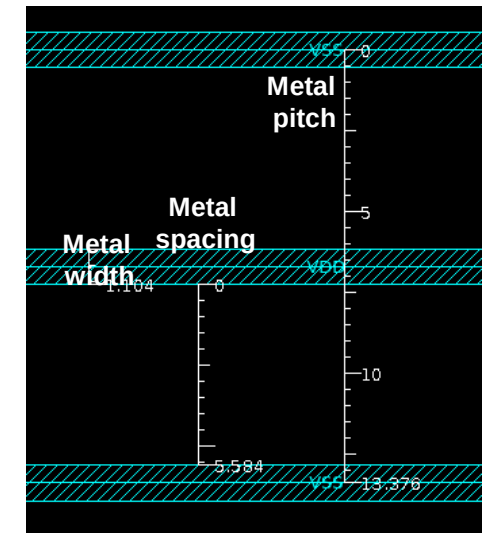
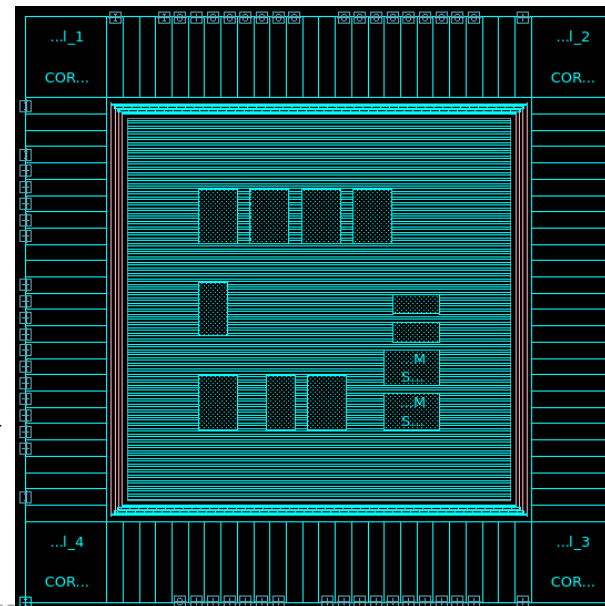


Strategy name

Via rule name

-tag option

Specifies a tag name to assign to all new vias and shapess created by this command



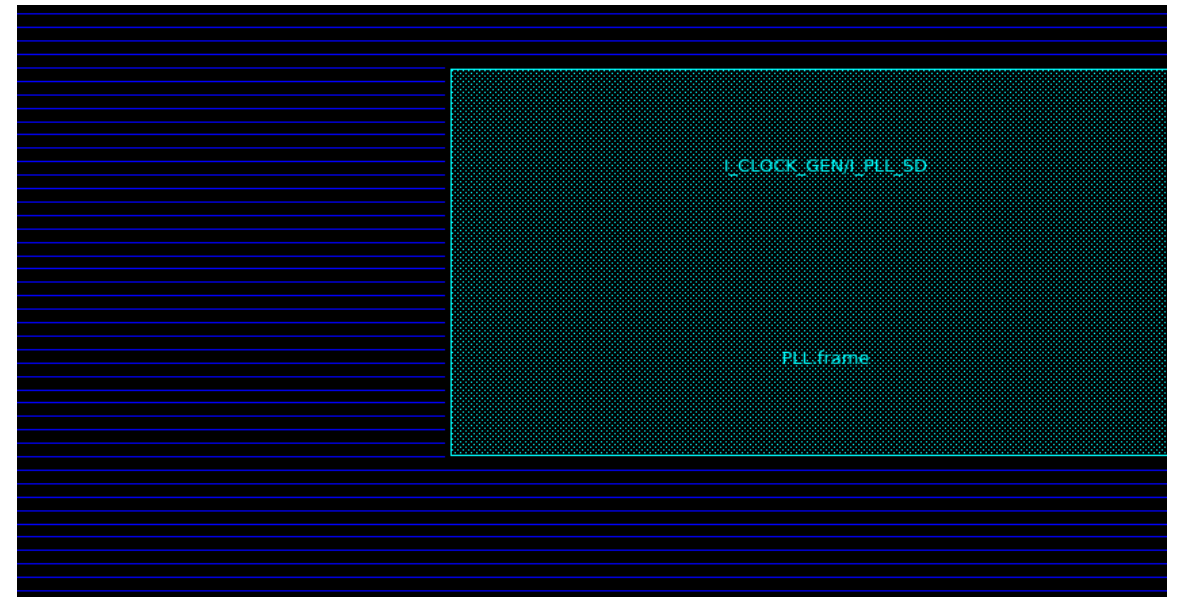
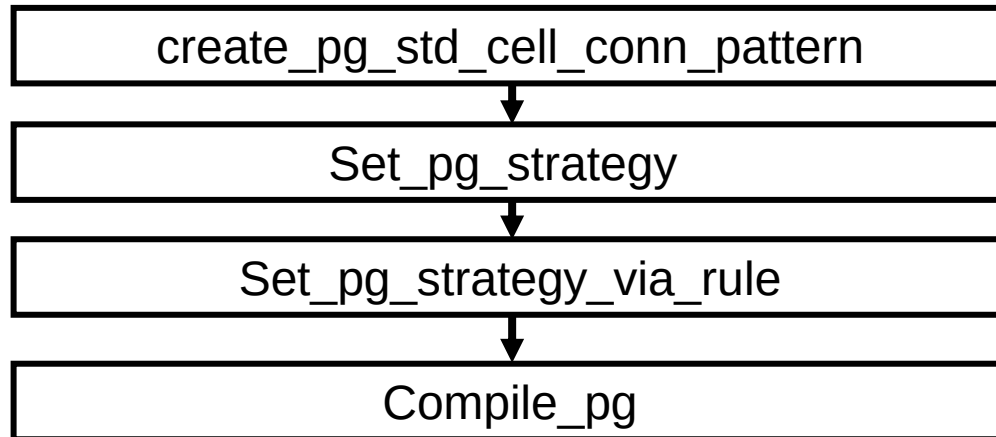
- ICC2 Command

```
compile_pg -strategies {METAL7_MAPPING_STRATEGY} -via_rule {M7_via} -tag MESH_7
```

# Introduction to Powerplan : Create STD Rail

- **Create STD Rail**

- STD Rail connect VDD and VSS to the STD cell
- STD Rails are created to tap power from Power stripes to STD cell Power/Ground Pins
- Step of create STD Rail



# Introduction to Powerplan : Create STD Rail

- **create\_pg\_std\_cell\_conn\_pattern**

- Creates a standard cell rail connection pattern
- This pattern can be instantiated in a design by set\_pg\_strategy command to create standard cell rails

- **Create\_pg\_std\_cell\_conn\_pattern**

create\_pg\_std\_cell\_conn\_pattern

\*pattern\_name std\_rail\_pattern

rail\_width 0.06

rail\_shift 3.344

layers M1

rail\_mask rail\_mask : top and bottom rail...

check\_std\_cell\_drc drc\_flag : check DRCs between...

mark\_as\_follow\_pin follow\_pin\_flag : mark generat...

parameters M1 0.06 3.344

(Fields with \* are required)

☐ Assign return to variable: result

create\_pg\_std\_cell\_conn\_pattern -rail\_width 0.06 \  
-rail\_shift 3.344 -layers M1 \  
-parameters {M1 0.06 3.344} std\_rail\_pattern

OK Cancel Apply Script Default Help ▼

Pattern name

Rail width

Specifies the shift distance of rails  
(The units are defined in the tech file)

Layer name

Specifies the list of parameters  
(Each parameter can be evaluated and used by  
other options in this command)

-parameters {layers} {rail\_width} {rail\_shift}

- ICC2 Command

```
create_pg_std_cell_conn_pattern std_rail_pattern \  
    -layers M1 -rail_width {0.06} \  
    -rail_shift {3.344} \  
    -parameters {M1 0.06 3.344}
```

# Introduction to Powerplan : Create STD Rail

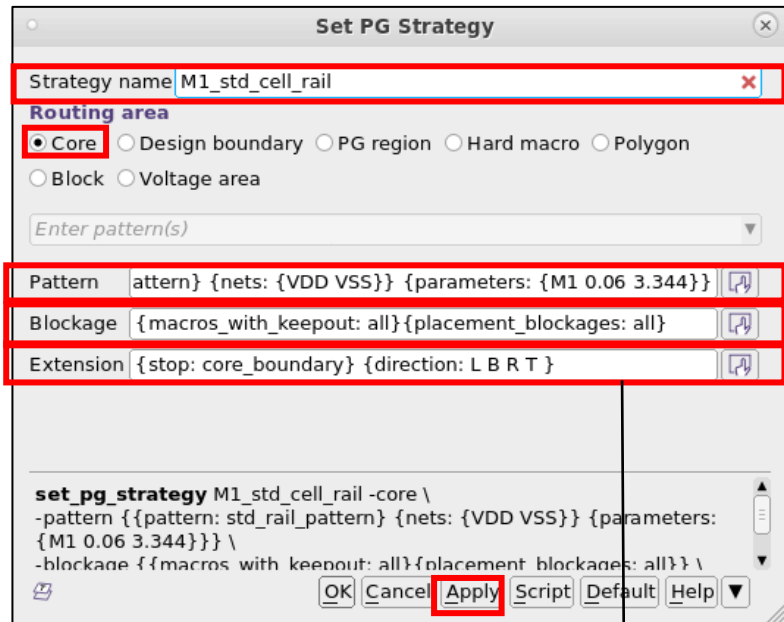
- **set\_pg\_strategy**

- Specifies the power ground network creation strategy

- **set\_pg\_strategy**

- **ICC2 Command**

```
set_pg_strategy M1_std_cell_rail \  
-core \  
-blockage {{macros_with_keepout: all}}{placement_blockages: all}} \  
-pattern {{pattern: std_rail_pattern}} {nets: {VDD VSS}} {parameters: {M1 0.06 3.344}} \  
-extension {{stop: core_boundary}} {direction: L B R T }}
```



## Strategy name

### -core option

Uses the core area as the power ground region

### -pattern option

- Specifies the power ground pattern to be instantiated in this strategy
- Pattern can be wire, composite, mesh, ring, macro connection, std cell rail connection or special pattern
- Only one pattern can be specified and this is a required option

```
{name: pattern_name}  
{nets: real_net_list}  
{parameters: var_list}  
{offset: {x_offset y_offset}}  
{offset_start: boundary | {x y}}
```

### -blockage option

- Specifies the routing blockage in the power ground network
- Blockages are separated by brace and contain the keywords nets;, layers;, and a target specification as follows;

```
{{nets: nets}  
{side : id_list}  
{layers : layer_list}  
{direction: directions}  
{stop: stop_target}}
```

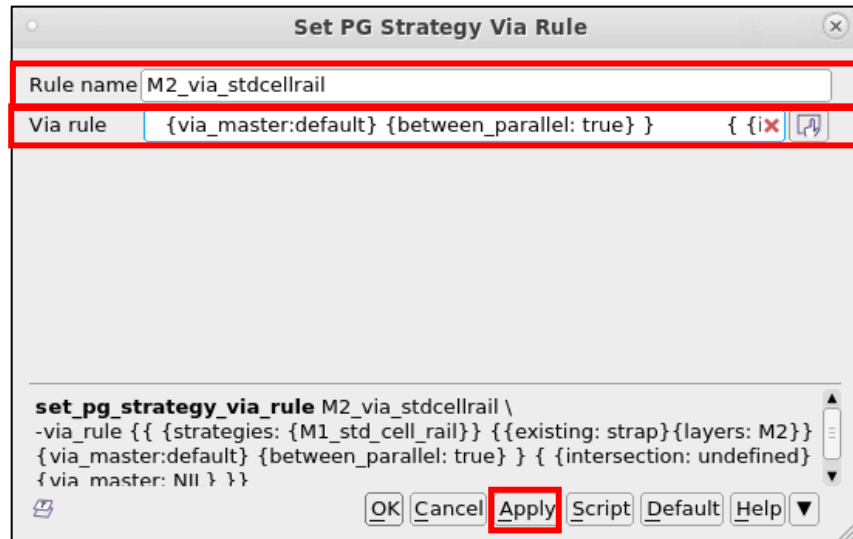
### -Extension option

- Specifies how a group of power and ground nets should extend beyond the power grid
- Each specification contains the keywords nets, side, layers, direction and stop as follows:

# Introduction to Powerplan : Create STD Rail

- **set\_pg\_strategy\_via\_rule**

- Specifies the via rule between strategies and collections of existing shapes
- **set\_pg\_strategy\_via\_rule**



Via rule name

## -via rule option

Specifies the via rule via\_rule\_spec between strategies, and also between the strategies and existing shapes

```
{intersection : all} {via_def} |  
{intersection : adjacent} {via_def} |  
list_of_filter_rules
```

- "{intersection : all}", where intersection is the keyword and all refers to all intersections between any two orthogonal shapes including straps, rings, standard cell rails created by strategies or any existing shapes in any different layers
- The second method is "{intersection : adjacent}", where intersection is the keyword and adjacent refers to all intersections between any two orthogonal shapes only in adjacent layers
- ```
{via_master : via_master_list}
```
- where via\_master is the keyword for both via masters and via master rules (Via masters include the via contact code defined in the technology file or user-defined via cells)
- Between\_parallel:true option create the via intersection between parallel shapes
- existing option is type of shape

- **ICC2 Command**

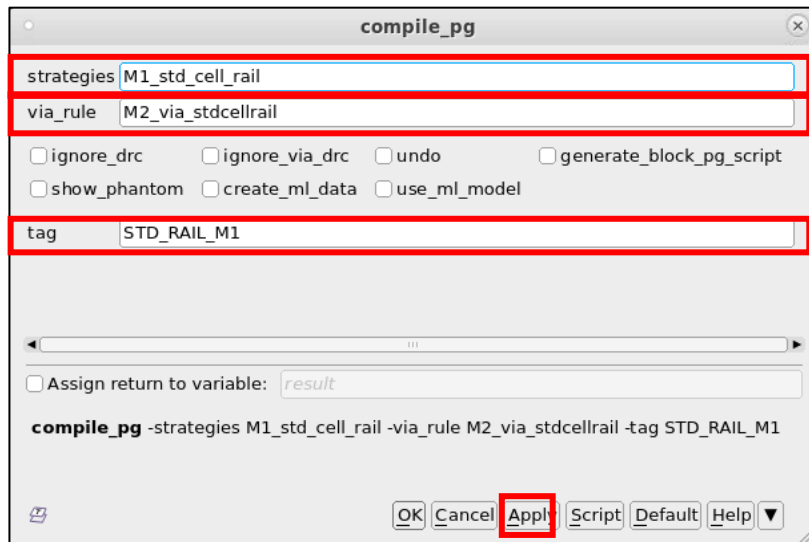
```
set_pg_strategy_via_rule M2_via_stdcellrail \  
-via_rule { \  
  { {strategies: {M1_std_cell_rail}} \  
    { {existing: strap} {layers: M2}} \  
    {via_master:default} {between_parallel: true} } \  
  { {intersection: undefined} {via_master: NIL} } }
```

# Introduction to Powerplan : Create STD Rail

- **compile\_pg**

- Creates a power and ground network including straps, rings, macro connections, standard cell connections and via connections based on the specified power ground strategies

- **compile\_pg**

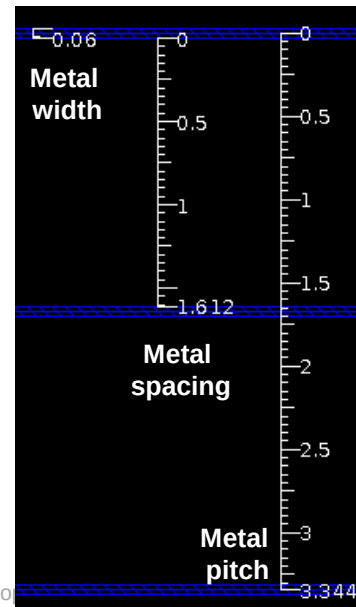


Strategy name

Via rule name

-tag option

Specifies a tag name to assign to all new vias and shapess created by this command



- ICC2 Command

```
compile_pg -strategies {M1_std_cell_rail} -via_rule {M2_via_stdcellrail} -tag {STD_RAIL_M1}
```

# WEEK 5 : LAB for Powerplan

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
< user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/03_powerplan/script
< user@linux script >> ls
01_design_setup_and_copy.tcl          05_PLL_manual_routing.tcl          09_create_std_rail.tcl          13_final_check_and_save.tcl
02_set_macro_and_create_blockage.tcl  06_Power_PAD_manual_routing.tcl    10_remove_setting_and_check_before_fix.tcl  full_script
03_create_core_ring.tcl              07_create_top_MESH.tcl            11_fix_DRC_and_create_via.tcl
04_create_Macro_and_PLL_ring.tcl      08_create_low_MESH.tcl            12_create_hard_blockage.tcl
```

- **Review 01\_design\_setup\_and\_copy.tcl**

```
# module load
#module load icc2/2021.06-SP5
# open icc2_shell
#icc2_shell -gui | tee -i ./03_powerplan.log

#set current step & before step
set step powerplan
set before_step floorplan

#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#open lib
open_lib ../02_floorplan/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../02_floorplan/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 02\_set\_macro\_and\_create\_placement\_blockage.tcl**

```
#####  
#remove constraints  
#####  
remove_pg_strategies -all  
remove_pg_patterns -all  
remove_pg_regions -all  
remove_pg_via_master_rules -all  
remove_pg_strategy_via_rules -all  
#remove_placement_blockage [get_placement_blockages -filter "blockage_type == hard"]  
  
#####  
#create_macro keepout margin  
#####  
  
#all Macros & PLL  
set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]  
#only Macros  
set only_macros "I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/REG_FILE_B_RAM I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/REG_FILE_A_RAM I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM"  
#only PLL  
set pll "I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD"
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 02\_set\_macro\_and\_create\_placement\_blockage.tcl**  
(create placement blockage - blocking stripes from being created during powerplan)

```
#####
#create_placement_blockage_for PLL
#####
set pll_bbox [get_attr [get_cells $pll] bbox] ;#set pll bbox (I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD)
set i 0

foreach x $pll_bbox {
set llx [expr [lindex $x 0 0] + 1.5]
set urx [expr [lindex $x 1 0] - 1.5]
set lly [expr [lindex $x 0 1] + 1.5]
set ury [expr [lindex $x 1 1] - 1.5]
set new_bboxes "${llx $lly} ${urx $ury}"
create_placement_blockage -name "pll_blk_hard_${i}" -type hard -boundary $new_bboxes ;# create placement blockages for PLL

incr i
}

#####
#create_placement_blockage_for macros
#####
set macro_bbox [get_attr [get_cells $only_macros] bbox] ;#set pll bbox (all_Macros except PLL)
set i 0

foreach x $macro_bbox {
set llx [expr [lindex $x 0 0] + 0.5]
set urx [expr [lindex $x 1 0] - 0.5]
set lly [expr [lindex $x 0 1] + 0.5]
set ury [expr [lindex $x 1 1] - 0.5]
set new_bboxes "${llx $lly} ${urx $ury}"
create_placement_blockage -name "mc_blk_hard_${i}" -type hard -boundary $new_bboxes ;# create placement blockages for Macros

incr i
}
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 03\_core\_ring.tcl**

```
#####
#create_core_Ring
#####

#create_pg_region CORE -polygon $core_bdry
set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10} ;# set via column 10 row 10

#create_ring_pattern {set vertical & horizontal (layer, width, spacing), corner bridge & via_rule}
create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 \
    -horizontal_width {5} -horizontal_spacing {2.188} \
    -vertical_layer M6 -vertical_width {4} -vertical_spacing {5.12} \
        -corner_bridge false \
        -via_rule { {intersection: adjacent} {via_master : pgvia_10x10} }

#set_pg_strategy
set_pg_strategy core_ring \
    -core \
    -pattern {{name: core_ring_pattern} \
        {nets: {VDD VSS VDD VSS}} {offset: {10 10}}} ;# set pg_strategy core area & pattern {strategy name, nets, offset}

#compile_pg
compile_pg -strategies {core_ring} -tag {CORE_RING} ;# compile_pg to create core_ring (tag CORE_RING)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 04\_create\_Macro\_and\_PLL\_ring.tcl (Macro Ring)**

```
#####
#create Ring pattern for Macro
#####

#create_Ring_pattern
create_pg_ring_pattern Macro_ring -horizontal_layer M5 \
-horizontal_width {1} -vertical_layer M6 -vertical_width {1} -corner_bridge false ;#create ring pattern (layer(vertical & horizontal), width, spacing, corner bridge)
#set_pg_strategy
set_pg_strategy ORCA_macro_ring -macros $only_macros \
-pattern { {pattern: Macro_ring} {nets: {VSS VDD}} {offset: {0.5 0.5}} } ;#set strategy for Macros / -pattern {pattern name} {net name} {offset}

#create_via_rule
set_pg_strategy_via_rule M_ring_vias -via_rule { \
  {{{strategies: {ORCA_macro_ring}} {layers: {M5}}} {existing: {strap }}{via_master: {default}}} \
  {{{strategies: {ORCA_macro_ring}} {layers: {M6}}} {existing: {strap }}{via_master: {default}}} \
} ;#create via rules for each layers

#compile_pg
compile_pg -strategies {ORCA_macro_ring} -via_rule {M_ring_vias} -tag {MACRO_RING} ;# compile_pg to create macro_ring (tag MACRO_RING)

#####
#create Macro Ring connection
#####

#create_macro_Ring_conn_pattern(vias)
create_pg_macro_conn_pattern Macro_ring_pin -pin_conn_type scattered_pin -layers {M5 M6} ;#create macro_ring pin connection pattern (M5 to M6)
#set_pg_strategy
set_pg_strategy Macro_pins_via -macros $only_macros \
-pattern { {pattern: Macro_ring_pin} {nets: {VSS VDD}} } ;#set strategy for Macros / -pattern {pattern name} {net name}

#compile_pg
compile_pg -strategies {Macro_pins_via} -tag {Macro_RING_con} ;# compile_pg to create Via & stripe (tag MACRO_RING_con)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 04\_create\_Macro\_and\_PLL\_ring.tcl (PLL\_SD Ring)**

```
#####
#create Ring pattern for PLL_SD
#####

#create_Ring_pattern
create_pg_ring_pattern PLL_SD_ring -horizontal_layer M5 \
-horizontal_width {1} -vertical_layer M6 -vertical_width {1} -corner_bridge false ;#create ring pattern (layer(vertical & horizontal), width, spacing, corner bridge)
#set_pg_strategy
set_pg_strategy ORCA_PLL_SD_ring -macros "I_CLOCK_GEN/I_PLL_SD" \
-pattern { {pattern: PLL_SD_ring} {nets: {AVSS DVDD2 AVDD2}} {offset: {0.5 0.5}} } ;#set strategy for PLL_SD / -pattern {pattern name} {net name} {offset}

#create_via_rule
set_pg_strategy_via_rule PLL_SD_ring_vias -via_rule { \
    {{strategies: {ORCA_PLL_SD_ring}} {layers: {M5}}} {existing: {strap }}{via_master: {default}}} \
    {{strategies: {ORCA_PLL_SD_ring}} {layers: {M6}}} {existing: {strap }}{via_master: {default}}} \
} ;#create via rules for each layers

#compile_pg
compile_pg -strategies {ORCA_PLL_SD_ring} -via_rule {PLL_SD_ring_vias} -tag {PLL_SD_RING} ;# compile_pg to create macro_ring (tag PLL_SD_RING)

#####
#create PLL Ring connection
#####

#create_macro_Ring_conn_pattern(vias)
create_pg_macro_conn_pattern pll_ring_pin -pin_conn_type scattered_pin -layers {M5 M6} ;#create macro_ring pin connection pattern (M5 to M6)
#set_pg_strategy
set_pg_strategy pll_pins_via -macros "I_CLOCK_GEN/I_PLL_SD" \
-pattern { {pattern: pll_ring_pin} {nets: {AVSS DVDD2 AVDD2}} } ;#set strategy for Macros / -pattern {pattern name} {net name}

#compile_pg
compile_pg -strategies {pll_pins_via} -tag {PLL_SD_RING_con} ;# compile_pg to create Via & stripe (tag PLL_SD_RING_con)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 04\_create\_Macro\_and\_PLL\_ring.tcl (PLL\_PCI Ring)**

```
#####
#create Ring pattern for PLL_SD
#####

#create_Ring_pattern
create_pg_ring_pattern PLL_SD_ring -horizontal layer M5 \
-horizontal_width {1} -vertical_layer M4 -vertical_width {1} -corner_bridge false ;#create ring pattern (layer(vertical & horizontal), width, spacing, corner bridge)
#set_pg_strategy
set_pg_strategy ORCA_PLL_SD_ring -macros "I CLOCK GEN/I PLL_SD" \
-pattern { {pattern: PLL_SD_ring} {nets: {AVSS DVDD2 AVDD2}} {offset: {0.5 0.5}} } ;#set strategy for PLL_SD / -pattern {pattern name} {net name} {offset}

#create_via_rule
set_pg_strategy_via_rule PLL_SD_ring_vias -via_rule { \
  {{{strategies: {ORCA_PLL_SD_ring}} {layers: {M5}}} {existing: {strap }}{via_master: {default}}} \
  {{{strategies: {ORCA_PLL_SD_ring}} {layers: {M4}}} {existing: {strap }}{via_master: {default}}} \
} ;#create via rules for each layers

#compile_pg
compile_pg -strategies {ORCA_PLL_SD_ring} -via_rule {PLL_SD_ring_vias} -tag {PLL_SD_RING} ;# compile_pg to create macro_ring (tag PLL_SD_RING)

#####
#create PLL Ring connection
#####

#create_macro_Ring_conn_pattern(vias)
create_pg_macro_conn_pattern pll_ring_pin -pin_conn_type scattered_pin -layers {M5 M6} ;#create macro_ring pin connection pattern (M5 to M6)
#set_pg_strategy
set_pg_strategy pll_pins_via -macros "I CLOCK GEN/I PLL_SD" \
-pattern { {pattern: pll_ring_pin} {nets: {AVSS DVDD2 AVDD2}} } ;#set strategy for Macros / -pattern {pattern name} {net name}

#compile_pg
compile_pg -strategies {pll_pins_via} -tag {PLL_SD_RING_con} ;# compile_pg to create Via & stripe (tag PLL_SD_RING_con)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts

- **Review 05\_PLL\_manual\_routing.tcl**

```
#####  
#PLL Power Manual Routing  
#####  
  
# Power was supplied by connecting PLL & Power PAD through manual routing with this script  
  
source ./user_script/PLL_power_manual_routing.tcl
```

- **Review PLL\_power\_manual\_routing.tcl (snap\_setting, AVDD)**

```
#set snap setting before create shape  
set_snap_setting -enabled {false}  
  
#####  
#AVDD  
#####  
  
#read def for manual routing  
read_def ./FP_data/PLL_AVDD.def  
  
#tag the AVDD shape  
change_selection [get_shapes -filter "net.name == AVDD"]  
set_attribute -objects [get_selection] -name tag -value PLL_PCI_AVDD ;#tag the shape (easy to find)  
  
#M7 M8 via connection AVDD  
set AVDD_shape [get_shapes -filter "tag == PLL_PCI_RING && layer.name == M6 || tag == PLL_PCI_AVDD"] ;#select net for create via  
set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10} ;#set via_master_rule (column 10 row 10)  
  
change_selection $AVDD_shape  
  
create_pg_vias -nets {AVDD} -to_types {lib_cell_pin_connect stripe} -from_layers M7 -to_layers M8 -via_masters {pgvia_10x10} -tag {PLL_PCI_AVDD} -shapes [get_selection] ;#create vias Manual routing(M7 to M8)  
  
#M6 M7 via connection AVDD  
set_pg_via_master_rule pgvia_5x5 -via_array_dimension {5 5} ;#set via_master_rule (column 5 row 5)  
  
create_pg_vias -nets {AVDD} -to_types {lib_cell_pin_connect stripe} -from_layers M4 -to_layers M7 -via_masters {pgvia_5x5} -tag {PLL_PCI_AVDD} -shapes [get_selection] ;#create vias Macro ring to manual routing (M6 to M7)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 05\_PLL\_manual\_routing.tcl (DVDD)**

```
#####  
#DVDD  
#####  
  
#read def for manual routing  
read_def ./FP_data/PLL_DVDD.def  
  
#tag the DVDD1 shape  
change_selection [get_shapes -filter "net.name == DVDD"]  
set_attribute -objects [get_selection] -name tag -value PLL_PCI_DVDD ;#tag the shape (easy to find)  
  
#M7_M8 via connection DVDD  
set DVDD_shape [get_shapes -filter "tag == PLL_PCI_RING && layer.name == M6 || tag == PLL_PCI_DVDD"] ;#select net for create via  
set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10} ;#set via_master_rule (column 10 row 10)  
  
change_selection $DVDD_shape  
  
create_pg_vias -nets {DVDD} -to_types {lib_cell_pin_connect stripe} -from_layers M7 -to_layers M8 -via_masters {pgvia_10x10} -tag {PLL_PCI_DVDD} -shapes [get_selection] ;#create vias Manual routing(M7 to M8)  
  
#M6_M7 via connection DVDD  
set_pg_via_master_rule pgvia_5x5 -via_array_dimension {5 5} ;#set via_master_rule (column 5 row 5)  
  
create_pg_vias -nets {DVDD} -to_types {lib_cell_pin_connect stripe} -from_layers M4 -to_layers M7 -via_masters {pgvia_5x5} -tag {PLL_PCI_DVDD} -shapes [get_selection] ;#create vias Macro_ring to manual routing (M6 to M7)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 05\_PLL\_manual\_routing.tcl (AVSS)**

```
#####  
#AVSS  
#####  
  
#read def for manual routing  
read_def ./FP_data/PLL_AVSS.def  
  
#tag the AVDD2 shape  
change_selection [get_shapes -filter "net.name == AVSS"]  
set_attribute -objects [get_selection] -name tag -value PLL_SD_AVSS ;#tag the shape (easy to find)  
  
#M7_M8 via connection AVSS  
set AVSS_shape [get_shapes -filter "tag == PLL_PCI_RING && layer.name == M6 || tag == PLL_SD_AVSS"] ;#select net for create via  
set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10} ;#set via_master_rule (column 10 row 10)  
  
change_selection $AVSS_shape  
  
create_pg_vias -nets {AVSS} -to_types {lib_cell_pin_connect stripe} -from_layers M7 -to_layers M8 -via_masters {pgvia_10x10} -tag {PLL_SD_AVSS} -shapes [get_selection] ;#create vias Manual routing(M7 to M8)  
  
#M6_M7 via connection AVSS  
set_pg_via_master_rule pgvia_5x5 -via_array_dimension {5 5} ;#set via_master_rule (column 5 row 5)  
  
create_pg_vias -nets {AVSS} -to_types {lib_cell_pin_connect stripe} -from_layers M4 -to_layers M7 -via_masters {pgvia_5x5} -tag {PLL_PCI_AVSS} -shapes [get_selection] ;#create vias Macro_ring to manual routing (M6 to M7)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 06\_Power\_Pad\_manual\_routing.tcl**

```
#read_def for manual routing
read_def ./FP_data/PAD_connection.def

#set attribute stripe

set M6_PAD [get_shapes -filter "width == 1.992 && layer.name == M6 "]
change_selection $M6_PAD
set_attribute -objects [get_selection] -name tag -value PAD_manual ;#tag the shape (easy to find)

set M7_PAD [get_shapes -filter "width == 1.992 && layer.name == M7 "]
change_selection $M7_PAD
set_attribute -objects [get_selection] -name tag -value PAD_manual ;#tag the shape (easy to find)

set M5_PAD [get_shapes -filter "width == 1.992 && layer.name == M5 "]
change_selection $M5_PAD
set_attribute -objects [get_selection] -name tag -value PAD_manual ;#tag the shape (easy to find)

#set via_master rule
set_pg_via_master_rule pgvia_8x8 -via_array_dimension {8 8} ;#set via_master_rule
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - Review 06\_Power\_Pad\_manual\_routing.tcl**

```
#####
#connect VDD net to PAD
#####
set M5_VDD [get_shapes -filter "tag == PAD_manual && owner.name == VDD && layer.name == M5 || tag == CORE_RING && owner.name == VDD"]
set M6_VDD [get_shapes -filter "tag == PAD_manual && owner.name == VDD && layer.name == M6 || tag == CORE_RING && owner.name == VDD"]
set M7_VDD [get_shapes -filter "tag == PAD_manual && owner.name == VDD && layer.name == M7 || tag == CORE_RING && owner.name == VDD"]

change_selection $M5_VDD
create_pg_vias -nets {VDD} -to_types {lib_cell_pin_connect ring} -from_layers M5 -to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]

change_selection $M6_VDD
create_pg_vias -nets {VDD} -to_types {lib_cell_pin_connect ring} -from_layers M6 -to_layers M7 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]

change_selection $M7_VDD
create_pg_vias -nets {VDD} -to_types {lib_cell_pin_connect ring} -from_layers M7 -to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
```

```
#####
#connect VSS net to PAD
#####
set M5_VSS [get_shapes -filter "tag == PAD_manual && owner.name == VSS && layer.name == M5 || tag == CORE_RING && owner.name == VSS"]
set M6_VSS [get_shapes -filter "tag == PAD_manual && owner.name == VSS && layer.name == M6 || tag == CORE_RING && owner.name == VSS"]
set M7_VSS [get_shapes -filter "tag == PAD_manual && owner.name == VSS && layer.name == M7 || tag == CORE_RING && owner.name == VSS"]

change_selection $M5_VSS
create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect ring} -from_layers M5 -to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]

change_selection $M6_VSS
create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect ring} -from_layers M6 -to_layers M7 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]

change_selection $M7_VSS
create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect ring} -from_layers M7 -to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 06\_Power\_Pad\_manual\_routing.tcl**

```
#####
#connect DVDD net to PAD
#####
set M5_DVDD [get_shapes -filter "owner.name == DVDD && layer.name == M8 || tag == PAD_manual && owner.name == DVDD"]

change_selection $M5_DVDD
create_pg_vias -nets {DVDD} -to_types {stripe ring} -from_layers M5 -to_layers M8 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]

#####
#connect AVDD net to PAD
#####
set M5_AVDD [get_shapes -filter "owner.name == AVDD && layer.name == M8 || tag == PAD_manual && owner.name == AVDD"]

change_selection $M5_AVDD
create_pg_vias -nets {AVDD} -to_types {stripe ring} -from_layers M5 -to_layers M8 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]

#####
#connect AVSS net to PAD
#####
set M5_AVSS [get_shapes -filter "owner.name == AVSS && layer.name == M8 || tag == PAD_manual && owner.name == AVSS"]
change_selection $M5_AVSS

create_pg_vias -nets {AVSS} -to_types {stripe ring} -from_layers M5 -to_layers M8 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts

- **Review 07\_create\_top\_MESH.tcl (Vertical M8)**

```
#####
#create_Power_MESH
#####

#create placement blockage before create MESH78 - because of PLL manual routing

set MESH_blk_bbox "{{1014.6780 716.0880} {1189.4520 737.6880}} {{1014.6780 662.1780} {1189.4520 683.7780}}}" ;set MESH(M7 M8) boundary

#create placement blockages to prevention overlap M7 & M8
set i 0
foreach x $MESH_blk_bbox {
  create_placement_blockage -name "MESH_pblk_${i}" -type hard -boundary $x

  incr i
}

#set M8 MESH boundary
set M8_MESH_bdry "{{250.5480 214.4400} {1189.4520 1225.5600}}"

#create_pg_reigion
create_pg_region MESH -polygon $M8_MESH_bdry

#create_M8_mesh_pattern
create_pg_mesh_pattern METAL8_PG_PATTERN -layers {{vertical_layer: M8} \
  {width: 4.64} \
  {spacing: 5.088} \
  {pitch: 19.456} \
  {trim: false}} \
  -parameters {4.64 19.456 5.088} ;# create mesh_pattern {name layer, width, spacing, pitch, trim}

#creat_M8_pg_strategy
set_pg_strategy METAL8_MAPPING_STRATEGY \
  -pg_regions MESH \
  -pattern {\
    {name: METAL8_PG_PATTERN} \
    {nets: {VDD VSS}} \
    {parameters: {4.64 19.456 5.088}} \
    {offset_start: {0 0}} \
  } \
  -blockage {placement_blockages: all} ;# create strategy {pattern name, net, parameters, offset}

#create_M8_via_rule
set_pg_strategy_via_rule M8_via \
  -via_rule {via_master: NIL} ;# create via rule {NIL - no via creation}

compile_pg -strategies {METAL8_MAPPING_STRATEGY} -via_rule {M8_via} -tag MESH_8 ;#compile_pg to create vertical MESH (tag MESH_8)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 07\_create\_top\_MESH.tcl (Horizontal M7)**

```
#create_M7_mesh_pattern
create_pg_mesh_pattern METAL7_PG_PATTERN -layers {{horizontal_layer: M7} \
    {width: 1.104} \
    {spacing: 5.584} \
    {pitch: 13.376} \
    {trim: false}} \
-parameters {1.104 13.376 5.584} ;# create mesh_pattern {name layer, width, spacing, pitch, trim}

#create_M7_MESH_strategy
set_pg_strategy METAL7_MAPPING_STRATEGY \
    -core \
    -pattern {\
        {name: METAL7_PG_PATTERN} \
        {nets: {VDD VSS}} \
        {parameters: {1.104 13.376 5.584}} \
        {offset_start: {0 0}}} \
    -blockage {placement_blockages: all} ;# create strategy {pattern name, net, parameters, offset}

#create_M7_via_rule
set_pg_strategy_via_rule M7_via \
    -via_rule {via_master: NIL} ;# create via rule {NIL - no via creation}

#compile_pg
compile_pg -strategies {METAL7_MAPPING_STRATEGY} -via_rule {M7_via} -tag MESH_7 ;#compile_pg to create horizontal MESH (tag MESH_7)

#remove placement blockages if you create MESH78 done
remove_placement_blockages [get_placement_blockages -filter "full_name =~ MESH_pblk_*"] ;#remove placement_blockage (MESH_pblk_1 MESH_pblk_2)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 08\_create\_low\_MESH.tcl (Vertical M2)**

```
#####  
#create_M2 Low MESH  
#####  
  
#create_M2_mesh_pattern  
create_pg_mesh_pattern P_m2_triple \  
    -layers { \  
        { {vertical_layer: M2} \  
          {width: 0.44 0.192 0.192} \  
          {spacing: 2.724 3.456} \  
          {pitch: 9.728} {offset: 1.216} {trim : false} } \  
        } ;# create mesh_pattern {name layer, width, spacing, pitch, trim}  
  
#create_M2_pg_strategy  
set_pg_strategy S_m2_vddvss \  
    -core \  
    -pattern { {name: P_m2_triple} {nets: {VDD VSS VSS}} {offset_start: {250.548 0}} } \  
    -blockage { {{nets: VDD VSS} {placement_blockages : all}}  
} ;# create strategy {pattern name, net, parameters, offset}  
  
#create_M2_via_rule  
set_pg_strategy_via_rule M2_via \  
    -via_rule {via_master: NIL} ;# create via rule {NIL - no via creation}  
  
#compile_pg  
compile_pg -strategies {S_m2_vddvss} -tag {low_MESH_2} \  
    -via_rule {M2_via} -drc no_check ;#compile_pg to create vertical low MESH (tag low_MESH_2)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 09\_create\_std\_rail.tcl (Horizontal M1)**

```
#####
#create STD_RAIL
#####

#create_M1_std_cell_conn_pattern
create_pg_std_cell_conn_pattern std_rail_pattern -layers M1 -rail_width {0.06} \
-rail_shift {3.344} \
-parameters {M1 0.06 3.344} ;#create std_cell conn pattern {layer, width, rail_shift}

#create_M1_pg_strategy
set_pg_strategy M1_std_cell_rail \
-core \
-blockage {{macros_with_keepout: all}{placement_blockages: all}} \
-pattern {{pattern: std_rail_pattern} {nets: {VDD VSS}} {parameters: {M1 0.06 3.344}}} \
-extension {{stop: outermost_ring} {direction: L B R T }} ;#create std_cell conn pattern {core area, blockage, pattern, extension}

#create_M1_via_rule
set_pg_strategy_via_rule M2_via_stdcellrail \
-via_rule { \
{ {strategies: {M1_std_cell_rail}} \
{{existing: strap}{layers: M2}} \
{via_master: default} {between_parallel: true} } \
{ {intersection: undefined}{via_master: NIL} }} ;# create via rule strategy name, existing, layer, via_master, between_parallel, {NIL - no via creation}

#compile_pg
compile_pg -strategies {M1_std_cell_rail} -via_rule {M2_via_stdcellrail} -tag {STD_RAIL_M1} ;#compile_pg to create STD Rail (tag STD_RAIL_M1)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 08\_create\_std\_rail.tcl (Horizontal M1)**

```
#####
#create_STD_RAIL
#####

#create_M1_std_cell_conn_pattern
create_pg_std_cell_conn_pattern std_rail_pattern -layers M1 -rail_width {0.06} \
                                           -rail_shift {3.344} \
                                           -parameters {M1 0.06 3.344} ;#create std_cell conn pattern {layer, width, rail_shift}

#create_M1_pg_strategy
set_pg_strategy M1_std_cell_rail \
    -core \
    -blockage {{macros_with_keepout: all}{placement_blockages: all}} \
    -pattern {{pattern: std_rail_pattern} {nets: {VDD VSS}} {parameters: {M1 0.06 3.344}}} \
    -extension {{stop: outermost_ring} {direction: L B R T }} ;#create std_cell conn pattern {core area, blockage, pattern, extension}

#create_M1_via_rule
set_pg_strategy_via_rule M2_via_stdcellrail \
    -via_rule { \
        { {strategies: {M1_std_cell_rail}} \
          {{existing: strap}{layers: M2}} \
          {via_master: default} {between_parallel: true} } \
        { {intersection: undefined}{via_master: NIL} }} ;# create via rule strategy name, existing, layer, via_master, between_parallel, {NIL - no via creation}

#compile_pg
compile_pg -strategies {M1_std_cell_rail} -via_rule {M2_via_stdcellrail} -tag {STD_RAIL_M1} ;#compile_pg to create STD Rail (tag STD_RAIL_M1)
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 10\_remove\_setting\_and\_check\_before\_fix.tcl**

```
#####  
#remove Settings (default value)  
#####  
  
remove_pg_strategies -all  
remove_pg_patterns -all  
remove_pg_regions -all  
remove_pg_via_master_rules -all  
remove_pg_strategy_via_rules -all  
  
#####  
#check_Powerplan  
#####  
  
#Drc check  
check_pg_drc  
  
#missing via check  
check_pg_missing_vias  
  
#connectivity check (exclude std cell)  
check_pg_connectivity -check_std_cell_pins none ;# std cells are not placed
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
- **Review 11\_drc\_fix\_and\_create\_via.tcl (Core ring, Macro ring, PLL ring, Power MESH)**

```
#####
#change & create via for fixing DRC01
#####

#create CoreRing M7 to M8 via
set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10} ;#set via_master_rule

set core_M7_M8_shape [get_shapes -filter "tag == CORE_RING && layer.name == M7 || tag == MESH_8"] ;#set CoreRing(M7) MESH(M8) shapes & selection
change_selection $core_M7_M8_shape

create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M7 \
               -to_layers M8 -via_masters {pgvia_10x10} -tag {core_M7_8_via} -shapes [get_selection] ;#create vias for CoreRing(M7) to MESH(M8)

#create MacroRing M6 to M7 via
set_pg_via_master_rule pgvia_3x3 -via_array_dimension {3 3} ;#set via_master_rule

set M6_M7_shape [get_shapes -filter "tag == MACRO_RING && layer.name == M6 || tag == MESH_7"] ;#set MacroRing(M6) MESH(M7) shapes & selection
change_selection $M6_M7_shape

create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M6 \
               -to_layers M7 -via_masters {pgvia_3x3} -tag {M6_7_via} -shapes [get_selection] ;#create vias for MacroRing(M6) to MESH(M7)

#create PLLRing M6 to M7 via
set M6_M7_shape [get_shapes -filter "tag == PLL_PCI_RING || tag == PLL_SD_RING || tag == MESH_7"] ;#set PLLRing(M6) MESH(M7) shapes & selection
change_selection $M6_M7_shape

create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M6 \
               -to_layers M7 -via_masters {pgvia_3x3} -tag {M6_7_via} -shapes [get_selection] ;#create vias for MacroRing(M6) to MESH(M7)

#create M7 to M8 via

set_pg_via_master_rule pgvia_5x8 -via_array_dimension {8 5} ;#set via_master_rule

set M7_M8_shape [get_shapes -filter "tag == MESH_8 || tag == MESH_7"] ;#set MESH(M7) MESH(M8) shapes & selection
change_selection $M7_M8_shape

create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M7 \
               -to_layers M8 -via_masters {pgvia_5x8} -tag {M7_8_via} -shapes [get_selection] ;#create vias for MacroRing(M7) to MESH(M8)

change_selection ;# deselect all
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 11\_drc\_fix\_and\_create\_via.tcl (STD rail to LOW MESH, Low MESH to Power MESH M7)**

```
#####
#change & create via for fixing DRC02
#####

#change via def M1 to M2 vias

set M1_M2_via [get_vias -filter "tag == STD_RAIL_M1"] ;#set StdRail(M1) LowMesh(M2) vias & selection
change_selection $M1_M2_via

set_attribute -objects [get_selection] -name via_def -value [get_via_defs -library [current_lib] VIA12SQ_C] ;#change via defs for StdRail(M1) to LowMesh(M2)
change_selection

#create M2 to M7 vias

set_pg_via_master_rule pgvia_2x4 -via_array_dimension {2 4}

set M2_M7_shape [get_shapes -filter "tag == low_MESH_2 || tag == MESH_7"] ;#set LowMesh(M2) MESH(M7) shapes & selection
change_selection $M2_M7_shape

create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M2 \
               -to_layers M7 -via_masters {pgvia_2x4} -tag {M2_7_via} -shapes [get_selection] ;#create vias for LowMesh(M2) to MESH(M7)

#change M2 to M7 vias columns
set ch_M2_via [get_vias -filter "tag == M2_7_via && owner.name == VSS"] ;#set LowMesh(M2) MESH(M7) vias & selection
change_selection $ch_M2_via
set_attribute -objects [get_selection] -name number_of_columns -value 1 ;#change via coluns for LowMesh(M2) to MESH(M7)
change_selection ;#deselect all

#remove placement blockage
remove_placement_blockage [get_placement_blockages -filter "blockage_type == hard"]
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 12\_create\_hard\_blockage.tcl (Macro & PLL)**

```
#remove placement_blockage
remove_placement_blockage [get_placement_blockages -filter "blockage_type == hard"]

#####
#create_placement_blockage for macros
#####
set macro_bbox [get_attr [get_cells $only_macros] bbox]
set i 0

foreach x $macro_bbox {
set llx [expr [lindex $x 0 0] - 2]
set urx [expr [lindex $x 1 0] + 2]
set lly [expr [lindex $x 0 1] - 2]
set ury [expr [lindex $x 1 1] + 2]
set new_bboxes "${llx $lly} ${urx $ury}"
create_placement_blockage -name "mc_blk_hard_${i}" -type hard -boundary $new_bboxes

incr i
} ;#create 3um placement blockage around the macro (Macro pblk bbox {llx-2 lly-2} {urx+2 ury+2})

#####
#create_placement_blockage for PLL
#####
set pll "I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD"
set pll_bbox [get_attr [get_cells $pll] bbox]
set i 0

foreach x $pll_bbox {
set llx [expr [lindex $x 0 0] - 2]
set urx [expr [lindex $x 1 0] + 2]
set lly [expr [lindex $x 0 1] - 2]
set ury [expr [lindex $x 1 1] + 2]
set new_bboxes "${llx $lly} ${urx $ury}"
create_placement_blockage -name "pll_blk_hard_${i}" -type hard -boundary $new_bboxes

incr i
} ;#create 4um placement blockage around the PLL (PLL pblk bbox {llx-2 lly-2} {urx+2 ury+2})
```

# Powerplan : Introduction to powerplan.tcl

- Review and check the variables in each scripts
  - **Review 13\_final\_check\_and\_save.tcl**

```
#####  
#check_Powerplan  
#####  
#Drc check  
check_pg_drc  
  
#missing via check  
check_pg_missing_via  
  
#connectivity check (exclude std cell)  
check_pg_connectivity -check_std_cell_pins none ;# std cells are not placed  
  
#save & exit  
  
save_lib -all  
exit
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 1. Invoke tool icc2\_shell

- **Recommend the icc2/2021.06-SP5 version**

Enter the command line by line in 03\_powerplan.tcl and check the result

>> icc2\_shell -gui | tee -i ./03\_powerplan.log ;# (| tee -i ./03\_powerplan.log) is the command to output log

```
< user@linux 03_powerplan >> icc2_shell -gui | tee -i ./03_powerplan.log

                                IC Compiler II (TM)

                                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

                                Copyright (c) 1988 - 2022 Synopsys, Inc.
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                                Inclusivity and Diversity" (Refer to article 000036315 at
                                https://solvnetplus.synopsys.com)

                                Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
                                icc2_shell>
```

## 2. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/01\_design\_setup\_and\_copy.tcl

- **set step & source design\_setup.tcl**

```
icc2_shell> set step powerplan
powerplan
icc2_shell> set before_step floorplan
floorplan
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```

# Powerplan : Use ICC2 (IC Compiler II tool)

2. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/01_design_setup_and_copy.tcl`

- open lib

```
icc2_shell> open_lib ../02_floorplan/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/02_floorplan/nlib/ORCA_floorplan.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_floorplan.nlib}
```

Check that ndm is properly applied each time you open the nlib

- Copy `ORCA_floorplan.nlib` to `ORCA_powerplan.nlib` & open `ORCA_powerplan.nlib`

```
icc2_shell> copy_lib -from ../02_floorplan/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
Information: Copying from library "ORCA_floorplan.nlib" to "./nlib/ORCA_powerplan.nlib".
{ORCA_powerplan.nlib}
icc2_shell> close_lib -all
Closing all libraries...
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

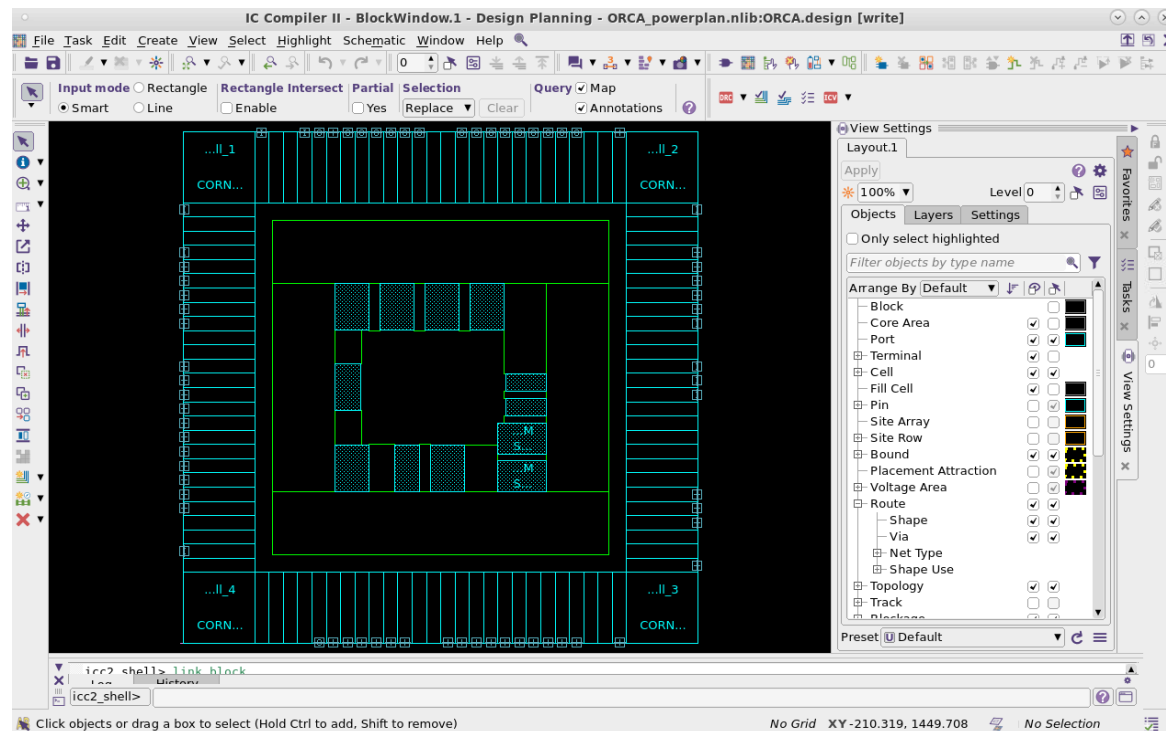
2. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/01_design_setup_and_copy.tcl`

- open block

```
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_powerplan.nlib:ORCA.design' in edit mode
Variable 1.104 is defined in -parameters option but not used by any pattern expression.
Variable 13.376 is defined in -parameters option but not used by any pattern expression.
Variable 5.584 is defined in -parameters option but not used by any pattern expression.
Warning: Some defined variables in -parameters option are not used by other options. (PGR-501)
{ORCA_powerplan.nlib:ORCA.design}
icc2_shell> link_block
Using libraries: ORCA_powerplan.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32
rvt saed32sram
Visiting block ORCA_powerplan.nlib:ORCA.design
Design 'ORCA' was successfully linked.
1
```

Check using libraries

- Check the copied block in the gui window



# Powerplan : Use ICC2 (IC Compiler II tool)

3. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/02_set_macro_and_create_placement_blockage.tcl`

- **Reset all settings & remove hard placement blockages before powerplan**

```
remove_pg_strategies -all
All strategies have been removed.
icc2_shell> remove_pg_patterns -all
All patterns have been removed.
icc2_shell> remove_pg_regions -all
No PG region is found.
icc2_shell> remove_pg_via_master_rules -all
All via def rules have been removed.
icc2_shell> remove_pg_strategy_via_rules -all
All strategy via rules have been removed.
icc2_shell> remove_placement_blockage [get_placement_blockages -filter "blockage_type == hard"]
Error: Nothing matched for blockage_list (SEL-005)
0
```

- **Create keepout margin for all Macros & PLL**

```
icc2_shell> set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]
{I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_
TOP/I_CONTEXT_MEM/CONTEXT_RAM2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_RISC_CORE/REG_FILE_B_RAM I_ORCA_TOP/I_RISC_CO
RE/REG_FILE_A_RAM I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FI
FO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM}
icc2_shell> set only_macros "I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_TOP/I_CONTEXT_M
EM/CONTEXT_RAM2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_RISC_CORE/REG_FILE_B_RAM I_ORCA_TOP/I_RISC_CORE/REG_FILE_A_R
AM I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM
I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM"
I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM2 I_ORCA_TOP/I_
CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_RISC_CORE/REG_FILE_B_RAM I_ORCA_TOP/I_RISC_CORE/REG_FILE_A_RAM I_ORCA_TOP/I_PCI_READ_FIFO
/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIF
O/SD_WFIFO_RAM
icc2_shell> set pll "I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD"
I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD
```

# Powerplan : Use ICC2 (IC Compiler II tool)

3. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/02_set_macro_and_create_placement_blockage.tcl`

- Check keepout margin before create placement blockage

The diagram illustrates the steps to verify the keepout margin for a specific cell in ICC2:

- Circuit View:** A large cell labeled `...CONTEXT_MEM/CONTEXT_RAM0` is shown within a `SRAM2RW64x32.frame`. A red box highlights the bottom-right corner of the cell.
- Context Menu:** A right-click menu is open, with the **Properties** option (labeled `Ctrl+R`) highlighted by a red box. A red arrow points from the highlighted area in the circuit view to this menu.
- Zoomed-in View:** A red arrow points from the zoomed-in view of the keepout margin to the **Properties** window. It shows a double-headed arrow indicating the width of the **Keep-out-margin**.
- Properties Window:** The **Properties** window for cell `I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0` is open. The **Basic** tab is selected. The **outer\_keepout\_margin\_hard** property is highlighted with a red box, showing a value of `1 1 1 1`. Other properties like **outer\_keepout\_margin\_soft** and **outer\_keepout\_margin\_hard\_macro** are set to `0 0 0 0`. The **ref\_phys\_block** is `saed32sram:SRAM2RW64x32.frame`.

- Open the properties window of the hard blockage to check that the keepout margin created correctly
- Zoom in and check if the size of the keepout margin is correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

## 3. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/02\_set\_macro\_and\_create\_placement\_blockage.tcl

- Create hard Placement\_blockage for all Macros & PLL

```
set pll_bbox [get_attr [get_cells $pll] bbox]
{{899.2350 707.9240} {1016.4580 760.0320}} {{899.2350 638.8160} {1016.4580 690.9240}}
icc2_shell> set i 0
0
icc2_shell>
icc2_shell> foreach x $pll_bbox {
? set llx [expr [lindex $x 0 0] + 1.5]
? set urx [expr [lindex $x 1 0] - 1.5]
? set lly [expr [lindex $x 0 1] + 1.5]
? set ury [expr [lindex $x 1 1] - 1.5]
? set new_bboxes "{$llx $lly} {$urx $ury}"
? create_placement_blockage -name "pll_blk_hard_${i}" -type hard -boundary $new_bboxes
?
? incr i
? }
icc2_shell> set macro_bbox [get_attr [get_cells $only_macros] bbox]
{{424.5430 881.7460} {521.4510 1015.1520}} {{550.4510 881.7460} {647.3590 1015.1520}} {{676.3590 881.
7460} {773.2670 1015.1520}} {{802.2670 881.7460} {899.1750 1015.1520}} {{877.7860 424.8460} {1015.459
0 514.4760}} {{877.7860 532.1860} {1015.4590 621.8160}} {{424.5430 655.1680} {497.3820 788.5740}} {{5
89.4510 424.8460} {662.2900 558.2520}} {{424.5430 424.8460} {521.4510 558.2520}} {{690.2900 424.8460}
{787.1980 558.2520}}
icc2_shell> set i 0
0
icc2_shell>
icc2_shell> foreach x $macro_bbox {
? set llx [expr [lindex $x 0 0] + 0.5]
? set urx [expr [lindex $x 1 0] - 0.5]
? set lly [expr [lindex $x 0 1] + 0.5]
? set ury [expr [lindex $x 1 1] - 0.5]
? set new_bboxes "{$llx $lly} {$urx $ury}"
? create_placement_blockage -name "mc_blk_hard_${i}" -type hard -boundary $new_bboxes
?
? incr i
? }
```

# Powerplan : Use ICC2 (IC Compiler II tool)

3. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/02_create_keepout_margin_and_blockage.tcl`

- Check placement blockage

The screenshot illustrates the process of checking a placement blockage in ICC2. The main window shows a chip layout with a central blockage area. A context menu is open over the blockage, with 'Properties' highlighted. A zoomed-in view of the blockage is shown at the bottom. The Properties window is open, showing details for 'mc\_blkg\_hard\_0'.

| Name               | Value                                    |
|--------------------|------------------------------------------|
| name               | mc blkg hard 0                           |
| full_name          | mc_blkg_hard_0                           |
| object_class       | placement_blockage                       |
| blockage_type      | hard                                     |
| blocked_percentage |                                          |
| bbox               | {425.0430 882.2460} {520.9510 1014.6520} |

- Open the properties window of the hard blockage to check that the name and type are created correctly
- Zoom in and check if the size of the hard blockage is correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

## 4. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/03\_create\_core\_ring.tcl

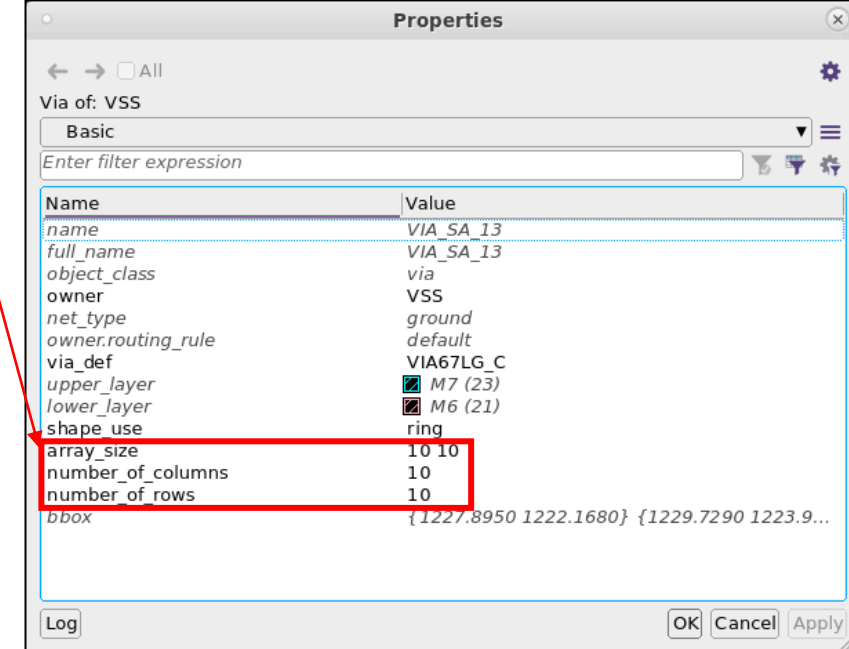
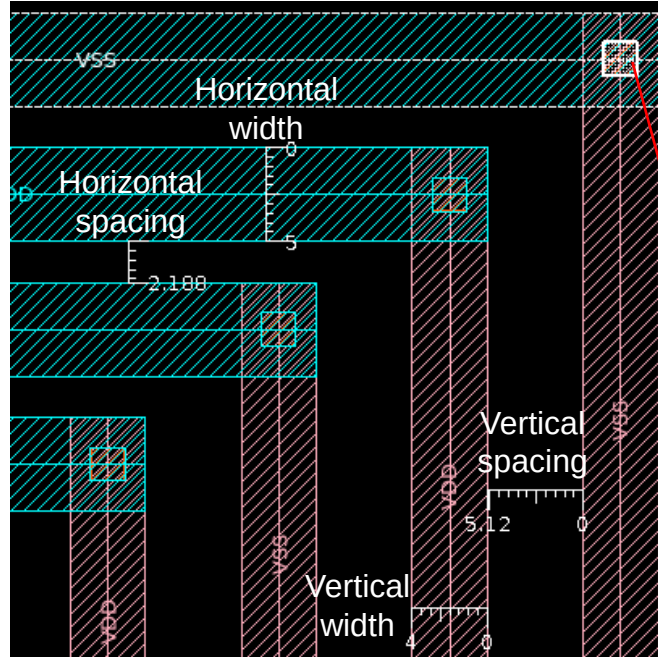
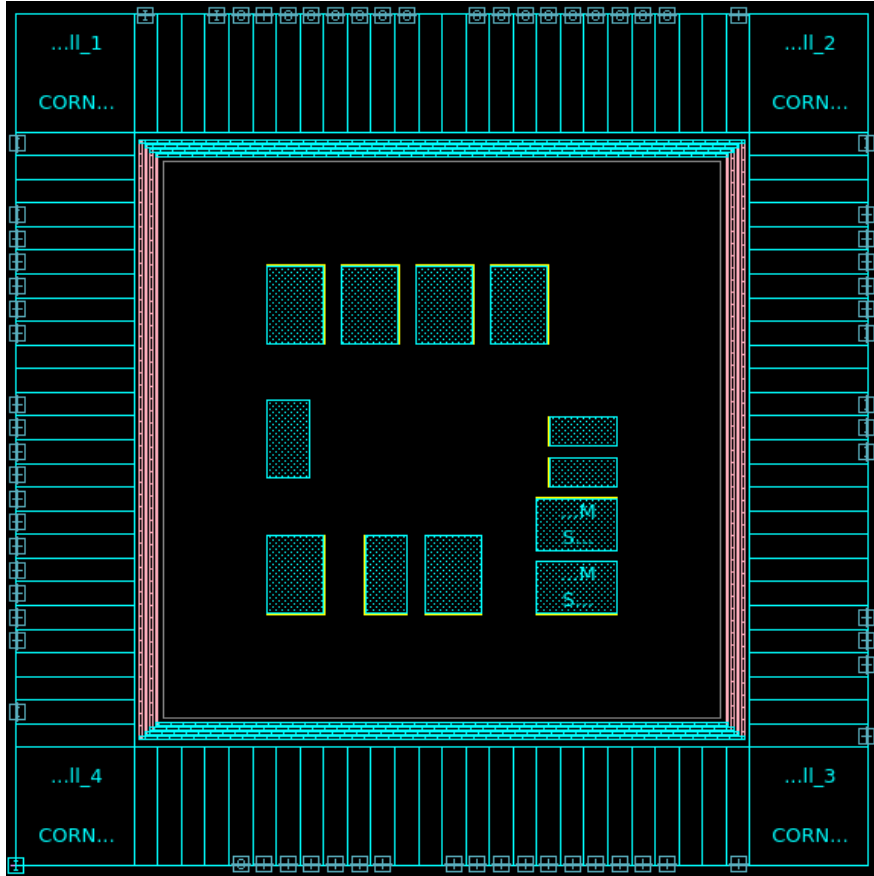
- Create Core Ring

```
icc2_shell> set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10}
Information: The command 'set_pg_via_master_rule' cleared the undo history. (UNDO-016)
Successfully set via def rule pgvia_10x10.
icc2_shell>
icc2_shell> create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 \
?   -horizontal_width {5} -horizontal_spacing {2.188} \
?   -vertical_layer M6 -vertical_width {4} -vertical_spacing {5.12} \
?   -corner_bridge false \
?   -via_rule { {intersection: adjacent} {via_master : pgvia_10x10} }
Successfully create PG ring pattern core_ring_pattern.
icc2_shell>
icc2_shell> set_pg_strategy core_ring \
?   -core \
?   -pattern {{name: core_ring_pattern} \
?   {nets: {VDD VSS VDD VSS}} {offset: {10 10}}}
Successfully set PG strategy core_ring.
icc2_shell>
icc2_shell> compile_pg -strategies {core_ring} -tag {CORE_RING}
Sanity check for inputs.
No strategy-level via rule is specified, the default rule will be applied.
Set tag to CORE_RING
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy core_ring.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Creating rings.
Creating via connection between strategies and existing shapes.
Via DRC checking runtime 0.00 seconds.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias inside strategy core ring.
Checking 16 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag CORE_RING to committed wires.
Committed 16 wires.
Committing wires takes 0.00 seconds.
Committed 16 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 0 seconds.
Successfully compiled PG.
Overall runtime: 0 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

4. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/03_create_core_ring.tcl`

- Check Core Ring



- Open the properties window of the Vias to check that the array size & (columns and rows) are created correctly
- Zoom in and check if the Horizontal & Vertical (width,spacing) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

- Create Macro Ring

```
icc2_shell> create_pg_ring_pattern Macro_ring -horizontal_layer M5 -horizontal_width {1} -vertical_layer M6 -vertical_width {1} -corner_bridge false
Successfully create PG ring pattern Macro_ring.
icc2_shell> set_pg_strategy ORCA_macro_ring -macros $only_macros -pattern { {pattern: Macro_ring} {nets: {VSS VDD}} {offset: {0.5 0.5}} }
Successfully set PG strategy ORCA_macro_ring.
icc2_shell>
icc2_shell> set_pg_strategy_via_rule M_ring_vias -via_rule { \
? {{{strategies: {ORCA_macro_ring}} {layers: {M5}}}} {existing: {strap }}{via_master: {default}}} \
? {{{strategies: {ORCA_macro_ring}} {layers: {M6}}}} {existing: {strap }}{via_master: {default}}} \
? }
Successfully set strategy via rule M_ring_vias.
icc2_shell>
icc2_shell> compile_pg -strategies {ORCA_macro_ring} -via_rule {M_ring_vias} -tag {MACRO_RING}
Sanity check for inputs.
Set tag to MACRO_RING
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy ORCA_macro_ring.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Creating rings.
Creating via connection between strategies and existing shapes.
Via DRC checking runtime 0.00 seconds.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias inside strategy ORCA_macro_ring.
Checking 80 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag MACRO_RING to committed wires.
Committed 80 wires.
Committing wires takes 0.00 seconds.
Committed 80 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 1 seconds.
Successfully compiled PG.
Overall runtime: 1 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

- Connect Macro Ring to Macro power pins

```
icc2_shell> create_pg_macro_conn_pattern Macro_ring_pin -pin_conn_type scattered_pin -layers {M5 M6}
Successfully create macro connection pattern Macro_ring_pin.
icc2_shell> set_pg_strategy Macro_pins_via -macros $only_macros -pattern { {pattern: Macro_ring_pin} {nets: {VSS VDD}} }
Successfully set PG strategy Macro_pins_via.
icc2_shell>
icc2_shell> compile_pg -strategies {Macro_pins_via} -tag {Macro_RING_con}
Sanity check for inputs.
No strategy-level via rule is specified, the default rule will be applied.
Set tag to Macro_RING_con
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy Macro_pins_via.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Connecting macros and pads.
Creating via connection between strategies and existing shapes.
Check and fix DRCs for 2686 stacked vias.
Checking DRC for 2686 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 100%
Via DRC checking runtime 2.00 seconds.
via connection runtime: 2 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias between strategies and existing shapes.
Checking 2686 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag Macro_RING_con to committed wires.
Committed 2686 wires.
Committing wires takes 0.00 seconds.
Committed 2686 vias.
Committed 0 wires for via creation.
Committing vias takes 1.00 seconds.
Overall PG creation runtime: 5 seconds.
Successfully compiled PG.
Overall runtime: 5 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

- Create PLL\_PCI Ring

```
icc2_shell> create_pg_ring_pattern PLL_PCI_ring -horizontal_layer M5 \
? -horizontal_width {1} -vertical_layer M4 -vertical_width {1} -corner_bridge false
Successfully create PG ring pattern PLL_PCI_ring.
icc2_shell> set_pg_strategy ORCA_PLL_PCI_ring -macros "I_CLOCK_GEN/I_PLL_PCI" \
? -pattern { {pattern: PLL_PCI_ring} {nets: {AVSS DVDD1 AVDD1}} {offset: {0.5 0.5}} }
Successfully set PG strategy ORCA_PLL_PCI_ring.
icc2_shell> set_pg_strategy_via_rule PLL_PCI_ring_vias -via_rule { \
? {{{strategies: {ORCA_PLL_PCI_ring}} {layers: {M5}}} {existing: {strap }}{via_master: {default}}} \
? {{{strategies: {ORCA_PLL_PCI_ring}} {layers: {M4}}} {existing: {strap }}{via_master: {default}}} \
? }
Successfully set strategy via rule PLL_PCI_ring_vias.
icc2_shell> compile_pg -strategies {ORCA_PLL_PCI_ring} -via_rule {PLL_PCI_ring_vias} -tag {PLL_PCI_RING}
Sanity check for inputs.
Set tag to PLL_PCI_RING
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy ORCA_PLL_PCI_ring.
Loading library and design information.
Number of Standard Cells: 9398
Number of Hard Macros: 12
Number of Pads: 104
Creating rings.
Creating via connection between strategies and existing shapes.
Via DRC checking runtime 0.00 seconds.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias inside strategy ORCA_PLL_PCI_ring.
Checking 12 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag PLL_PCI_RING to committed wires.
Committed 12 wires.
Committing wires takes 0.00 seconds.
Committed 12 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 1 seconds.
Successfully compiled PG.
Overall runtime: 1 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

- Connect PLL\_PCI Ring to Macro power pins

```
icc2_shell> create_pg_macro_conn_pattern pll_ring_pin -pin_conn_type scattered_pin -layers {M5 M6}
Successfully create macro connection pattern pll_ring_pin.
icc2_shell> set_pg_strategy pll_pins_via -macros "I_CLOCK_GEN/I_PLL_PCI" -pattern { {pattern: pll_ring_pin} {nets: {AVSS
DVDD1 AVDD1}} }
Successfully set PG strategy pll_pins_via.
icc2_shell>
icc2_shell> compile_pg -strategies {pll_pins_via} -tag {PLL_PCI_RING_con}
Sanity check for inputs.
No strategy-level via rule is specified, the default rule will be applied.
Set tag to PLL_PCI_RING_con
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy pll_pins_via.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Connecting macros and pads.
Creating via connection between strategies and existing shapes.
Check and fix DRCs for 3 stacked vias.
Checking DRC for 3 stacked vias:0% 30% 65% 100%
Via DRC checking runtime 0.00 seconds.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias between strategies and existing shapes.
Checking 3 stacked vias:0% 30% 60% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag PLL_PCI_RING_con to committed wires.
Committed 3 wires.
Committing wires takes 0.00 seconds.
Committed 3 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 0 seconds.
Successfully compiled PG.
Overall runtime: 0 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

- Create PLL\_SD Ring

```
icc2_shell> create_pg_ring_pattern PLL_SD_ring -horizontal_layer M5 \
? -horizontal_width {1} -vertical_layer M4 -vertical_width {1} -corner_bridge false
Successfully create PG ring pattern PLL_SD_ring.
icc2_shell> set_pg_strategy ORCA_PLL_SD_ring -macros "I_CLOCK_GEN/I_PLL_SD" \
? -pattern { {pattern: PLL_SD_ring} {nets: {AVSS DVDD2 AVDD2}} {offset: {0.5 0.5}} }
Successfully set PG strategy ORCA_PLL_SD_ring.
icc2_shell> set_pg_strategy_via_rule PLL_SD_ring_vias -via_rule { \
? {{{strategies: {ORCA_PLL_SD_ring}}} {layers: {M5}}} {existing: {strap }}{via_master: {default}}} \
? {{{strategies: {ORCA_PLL_SD_ring}}} {layers: {M4}}} {existing: {strap }}{via_master: {default}}} \
? }
Successfully set strategy via rule PLL_SD_ring_vias.
icc2_shell> compile_pg -strategies {ORCA_PLL_SD_ring} -via_rule {PLL_SD_ring_vias} -tag {PLL_SD_RING}
Sanity check for inputs.
Set tag to PLL_SD_RING
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy ORCA_PLL_SD_ring.
Loading library and design information.
Number of Standard Cells: 9398
Number of Hard Macros: 12
Number of Pads: 104
Creating rings.
Creating via connection between strategies and existing shapes.
Via DRC checking runtime 0.00 seconds.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias inside strategy ORCA_PLL_SD_ring.
Checking 12 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag PLL_SD_RING to committed wires.
Committed 12 wires.
Committing wires takes 0.00 seconds.
Committed 12 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 1 seconds.
Successfully compiled PG.
Overall runtime: 1 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 5. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

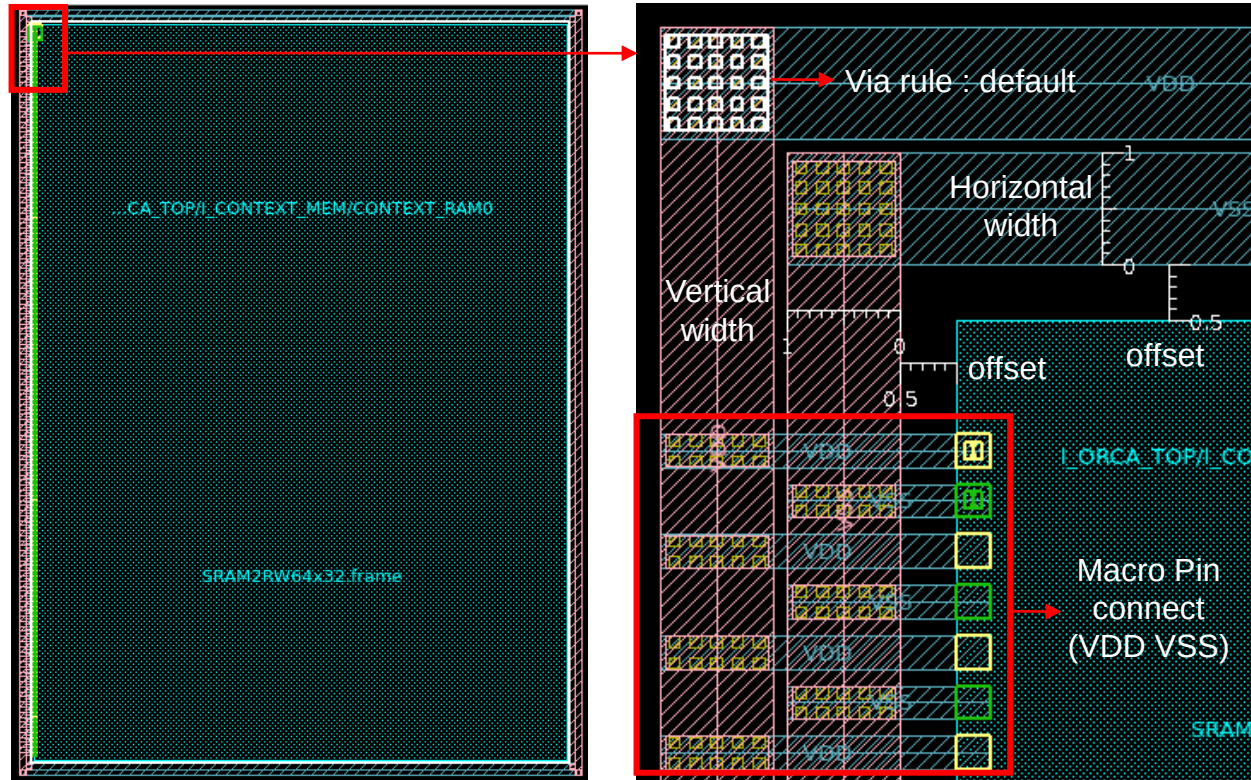
- Connect PLL\_PCI Ring to Macro power pins

```
icc2_shell> create_pg_macro_conn_pattern pll_ring_pin -pin_conn_type scattered_pin -layers {M5 M6}
Pattern pll_ring_pin exists and is re-defined.
Successfully create macro connection pattern pll_ring_pin.
icc2_shell> set_pg_strategy pll_pins_via -macros "I_CLOCK_GEN/I_PLL_SD" -pattern { {pattern: pll_ring_pin} {nets: {AVSS
DVDD2 AVDD2}} }
Strategy pll_pins_via exists and is re-defined for the current design.
Successfully set PG strategy pll_pins_via.
icc2_shell> compile_pg -strategies {pll_pins_via} -tag {PLL_SD_RING_con}
Sanity check for inputs.
No strategy-level via rule is specified, the default rule will be applied.
Set tag to PLL_SD_RING_con
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy pll_pins_via.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Connecting macros and pads.
Creating via connection between strategies and existing shapes.
Check and fix DRCs for 3 stacked vias.
Checking DRC for 3 stacked vias:0% 30% 65% 100%
Via DRC checking runtime 0.00 seconds.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias between strategies and existing shapes.
Checking 3 stacked vias:0% 30% 60% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag PLL_SD_RING_con to committed wires.
Committed 3 wires.
Committing wires takes 0.00 seconds.
Committed 3 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 0 seconds.
Successfully compiled PG.
Overall runtime: 0 seconds.
1
```

# Powerplan : Use ICC2 (IC Compiler II tool)

4. Icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/04\_create\_Macro\_and\_PLL\_ring.tcl

- Check Macro Ring

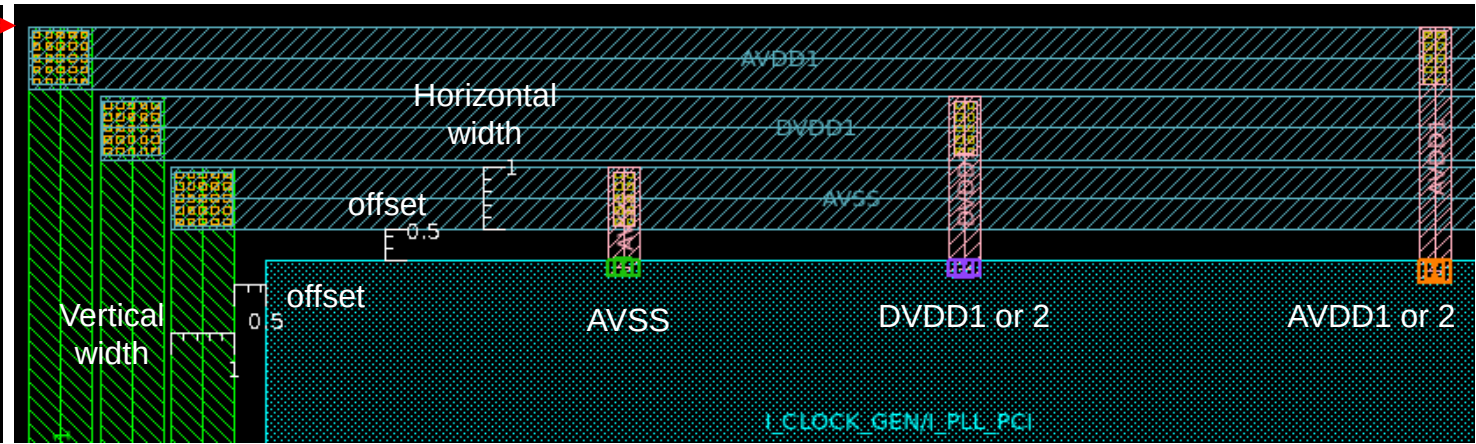
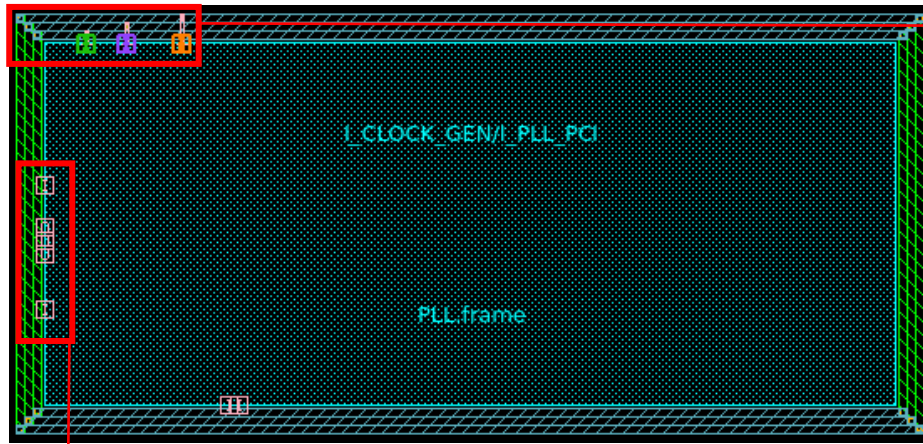


- Zoom in and check Macro Ring and Macro Power pins (VDD, VSS) are correctly connected
- Zoom in and check if the Horizontal & Vertical (width,offset) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

4. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/04_create_Macro_and_PLL_ring.tcl`

- Check PLL Ring



- Because the pins on the PLL are M6, M4 was used for the vertical layer of the PLL ring considering routing

- Zoom in and check PLL Ring and PLL Power pins (AVDD1, AVDD2, DVDD1, DVDD2, AVSS) are correctly connected
- Zoom in and check if the Horizontal & Vertical (width,offset) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

- Connect PLL power pin to Power PAD (AVDD)

```
icc2_shell> set_snap_setting -enabled {false}
icc2_shell> read_def ./FP_data/PLL_AVDD.def
Information: Loading DEF file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/03_powerplan/FP_data/PLL_AVDD.def' (FILE-007)
Information: Reading 'VERSION' statement. (DEFR-016)
Information: Reading 'DIVIDERCHAR' statement. (DEFR-016)
Information: Reading 'BUSBITCHARS' statement. (DEFR-016)
Information: Reading 'DESIGN' statement. (DEFR-016)
Information: Reading 'UNITS' statement. (DEFR-016)
Information: Reading 'SPECIALNETS' section. (DEFR-016)
-----
Number of Processed/Read DEF Constructs
-----
VERSION                : 1/1
DIVIDERCHAR             : 1/1
BUSBITCHARS            : 1/1
DESIGN                  : 1/1
UNITS                   : 1/1
SPECIALNETS            : 1/1
1
icc2_shell> change_selection [get_shapes -filter "net.name == AVDD"]
icc2_shell> set_attribute -objects [get_selection] -name tag -value PLL_PCI_AVDD
Warning: Attribute 'tag' has not been defined for the terminal object class. (ATTR-1)
{PATH_25_38 PATH_25_37 PATH_25_36 PATH_25_35 PATH_25_34 PATH_25_33 PATH_25_32 PATH_25_31 PATH_25_30 PATH_25_29 PATH_25_28 PATH_25_27 PATH_25_26 PATH_25_25 PATH_25_24 PATH_25_23 PATH_25_22 PATH_25_21 PATH_25_20 PATH_25_19 PATH_25_18 PATH_25_17 PATH_25_16 PATH_25_15 PATH_23_15 PATH_25_13 PATH_23_14 PATH_25_12 PATH_23_13 PATH_25_11 PATH_23_12 PATH_25_10 PATH_23_11 PATH_25_9 PATH_23_10 PATH_25_8 PATH_23_9 PATH_25_7 PATH_23_8 PATH_25_6 PATH_23_7 PATH_25_5 PATH_23_6 PATH_25_4 PATH_25_40 PATH_23_1 PATH_25_39 PATH_23_0 PATH_25_41 PATH_23_2 PATH_25_0 PATH_25_42 PATH_23_3 PATH_25_1 PATH_23_4 PATH_25_2 PATH_23_5 PATH_25_3 PATH_23_16 PATH_25_14 PATH_23_17 PATH_23_18 PATH_23_19 PATH_23_20}
icc2_shell> set AVDD_shape [get_shapes -filter "tag == PLL_PCI_RING && layer.name == M6 || tag == PLL_PCI_AVDD"]
{PATH_23_0 PATH_23_1 PATH_23_2 PATH_23_3 PATH_23_4 PATH_23_5 PATH_23_6 PATH_23_7 PATH_23_8 PATH_23_9 PATH_23_10 PATH_23_11 PATH_23_12 PATH_23_13 PATH_23_14 PATH_23_15 PATH_23_16 PATH_23_17 PATH_23_18 PATH_23_19 PATH_23_20 PATH_25_0 PATH_25_1 PATH_25_2 PATH_25_3 PATH_25_4 PATH_25_5 PATH_25_6 PATH_25_7 PATH_25_8 PATH_25_9 PATH_25_10 PATH_25_11 PATH_25_12 PATH_25_13 PATH_25_14 PATH_25_15 PATH_25_16 PATH_25_17 PATH_25_18 PATH_25_19 PATH_25_20 PATH_25_21 PATH_25_22 PATH_25_23 PATH_25_24 PATH_25_25 PATH_25_26 PATH_25_27 PATH_25_28 PATH_25_29 PATH_25_30 PATH_25_31 PATH_25_32 PATH_25_33 PATH_25_34 PATH_25_35 PATH_25_36 PATH_25_37 PATH_25_38 PATH_25_39 PATH_25_40 PATH_25_41 PATH_25_42}
icc2_shell> set_pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10}
Information: The command 'set_pg_via_master_rule' cleared the undo history. (UNDO-016)
Successfully set via def rule pgvia_10x10.
icc2_shell> change_selection $AVDD_shape
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

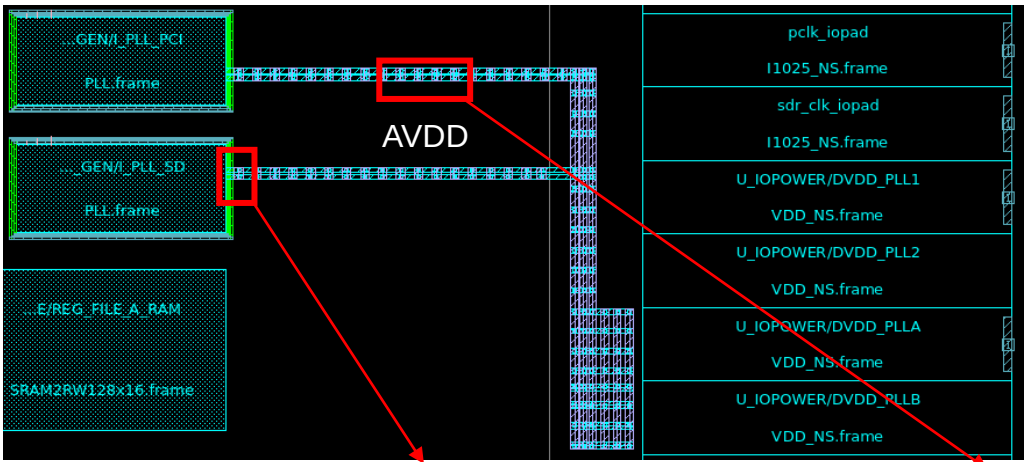
- Connect PLL power pin to Power PAD (AVDD)

```
icc2_shell> create_pg_vias -nets {AVDD} -to types {lib_cell_pin_connect stripe} -from_layers M7 -to_layers M8 -via_masters
{pgvia_10x10} -tag {PLL_PCI_AVDD} -shapes [get_selection] ;#create vias Manual routing(M7 to M8)
Set tag to PLL_PCI_AVDD
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 74 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 161 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 161 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 161 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 161 vias.
Setting tag PLL_PCI_AVDD to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 0 seconds.
Successfully created PG Vias.
1
icc2_shell>
icc2_shell> #M6_M7 via connection AVDD
icc2_shell> set_pg_via_master_rule pgvia_5x5 -via_array_dimension {5 5} ;#set via_master_rule (column 5 row 5)
Successfully set via def rule pgvia_5x5.
icc2_shell>
icc2_shell> create_pg_vias -nets {AVDD} -to types {lib_cell_pin_connect stripe} -from_layers M4 -to_layers M7 -via_masters
{pgvia_5x5} -tag {PLL_PCI_AVDD} -shapes [get_selection] ;#create vias Macro_ring to manual routing (M6 to M7)
Set tag to PLL_PCI_AVDD
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 74 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 4 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 4 stacked vias:0% 25% 50% 75% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 4 stacked vias:0% 20% 50% 70% 100%
Committed 12 vias.
Setting tag PLL_PCI_AVDD to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```

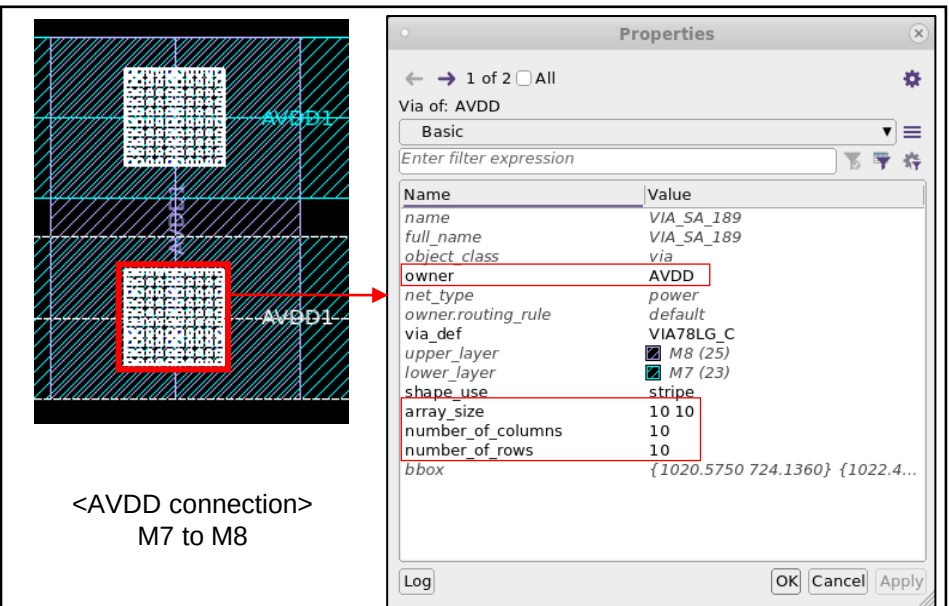
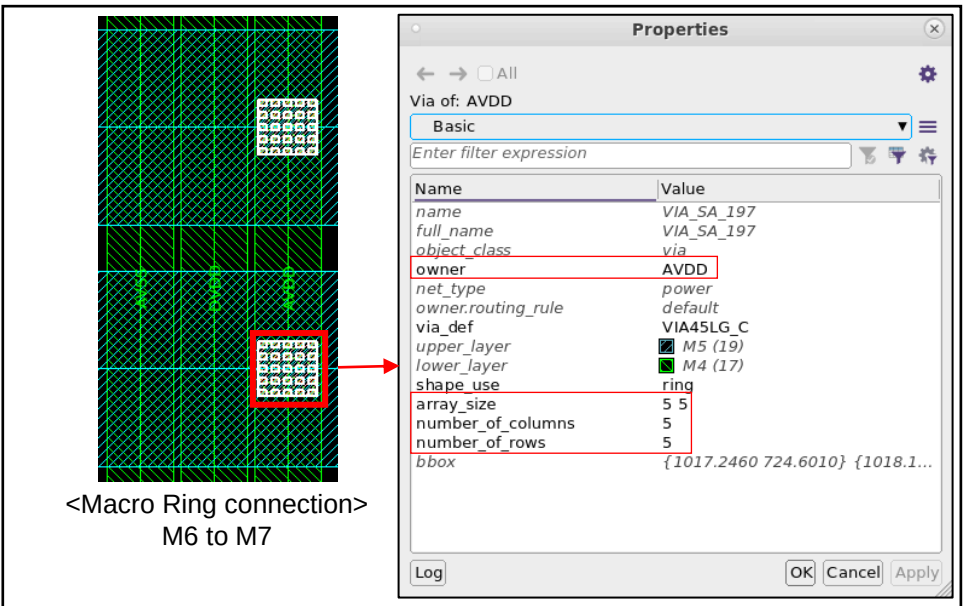
# Powerplan : Use ICC2 (IC Compiler II tool)

6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

- Check AVDD after create shape & vias



- Zoom in and check Macro Ring to AVDD stripes are correctly connected
- Zoom in and check AVDD stripe M7 to M8 are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created



# Powerplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/05_PLL_manual_routing.tcl`

- Connect PLL power pin to Power PAD (DVDD)

```
icc2_shell> read_def ./FP_data/PLL_DVDD.def
Information: Loading DEF file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/03_powerplan/FP_data/PLL_DVDD.def' (FILE-007)
Information: Reading 'VERSION' statement. (DEFR-016)
Information: Reading 'DIVIDERCHAR' statement. (DEFR-016)
Information: Reading 'BUSBITCHARS' statement. (DEFR-016)
Information: Reading 'DESIGN' statement. (DEFR-016)
Information: Reading 'UNITS' statement. (DEFR-016)
Information: Reading 'SPECIALNETS' section. (DEFR-016)
-----
Number of Processed/Read DEF Constructs
-----
VERSION                : 1/1
DIVIDERCHAR            : 1/1
BUSBITCHARS            : 1/1
DESIGN                 : 1/1
UNITS                 : 1/1
SPECIALNETS           : 1/1
1
icc2_shell> change_selection [get_shapes -filter "net.name == DVDD"]
icc2_shell> set_attribute -objects [get_selection] -name tag -value PLL_PCI_DVDD
Warning: Attribute 'tag' has not been defined for the terminal object class. (ATTR-1)
{PATH_25_42 PATH_25_41 PATH_25_40 PATH_25_39 PATH_25_38 PATH_25_37 PATH_25_36 PATH_25_35 PATH_25_34 PATH_25_33 PATH_25_32 PATH_25_31 PATH_25_30 PATH_25_29 PATH_25_28 PATH_25_27 PATH_25_26 PATH_25_25 PATH_25_24 PATH_25_23 PATH_25_22 PATH_25_21 PATH_25_20 PATH_25_19 PATH_25_18 PATH_25_17 PATH_25_16 PATH_25_15 PATH_25_14 PATH_25_13 PATH_25_12 PATH_25_11 PATH_25_10 PATH_25_9 PATH_25_8 PATH_25_7 PATH_25_6 PATH_25_5 PATH_25_4 PATH_25_3 PATH_25_2 PATH_25_1 PATH_25_0 PATH_24_4 PATH_24_3 PATH_24_2 PATH_24_1 PATH_24_0 PATH_23_4 PATH_23_3 PATH_23_2 PATH_23_1 PATH_23_0 PATH_22_4 PATH_22_3 PATH_22_2 PATH_22_1 PATH_22_0 PATH_21_4 PATH_21_3 PATH_21_2 PATH_21_1 PATH_21_0 PATH_20_4 PATH_20_3 PATH_20_2 PATH_20_1 PATH_20_0 PATH_19_4 PATH_19_3 PATH_19_2 PATH_19_1 PATH_19_0 PATH_18_4 PATH_18_3 PATH_18_2 PATH_18_1 PATH_18_0 PATH_17_4 PATH_17_3 PATH_17_2 PATH_17_1 PATH_17_0 PATH_16_4 PATH_16_3 PATH_16_2 PATH_16_1 PATH_16_0 PATH_15_4 PATH_15_3 PATH_15_2 PATH_15_1 PATH_15_0 PATH_14_4 PATH_14_3 PATH_14_2 PATH_14_1 PATH_14_0 PATH_13_4 PATH_13_3 PATH_13_2 PATH_13_1 PATH_13_0 PATH_12_4 PATH_12_3 PATH_12_2 PATH_12_1 PATH_12_0 PATH_11_4 PATH_11_3 PATH_11_2 PATH_11_1 PATH_11_0 PATH_10_4 PATH_10_3 PATH_10_2 PATH_10_1 PATH_10_0 PATH_9_4 PATH_9_3 PATH_9_2 PATH_9_1 PATH_9_0 PATH_8_4 PATH_8_3 PATH_8_2 PATH_8_1 PATH_8_0 PATH_7_4 PATH_7_3 PATH_7_2 PATH_7_1 PATH_7_0 PATH_6_4 PATH_6_3 PATH_6_2 PATH_6_1 PATH_6_0 PATH_5_4 PATH_5_3 PATH_5_2 PATH_5_1 PATH_5_0 PATH_4_4 PATH_4_3 PATH_4_2 PATH_4_1 PATH_4_0 PATH_3_4 PATH_3_3 PATH_3_2 PATH_3_1 PATH_3_0 PATH_2_4 PATH_2_3 PATH_2_2 PATH_2_1 PATH_2_0 PATH_1_4 PATH_1_3 PATH_1_2 PATH_1_1 PATH_1_0 PATH_0_4 PATH_0_3 PATH_0_2 PATH_0_1 PATH_0_0}
icc2_shell> set DVDD_shape [get_shapes -filter "tag == PLL_PCI_RING && layer.name == M6 || tag == PLL_PCI_DVDD"] ;
{PATH_19_19 PATH_19_20 PATH_19_21 PATH_19_22 PATH_19_23 PATH_19_24 PATH_19_25 PATH_19_26 PATH_19_27 PATH_19_28 PATH_19_29 PATH_19_30 PATH_19_31 PATH_19_32 PATH_19_33 PATH_19_34 PATH_19_35 PATH_19_36 PATH_19_37 PATH_19_38 PATH_19_39 PATH_19_40 PATH_19_41 PATH_19_42 PATH_18_19 PATH_18_20 PATH_18_21 PATH_18_22 PATH_18_23 PATH_18_24 PATH_18_25 PATH_18_26 PATH_18_27 PATH_18_28 PATH_18_29 PATH_18_30 PATH_18_31 PATH_18_32 PATH_18_33 PATH_18_34 PATH_18_35 PATH_18_36 PATH_18_37 PATH_18_38 PATH_18_39 PATH_18_40 PATH_18_41 PATH_18_42 PATH_17_19 PATH_17_20 PATH_17_21 PATH_17_22 PATH_17_23 PATH_17_24 PATH_17_25 PATH_17_26 PATH_17_27 PATH_17_28 PATH_17_29 PATH_17_30 PATH_17_31 PATH_17_32 PATH_17_33 PATH_17_34 PATH_17_35 PATH_17_36 PATH_17_37 PATH_17_38 PATH_17_39 PATH_17_40 PATH_17_41 PATH_17_42 PATH_16_19 PATH_16_20 PATH_16_21 PATH_16_22 PATH_16_23 PATH_16_24 PATH_16_25 PATH_16_26 PATH_16_27 PATH_16_28 PATH_16_29 PATH_16_30 PATH_16_31 PATH_16_32 PATH_16_33 PATH_16_34 PATH_16_35 PATH_16_36 PATH_16_37 PATH_16_38 PATH_16_39 PATH_16_40 PATH_16_41 PATH_16_42 PATH_15_19 PATH_15_20 PATH_15_21 PATH_15_22 PATH_15_23 PATH_15_24 PATH_15_25 PATH_15_26 PATH_15_27 PATH_15_28 PATH_15_29 PATH_15_30 PATH_15_31 PATH_15_32 PATH_15_33 PATH_15_34 PATH_15_35 PATH_15_36 PATH_15_37 PATH_15_38 PATH_15_39 PATH_15_40 PATH_15_41 PATH_15_42 PATH_14_19 PATH_14_20 PATH_14_21 PATH_14_22 PATH_14_23 PATH_14_24 PATH_14_25 PATH_14_26 PATH_14_27 PATH_14_28 PATH_14_29 PATH_14_30 PATH_14_31 PATH_14_32 PATH_14_33 PATH_14_34 PATH_14_35 PATH_14_36 PATH_14_37 PATH_14_38 PATH_14_39 PATH_14_40 PATH_14_41 PATH_14_42 PATH_13_19 PATH_13_20 PATH_13_21 PATH_13_22 PATH_13_23 PATH_13_24 PATH_13_25 PATH_13_26 PATH_13_27 PATH_13_28 PATH_13_29 PATH_13_30 PATH_13_31 PATH_13_32 PATH_13_33 PATH_13_34 PATH_13_35 PATH_13_36 PATH_13_37 PATH_13_38 PATH_13_39 PATH_13_40 PATH_13_41 PATH_13_42 PATH_12_19 PATH_12_20 PATH_12_21 PATH_12_22 PATH_12_23 PATH_12_24 PATH_12_25 PATH_12_26 PATH_12_27 PATH_12_28 PATH_12_29 PATH_12_30 PATH_12_31 PATH_12_32 PATH_12_33 PATH_12_34 PATH_12_35 PATH_12_36 PATH_12_37 PATH_12_38 PATH_12_39 PATH_12_40 PATH_12_41 PATH_12_42 PATH_11_19 PATH_11_20 PATH_11_21 PATH_11_22 PATH_11_23 PATH_11_24 PATH_11_25 PATH_11_26 PATH_11_27 PATH_11_28 PATH_11_29 PATH_11_30 PATH_11_31 PATH_11_32 PATH_11_33 PATH_11_34 PATH_11_35 PATH_11_36 PATH_11_37 PATH_11_38 PATH_11_39 PATH_11_40 PATH_11_41 PATH_11_42 PATH_10_19 PATH_10_20 PATH_10_21 PATH_10_22 PATH_10_23 PATH_10_24 PATH_10_25 PATH_10_26 PATH_10_27 PATH_10_28 PATH_10_29 PATH_10_30 PATH_10_31 PATH_10_32 PATH_10_33 PATH_10_34 PATH_10_35 PATH_10_36 PATH_10_37 PATH_10_38 PATH_10_39 PATH_10_40 PATH_10_41 PATH_10_42 PATH_9_19 PATH_9_20 PATH_9_21 PATH_9_22 PATH_9_23 PATH_9_24 PATH_9_25 PATH_9_26 PATH_9_27 PATH_9_28 PATH_9_29 PATH_9_30 PATH_9_31 PATH_9_32 PATH_9_33 PATH_9_34 PATH_9_35 PATH_9_36 PATH_9_37 PATH_9_38 PATH_9_39 PATH_9_40 PATH_9_41 PATH_9_42 PATH_8_19 PATH_8_20 PATH_8_21 PATH_8_22 PATH_8_23 PATH_8_24 PATH_8_25 PATH_8_26 PATH_8_27 PATH_8_28 PATH_8_29 PATH_8_30 PATH_8_31 PATH_8_32 PATH_8_33 PATH_8_34 PATH_8_35 PATH_8_36 PATH_8_37 PATH_8_38 PATH_8_39 PATH_8_40 PATH_8_41 PATH_8_42 PATH_7_19 PATH_7_20 PATH_7_21 PATH_7_22 PATH_7_23 PATH_7_24 PATH_7_25 PATH_7_26 PATH_7_27 PATH_7_28 PATH_7_29 PATH_7_30 PATH_7_31 PATH_7_32 PATH_7_33 PATH_7_34 PATH_7_35 PATH_7_36 PATH_7_37 PATH_7_38 PATH_7_39 PATH_7_40 PATH_7_41 PATH_7_42 PATH_6_19 PATH_6_20 PATH_6_21 PATH_6_22 PATH_6_23 PATH_6_24 PATH_6_25 PATH_6_26 PATH_6_27 PATH_6_28 PATH_6_29 PATH_6_30 PATH_6_31 PATH_6_32 PATH_6_33 PATH_6_34 PATH_6_35 PATH_6_36 PATH_6_37 PATH_6_38 PATH_6_39 PATH_6_40 PATH_6_41 PATH_6_42 PATH_5_19 PATH_5_20 PATH_5_21 PATH_5_22 PATH_5_23 PATH_5_24 PATH_5_25 PATH_5_26 PATH_5_27 PATH_5_28 PATH_5_29 PATH_5_30 PATH_5_31 PATH_5_32 PATH_5_33 PATH_5_34 PATH_5_35 PATH_5_36 PATH_5_37 PATH_5_38 PATH_5_39 PATH_5_40 PATH_5_41 PATH_5_42 PATH_4_19 PATH_4_20 PATH_4_21 PATH_4_22 PATH_4_23 PATH_4_24 PATH_4_25 PATH_4_26 PATH_4_27 PATH_4_28 PATH_4_29 PATH_4_30 PATH_4_31 PATH_4_32 PATH_4_33 PATH_4_34 PATH_4_35 PATH_4_36 PATH_4_37 PATH_4_38 PATH_4_39 PATH_4_40 PATH_4_41 PATH_4_42 PATH_3_19 PATH_3_20 PATH_3_21 PATH_3_22 PATH_3_23 PATH_3_24 PATH_3_25 PATH_3_26 PATH_3_27 PATH_3_28 PATH_3_29 PATH_3_30 PATH_3_31 PATH_3_32 PATH_3_33 PATH_3_34 PATH_3_35 PATH_3_36 PATH_3_37 PATH_3_38 PATH_3_39 PATH_3_40 PATH_3_41 PATH_3_42 PATH_2_19 PATH_2_20 PATH_2_21 PATH_2_22 PATH_2_23 PATH_2_24 PATH_2_25 PATH_2_26 PATH_2_27 PATH_2_28 PATH_2_29 PATH_2_30 PATH_2_31 PATH_2_32 PATH_2_33 PATH_2_34 PATH_2_35 PATH_2_36 PATH_2_37 PATH_2_38 PATH_2_39 PATH_2_40 PATH_2_41 PATH_2_42 PATH_1_19 PATH_1_20 PATH_1_21 PATH_1_22 PATH_1_23 PATH_1_24 PATH_1_25 PATH_1_26 PATH_1_27 PATH_1_28 PATH_1_29 PATH_1_30 PATH_1_31 PATH_1_32 PATH_1_33 PATH_1_34 PATH_1_35 PATH_1_36 PATH_1_37 PATH_1_38 PATH_1_39 PATH_1_40 PATH_1_41 PATH_1_42 PATH_0_19 PATH_0_20 PATH_0_21 PATH_0_22 PATH_0_23 PATH_0_24 PATH_0_25 PATH_0_26 PATH_0_27 PATH_0_28 PATH_0_29 PATH_0_30 PATH_0_31 PATH_0_32 PATH_0_33 PATH_0_34 PATH_0_35 PATH_0_36 PATH_0_37 PATH_0_38 PATH_0_39 PATH_0_40 PATH_0_41 PATH_0_42}
icc2_shell> set pg_via_master_rule pgvia_10x10 -via_array_dimension {10 10} ;
Information: The command 'set_pg_via_master_rule' cleared the undo history. (UNDO-016)
Successfully set via def rule pgvia_10x10.
icc2_shell> change_selection $DVDD_shape
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

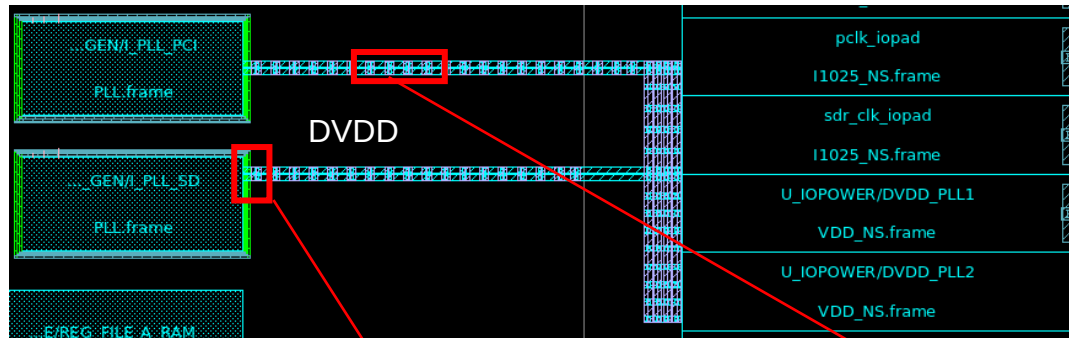
- Connect PLL power pin to Power PAD (DVDD)

```
icc2_shell> create_pg_vias -nets {DVDD} -to_types {lib_cell_pin_connect stripe} -from_layers M7 -to_layers M8 -via_masters {pgvia_10x10} -tag {PLL_PCI_DVDD} -shapes [get_selection]
Set tag to PLL_PCI_DVDD
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 69 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 142 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 142 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 142 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 142 vias.
Setting tag PLL_PCI_DVDD to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 0 seconds.
Successfully created PG Vias.
1
icc2_shell> set_pg_via_master_rule pgvia_5x5 -via_array_dimension {5 5} ;#set via_master_rule (column 5 row 5)
Successfully set via def rule pgvia_5x5.
icc2_shell>
icc2_shell> create_pg_vias -nets {DVDD} -to_types {lib_cell_pin_connect stripe} -from_layers M4 -to_layers M7 -via_masters {pgvia_5x5} -tag {PLL_PCI_DVDD} -shapes [get_selection]
Set tag to PLL_PCI_DVDD
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 69 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 4 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 4 stacked vias:0% 25% 50% 75% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 4 stacked vias:0% 20% 50% 70% 100%
Committed 12 vias.
Setting tag PLL_PCI_DVDD to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```

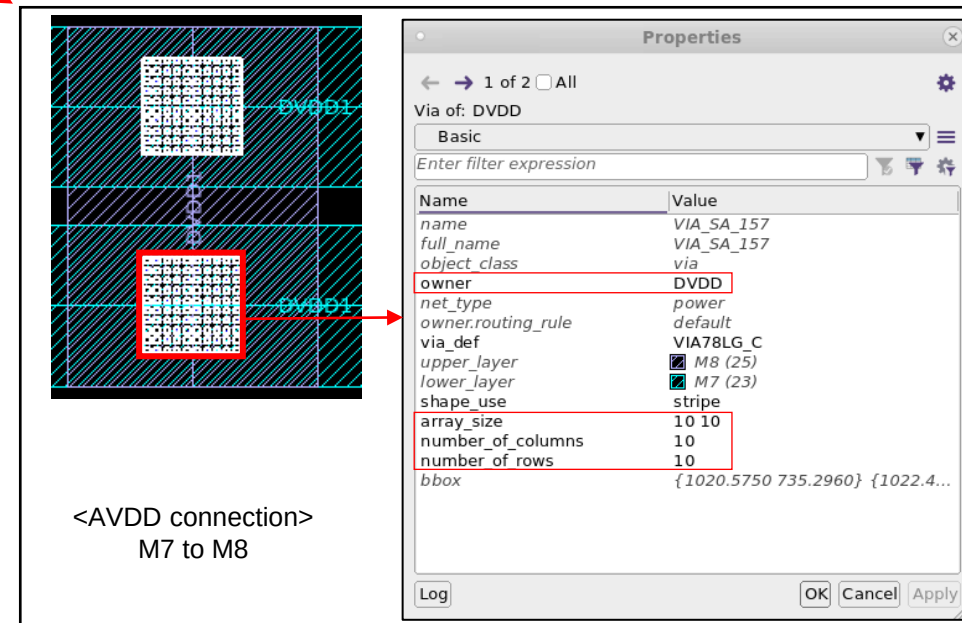
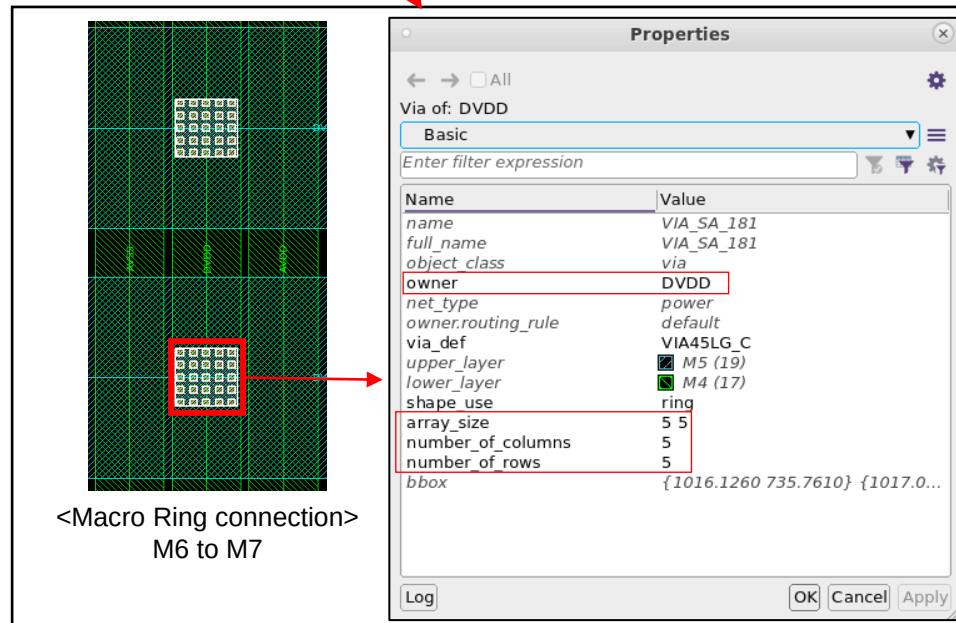
# Powerplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/05_PLL_manual_routing.tcl`

- Check DVDD after create shape & vias



- Zoom in and check Macro Ring to DVDD stripes are correctly connected
- Zoom in and check DVDD stripe M7 to M8 are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created



# Powerplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

- Connect PLL power pin to Power PAD (AVSS)

```
icc2_shell> read_def ./FP_data/PLL_AVSS.def
Information: Loading DEF file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/03_powerplan/FP_data/PLL_AVSS.def' (FILE-007)
Information: Reading 'VERSION' statement. (DEFR-016)
Information: Reading 'DIVIDERCHAR' statement. (DEFR-016)
Information: Reading 'BUSBITCHARS' statement. (DEFR-016)
Information: Reading 'DESIGN' statement. (DEFR-016)
Information: Reading 'UNITS' statement. (DEFR-016)
Information: Reading 'SPECIALNETS' section. (DEFR-016)
Information: Reading 'NETS' section. (DEFR-016)

-----
Number of Processed/Read DEF Constructs
-----
VERSION                : 1/1
DIVIDERCHAR            : 1/1
BUSBITCHARS            : 1/1
DESIGN                 : 1/1
UNITS                  : 1/1
SPECIALNETS           : 1/1
NETS                   : 1/1
1
icc2_shell>
icc2_shell> change_selection [get_shapes -filter "net.name == AVSS"]
icc2_shell> set_attribute -objects [get_selection] -name tag -value PLL_SD_AVSS ;#tag the shape (easy to find)
Warning: Attribute 'tag' has not been defined for the terminal object class. (ATTR-1)
{PATH 25 15 PATH 25 14 PATH 25 12 PATH 25 11 PATH 25 10 PATH 25 9 PATH 25 8 PATH 25 7 PATH 25 6 PATH 25 5 PATH 25 4 PATH 25 3 PATH 25 2 PATH 25 1 PATH 25 0 PATH 25 13 PATH 21 0 PATH 17 7 PATH 17 6 PATH 17 1 PATH 17 0 PATH 25 40 PATH 23 24 PATH 25 39 PATH 23 23 PATH 25 38 PATH 23 22 PATH 25 37 PATH 23 21 PATH 25 34 PATH 23 18 PATH 25 33 PATH 23 17 PATH 25 32 PATH 23 16 PATH 25 31 PATH 23 15 PATH 25 30 PATH 23 14 PATH 25 29 PATH 23 13 PATH 25 28 PATH 23 12 PATH 25 27 PATH 23 11 PATH 25 26 PATH 23 10 PATH 25 25 PATH 23 9 PATH 25 24 PATH 23 8 PATH 25 23 PATH 23 7 PATH 25 22 PATH 23 6 PATH 25 21 PATH 23 5 PATH 25 20 PATH 23 4 PATH 25 19 PATH 23 3 PATH 25 18 PATH 23 2 PATH 25 17 PATH 23 1 PATH 25 16 PATH 21 3 PATH 23 0 PATH 23 26 PATH 19 24 PATH 25 41 PATH 23 25 PATH 19 23 PATH 25 36 PATH 23 20 PATH 19 18 PATH 25 35 PATH 23 19 PATH 19 17}
icc2_shell> set AVSS_shape [get_shapes -filter "tag == PLL_PCI_RING && layer.name == M6 || tag == PLL_SD_AVSS"] ;#select net for create via
{PATH 19 17 PATH 19 18 PATH 19 23 PATH 19 24 PATH 23 0 PATH 23 1 PATH 23 2 PATH 23 3 PATH 23 4 PATH 23 5 PATH 23 6 PATH 23 7 PATH 23 8 PATH 23 9 PATH 23 10 PATH 23 11 PATH 23 12 PATH 23 13 PATH 23 14 PATH 23 15 PATH 23 16 PATH 23 17 PATH 23 18 PATH 23 19 PATH 23 20 PATH 23 21 PATH 23 22 PATH 23 23 PATH 23 24 PATH 23 25 PATH 23 26 PATH 17 0 PATH 17 1 PATH 17 6 PATH 17 7 PATH 21 0 PATH 21 3 PATH 25 0 PATH 25 1 PATH 25 2 PATH 25 3 PATH 25 4 PATH 25 5 PATH 25 6 PATH 25 7 PATH 25 8 PATH 25 9 PATH 25 10 PATH 25 11 PATH 25 12 PATH 25 13 PATH 25 14 PATH 25 15 PATH 25 16 PATH 25 17 PATH 25 18 PATH 25 19 PATH 25 20 PATH 25 21 PATH 25 22 PATH 25 23 PATH 25 24 PATH 25 25 PATH 25 26 PATH 25 27 PATH 25 28 PATH 25 29 PATH 25 30 PATH 25 31 PATH 25 32 PATH 25 33 PATH 25 34 PATH 25 35 PATH 25 36 PATH 25 37 PATH 25 38 PATH 25 39 PATH 25 40 PATH 25 41}
icc2_shell> set_pg_via_master_rule pgvia 10x10 -via_array_dimension {10 10} ;#set via master_rule (column 10 row 10)
Information: The command 'set_pg_via_master_rule' cleared the undo history. (UNDO-016)
Successfully set via def rule pgvia 10x10.
icc2_shell>
icc2_shell> change_selection $AVSS_shape
```

# Powerplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

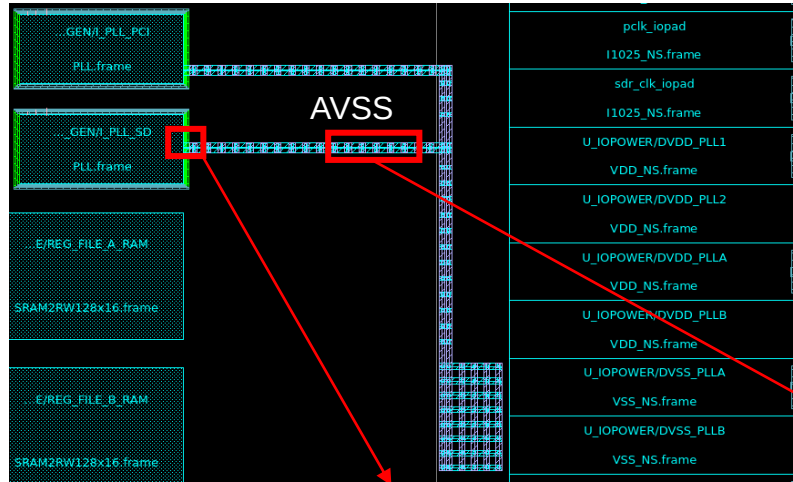
- Connect PLL power pin to Power PAD (AVSS)

```
icc2_shell> create_pg_vias -nets {AVSS} -to_types {lib_cell_pin_connect stripe} -from_layers M7 -to_layers M8 -via_masters {pgvia_10x10} -tag {PLL_SD_AVSS} -shapes [get_selection]
Set tag to PLL_SD_AVSS
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 79 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 158 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 158 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 158 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 158 vias.
Setting tag PLL_SD_AVSS to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
icc2_shell> set_pg_via_master_rule pgvia_5x5 -via_array_dimension {5 5}
Successfully set via def rule pgvia_5x5.
icc2_shell> create_pg_vias -nets {AVSS} -to_types {lib_cell_pin_connect stripe} -from_layers M4 -to_layers M7 -via_masters {pgvia_5x5} -tag {PLL_PCI_AVSS} -shapes [get_selection]
Set tag to PLL_PCI_AVSS
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 79 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 4 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 4 stacked vias:0% 25% 50% 75% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 4 stacked vias:0% 20% 50% 70% 100%
Committed 12 vias.
Setting tag PLL_PCI_AVSS to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```

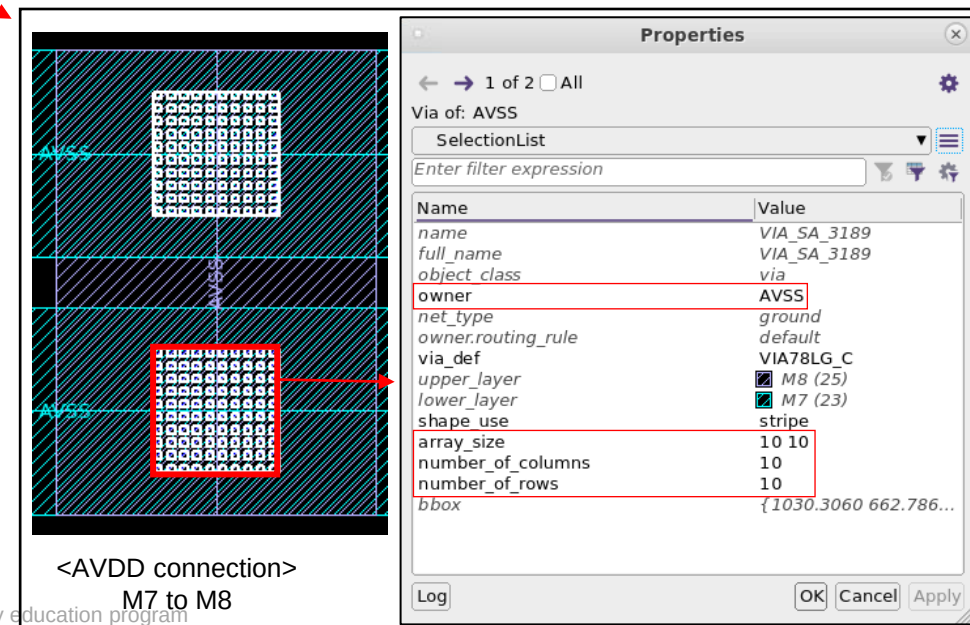
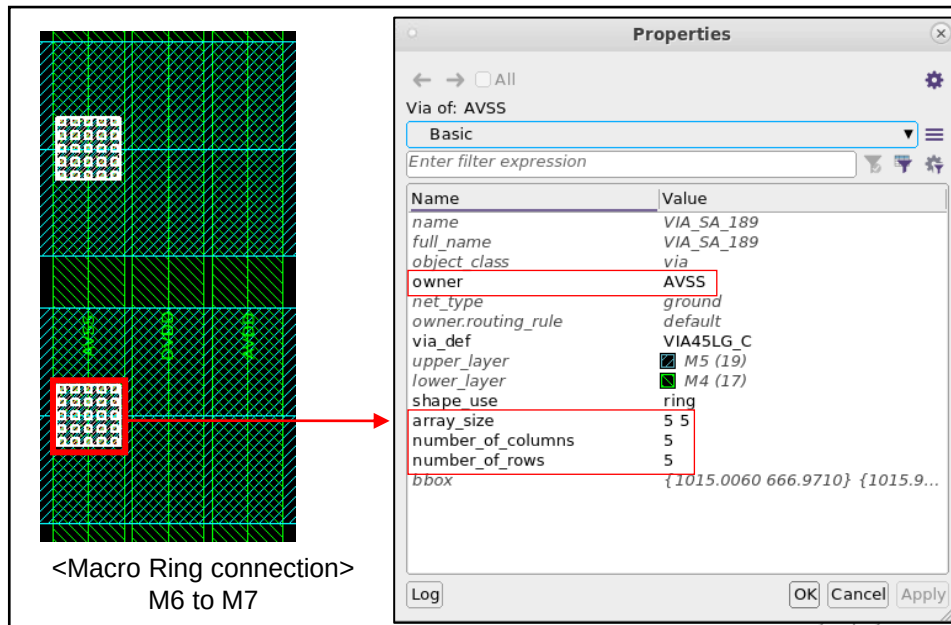
# Powerplan : Use ICC2 (IC Compiler II tool)

6. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/05_PLL_manual_routing.tcl`

- Check AVSS after create shape & vias



- Zoom in and check Macro Ring to AVSS stripes are correctly connected
- Zoom in and check AVSS stripe M7 to M8 are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created

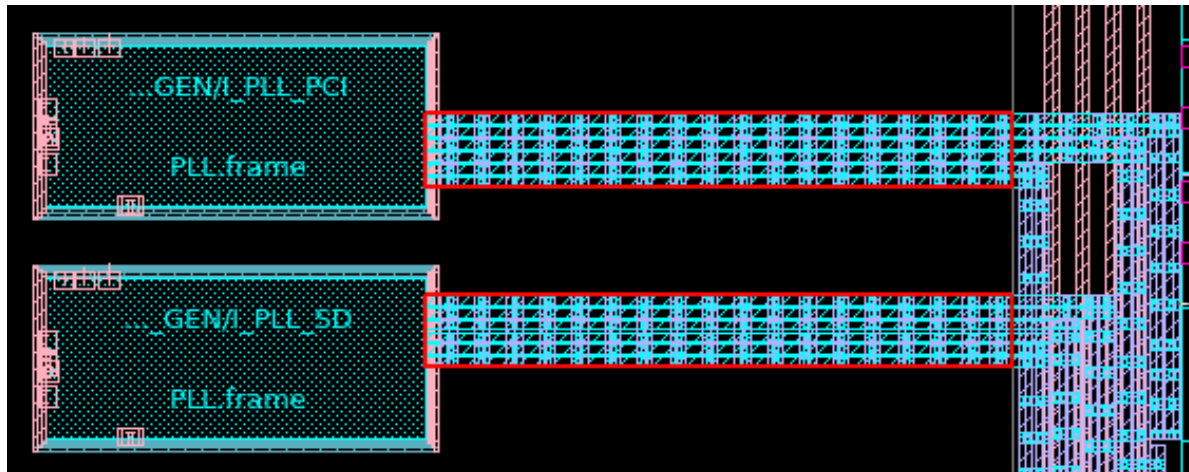


# Powerplan : Use ICC2 (IC Compiler II tool)

## 6. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/05\_PLL\_manual\_routing.tcl

- Create placement blockages before create Power MESH

```
icc2_shell> set MESH_blk_bbox "{{1014.6780 716.0880} {1189.4520 737.6880}} {{1014.6780 662.1780} {1189.4520 683.7780}}}"  
icc2_shell>  
icc2_shell> set i 0  
0  
icc2_shell> foreach x $MESH_blk_bbox {  
? create_placement_blockage -name "MESH_pblk_${i}" -type hard -boundary $x  
?  
? incr i  
? }  
}
```



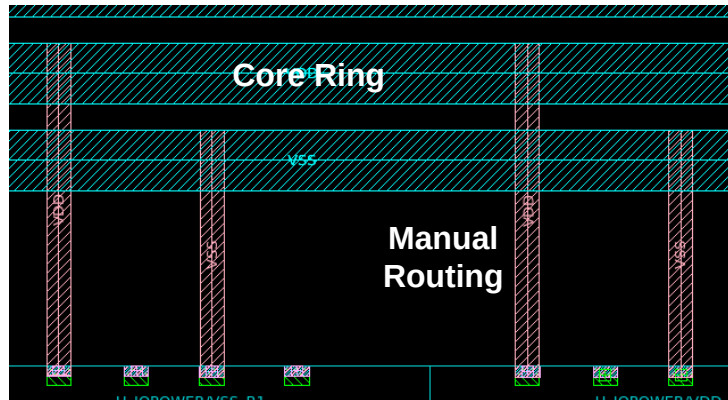
- Top MESH used M7, M8 layers
- M7, M8 Layer was also used for PLL to PAD connections, placement blockage was created in the area to avoid overlap during create Top MESH

# Powerplan : Use ICC2 (IC Compiler II tool)

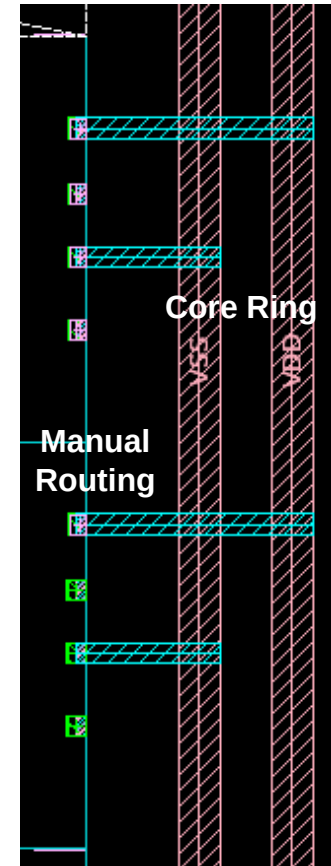
7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Read def about manual routing for P/G PAD connection

```
icc2_shell> read_def ./FP_data/PAD_connection.def
Information: Loading DEF file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/03_powerplan/FP_data/PAD_connection.def' (FILE-007)
Information: Reading 'VERSION' statement. (DEFR-016)
Information: Reading 'DIVIDERCHAR' statement. (DEFR-016)
Information: Reading 'BUSBITCHARS' statement. (DEFR-016)
Information: Reading 'DESIGN' statement. (DEFR-016)
Information: Reading 'UNITS' statement. (DEFR-016)
Information: Reading 'SPECIALNETS' section. (DEFR-016)
Information: Reading 'NETS' section. (DEFR-016)
-----
Number of Processed/Read DEF Constructs
-----
VERSION                : 1/1
DIVIDERCHAR            : 1/1
BUSBITCHARS            : 1/1
DESIGN                 : 1/1
UNITS                  : 1/1
SPECIALNETS           : 5/5
NETS                   : 4/4
1
```



Horizontal  
Ring & PAD co  
nnection



Vertical  
Ring & PAD co  
nnection

- Horizontal (M5 M7) Vertical (M6)
- Connect the P/G pins and core rings of the Power/Ground PADs

# Powerplan : Use ICC2 (IC Compiler II tool)

## 7. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/06\_Power\_PAD\_manual\_routing.tcl

- Set M6\_PAD (M6 Vertical manual routing)

```
icc2_shell> set M6_PAD [get_shapes -filter "width == 1.992 && layer.name == M6 "]
{PATH_21_96 PATH_21_97 PATH_21_98 PATH_21_99 PATH_21_100 PATH_21_101 PATH_21_102 PATH_21_103 PATH_21_104 PATH_21_105 PATH_21_106 P
ATH_21_107 PATH_21_108 PATH_21_109 PATH_21_110 PATH_21_111 PATH_21_112 PATH_21_113 PATH_21_114 PATH_21_115 PATH_21_116 PATH_21_117
PATH_21_118 PATH_21_119}
icc2_shell> change_selection $M6_PAD
icc2_shell> set_attribute -objects [get_selection] -name tag -value PAD_manual
{PATH_21_119 PATH_21_118 PATH_21_117 PATH_21_116 PATH_21_115 PATH_21_114 PATH_21_113 PATH_21_112 PATH_21_101 PATH_21_100 PATH_21_9
9 PATH_21_98 PATH_21_96 PATH_21_97 PATH_21_102 PATH_21_103 PATH_21_104 PATH_21_105 PATH_21_106 PATH_21_107 PATH_21_108 PATH_21_109
PATH_21_110 PATH_21_111}
```

- Set M7\_PAD (M7 Horizontal manual routing)

```
icc2_shell> set M7_PAD [get_shapes -filter "width == 1.992 && layer.name == M7 "]
{PATH_23_16 PATH_23_17 PATH_23_18 PATH_23_19 PATH_23_20 PATH_23_21 PATH_23_22 PATH_23_23 PATH_23_24 PATH_23_25 PATH_23_26 PATH_23_
27 PATH_23_28 PATH_23_29 PATH_23_30 PATH_23_31 PATH_23_32 PATH_23_33 PATH_23_34 PATH_23_35 PATH_23_36 PATH_23_37 PATH_23_38 PATH_2
3_39}
icc2_shell> change_selection $M7_PAD
icc2_shell> set_attribute -objects [get_selection] -name tag -value PAD_manual
{PATH_23_39 PATH_23_38 PATH_23_37 PATH_23_36 PATH_23_35 PATH_23_34 PATH_23_33 PATH_23_32 PATH_23_31 PATH_23_30 PATH_23_29 PATH_23_
28 PATH_23_27 PATH_23_26 PATH_23_25 PATH_23_24 PATH_23_23 PATH_23_22 PATH_23_21 PATH_23_20 PATH_23_19 PATH_23_18 PATH_23_17 PATH_2
3_16}
```

- Set M5\_PAD (M5 Horizontal manual routing)

```
icc2_shell> set M5_PAD [get_shapes -filter "width == 1.992 && layer.name == M5 "]
{PATH_19_97 PATH_19_98 PATH_19_99 PATH_19_100 PATH_19_101 PATH_19_102 PATH_19_103 PATH_19_104 PATH_19_105 PATH_19_106 PATH_19_107
PATH_19_108}
icc2_shell> change_selection $M5_PAD
icc2_shell> set_attribute -objects [get_selection] -name tag -value PAD_manual
{PATH_19_108 PATH_19_107 PATH_19_106 PATH_19_105 PATH_19_104 PATH_19_103 PATH_19_102 PATH_19_101 PATH_19_100 PATH_19_99 PATH_19_98
PATH_19_97}
```

- Set pg\_via\_master\_rule

```
icc2_shell> set_pg_via_master_rule pgvia_8x8 -via_array_dimension {8 8}
Information: The command 'set_pg_via_master_rule' cleared the undo history. (UNDO-016)
Successfully set via def rule pgvia_8x8.
```

# Powerplan : Use ICC2 (IC Compiler II tool)

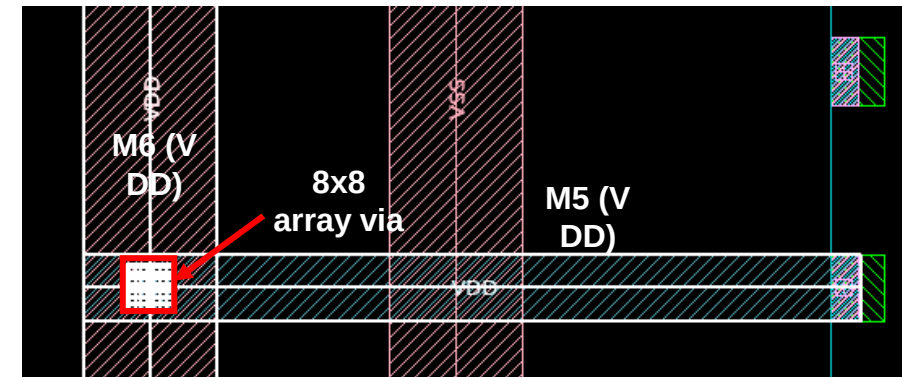
## 7. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/06\_Power\_PAD\_manual\_routing.tcl

- Set M5 ~ M7 VDD manual routing

```
icc2_shell> set M5_VDD [get_shapes -filter "tag == PAD_manual && owner.name == VDD && layer.name == M5 || tag == CORE_RING && owner.name == VDD"]
{PATH_19_97 PATH_19_98 PATH_23_0 PATH_23_1 PATH_23_4 PATH_23_5 PATH_21_0 PATH_21_1 PATH_21_4 PATH_21_5}
icc2_shell> set M6_VDD [get_shapes -filter "tag == PAD_manual && owner.name == VDD && layer.name == M6 || tag == CORE_RING && owner.name == VDD"]
{PATH_23_0 PATH_23_1 PATH_23_4 PATH_23_5 PATH_21_0 PATH_21_1 PATH_21_4 PATH_21_5 PATH_21_96 PATH_21_97 PATH_21_98 PATH_21_99 PATH_21_100 PATH_21_101 PATH_21_102 PATH_21_103 PATH_21_104 PATH_21_105 PATH_21_106 PATH_21_107}
icc2_shell> set M7_VDD [get_shapes -filter "tag == PAD_manual && owner.name == VDD && layer.name == M7 || tag == CORE_RING && owner.name == VDD"]
{PATH_23_0 PATH_23_1 PATH_23_4 PATH_23_5 PATH_23_16 PATH_23_17 PATH_23_18 PATH_23_19 PATH_23_20 PATH_23_21 PATH_23_22 PATH_23_23 PATH_23_24 PATH_23_25 PATH_23_26 PATH_23_27 PATH_21_0 PATH_21_1 PATH_21_4 PATH_21_5}
```

- Create\_pg\_vias for M5 VDD manual routing

```
icc2_shell> change_selection $M5_VDD
icc2_shell> create_pg_vias -nets {VDD} -to_types {lib_cell_pin_connect ring} -from_layers M5 -to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
Set tag to PLL_manual
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 10 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 2 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 2 stacked vias:0% 50% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 2 stacked vias:0% 50% 100%
Committed 2 vias.
Setting tag PLL_manual to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 0 seconds.
Successfully created PG Vias.
1
```

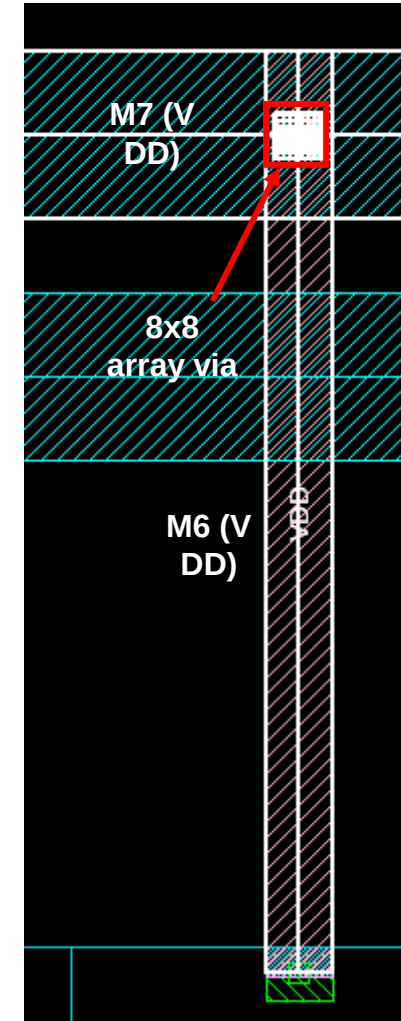


# Powerplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Create\_pg\_vias for M6 VDD manual routing

```
change_selection $M6_VDD
icc2_shell> create_pg_vias -nets {VDD} -to_types {lib_cell_pin_connect ring} -from_layers M6 -to_layers M7 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
Set tag to PLL_manual
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 20 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 12 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 12 stacked vias:0% 5% 15% 25% 30% 40% 50% 55% 65% 75% 80% 90% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 12 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 12 vias.
Setting tag PLL_manual to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```

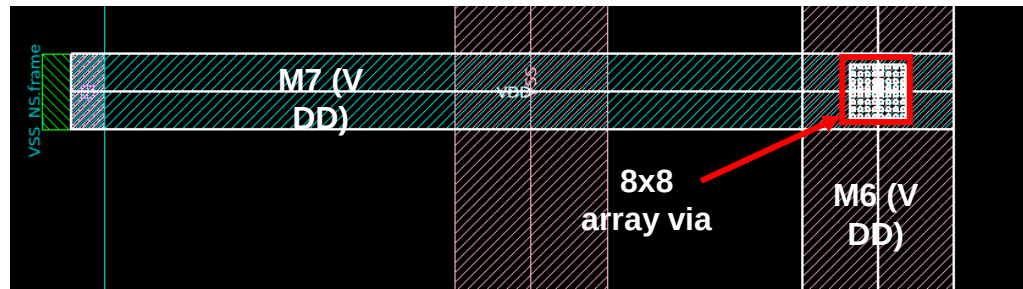


# Powerplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Create\_pg\_vias for M7 VDD manual routing

```
icc2_shell> change_selection $M7_VDD
icc2_shell> create_pg_vias -nets {VDD} -to_types {lib_cell_pin_connect ring} -from_layers M7 -to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
Set tag to PLL_manual
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 20 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 12 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 12 stacked vias:0% 5% 15% 25% 30% 40% 50% 55% 65% 75% 80% 90% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 12 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 12 vias.
Setting tag PLL_manual to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 0 seconds.
Successfully created PG Vias.
1
```



# Powerplan : Use ICC2 (IC Compiler II tool)

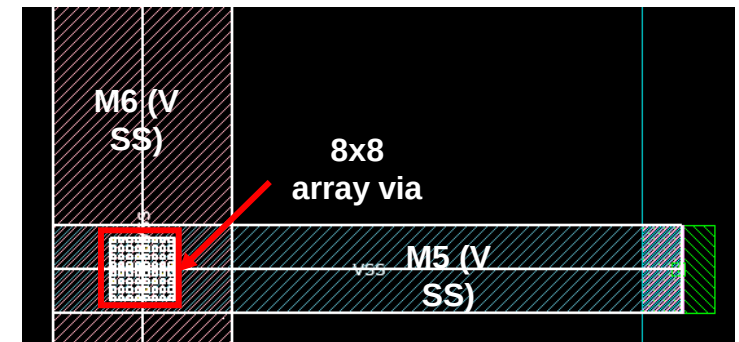
## 7. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/06\_Power\_PAD\_manual\_routing.tcl

- Set M5 ~ M7 VSS manual routing

```
icc2_shell> set M5_VSS [get_shapes -filter "tag == PAD_manual && owner.name == VSS && layer.name == M5 || tag == CORE_RING && owner.name == VSS"]  
{PATH_19_103 PATH_19_104 PATH_19_105 PATH_19_106 PATH_23_2 PATH_23_3 PATH_23_6 PATH_23_7 PATH_21_2 PATH_21_3 PATH_21_6 PATH_21_7}  
icc2_shell> set M6_VSS [get_shapes -filter "tag == PAD_manual && owner.name == VSS && layer.name == M6 || tag == CORE_RING && owner.name == VSS"]  
{PATH_23_2 PATH_23_3 PATH_23_6 PATH_23_7 PATH_21_2 PATH_21_3 PATH_21_6 PATH_21_7 PATH_21_108 PATH_21_109 PATH_21_110 PATH_21_111 PATH_21_112 PATH_21_113 PATH_21_114 PATH_21_115 PATH_21_116 PATH_21_117 PATH_21_118 PATH_21_119}  
icc2_shell> set M7_VSS [get_shapes -filter "tag == PAD_manual && owner.name == VSS && layer.name == M7 || tag == CORE_RING && owner.name == VSS"]  
{PATH_23_2 PATH_23_3 PATH_23_6 PATH_23_7 PATH_23_28 PATH_23_29 PATH_23_30 PATH_23_31 PATH_23_32 PATH_23_33 PATH_23_34 PATH_23_35 PATH_23_36 PATH_23_37 PATH_23_38 PATH_23_39 PATH_21_2 PATH_21_3 PATH_21_6 PATH_21_7}
```

- Create\_pg\_vias for M5 VSS manual routing

```
icc2_shell> change_selection $M5_VSS  
icc2_shell> create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect ring} -from_layers M5  
-to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]  
Set tag to PLL_manual  
Query intersections and create vias.  
Object query runtime: 0 seconds.  
Determining intersections for 12 shapes starts  
Intersection detection runtime: 0 seconds.  
Instantiating vias for 4 intersections starts  
Via instantiation runtime: 0 seconds.  
Starting via DRC checking ...  
Checking DRC for 4 stacked vias:0% 25% 50% 75% 100%  
Overall via DRC checking runtime: 0 seconds.  
Checking dangling/floating vias.  
Start iteration 1: Checking 4 stacked vias:0% 20% 50% 70% 100%  
Committed 4 vias.  
Setting tag PLL_manual to committed wires.  
Committing vias takes 0.00 seconds.  
Overall runtime: 0 seconds.  
Successfully created PG Vias.  
1
```

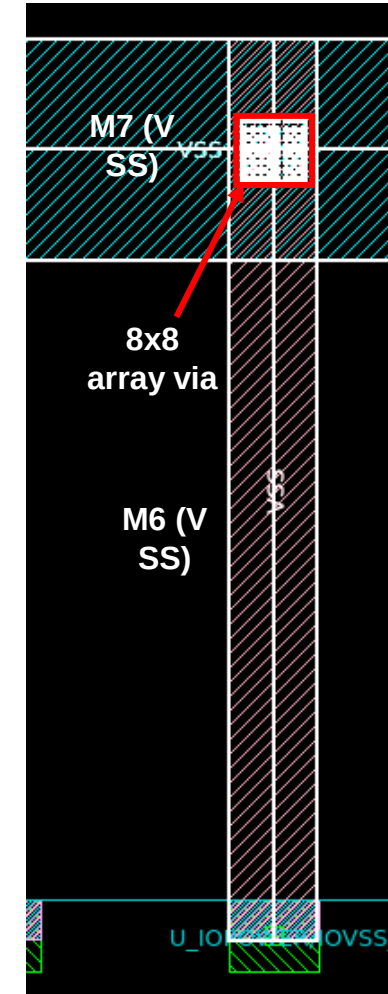


# Powerplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Create\_pg\_vias for M6 VSS manual routing

```
change_selection $M6_VSS
icc2_shell> create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect ring} -from_layers M6
-to_layers M7 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
Set tag to PLL_manual
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 20 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 12 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 12 stacked vias:0% 5% 15% 25% 30% 40% 50% 55% 65% 75% 80% 90% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 12 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 12 vias.
Setting tag PLL_manual to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```

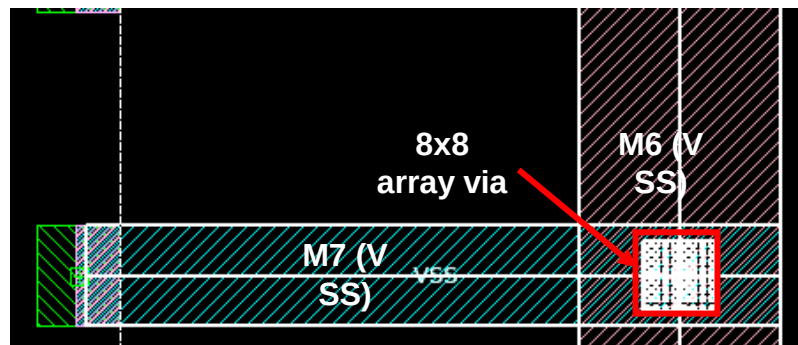


# Powerplan : Use ICC2 (IC Compiler II tool)

7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Create\_pg\_vias for M7 VSS manual routing

```
icc2_shell> change_selection $M7_VSS
icc2_shell> create_pg_vias -nets {VSS} -to_types {lib_cell_pin_connect ring} -from_layers M7
-to_layers M6 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]
Set tag to PLL_manual
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 20 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 12 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 12 stacked vias:0% 5% 15% 25% 30% 40% 50% 55% 65% 75% 80% 90% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 12 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 12 vias.
Setting tag PLL_manual to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```



# Powerplan : Use ICC2 (IC Compiler II tool)

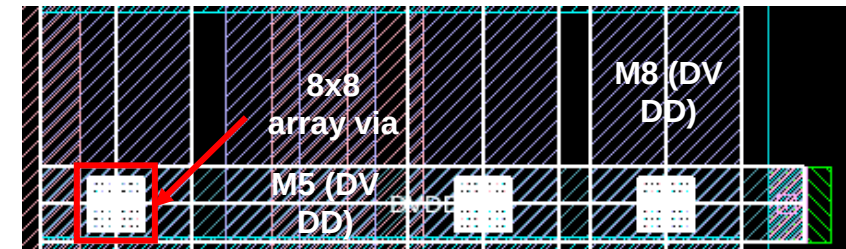
7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Set M5\_DVDD manual routing for PLL to PAD

```
icc2_shell> set M5_DVDD [get_shapes -filter "owner.name == DVDD && layer.name == M8 || tag == PAD_manual && owner.name == DVDD"]  
{PATH_25_43 PATH_25_44 PATH_25_45 PATH_25_46 PATH_25_47 PATH_25_48 PATH_25_49 PATH_25_50 PATH_25_51 PATH_25_52 PATH_25_53 PATH_25_54 PATH_25_55 PATH_25_56 PATH_25_57 PATH_25_58 PATH_25_59 PATH_25_60 PATH_25_61 PATH_25_62 PATH_25_63 PATH_25_64 PATH_25_65 PATH_25_66 PATH_25_67 PATH_25_68 PATH_25_69 PATH_25_70 PATH_25_71 PATH_25_72 PATH_25_73 PATH_25_74 PATH_25_75 PATH_25_76 PATH_25_77 PATH_25_78 PATH_25_79 PATH_25_80 PATH_25_81 PATH_25_82 PATH_25_83 PATH_25_84 PATH_25_85}
```

- Create\_pg\_vias for M5 DVDD manual routing

```
icc2_shell> create_pg_vias -nets {DVDD} -to_types {stripe ring} -from_layers M5 -to_layers M8 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]  
Set tag to PLL_manual  
Query intersections and create vias.  
Object query runtime: 0 seconds.  
Determining intersections for 45 shapes starts  
Intersection detection runtime: 0 seconds.  
Instantiating vias for 8 intersections starts  
Via instantiation runtime: 0 seconds.  
Starting via DRC checking ...  
Checking DRC for 8 stacked vias:0% 10% 25% 35% 50% 60% 75% 85% 100%  
Overall via DRC checking runtime: 0 seconds.  
Checking dangling/floating vias.  
Start iteration 1: Checking 8 stacked vias:0% 10% 20% 30% 50% 60% 70% 80% 100%  
Start iteration 2: Checking 6 stacked vias:0% 10% 30% 50% 60% 80% 100%  
Committed 18 vias.  
Setting tag PLL_manual to committed wires.  
Committing vias takes 0.00 seconds.  
Overall runtime: 0 seconds.  
Successfully created PG Vias.  
1
```



# Powerplan : Use ICC2 (IC Compiler II tool)

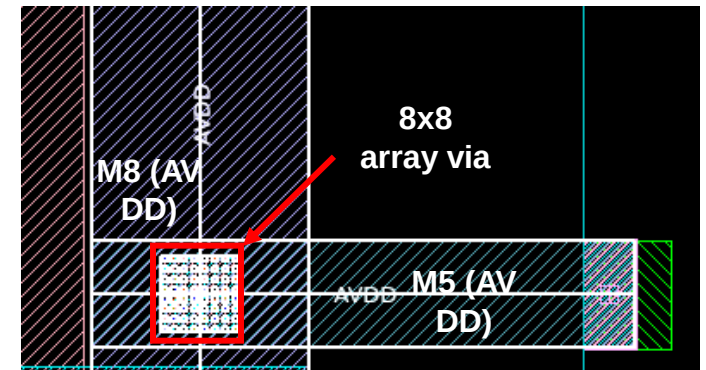
7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Set M5\_AVDD manual routing for PLL to PAD

```
icc2_shell> set M5_AVDD [get_shapes -filter "owner.name == AVDD && layer.name == M8 || tag == PAD_manual && owner.name == AVDD"]  
{PATH_19_2759 PATH_19_2760 PATH_25_0 PATH_25_1 PATH_25_2 PATH_25_3 PATH_25_4 PATH_25_5 PATH_25_6 PATH_25_7 PATH_25_8 PATH_25_9 PATH_25_10 PATH_25_11 PATH_25_12 PATH_25_13 PATH_25_14 PATH_25_15 PATH_25_16 PATH_25_17 PATH_25_18 PATH_25_19 PATH_25_20 PATH_25_21 PATH_25_22 PATH_25_23 PATH_25_24 PATH_25_25 PATH_25_26 PATH_25_27 PATH_25_28 PATH_25_29 PATH_25_30 PATH_25_31 PATH_25_32 PATH_25_33 PATH_25_34 PATH_25_35 PATH_25_36 PATH_25_37 PATH_25_38 PATH_25_39 PATH_25_40 PATH_25_41 PATH_25_42}
```

- Create\_pg\_vias for M5 AVDD manual routing

```
icc2_shell> change_selection $M5_AVDD  
icc2_shell> create_pg_vias -nets {AVDD} -to_types {stripe ring} -from_layers M5 -to_layers M8 -via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]  
Set tag to PLL_manual  
Query intersections and create vias.  
Object query runtime: 0 seconds.  
Determining intersections for 45 shapes starts  
Intersection detection runtime: 0 seconds.  
Instantiating vias for 2 intersections starts  
Via instantiation runtime: 0 seconds.  
Starting via DRC checking ...  
Checking DRC for 2 stacked vias:0% 50% 100%  
Overall via DRC checking runtime: 0 seconds.  
Checking dangling/floating vias.  
Start iteration 1: Checking 2 stacked vias:0% 50% 100%  
Committed 6 vias.  
Setting tag PLL_manual to committed wires.  
Committing vias takes 0.00 seconds.  
Overall runtime: 0 seconds.  
Successfully created PG Vias.  
1
```



# Powerplan : Use ICC2 (IC Compiler II tool)

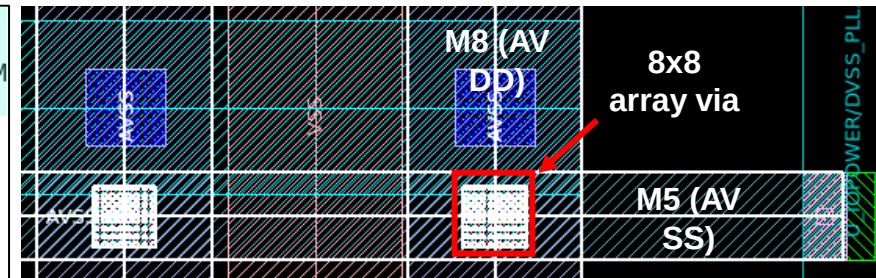
7. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/06_Power_PAD_manual_routing.tcl`

- Set M5\_AVSS manual routing for PLL to PAD

```
icc2_shell> set M5_AVSS [get_shapes -filter "owner.name == AVSS && layer.name == M5 || tag == PAD_manual && owner.name == AVSS"]  
{PATH_19_2757 PATH_19_2758 PATH_25_86 PATH_25_87 PATH_25_88 PATH_25_89 PATH_25_90 PATH_25_91  
PATH_25_92 PATH_25_93 PATH_25_94 PATH_25_95 PATH_25_96 PATH_25_97 PATH_25_98 PATH_25_99 PATH_25_100  
PATH_25_101 PATH_25_102 PATH_25_103 PATH_25_104 PATH_25_105 PATH_25_106 PATH_25_107 PATH_25_108  
PATH_25_109 PATH_25_110 PATH_25_111 PATH_25_112 PATH_25_113 PATH_25_114 PATH_25_115  
PATH_25_116 PATH_25_117 PATH_25_118 PATH_25_119 PATH_25_120 PATH_25_121 PATH_25_122 PATH_25_123  
PATH_25_124 PATH_25_125 PATH_25_126 PATH_25_127}
```

- Create\_pg\_vias for M5 AVSS manual routing

```
icc2_shell> change_selection $M5_AVSS  
icc2_shell>  
icc2_shell> create_pg_vias -nets {AVSS} -to_types {stripe ring} -from_layers M5 -to_layers M8  
-via_masters {pgvia_8x8} -tag {PLL_manual} -shapes [get_selection]  
Set tag to PLL_manual  
Query intersections and create vias.  
Object query runtime: 0 seconds.  
Determining intersections for 44 shapes starts  
Intersection detection runtime: 0 seconds.  
Instantiating vias for 8 intersections starts  
Via instantiation runtime: 0 seconds.  
Starting via DRC checking ...  
Checking DRC for 8 stacked vias:0% 10% 25% 35% 50% 60% 75% 85% 100%  
Overall via DRC checking runtime: 0 seconds.  
Checking dangling/floating vias.  
Start iteration 1: Checking 8 stacked vias:0% 10% 20% 30% 50% 60% 70% 80% 100%  
Committed 24 vias.  
Setting tag PLL_manual to committed wires.  
Committing vias takes 0.00 seconds.  
Overall runtime: 1 seconds.  
Successfully created PG Vias.  
1
```

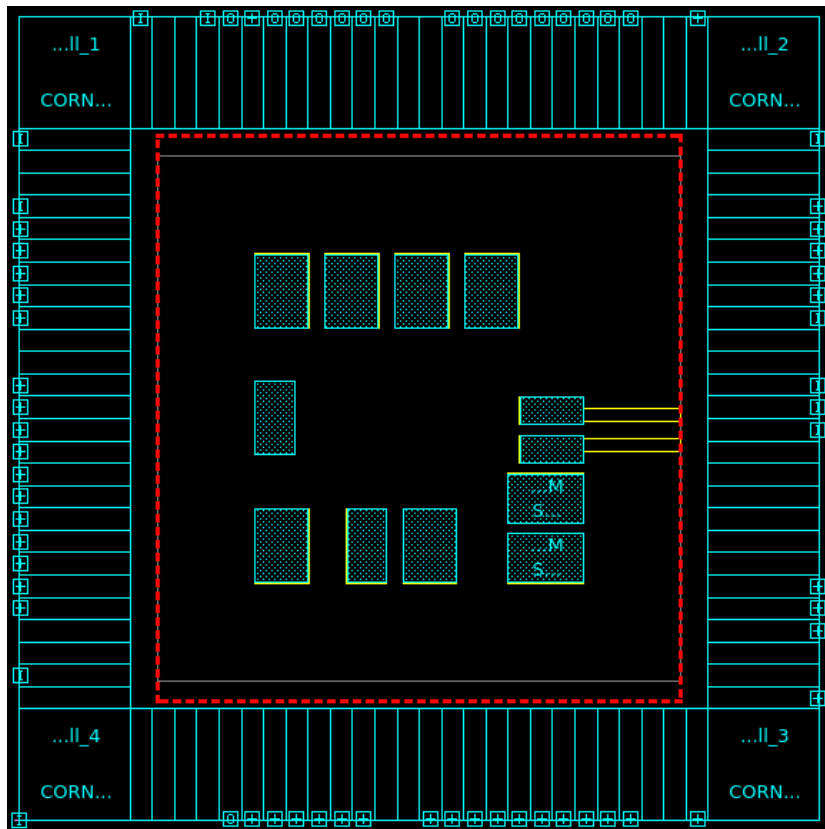


# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

- Create M8(vertical) pg\_region

```
icc2_shell> set M8_MESH_bdry "{250.5480 214.4400} {1189.4520 1225.5600}"  
{250.5480 214.4400} {1189.4520 1225.5600}  
icc2_shell> create_pg_region MESH -polygon $M8_MESH_bdry  
Information: The command 'create_pg_region' cleared the undo history. (UNDO-016)  
Define region based on polygon shape.  
Create region MESH  
{MESH}
```



- To supply power to the core area from the core ring, the top MESH and the core ring must be connected to each other
- Create the pg\_region including the area to which the MESH (M8) and Core Ring are to be connected

# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

- Create M8(vertical) MESH (pattern, strategy, via\_rule)

```
icc2_shell> #create M8 mesh pattern
icc2_shell> create_pg_mesh_pattern METAL8_PG_PATTERN -layers {{vertical_layer: M8} \
? {width: 4.64}\
? {spacing: 5.088} \
? {pitch: 19.456} \
? {trim: false}} \
? -parameters {4.64 19.456 5.088}
Variable 4.64 is defined in -parameters option but not used by any pattern expression.
Variable 19.456 is defined in -parameters option but not used by any pattern expression.
Variable 5.088 is defined in -parameters option but not used by any pattern expression.
Warning: Some defined variables in -parameters option are not used by other options. (PGR-501)
Successfully create mesh pattern METAL8_PG_PATTERN.
1
icc2_shell>
icc2_shell> #creat M8_pg_strategy
icc2_shell> set_pg_strategy METAL8_MAPPING_STRATEGY \
? -pg_regions MESH \
? -pattern {\
? {name: METAL8_PG_PATTERN} \
? {nets: {VDD VSS}} \
? {parameters: {4.64 19.456 5.088}} \
? {offset_start: {0 0}} \
? -blockage {placement_blockages: all}
Successfully set PG strategy METAL8_MAPPING_STRATEGY.
icc2_shell>
icc2_shell> #create_M8_via_rule
icc2_shell>
icc2_shell> set_pg_strategy_via_rule M8_via \
? -via_rule {via_master: NIL}
Only via_master is specified in the rule, the via_master will be applied to all intersections.
Please specified intersection rule or filtering rule if this is not desired.
Successfully set strategy via rule M8 via.
```

Via is not created using -via\_rule {via\_master: NIL} option  
(DRC prevention)

# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

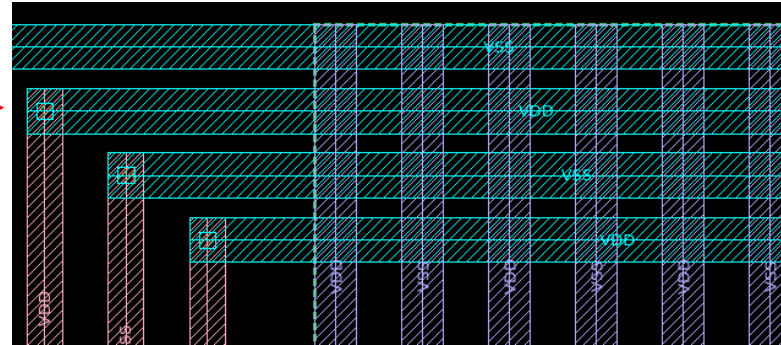
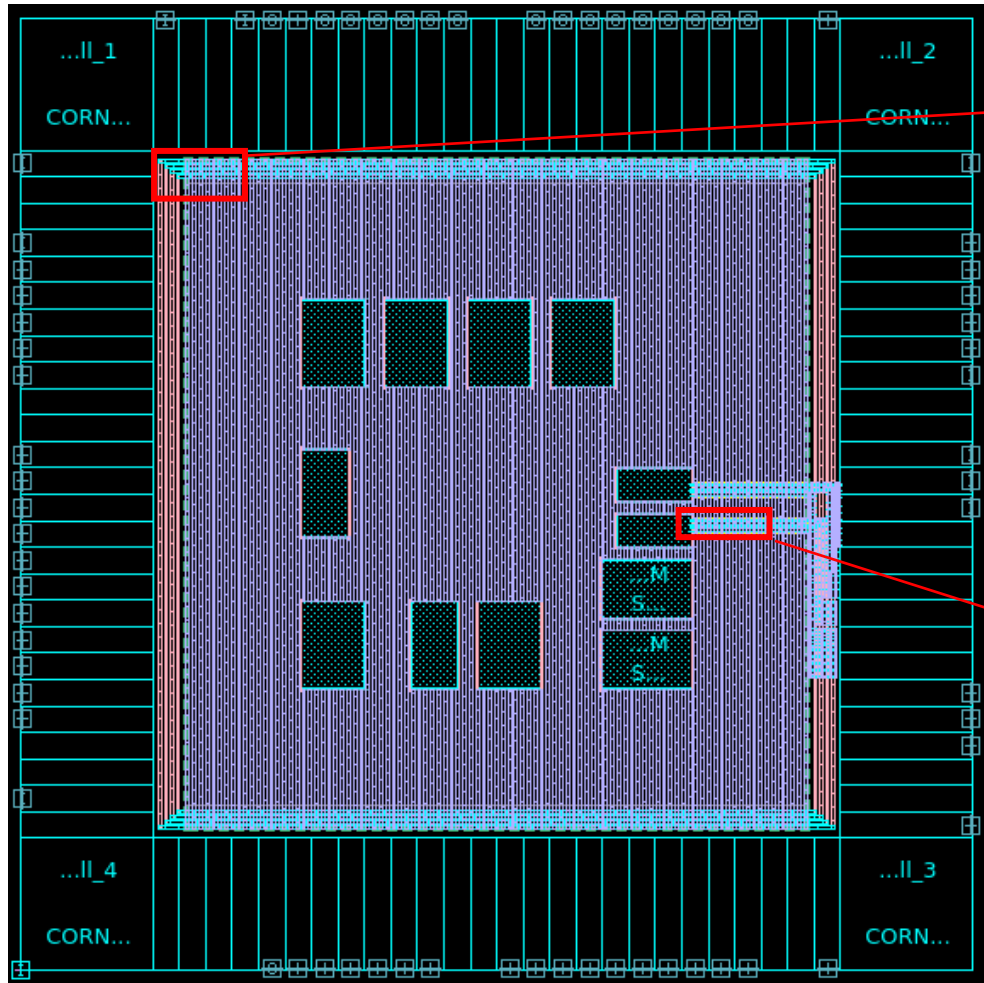
- Create M8(vertical) MESH (compile\_pg)

```
icc2_shell> compile_pg -strategies {METAL8_MAPPING_STRATEGY} -via_rule {M8_via} -tag MESH_8
Sanity check for inputs.
Set tag to MESH_8
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy METAL8_MAPPING_STRATEGY.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Creating straps and vias in power plan.
Creating wire shapes for strategies METAL8_MAPPING_STRATEGY .
Creating wire shapes runtime: 0 seconds
Blockage cutting and DRC fixing for wire shapes for strategies METAL8_MAPPING_STRATEGY .
Check and fix DRC for 262 wires for strategy METAL8_MAPPING_STRATEGY.
Checking DRC for 262 wires:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Creating 262 wires after DRC fixing.
Wire DRC checking runtime 0.00 seconds.
Creating via shapes for strategies METAL8_MAPPING_STRATEGY .
Working on strategy METAL8_MAPPING_STRATEGY.
Number of detected intersections: 0
Total runtime of via shapes creation: 0 seconds
Runtime of via DRC checking for strategy METAL8_MAPPING_STRATEGY: 0.00 seconds.
Creating via connection between strategies and existing shapes.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag MESH_8 to committed wires.
Committed 262 wires.
Committing wires takes 0.00 seconds.
Committed 0 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 0 seconds.
Successfully compiled PG.
Overall runtime: 0 seconds.
1
```

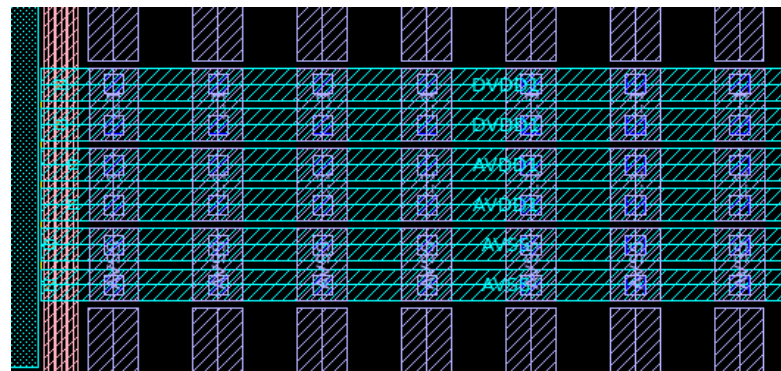
# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

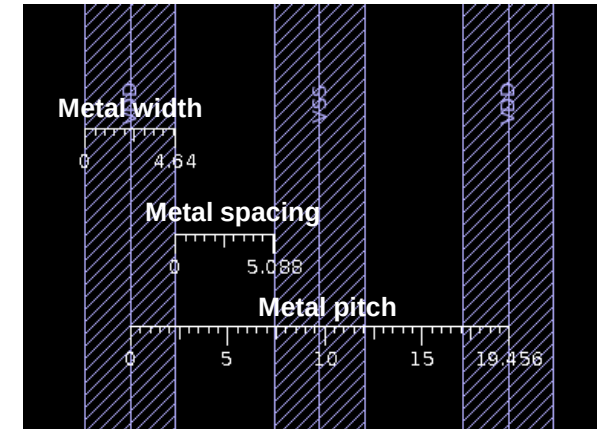
- Check M8(vertical) MESH after created



Check that M8 MESH is created to be connected core ring



Check M8 MESH is correctly created so that it does not overlap with PLL power routing



Check Metal (width, spacing, pitch)

# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

- Create M7(horizontal) MESH (pattern, strategy, via\_rule)

```
icc2_shell> #create_M7_mesh_pattern
icc2_shell>
icc2_shell> create_pg_mesh_pattern METAL7_PG_PATTERN -layers {{horizontal_layer: M7}} \
? {width: 1.104}\
? {spacing: 5.584}\
? {pitch: 13.376}\
? {trim: false}} \
? -parameters {1.104 13.376 5.584}
Variable 1.104 is defined in -parameters option but not used by any pattern expression.
Variable 13.376 is defined in -parameters option but not used by any pattern expression.
Variable 5.584 is defined in -parameters option but not used by any pattern expression.
Warning: Some defined variables in -parameters option are not used by other options. (PGR-501)
Successfully create mesh pattern METAL7_PG_PATTERN.
1
icc2_shell>
icc2_shell> #create_M7_MESH_strategy
icc2_shell>
icc2_shell> set_pg_strategy METAL7_MAPPING_STRATEGY \
? -core \
? -pattern {\
? {name: METAL7_PG_PATTERN} \
? {nets: {VDD VSS}} \
? {parameters: {1.104 13.376 5.584}} \
? {offset start: {0 0}} \
? -blockage {placement_blockages: all}
Successfully set PG strategy METAL7_MAPPING_STRATEGY.
icc2_shell>
icc2_shell> #create_M7_via_rule
icc2_shell>
icc2_shell> set_pg_strategy_via_rule M7_via \
? -via_rule {via_master: NIL}
Only via_master is specified in the rule, the via_master will be applied to all intersections.
Please specified intersection rule or filtering rule if this is not desired.
Successfully set strategy via rule M7 via.
```

Via is not created using -via\_rule {via\_master: NIL} option  
(DRC prevention)

# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

- Create M8(vertical) MESH (compile\_pg) & remove placement\_blockages (Blockages previously created to prevent overlap)

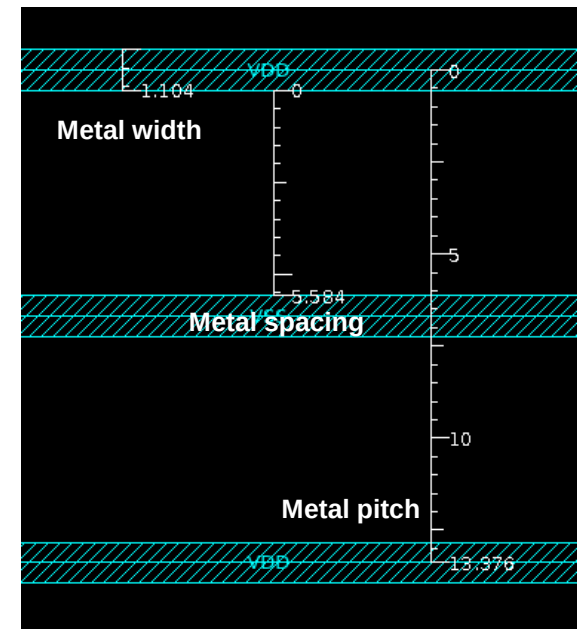
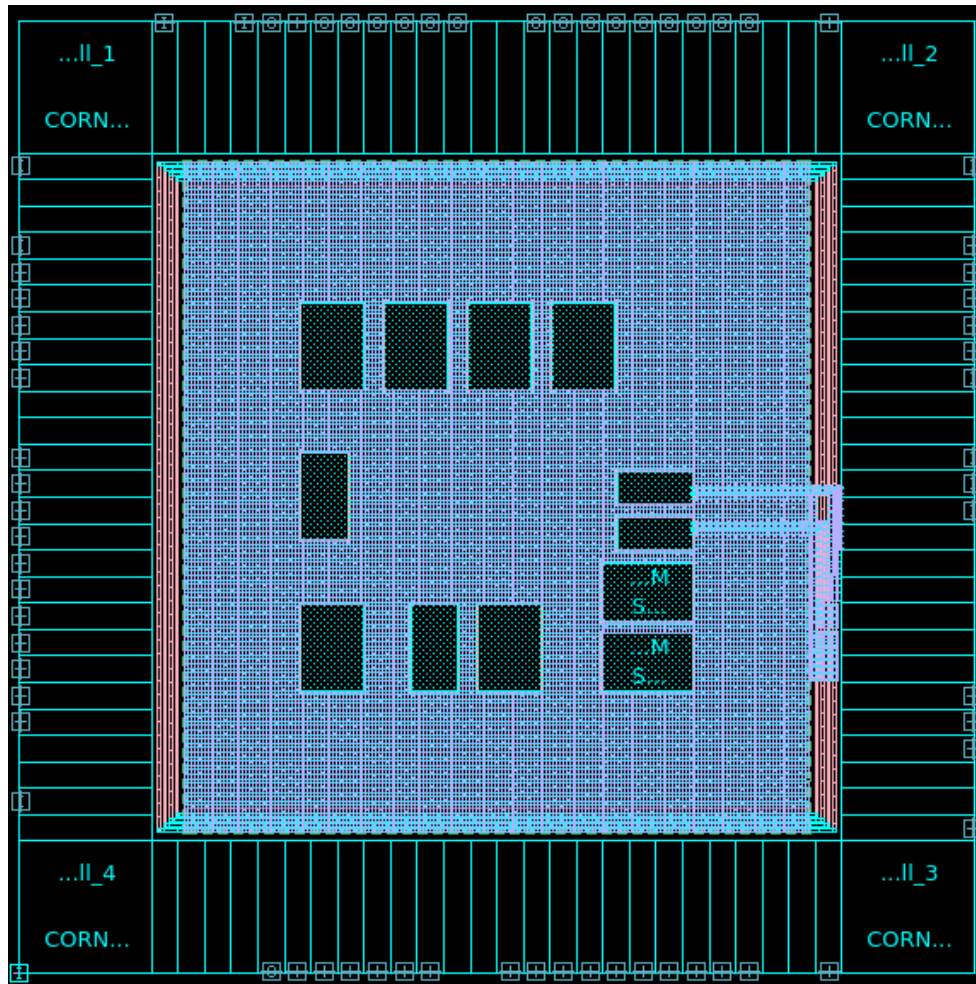
```
icc2_shell> compile_pg -strategies {METAL7_MAPPING_STRATEGY} -via_rule {M7_via} -tag MESH_7
Sanity check for inputs.
Set tag to MESH_7
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy METAL7_MAPPING_STRATEGY.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Creating straps and vias in power plan.
Creating wire shapes for strategies METAL7_MAPPING_STRATEGY .
Creating wire shapes runtime: 0 seconds
Blockage cutting and DRC fixing for wire shapes for strategies METAL7_MAPPING_STRATEGY .
Check and fix DRC for 334 wires for strategy METAL7_MAPPING_STRATEGY.
Checking DRC for 334 wires:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Creating 334 wires after DRC fixing.
Wire DRC checking runtime 0.00 seconds.
Creating via shapes for strategies METAL7_MAPPING_STRATEGY .
Working on strategy METAL7_MAPPING_STRATEGY.
Number of detected intersections: 0
Total runtime of via shapes creation: 0 seconds
Runtime of via DRC checking for strategy METAL7_MAPPING_STRATEGY: 0.00 seconds.
Creating via connection between strategies and existing shapes.
via connection runtime: 0 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag MESH_7 to committed wires.
Committed 334 wires.
Committing wires takes 0.00 seconds.
Committed 0 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 0 seconds.
Successfully compiled PG.
Overall runtime: 0 seconds.
1
icc2_shell> remove_placement_blockages [get_placement_blockages -filter "full_name =~ MESH_pblk_*"]
2
```

Remove Blockages previously created to prevent overlap

# Powerplan : Use ICC2 (IC Compiler II tool)

8. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/07_create_top_MESH.tcl`

- Check M7(horizontal) MESH after created



Check Metal (width, spacing, pitch)

# Powerplan : Use ICC2 (IC Compiler II tool)

9. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/08_create_low_MESH.tcl`

- Create M2(vertical) low MESH (pattern, strategy, via\_rule)

```
icc2_shell> #create_M2_mesh_pattern
icc2_shell> create_pg_mesh_pattern P_m2_triple \
? -layers { \
? { {vertical_layer: M2} \
? {width: 0.44 0.192 0.192} \
? {spacing: 2.724 3.456} \
? {pitch: 9.728} {offset: 1.216} {trim : false} } \
? }
Pattern P_m2_triple exists and is re-defined.
Successfully create mesh pattern P_m2_triple.
1
icc2_shell> #create_M2_pg_strategy
icc2_shell> set_pg_strategy S_m2_vddvss \
? -core \
? -pattern { {name: P_m2_triple} {nets: {VDD VSS VSS}} {offset_start: {250.548 0}} } \
? -blockage { {nets: VDD VSS} {placement_blockages : all}}
? }
Strategy S_m2_vddvss exists and is re-defined for the current design.
Successfully set PG strategy S_m2_vddvss.
icc2_shell>
icc2_shell> #create_M2_via_rule
icc2_shell> set_pg_strategy_via_rule M2 via \
? -via_rule {via_master: NIL}
Only via_master is specified in the rule, the via_master will be applied to all intersections.
Please specified intersection rule or filtering rule if this is not desired.
Successfully set strategy via rule M2_via.
```

Via is not created using -via\_rule {via\_master: NIL} option  
(DRC prevention)

# Powerplan : Use ICC2 (IC Compiler II tool)

9. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/08_create_low_MESH.tcl`

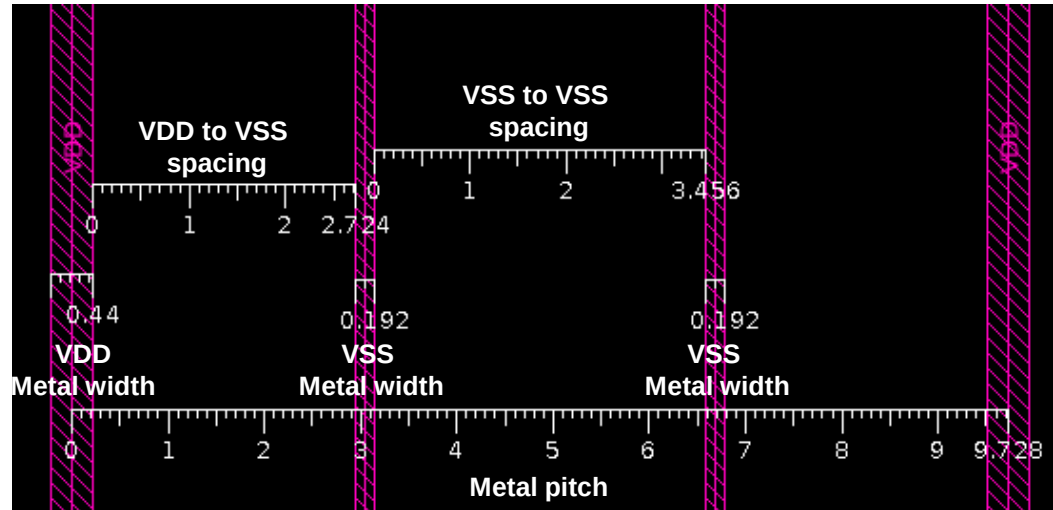
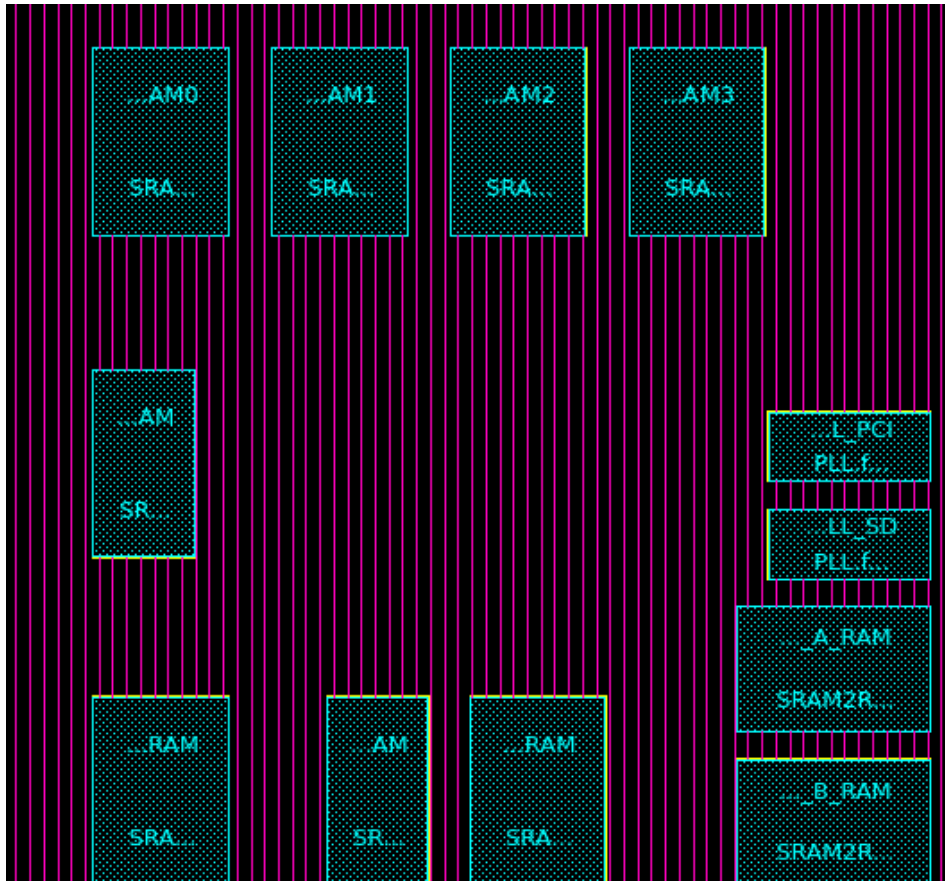
- Create M2(vertical) low MESH (compile\_pg)

```
icc2_shell> compile_pg -strategies {S_m2_vddvss} -tag {low_MESH_2} -via_rule {M2_via} -drc no_check
Sanity check for inputs.
Set tag to low_MESH_2
DRC checking and fixing will be skipped for created shapes.
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy S_m2_vddvss.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Creating straps and vias in power plan.
Creating wire shapes for strategies S_m2_vddvss .
Creating wire shapes runtime: 0 seconds
Blockage cutting and DRC fixing for wire shapes for strategies S_m2_vddvss .
Creating via shapes for strategies S_m2_vddvss .
Working on strategy S_m2_vddvss.
Number of detected intersections: 0
Total runtime of via shapes creation: 0 seconds
Creating via connection between strategies and existing shapes.
via connection runtime: 0 seconds.
Committing wires and vias.
Setting tag low_MESH_2 to committed wires.
Committed 666 wires.
Committing wires takes 0.00 seconds.
Committed 0 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 1 seconds.
Successfully compiled PG.
Overall runtime: 1 seconds.
1
```

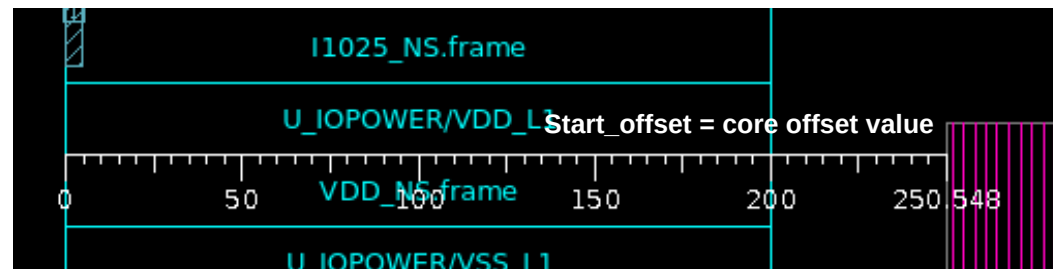
# Powerplan : Use ICC2 (IC Compiler II tool)

9. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/08_create_low_MESH.tcl`

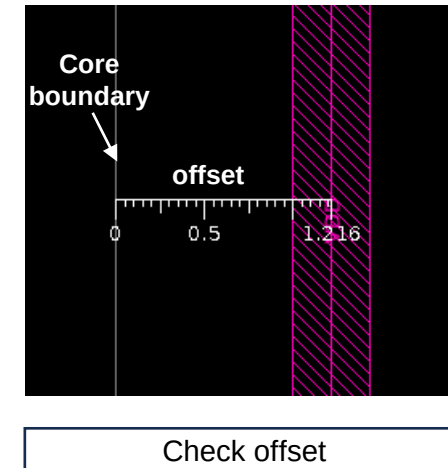
- Check M2(vertical) low MESH after created



Check Metal (width, spacing, pitch)



Check start\_offset



# Powerplan : Use ICC2 (IC Compiler II tool)

10. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/09_create_std_rail.tcl`

- Create M1(horizontal) STD rail (pattern, strategy, via\_rule)

```
icc2_shell> #set via master rule for connect M1 to M2
icc2_shell> set_pg_via_master_rule pgvia_1x2 -via_array_dimension {2 1}
Successfully set via def rule pgvia_1x2.
icc2_shell>
icc2_shell> #create M1_std_cell_conn_pattern
icc2_shell> create_pg_std_cell_conn_pattern std_rail_pattern -layers M1 -rail_width {0.06} -rail_shift {3.344} -parameters {M1 0.06 3.344}
Variable M1 is defined in -parameters option but not used by other option.
Variable 0.06 is defined in -parameters option but not used by other option.
Variable 3.344 is defined in -parameters option but not used by other option.
Warning: Some defined variables in -parameters option are not used by other options. (PGR-501)
Successfully create standard cell rail pattern std_rail_pattern.
icc2_shell>
icc2_shell> #create M1_pg_strategy
icc2_shell> set_pg_strategy M1_std_cell_rail \
? -core \
? -blockage {{macros_with_keepout: all}{placement_blockages: all}} \
? -pattern {{pattern: std_rail_pattern} {nets: {VDD VSS}} {parameters: {M1 0.06 3.344}}} \
? -extension {{stop: core_boundary} {direction: L B R T }}
Successfully set PG strategy M1_std_cell_rail.
icc2_shell>
icc2_shell> #create M1_via_rule
icc2_shell> set_pg_strategy_via_rule M2_via_stdcellrail \
? -via_rule { \
?   { {strategies: {M1_std_cell_rail}} \
?     {{existing: strap}{layers: M2}} \
?     {via_master: pgvia_1x2} {between_parallel: true} } \
?     { {intersection: undefined}{via_master: NIL} }}
Successfully set strategy via rule M2 via stdcellrail.
```

Create via between M1 and M2 only

# Powerplan : Use ICC2 (IC Compiler II tool)

10. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/09_create_std_rail.tcl`

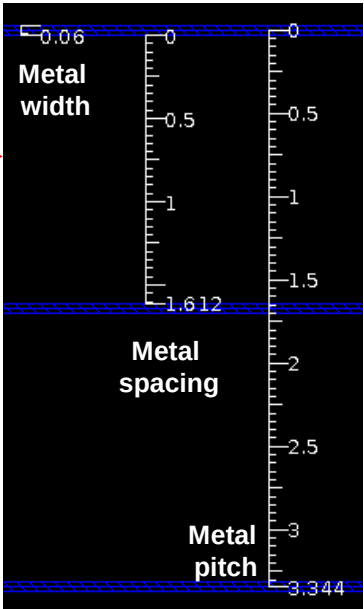
- Create M1(horizontal) STD rail (compile\_pg)

```
icc2_shell> compile_pg -strategies {M1_std_cell_rail} -via_rule {M2_via_stdcellrail} -tag {STD_RAIL_M1}
Sanity check for inputs.
Set tag to STD_RAIL_M1
Automatic PG net connection through connect_pg_net is disabled.
Updating PG strategies.
Updating strategy M1_std_cell_rail.
Loading library and design information.
Number of Standard Cells: 0
Number of Hard Macros: 12
Number of Pads: 104
Creating standard cell rails.
Creating standard cell rails for strategy M1_std_cell_rail.
DRC checking and fixing for standard cell rail strategy M1_std_cell_rail.
Checking DRC for 1369 wires:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Creating 1369 wires after DRC fixing.
Wire DRC checking runtime 1.00 seconds.
Creating via connection between strategies and existing shapes.
Check and fix DRCs for 69266 stacked vias.
Checking DRC for 69266 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 100%
Via DRC checking runtime 15.00 seconds.
Via connection runtime: 15 seconds.
Removing dangling/floating wire/vias after DRC check.
Start iteration 1:
Checking potential dangling/floating power plan wires.
Checking dangling/floating vias between strategies and existing shapes.
Checking 69266 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Finish removing all dangling or floating wires and vias.
Committing wires and vias.
Setting tag STD_RAIL_M1 to committed wires.
Committed 1369 wires.
Committing wires takes 0.00 seconds.
Committed 69266 vias.
Committed 0 wires for via creation.
Committing vias takes 0.00 seconds.
Overall PG creation runtime: 16 seconds.
Successfully compiled PG.
Overall runtime: 16 seconds.
1
```

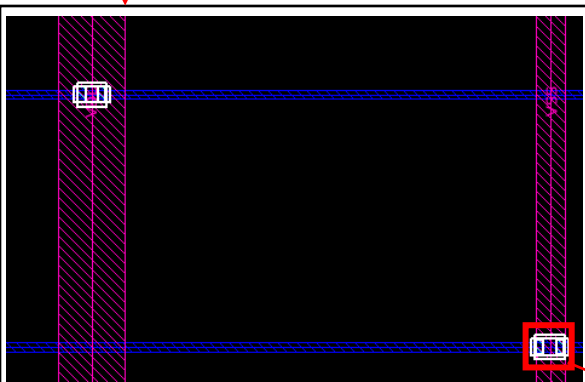
# Powerplan : Use ICC2 (IC Compiler II tool)

10. lcc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/09\_create\_std\_rail.tcl

- Check M1(horizontal) STD rail after created



Check Metal  
(width, spacing, pitch)



Check M1 to M2 vias  
(array\_size, columns, rows)

| Properties              |                       |
|-------------------------|-----------------------|
| ← → 1 of 3 All          |                       |
| Via of: VSS             |                       |
| Basic                   |                       |
| Enter filter expression |                       |
| Name                    | Value                 |
| name                    | VIA_SA_64221          |
| full_name               | VIA_SA_64221          |
| object_class            | via                   |
| owner                   | VSS                   |
| net_type                | ground                |
| owner.routing_rule      | default               |
| via_def                 | VIA12BAR_C            |
| upper_layer             | M2 (13)               |
| lower_layer             | M1 (11)               |
| shape_use               | lib_cell_pin_connect  |
| array_size              | 1 2                   |
| number_of_columns       | 2                     |
| number_of_rows          | 1                     |
| bbox                    | {422.9810 1045.124... |

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- `remove_pg(strategy, pattern, region, via_master_rule, strategy_via_rule)`

```
icc2_shell> remove_pg_strategies -all
All strategies have been removed.
icc2_shell> remove_pg_patterns -all
All patterns have been removed.
icc2_shell> remove_pg_regions -all
No PG region is found.
icc2_shell> remove_pg_via_master_rules -all
All via def rules have been removed.
icc2_shell> remove_pg_strategy_via_rules -all
All strategy via rules have been removed.
```

- `Check_pg_drc`

Verifies and reports technology routing design rules violations and illegal overlaps of PG net objects

```
icc2_shell> check_pg_drc
Command check_pg_drc started at Wed Oct 4 15:05:11 2023
Command check_pg_drc finished at Wed Oct 4 15:05:17 2023
CPU usage for check_pg_drc: 5.77 seconds ( 0.00 hours)
Elapsed time for check_pg_drc: 6.08 seconds ( 0.00 hours)
No errors found.
```

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- `Check_pg_missing_vias`

Checks missing vias within the power and ground network

```
icc2_shell> check_pg_missing_vias
Check net AVDD vias...
Number of missing vias: 0
Checking net AVDD vias took 0 seconds.
Check net AVSS vias...
Number of missing vias: 0
Checking net AVSS vias took 0 seconds.
Check net DVDD vias...
Number of missing vias: 0
Checking net DVDD vias took 0 seconds.
Check net VDD vias...
Number of missing vias on VIA6 layer: 198
Number of missing vias on VIA7 layer: 3064
Total number of missing vias: 3262
Checking net VDD vias took 0 seconds.
Check net VDDIO vias...
Number of missing vias: 0
Checking net VDDIO vias took 0 seconds.
Check net VSS vias...
Number of missing vias on VIA6 layer: 188
Number of missing vias on VIA7 layer: 2994
Total number of missing vias: 3182
Checking net VSS vias took 1 seconds.
Check net VSSIO vias...
Number of missing vias: 0
Checking net VSSIO vias took 0 seconds.
Overall runtime: 1 seconds.
```

Missing via occurred in VDD, VSS net because it did not create via during MESH creation

Checks missing vias within the power and ground network

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- **Check\_pg\_connectivity** (exclude std cell – because std cells are not placed) - Verifies physical connectivity of the power ground network

```
icc2_shell> check_pg_connectivity -check_std_cell_pins none
Loading cell instances...
Number of Standard Cells: 0
Number of Macro Cells: 12
Number of IO Pad Cells: 108
Number of Blocks: 0
Loading P/G wires and vias...
Number of AVDD Wires: 74
Number of AVDD Vias: 183
Number of AVDD Terminals: 1
*****Verify net AVDD connectivity*****
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 2
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of AVSS Wires: 79
Number of AVSS Vias: 180
Number of AVSS Terminals: 1
*****Verify net AVSS connectivity*****
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 2
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of DVDD Wires: 69
Number of DVDD Vias: 164
Number of DVDD Terminals: 1
*****Verify net DVDD connectivity*****
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 2
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
```

Ignore Number of floating  
I/O pads & Terminals

```
Loading cell instances...
Loading P/G wires and vias...
Number of VDD Wires: 2600
Number of VDD Vias: 24507
Number of VDD Terminals: 1
*****Verify net VDD connectivity*****
  Number of floating wires: 1691
  Number of floating vias: 1392
  Number of floating std cells: 0
  Number of floating hard macros: 10
  Number of floating I/O pads: 108
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VDDIO Wires: 0
Number of VDDIO Vias: 0
Number of VDDIO Terminals: 1
*****Verify net VDDIO connectivity*****
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 106
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 2811
Number of VSS Vias: 47511
Number of VSS Terminals: 1
*****Verify net VSS connectivity*****
  Number of floating wires: 1687
  Number of floating vias: 1390
  Number of floating std cells: 0
  Number of floating hard macros: 10
  Number of floating I/O pads: 108
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
```

VDD is not connected  
because it did not create  
via while creating MESH

VSS is not connected  
because it did not create  
via while creating MESH

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `lcc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- **Check\_pg\_connectivity** (exclude std cell – because std cells are not placed) - Verifies physical connectivity of the power ground network

```
Loading cell instances...
Loading P/G wires and vias...
Number of VSSIO Wires: 0
Number of VSSIO Vias: 0
Number of VSSIO Terminals: 1
*****Verify net VSSIO connectivity*****
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 106
  Number of floating terminals: 1
  Number of floating hierarchical blocks: 0
*****
Overall runtime: 0 seconds.
{PATH_19_17 PATH_19_18 PATH_19_19 PATH_19_20 PATH_19_21 PATH_19_22 PATH_19_23 PATH_19_24 PATH_19_25 PATH_19_26 PATH_19_27 PA
TH_19_28 PATH_19_29 PATH_19_30 PATH_19_31 PATH_19_32 PATH_19_33 PATH_19_34 PATH_19_35 PATH_19_36 PATH_19_37 PATH_19_38 PATH
19_39 PATH_19_40 PATH_19_41 PATH_19_42 PATH_19_43 PATH_19_44 PATH_19_45 PATH_19_46 PATH_19_47 PATH_19_48 PATH_19_49 PATH_19
50 PATH_19_51 PATH_19_52 PATH_19_53 PATH_19_54 PATH_19_55 PATH_19_56 RECT_19_57 RECT_19_58 RECT_19_59 RECT_19_60 RECT_19_61
RECT_19_62 RECT_19_63 RECT_19_64 RECT_19_65 RECT_19_66 RECT_19_67 RECT_19_68 RECT_19_69 RECT_19_70 RECT_19_71 RECT_19_72 PAT
H_19_73 PATH_19_74 PATH_19_75 PATH_19_76 PATH_19_77 PATH_19_78 PATH_19_79 PATH_19_80 PATH_19_81 PATH_19_82 PATH_19_83 PATH_1
9_84 PATH_19_85 PATH_19_86 PATH_19_87 PATH_19_88 PATH_19_89 PATH_19_90 PATH_19_91 PATH_19_92 PATH_19_93 PATH_19_94 PATH_19_9
5 PATH_19_96 PATH_19_97 PATH_19_98 PATH_19_99 PATH_19_100 PATH_19_101 PATH_19_102 PATH_19_103 PATH_19_104 PATH_19_105 PATH_1
9_106 PATH_19_107 PATH_19_108 PATH_19_109 PATH_19_110 PATH_19_111 PATH_19_112 PATH_19_113 PATH_19_114 PATH_19_115 PATH_19_11
6 ...}
```

Ignore (VDDIO, VSSIO)

Number of floating I/O pads

Floating net list

# Powerplan : Use ICC2 (IC Compiler II tool) Error Browser

- The tool provides several verification and checking processes that you can use to verify your design and save resulting error data on disk (Use the "Error Browser" window to display details about these errors)

## <Error Browser Window Overview>

The Error Browser window displays error information in a structured manner. It includes a menu bar (File, Errors, Select, Highlight, Help) and a toolbar for error grouping and options. The main area is divided into two panes: an Error Hierarchy Tree on the left and an Error List on the right. The Error Hierarchy Tree shows a tree structure of error sets, including context\_mem:ORCA.design, ORCA\_dppinassgn.err, Pin Alignment, and Pin Short. The Error List displays a table of errors with columns for ID, Status, Color, Type, Layer, Pin Type, Magnitude, Information, and Summary. The bottom pane shows detailed information for a selected error, including layer, pin type, type summary, object information, mis-alignment distance, and pin locations. Command buttons (Clear List, Fixed) and options (Show, Follow, zoom, 1.0, Dim) are located at the bottom.

| ErrorSet                | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|-------------------------|-------|---------|-------|---------|----------|----------|
| context_mem:ORCA.design | 178   | 178     | 0     | 0       | 0        | 178      |
| ORCA_dppinassgn.err     | 178   | 178     | 0     | 0       | 0        | 178      |
| Pin Alignment           | 177   | 177     | 0     | 0       | 0        | 177      |
| METAL2 (18)-METAL2 (18) | 93    | 93      | 0     | 0       | 0        |          |
| METAL2 (18)-METAL3 (22) | 59    | 59      | 0     | 0       | 0        |          |
| METAL2 (18)-METAL4 (26) | 2     | 2       | 0     | 0       | 0        |          |
| METAL3 (22)-METAL3 (22) | 23    | 23      | 0     | 0       | 0        |          |
| Pin Short               | 1     | 1       | 0     | 0       | 0        | 1        |

| #  | ID | Status | Color | Type          | Layer           | Pin Type | Magnitude | Information          | Summary                       |
|----|----|--------|-------|---------------|-----------------|----------|-----------|----------------------|-------------------------------|
| 0  | 6  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 0.3450    | Pin A2[5] on cell... | Pin Alignment: status = error |
| 1  | 7  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 7.1050    | Pin A2[6] on cell... | Pin Alignment: status = error |
| 2  | 8  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 46.4750   | Pin A2[0] on cell... | Pin Alignment: status = error |
| 3  | 9  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 51.7950   | Pin A2[1] on cell... | Pin Alignment: status = error |
| 4  | 0  |        |       | Pin Short     | METAL2 (18)     | Normal   |           | Pin RegPort_B[2...   | Pin Short: status = error     |
| 5  | 1  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 194.1520  | Pin A2[0] on cell... | Pin Alignment: status = error |
| 6  | 2  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 185.1320  | Pin A2[1] on cell... | Pin Alignment: status = error |
| 7  | 3  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 175.2920  | Pin A2[2] on cell... | Pin Alignment: status = error |
| 8  | 4  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 169.1420  | Pin A2[3] on cell... | Pin Alignment: status = error |
| 9  | 5  |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 181.0320  | Pin A2[4] on cell... | Pin Alignment: status = error |
| 10 | 10 |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 65.8550   | Pin A2[2] on cell... | Pin Alignment: status = error |
| 11 | 11 |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 61.6150   | Pin A2[3] on cell... | Pin Alignment: status = error |
| 12 | 12 |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 73.2350   | Pin A2[4] on cell... | Pin Alignment: status = error |
| 13 | 13 |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 67.9050   | Pin A2[5] on cell... | Pin Alignment: status = error |
| 14 | 14 |        |       | Pin Alignment | METAL2 (18)-... | Normal   | 95.8750   | Pin A2[6] on cell... | Pin Alignment: status = error |

0: Layer: METAL2 (18)-METAL3 (22) Type: Pin Alignment  
Pin type: Normal  
Type Summary : Pin Alignment: status = error  
Obj Info : Pin A2[5] on cell REG\_FILE\_A\_RAM is mis-aligned with pin Addr\_A[5] on cell I\_REG\_FILE  
Mis-alignment distance is: 0.345 um  
Pin1: I\_ORCA\_TOP/I\_RISC\_CORE/I\_REG\_FILE/REG\_FILE\_A\_RAM/A2[5]  
Pin2: I\_ORCA\_TOP/I\_RISC\_CORE/I\_REG\_FILE/Addr\_A[5]

Command Buttons: Clear List, Fixed  
Options: Show: all, Follow: zoom, 1.0, Dim

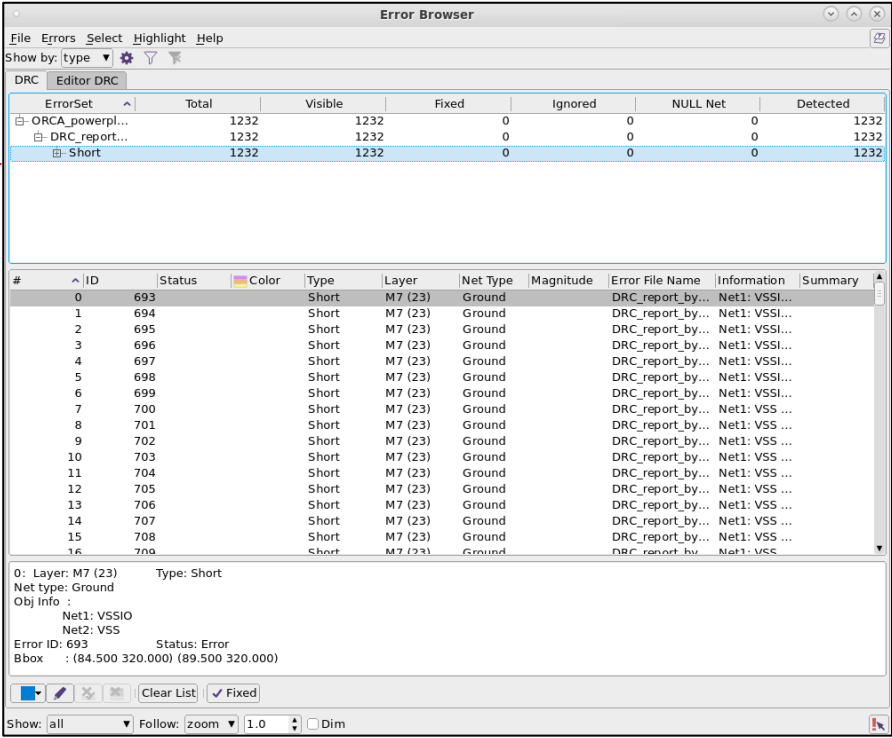
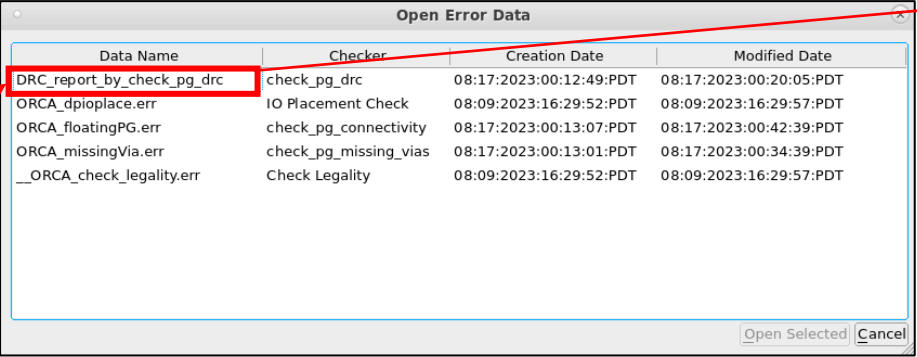
## <A typical use flow for debugging violations>

1. Verify the design and save the results on disk.
2. Open the saved data in the error browser.
3. Select a group of errors that you want to inspect. You can group errors by either the error type or the layer associated with the errors.  
  
Alternatively, you can choose a group of errors by their physical location in the design. For details, see [Analyzing Errors by Area](#) below.
4. Analyze and debug the errors interactively by selecting them in the error list and zooming or panning as needed to view them in the layout view.
5. Mark errors as fixed after you have inspected and fixed them.
6. Optionally ignore fixed errors by hiding them from view.

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error



In the gui window, you can open the Error browser view- Error\_Browser or press Ctrl+Shift+E hotkey

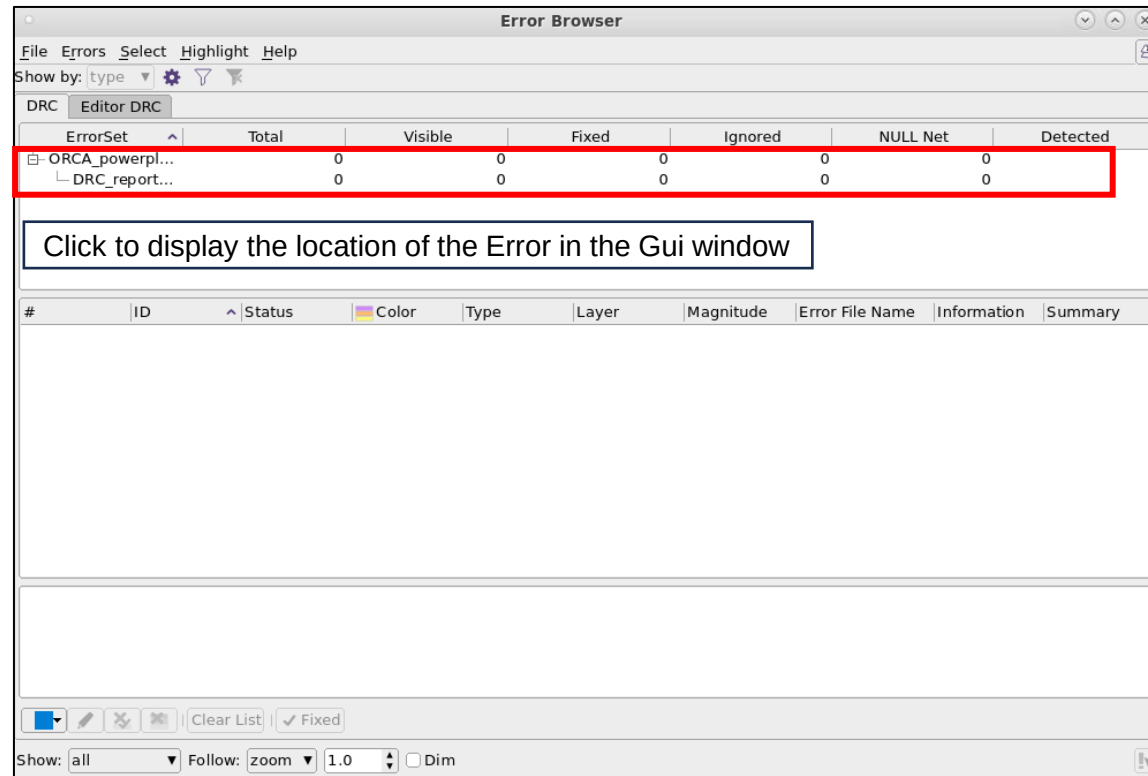
When the window opens, you can open the Error browser by checking the data name of the previously checked item, finding the required item, and clicking it

Use the "Error Browser" window to display details about these errors

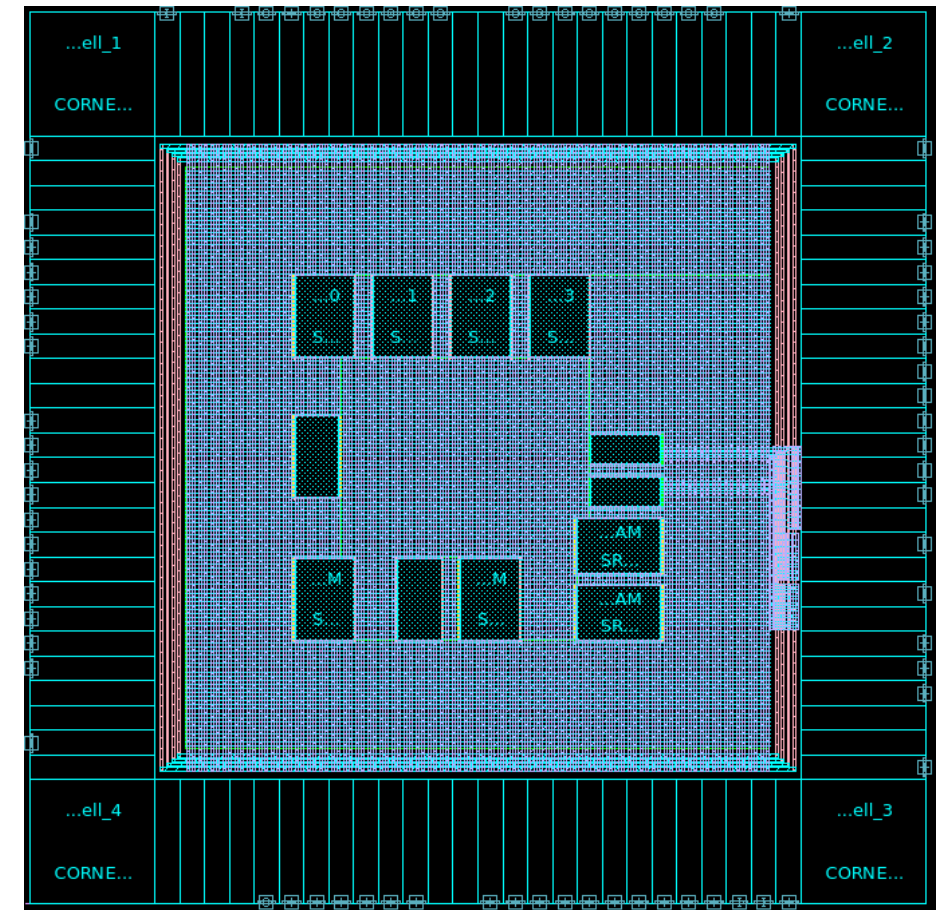
# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (check\_pg\_DRC)



❖ No Errors in check\_pg\_drc



# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (check\_pg\_missing\_vias)

Click to display the location of the Error in the Gui window

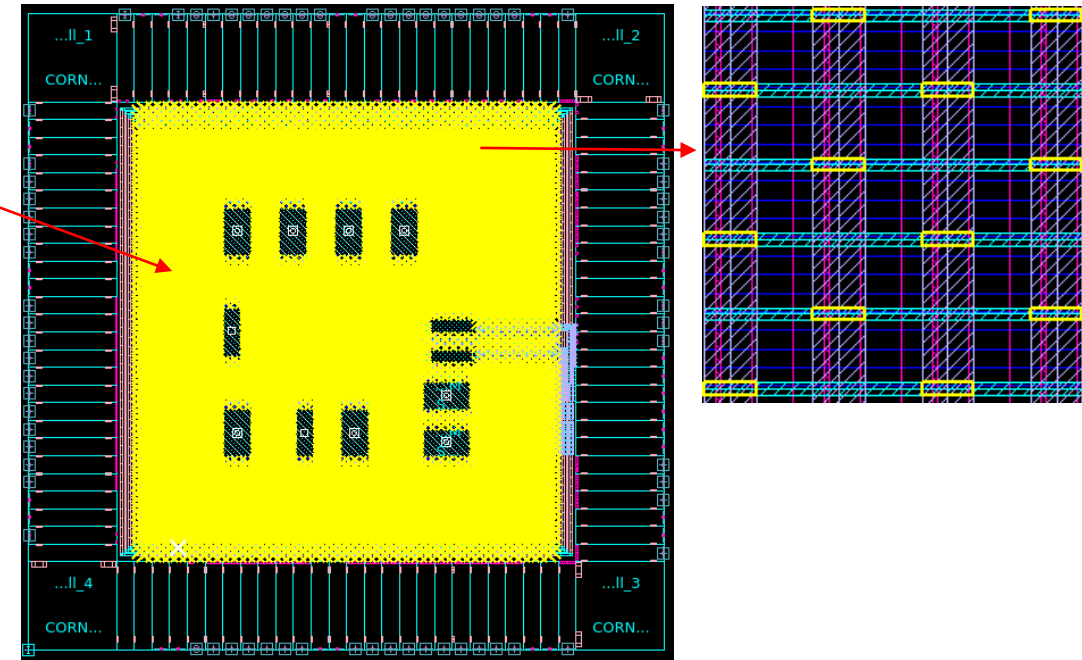
| ErrorSet        | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|-----------------|-------|---------|-------|---------|----------|----------|
| ORCA_powerpl... | 6444  | 6444    | 0     | 0       | 0        | 6444     |
| ORCA missi...   | 6444  | 6444    | 0     | 0       | 0        | 6444     |
| default         | 6444  | 6444    | 0     | 0       | 0        | 6444     |

| #  | ID   | Status | Color | Type    | Layer     | Net Type | Magnitude | Error File Name | Information     | Summary      |
|----|------|--------|-------|---------|-----------|----------|-----------|-----------------|-----------------|--------------|
| 0  | 6111 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 1  | 6112 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 2  | 6113 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 3  | 6114 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 4  | 6115 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 5  | 6116 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 6  | 6117 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 7  | 6118 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 8  | 6119 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 9  | 6120 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 10 | 6121 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 11 | 6122 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 12 | 6123 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 13 | 6124 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 14 | 6125 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 15 | 6126 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |
| 16 | 6127 |        |       | default | VIA7 (24) | Ground   |           | ORCA_missing... | Missing via ... | Status = ... |

0: Layer: VIA7 (24) Type: default  
Net type: Ground  
Type Summary : Status = error  
Obj Info : Missing via of net VSS on layer VIA7 from via rule:default  
Net: VSS  
Error ID: 6111 Status: Error  
Bbox : (338.160 228.816) (342.800 233.816)

Clear List Fixed

Show: all Follow: zoom 1.0 Dim



- ❖ This violation before VIA insertion and resolve this violation after VIA insertion
- ❖ Violation occurs on the corresponding VDD and VSS Stripe because no VIA was created except for the M1 and M2 connections when creating the MESH

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (check\_pg\_connectivity)

**Error Browser**

File Errors Select Highlight Help

Show by: type

DRC Editor DRC

| ErrorSet            | Total       | Visible     | Fixed    | Ignored  | NULL Net | Detected    |
|---------------------|-------------|-------------|----------|----------|----------|-------------|
| ORCA_powerpl...     | 34849       | 34849       | 0        | 0        | 0        | 34849       |
| ORCA floati...      | 34849       | 34849       | 0        | 0        | 0        | 34849       |
| <b>Floating ...</b> | <b>4354</b> | <b>4354</b> | <b>0</b> | <b>0</b> | <b>0</b> | <b>4354</b> |
| Floating ...        | 24328       | 24328       | 0        | 0        | 0        | 24328       |
| Floating ...        | 7           | 7           | 0        | 0        | 0        | 7           |
| Floating ...        | 2782        | 2782        | 0        | 0        | 0        | 2782        |
| Floating ...        | 3378        | 3378        | 0        | 0        | 0        | 3378        |

| #  | ID    | Status | Color | Type          | Layer   | Cell Type | Magnitude | Error File Name   | Information     | Summary      |
|----|-------|--------|-------|---------------|---------|-----------|-----------|-------------------|-----------------|--------------|
| 0  | 18963 |        |       | Floating h... | M4 (17) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 1  | 18964 |        |       | Floating h... | M5 (19) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 2  | 18965 |        |       | Floating h... | M1 (11) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 3  | 18966 |        |       | Floating h... | M2 (13) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 4  | 18967 |        |       | Floating h... | M3 (15) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 5  | 18968 |        |       | Floating h... | M4 (17) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 6  | 18969 |        |       | Floating h... | M5 (19) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 7  | 18970 |        |       | Floating h... | M1 (11) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 8  | 18971 |        |       | Floating h... | M2 (13) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 9  | 18972 |        |       | Floating h... | M3 (15) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 10 | 18973 |        |       | Floating h... | M4 (17) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 11 | 18974 |        |       | Floating h... | M5 (19) | Macro     |           | ORCA_floatingP... | Floating pin... | Status = ... |

0: Layer: M4 (17) Type: Floating hard macro pin violation  
Cell type: Macro

Type Summary : Status = error  
Obj Info : Floating pin I\_ORCA\_TOP/I\_CONTEXT\_MEM/CONTEXT\_RAM1/I2[18] is not connected to net VSS  
Cell: I\_ORCA\_TOP/I\_CONTEXT\_MEM/CONTEXT\_RAM1

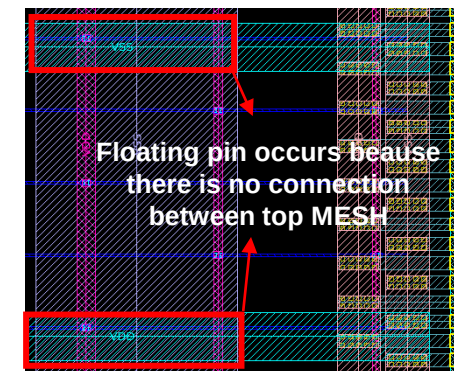
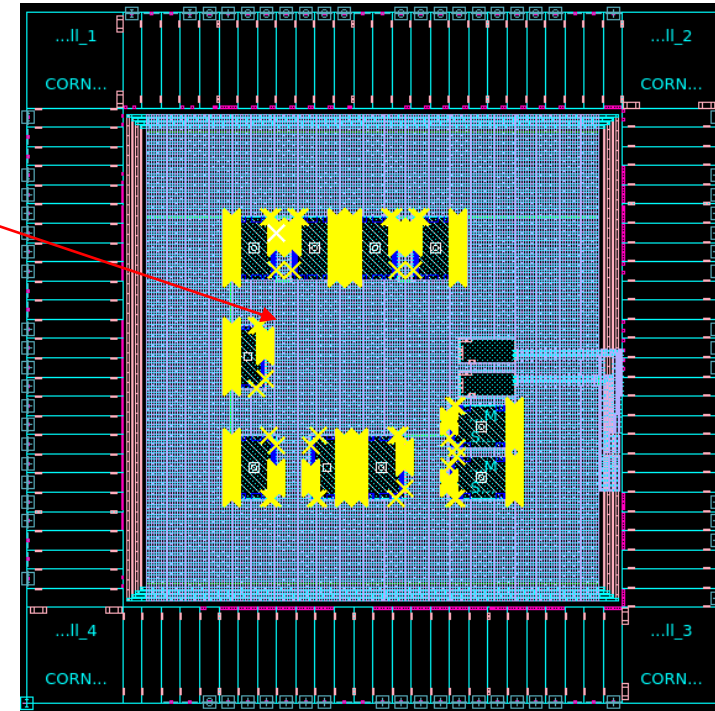
Error ID: 18963 Status: Error

Clear List Fixed

Show: all Follow: zoom 1.0 Dim

Click to display the location of the Error in the Gui window

- ❖ This violation before VIA insertion and resolve this violation after VIA insertion
- ❖ Violation caused by a lack of connection between Macro Ring to top MESH (Therefore, power is not supplied to the Macro power pins)



# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (check\_pg\_connectivity)

Error Browser

Show by: type

DRC Editor DRC

| ErrorSet        | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|-----------------|-------|---------|-------|---------|----------|----------|
| ORCA_powerpl... | 34849 | 34849   | 0     | 0       | 0        | 34849    |
| ORCA_floati...  | 34849 | 34849   | 0     | 0       | 0        | 34849    |
| Floating ...    | 4354  | 4354    | 0     | 0       | 0        | 4354     |
| Floating ...    | 24328 | 24328   | 0     | 0       | 0        | 24328    |
| Floating ...    | 7     | 7       | 0     | 0       | 0        | 7        |
| Floating ...    | 2782  | 2782    | 0     | 0       | 0        | 2782     |
| Floating ...    | 3378  | 3378    | 0     | 0       | 0        | 3378     |

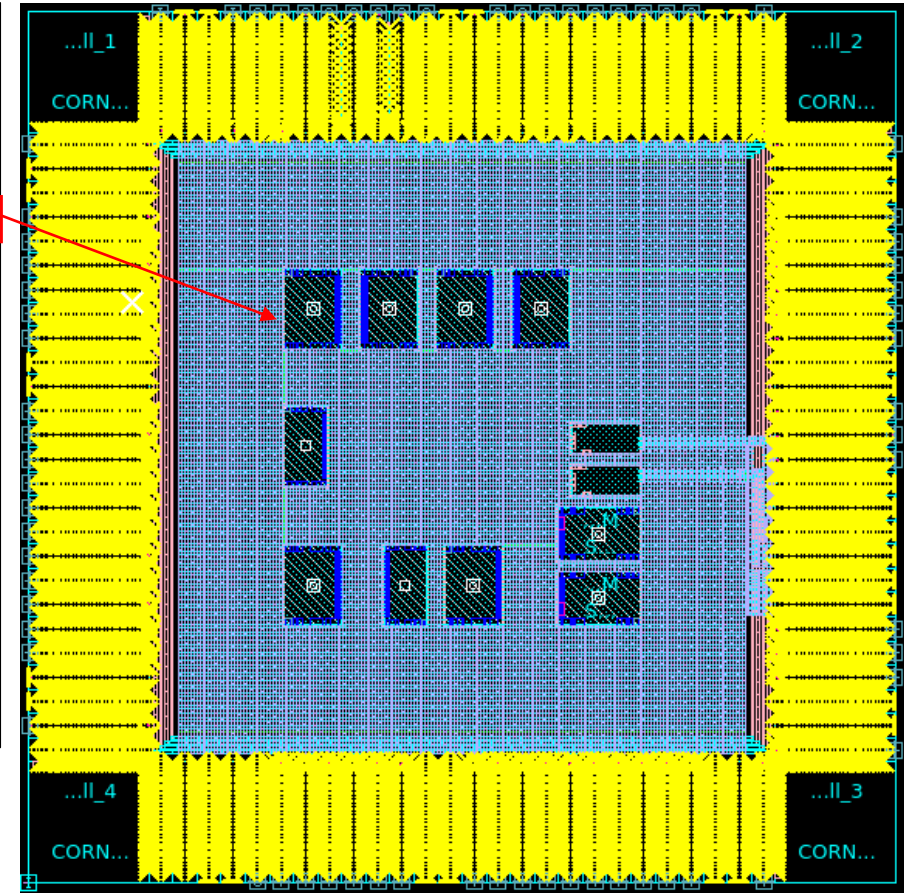
| #  | ID    | Status | Color | Type          | Layer   | Cell Type | Magnitude | Error File Name   | Information     | Summary      |
|----|-------|--------|-------|---------------|---------|-----------|-----------|-------------------|-----------------|--------------|
| 0  | 30051 |        |       | Floating p... | M6 (21) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 1  | 30052 |        |       | Floating p... | M6 (21) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 2  | 30053 |        |       | Floating p... | M6 (21) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 3  | 30054 |        |       | Floating p... | M6 (21) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 4  | 30055 |        |       | Floating p... | M6 (21) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 5  | 30056 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 6  | 30057 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 7  | 30058 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 8  | 30059 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 9  | 30060 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 10 | 30061 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 11 | 30062 |        |       | Floating p... | M7 (23) | IO        |           | ORCA_floatingP... | Floating pin... | Status = ... |

0: Layer: M6 (21) Type: Floating pad pin violation  
Cell type: IO  
Type Summary : Status = error  
Obj Info : Floating pin SDRAM\_DQ\_Bus\_12\_sdram\_DQ\_iopad/VSSIO is not connected to net VSSIO  
Cell: SDRAM\_DQ\_Bus\_12\_sdram\_DQ\_iopad  
Error ID: 30051 Status: Error

Clear List Fixed

Show: all Follow: zoom 1.0 Dim

Click to display the location of the Error in the Gui window

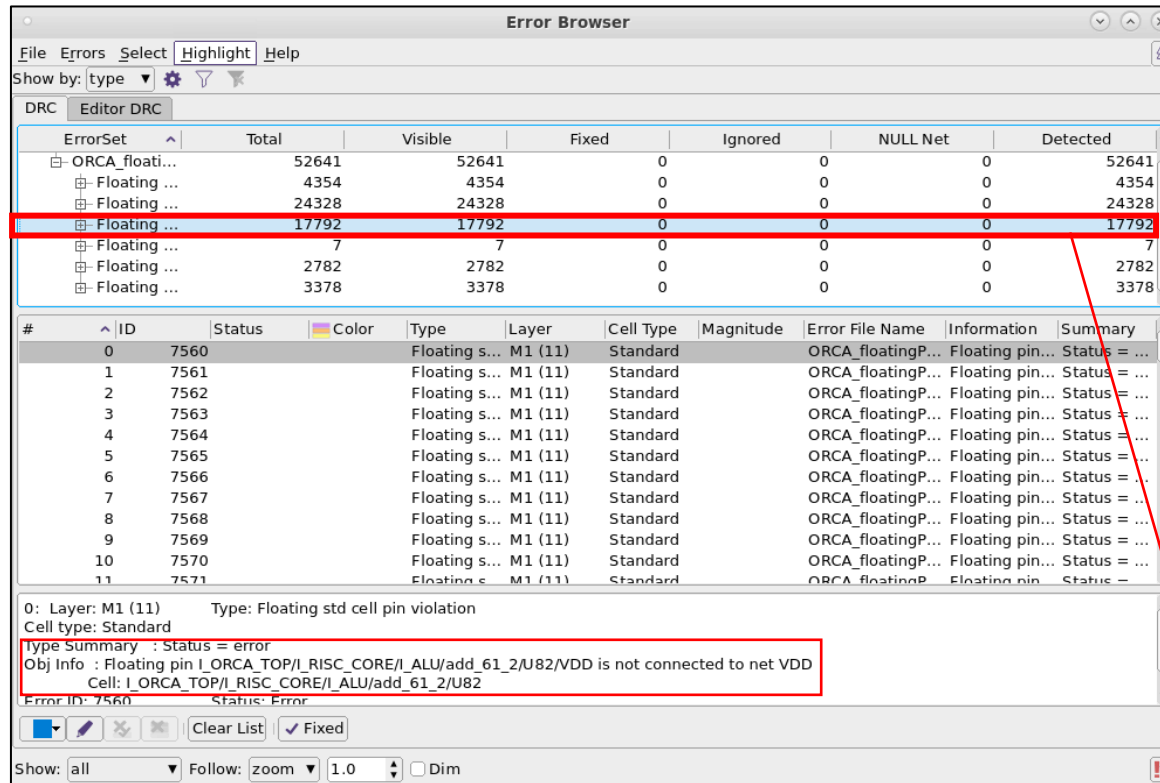


❖ Ignore IO Pad & terminal violation because of Power/Ground PAD ISSUE ( Power/Ground pad have problem for pin connection)

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (If you not use `-check_std_cell_pins none` option in `check_pg_connectivity`)

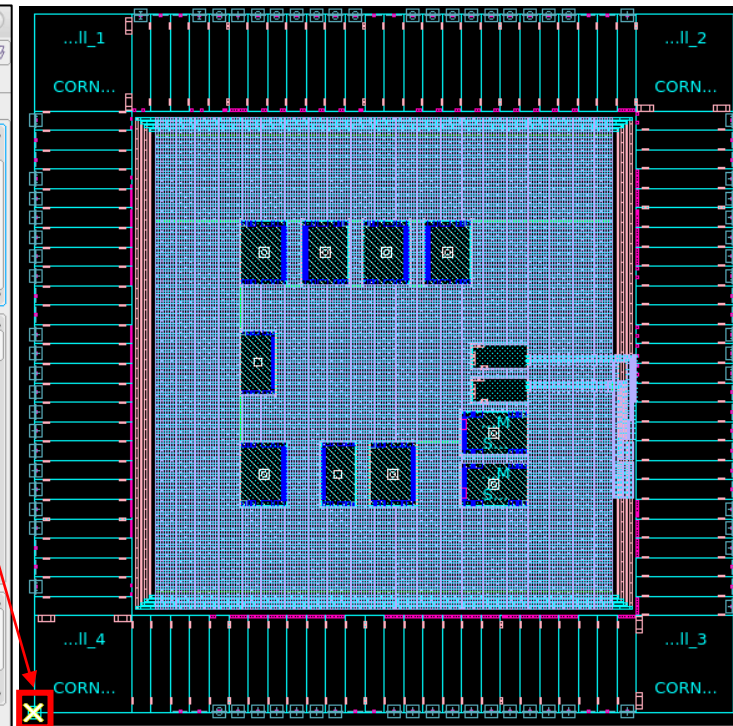


| ErrorSet       | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|----------------|-------|---------|-------|---------|----------|----------|
| ORCA_floati... | 52641 | 52641   | 0     | 0       | 0        | 52641    |
| Floating ...   | 4354  | 4354    | 0     | 0       | 0        | 4354     |
| Floating ...   | 24328 | 24328   | 0     | 0       | 0        | 24328    |
| Floating ...   | 17792 | 17792   | 0     | 0       | 0        | 17792    |
| Floating ...   | 7     | 7       | 0     | 0       | 0        | 7        |
| Floating ...   | 2782  | 2782    | 0     | 0       | 0        | 2782     |
| Floating ...   | 3378  | 3378    | 0     | 0       | 0        | 3378     |

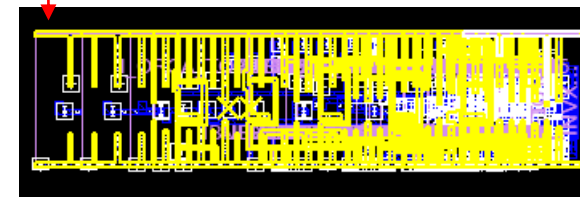
| #  | ID   | Status | Color | Type          | Layer   | Cell Type | Magnitude | Error File Name   | Information     | Summary      |
|----|------|--------|-------|---------------|---------|-----------|-----------|-------------------|-----------------|--------------|
| 0  | 7560 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 1  | 7561 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 2  | 7562 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 3  | 7563 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 4  | 7564 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 5  | 7565 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 6  | 7566 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 7  | 7567 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 8  | 7568 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 9  | 7569 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 10 | 7570 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |
| 11 | 7571 |        |       | Floating s... | M1 (11) | Standard  |           | ORCA_floatingP... | Floating pin... | Status = ... |

0: Layer: M1 (11) Type: Floating std cell pin violation  
Cell type: Standard  
Type Summary : Status = error  
Obj Info : Floating pin I\_ORCA\_TOP/I\_RISC\_CORE/I\_ALU/add\_61\_2/U82/VDD is not connected to net VDD  
Cell: I\_ORCA\_TOP/I\_RISC\_CORE/I\_ALU/add\_61\_2/U82  
Error ID: 7560 Status: Error  
Clear List Fixed



Click to display the location of the Error in the Gui window

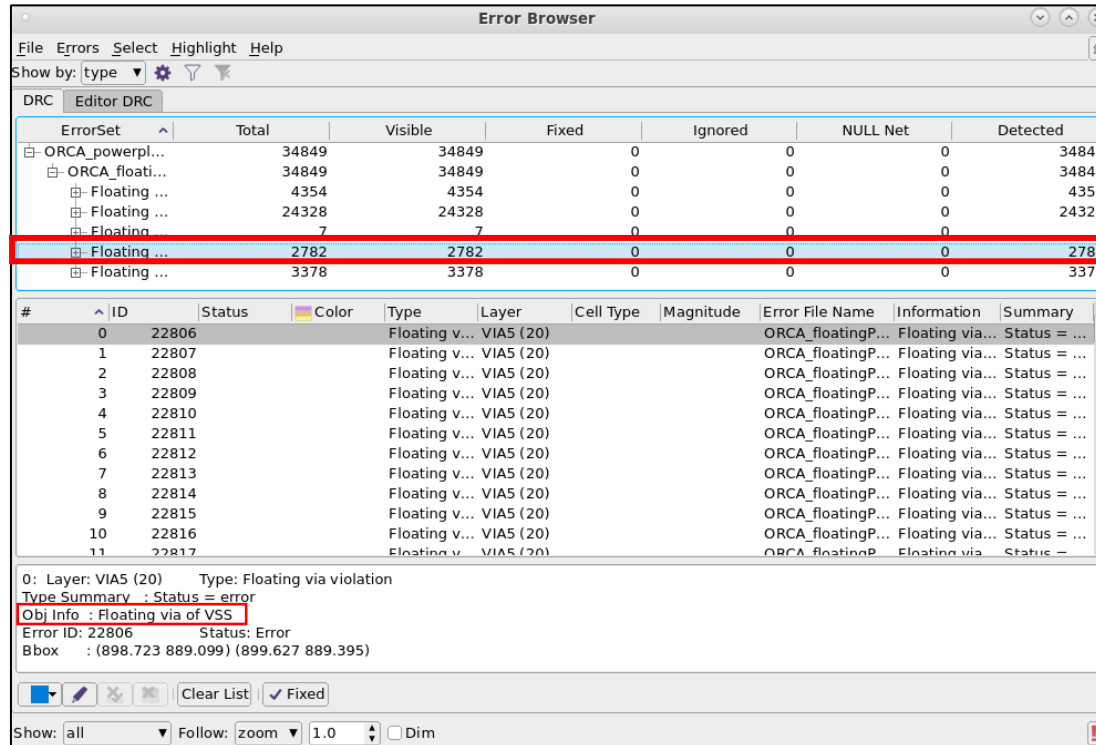
❖ Connectivity violation caused by STD cells are not yet placed  
(All STD cells VDD, VSS pins)



# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (check\_pg\_connectivity)



| ErrorSet            | Total       | Visible     | Fixed    | Ignored  | NULL Net | Detected    |
|---------------------|-------------|-------------|----------|----------|----------|-------------|
| ORCA_powerpl...     | 34849       | 34849       | 0        | 0        | 0        | 34849       |
| ORCA_floati...      | 34849       | 34849       | 0        | 0        | 0        | 34849       |
| Floating ...        | 4354        | 4354        | 0        | 0        | 0        | 4354        |
| Floating ...        | 24328       | 24328       | 0        | 0        | 0        | 24328       |
| Floating ...        | 7           | 7           | 0        | 0        | 0        | 7           |
| <b>Floating ...</b> | <b>2782</b> | <b>2782</b> | <b>0</b> | <b>0</b> | <b>0</b> | <b>2782</b> |
| Floating ...        | 3378        | 3378        | 0        | 0        | 0        | 3378        |

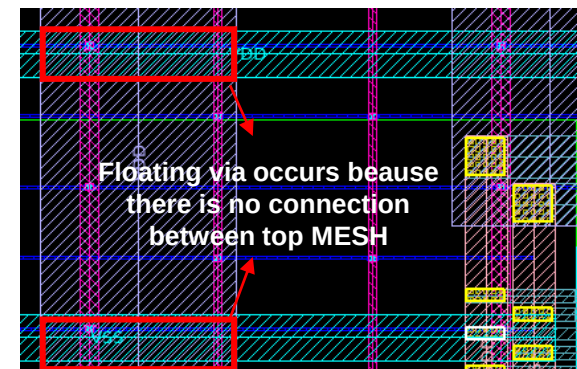
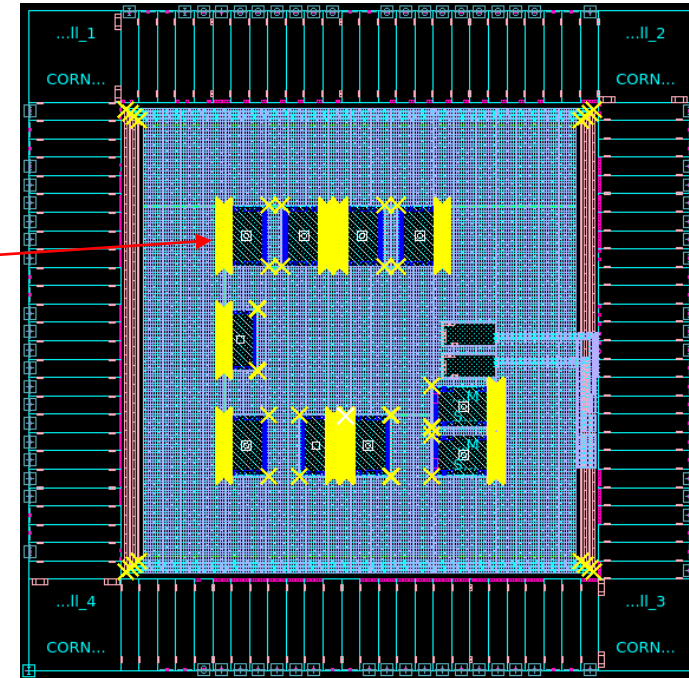
  

| #  | ID    | Status | Color | Type          | Layer     | Cell Type         | Magnitude | Error File Name   | Information     | Summary      |
|----|-------|--------|-------|---------------|-----------|-------------------|-----------|-------------------|-----------------|--------------|
| 0  | 22806 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 1  | 22807 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 2  | 22808 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 3  | 22809 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 4  | 22810 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 5  | 22811 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 6  | 22812 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 7  | 22813 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 8  | 22814 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 9  | 22815 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 10 | 22816 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |
| 11 | 22817 |        |       | Floating v... | VIA5 (20) | ORCA_floatingP... |           | ORCA_floatingP... | Floating via... | Status = ... |

0: Layer: VIA5 (20) Type: Floating via violation  
Type Summary : Status = error  
Obj Info : Floating via of VSS  
Error ID: 22806 Status: Error  
Bbox : (898.723 889.099) (899.627 889.395)

Clear List Fixed

Show: all Follow: zoom 1.0 Dim



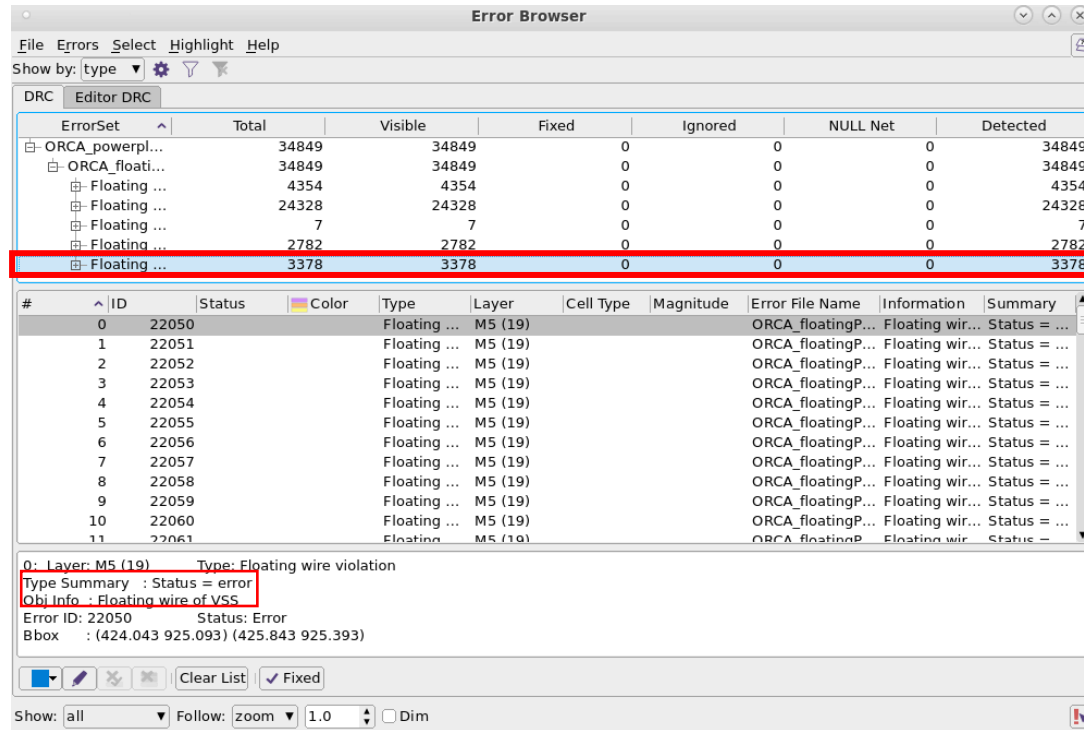
Click to display the location of the Error in the Gui window

- ❖ This violation occurred due to floating vias because the Core Ring to Core Ring, Macro Ring, to Macro power pins are connected but not connected with top MESH

# Powerplan : Use ICC2 (IC Compiler II tool)

11. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/10_remove_setting_and_check_before_fix.tcl`

- Open the Error Browser & analyzed Error (check\_pg\_connectivity)

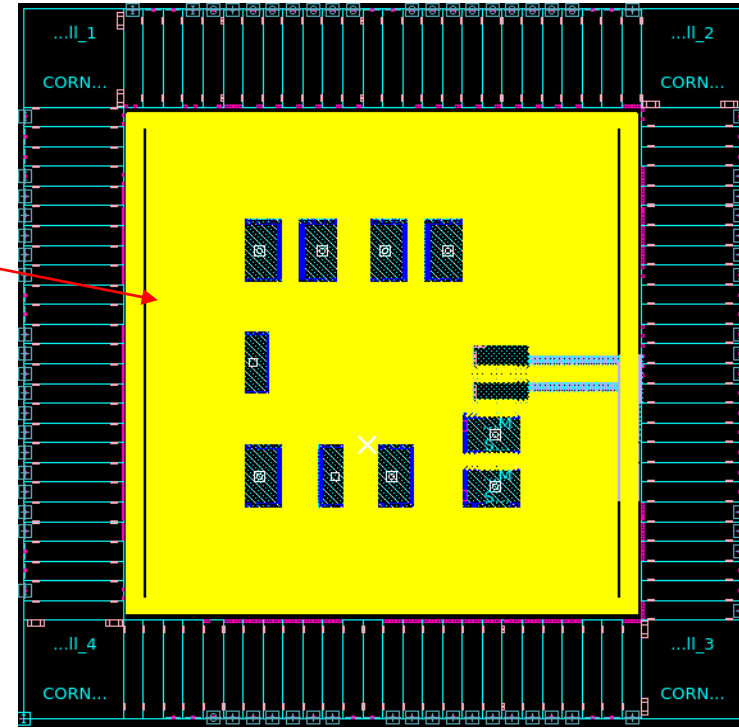


| ErrorSet        | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|-----------------|-------|---------|-------|---------|----------|----------|
| ORCA_powerpl... | 34849 | 34849   | 0     | 0       | 0        | 34849    |
| ORCA_floati...  | 34849 | 34849   | 0     | 0       | 0        | 34849    |
| Floating ...    | 4354  | 4354    | 0     | 0       | 0        | 4354     |
| Floating ...    | 24328 | 24328   | 0     | 0       | 0        | 24328    |
| Floating ...    | 7     | 7       | 0     | 0       | 0        | 7        |
| Floating ...    | 2782  | 2782    | 0     | 0       | 0        | 2782     |
| Floating ...    | 3378  | 3378    | 0     | 0       | 0        | 3378     |

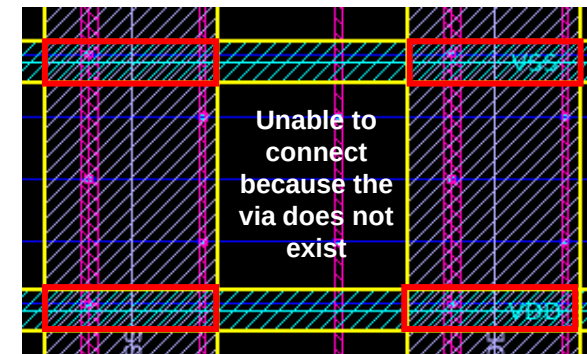
| #  | ID    | Status | Color | Type         | Layer   | Cell Type | Magnitude | Error File Name   | Information     | Summary      |
|----|-------|--------|-------|--------------|---------|-----------|-----------|-------------------|-----------------|--------------|
| 0  | 22050 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 1  | 22051 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 2  | 22052 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 3  | 22053 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 4  | 22054 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 5  | 22055 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 6  | 22056 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 7  | 22057 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 8  | 22058 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 9  | 22059 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 10 | 22060 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |
| 11 | 22061 |        |       | Floating ... | M5 (19) |           |           | ORCA_floatingP... | Floating wir... | Status = ... |

0: Layer: M5 (19) Type: Floating wire violation  
Type Summary : Status = error  
Obj Info : Floating wire of VSS  
Error ID: 22050 Status: Error  
Bbox : (424.043 925.093) (425.843 925.393)



Click to display the location of the Error in the Gui window

- ❖ This violation before VIA insertion and resolve this violation after VIA insertion
- ❖ Violation occurs on the corresponding VDD and VSS Stripe because no VIA was created except for the M1 and M2 connections when creating the MESH

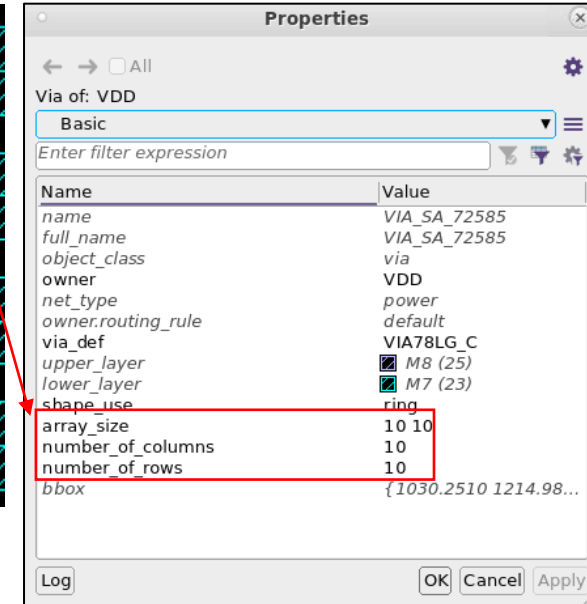
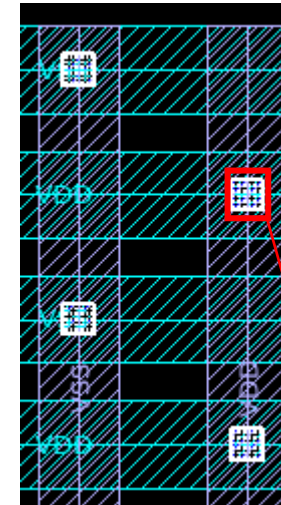


# Powerplan : Use ICC2 (IC Compiler II tool)

## 12. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/11\_fix\_DRC\_and\_create\_via.tcl

- Create Vias (Core Ring (M7) to top MESH (M8))

```
icc2_shell> set_pg_via master_rule pgvia_10x10 -via_array_dimension {10 10}
Successfully set via def rule pgvia_10x10.
icc2_shell> set_core M7 M8 shape [get_shapes -filter "tag == CORE RING && layer.name == M7 || tag == MESH 8"]
{PATH_23_0 PATH_23_1 PATH_23_2 PATH_23_3 PATH_23_4 PATH_23_5 PATH_23_6 PATH_23_7 PATH_25_136 PATH_25_137 PATH_25_138 PATH_25_139 PATH_25_140 PATH_25_141 PATH_25_142 PATH_25_143 PATH_25_144 PATH_25_145 PATH_25_146 PATH_25_147 PATH_25_148 PATH_25_149 PATH_25_150 PATH_25_151 PATH_25_152 PATH_25_153 PATH_25_154 PATH_25_155 PATH_25_156 PATH_25_157 PATH_25_158 PATH_25_159 PATH_25_160 PATH_25_161 PATH_25_162 PATH_25_163 PATH_25_164 PATH_25_165 PATH_25_166 PATH_25_167 PATH_25_168 PATH_25_169 PATH_25_170 PATH_25_171 PATH_25_172 PATH_25_173 PATH_25_174 PATH_25_175 PATH_25_176 PATH_25_177 PATH_25_178 PATH_25_179 PATH_25_180 PATH_25_181 PATH_25_182 PATH_25_183 PATH_25_184 PATH_25_185 PATH_25_186 PATH_25_187 PATH_25_188 PATH_25_189 PATH_25_190 PATH_25_191 PATH_25_192 PATH_25_193 PATH_25_194 PATH_25_195 PATH_25_196 PATH_25_197 PATH_25_198 PATH_25_199 PATH_25_200 PATH_25_201 PATH_25_202 PATH_25_203 PATH_25_204 PATH_25_205 PATH_25_206 PATH_25_207 PATH_25_208 PATH_25_209 PATH_25_210 PATH_25_211 PATH_25_212 PATH_25_213 PATH_25_214 PATH_25_215 PATH_25_216 PATH_25_217 PATH_25_218 PATH_25_219 PATH_25_220 PATH_25_221 PATH_25_222 PATH_25_223 PATH_25_224 PATH_25_225 PATH_25_226 PATH_25_227 ...}
icc2_shell> change_selection $core M7 M8 shape
icc2_shell> create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M7 \
? -to_layers M8 -via_masters {pgvia_10x10} -tag {core_M7_8_via} -shapes [get_selection]
Set tag to core_M7_8_via
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 270 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 388 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 388 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 388 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 388 vias.
Setting tag core_M7_8_via to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 0 seconds.
Successfully created PG Vias.
1
```



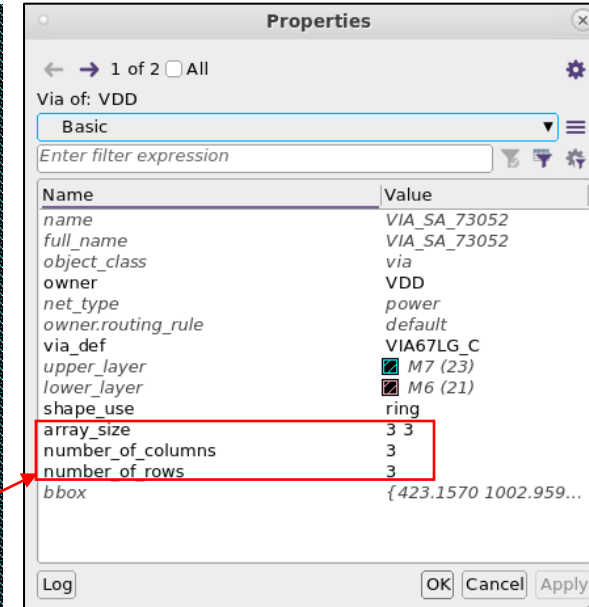
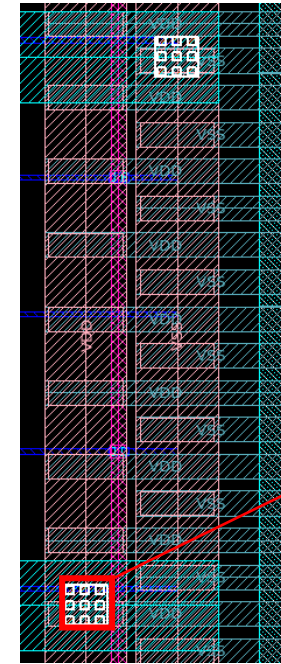
- Zoom in and check Core Ring (M7) to top MESH (M8) stripes are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

12. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/11_fix_DRC_and_create_via.tcl`

- Create Vias (Macro Ring (M6) to top MESH (M7))

```
icc2_shell> set_pg_via_master_rule pgvia_3x3 -via_array_dimension {3 3}
Successfully set via def rule pgvia 3x3.
icc2_shell> set M6 M7_shape [get_shapes -filter "tag == MACRO RING && layer.name == M6 || tag == MESH 7"]
{PATH_23_89 PATH_23_90 PATH_23_91 PATH_23_92 PATH_23_93 PATH_23_94 PATH_23_95 PATH_23_96 PATH_23_97 PATH_23_98 PA
TH_23_99 PATH_23_100 PATH_23_101 PATH_23_102 PATH_23_103 PATH_23_104 PATH_23_105 PATH_23_106 PATH_23_107 PATH_23
108 PATH_23_109 PATH_23_110 PATH_23_111 PATH_23_112 PATH_23_113 PATH_23_114 PATH_23_115 PATH_23_116 PATH_23_117 P
ATH_23_118 PATH_23_119 PATH_23_120 PATH_23_121 PATH_23_122 PATH_23_123 PATH_23_124 PATH_23_125 PATH_23_126 PATH_2
3_127 PATH_23_128 PATH_23_129 PATH_23_130 PATH_23_131 PATH_23_132 PATH_23_133 PATH_23_134 PATH_23_135 PATH_23_136
PATH_23_137 PATH_23_138 PATH_23_139 PATH_23_140 PATH_23_141 PATH_23_142 PATH_23_143 PATH_23_144 PATH_23_145 PATH
23_146 PATH_23_147 PATH_23_148 PATH_23_149 PATH_23_150 PATH_23_151 PATH_23_152 PATH_23_153 PATH_23_154 PATH_23_1
55 PATH_23_156 PATH_23_157 PATH_23_158 PATH_23_159 PATH_23_160 PATH_23_161 PATH_23_162 PATH_23_163 PATH_23_164 PA
TH_23_165 PATH_23_166 PATH_23_167 PATH_23_168 PATH_23_169 PATH_23_170 PATH_23_171 PATH_23_172 PATH_23_173 PATH_23
174 PATH_23_175 PATH_23_176 PATH_23_177 PATH_23_178 PATH_23_179 PATH_23_180 PATH_23_181 PATH_23_182 PATH_23_183
PATH_23_184 PATH_23_185 PATH_23_186 PATH_23_187 PATH_23_188 ...}
icc2_shell> change_selection $M6 M7_shape
icc2_shell> create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M6 \
? -to_layers M7 -via_masters {pgvia_3x3} -tag {M6_7_via} -shapes [get_selection]
Set tag to M6_7_via
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 374 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 386 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 386 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95% 1
00%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 386 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 386 vias.
Setting tag M6_7_via to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 0 seconds.
Successfully created PG Vias.
1
```



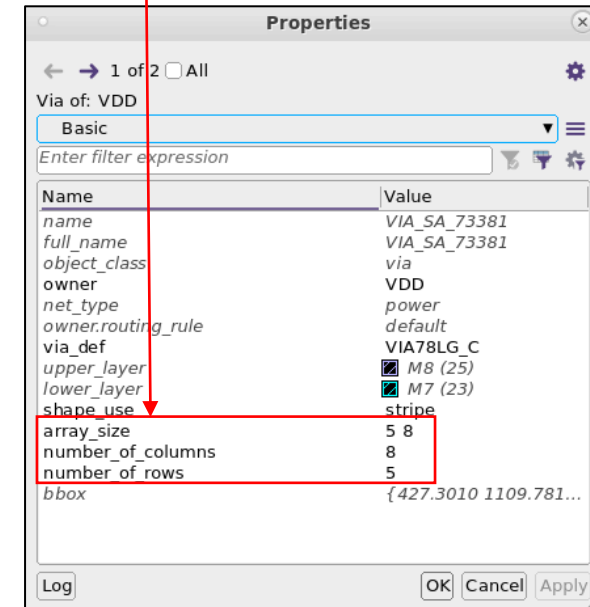
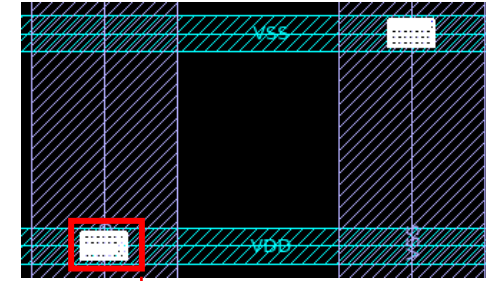
- Zoom in and check Macro Ring (M6) to top MESH (M7) stripes are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

12. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/11_fix_DRC_and_create_via.tcl`

- Create Vias (top MESH (M7) to top MESH (M8))

```
icc2_shell> set_pg_via_master_rule pgvia_5x8 -via_array_dimension {8 5}
Successfully set via def rule pgvia_5x8.
icc2_shell> set M7_M8_shape [get_shapes -filter "tag == MESH_8 || tag == MESH_7"]
{PATH_23_89 PATH_23_90 PATH_23_91 PATH_23_92 PATH_23_93 PATH_23_94 PATH_23_95 PATH_23_96 PATH_23_97 PATH_23_98 PA
TH_23_99 PATH_23_100 PATH_23_101 PATH_23_102 PATH_23_103 PATH_23_104 PATH_23_105 PATH_23_106 PATH_23_107 PATH_23
108 PATH_23_109 PATH_23_110 PATH_23_111 PATH_23_112 PATH_23_113 PATH_23_114 PATH_23_115 PATH_23_116 PATH_23_117 P
ATH_23_118 PATH_23_119 PATH_23_120 PATH_23_121 PATH_23_122 PATH_23_123 PATH_23_124 PATH_23_125 PATH_23_126 PATH_2
3_127 PATH_23_128 PATH_23_129 PATH_23_130 PATH_23_131 PATH_23_132 PATH_23_133 PATH_23_134 PATH_23_135 PATH_23_136
PATH_23_137 PATH_23_138 PATH_23_139 PATH_23_140 PATH_23_141 PATH_23_142 PATH_23_143 PATH_23_144 PATH_23_145 PATH
23_146 PATH_23_147 PATH_23_148 PATH_23_149 PATH_23_150 PATH_23_151 PATH_23_152 PATH_23_153 PATH_23_154 PATH_23_1
55 PATH_23_156 PATH_23_157 PATH_23_158 PATH_23_159 PATH_23_160 PATH_23_161 PATH_23_162 PATH_23_163 PATH_23_164 PA
TH_23_165 PATH_23_166 PATH_23_167 PATH_23_168 PATH_23_169 PATH_23_170 PATH_23_171 PATH_23_172 PATH_23_173 PATH_23
174 PATH_23_175 PATH_23_176 PATH_23_177 PATH_23_178 PATH_23_179 PATH_23_180 PATH_23_181 PATH_23_182 PATH_23_183
PATH_23_184 PATH_23_185 PATH_23_186 PATH_23_187 PATH_23_188 ...}
icc2_shell> change_selection $M7_M8_shape
icc2_shell> create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M7 \
? -to_layers M8 -via_masters {pgvia_5x8} -tag {M7_8_via} -shapes [get_selection]
Set tag to M7_8_via
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 596 shapes starts
Intersection detection runtime: 1 seconds.
Instantiating vias for 5670 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 5670 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95%
100%
Overall via DRC checking runtime: 0 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 5670 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 5670 vias.
Setting tag M7_8_via to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 1 seconds.
Successfully created PG Vias.
1
```



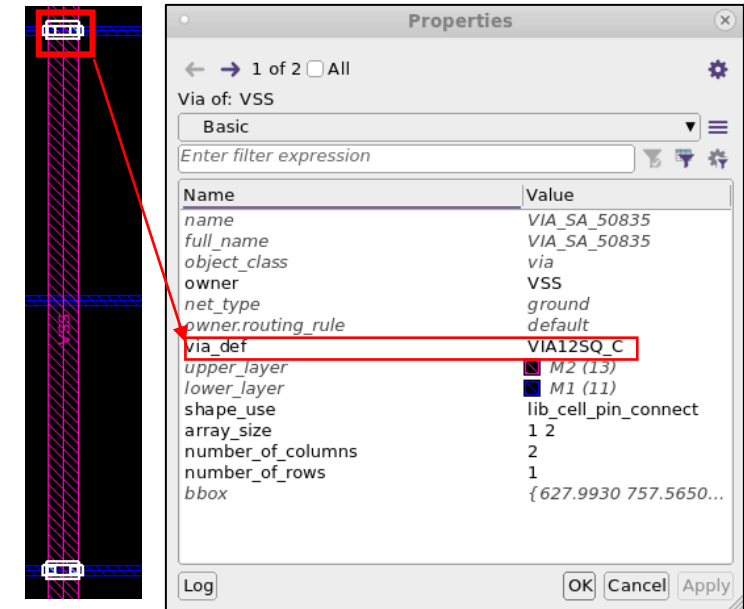
- Zoom in and check top MESH (M7) to top MESH (M8) stripes are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

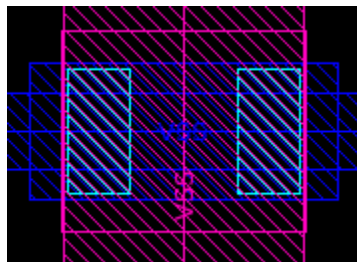
12. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/11_fix_DRC_and_create_via.tcl`

- Change Via\_def (STD\_Rail (M1) to low MESH (M2) vias)

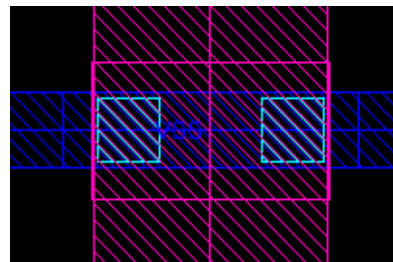
```
icc2_shell> set M1 M2 via [get vias -filter "tag == STD_RAIL_M1"]
{VIA_SA_3274 VIA_SA_3275 VIA_SA_3276 VIA_SA_3277 VIA_SA_3278 VIA_SA_3279 VIA_SA_3280 VIA_SA_3281 VIA_SA_3282 VIA
SA_3283 VIA_SA_3284 VIA_SA_3285 VIA_SA_3286 VIA_SA_3287 VIA_SA_3288 VIA_SA_3289 VIA_SA_3290 VIA_SA_3291 VIA_SA_32
92 VIA_SA_3293 VIA_SA_3294 VIA_SA_3295 VIA_SA_3296 VIA_SA_3297 VIA_SA_3298 VIA_SA_3299 VIA_SA_3300 VIA_SA_3301 VI
A_SA_3302 VIA_SA_3303 VIA_SA_3304 VIA_SA_3305 VIA_SA_3306 VIA_SA_3307 VIA_SA_3308 VIA_SA_3309 VIA_SA_3310 VIA_SA
3311 VIA_SA_3312 VIA_SA_3313 VIA_SA_3314 VIA_SA_3315 VIA_SA_3316 VIA_SA_3317 VIA_SA_3318 VIA_SA_3319 VIA_SA_3320
VIA_SA_3321 VIA_SA_3322 VIA_SA_3323 VIA_SA_3324 VIA_SA_3325 VIA_SA_3326 VIA_SA_3327 VIA_SA_3328 VIA_SA_3329 VIA_S
A_3330 VIA_SA_3331 VIA_SA_3332 VIA_SA_3333 VIA_SA_3334 VIA_SA_3335 VIA_SA_3336 VIA_SA_3337 VIA_SA_3338 VIA_SA_333
9 VIA_SA_3340 VIA_SA_3341 VIA_SA_3342 VIA_SA_3343 VIA_SA_3344 VIA_SA_3345 VIA_SA_3346 VIA_SA_3347 VIA_SA_3348 VIA
SA_3349 VIA_SA_3350 VIA_SA_3351 VIA_SA_3352 VIA_SA_3353 VIA_SA_3354 VIA_SA_3355 VIA_SA_3356 VIA_SA_3357 VIA_SA_3
358 VIA_SA_3359 VIA_SA_3360 VIA_SA_3361 VIA_SA_3362 VIA_SA_3363 VIA_SA_3364 VIA_SA_3365 VIA_SA_3366 VIA_SA_3367 V
IA_SA_3368 VIA_SA_3369 VIA_SA_3370 VIA_SA_3371 VIA_SA_3372 VIA_SA_3373 ...}
icc2_shell> change selection $M1 M2 via
icc2_shell> set_attribute -objects [get_selection] -name via_def -value [get_via_defs -library [current_lib] VIA1
25Q_C]
{VIA_SA_68849 VIA_SA_68848 VIA_SA_68847 VIA_SA_68846 VIA_SA_68845 VIA_SA_68844 VIA_SA_68843 VIA_SA_68842 VIA_SA_6
8841 VIA_SA_68840 VIA_SA_68839 VIA_SA_68838 VIA_SA_68837 VIA_SA_68836 VIA_SA_68835 VIA_SA_68834 VIA_SA_68833 VIA
SA_68832 VIA_SA_68831 VIA_SA_68830 VIA_SA_68829 VIA_SA_68828 VIA_SA_68827 VIA_SA_68826 VIA_SA_68825 VIA_SA_68824
VIA_SA_68823 VIA_SA_68747 VIA_SA_68746 VIA_SA_68745 VIA_SA_68744 VIA_SA_68743 VIA_SA_68742 VIA_SA_68741 VIA_SA_68
740 VIA_SA_68739 VIA_SA_68738 VIA_SA_68737 VIA_SA_68736 VIA_SA_68735 VIA_SA_68734 VIA_SA_68733 VIA_SA_68732 VIA_S
A_68731 VIA_SA_68730 VIA_SA_68729 VIA_SA_68728 VIA_SA_68727 VIA_SA_68726 VIA_SA_68725 VIA_SA_68724 VIA_SA_68723 V
IA_SA_68722 VIA_SA_68721 VIA_SA_68645 VIA_SA_68644 VIA_SA_68643 VIA_SA_68642 VIA_SA_68641 VIA_SA_68640 VIA_SA_686
39 VIA_SA_68638 VIA_SA_68637 VIA_SA_68636 VIA_SA_68635 VIA_SA_68634 VIA_SA_68633 VIA_SA_68632 VIA_SA_68631 VIA_SA
68630 VIA_SA_68629 VIA_SA_68628 VIA_SA_68627 VIA_SA_68626 VIA_SA_68625 VIA_SA_68624 VIA_SA_68623 VIA_SA_68622 VI
A_SA_68621 VIA_SA_68620 VIA_SA_68619 VIA_SA_68543 VIA_SA_68542 VIA_SA_68541 VIA_SA_68540 VIA_SA_68539 VIA_SA_6853
8 VIA_SA_68537 VIA_SA_68536 VIA_SA_68535 VIA_SA_68534 VIA_SA_68533 VIA_SA_68532 VIA_SA_68531 VIA_SA_68530 VIA_SA
68529 VIA_SA_68528 VIA_SA_68527 VIA_SA_68526 VIA_SA_68525 ...}
```



<Before changing>



<After changing>



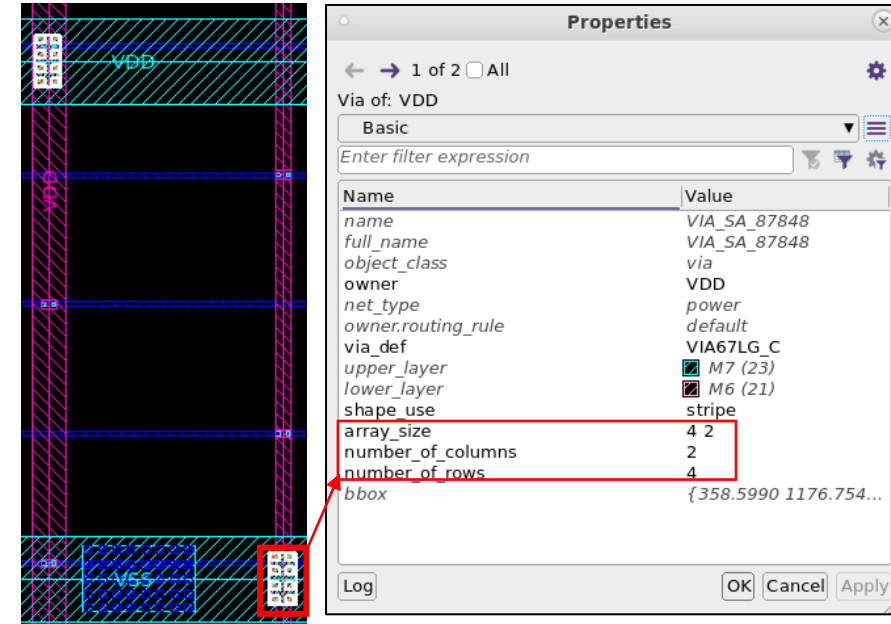
- Zoom in and check STD\_Rail (M1) to low MESH (M7) stripes are correctly changed
- Open the properties window of the vias to check that via\_def is correctly changed

# Powerplan : Use ICC2 (IC Compiler II tool)

12. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/11_fix_DRC_and_create_via.tcl`

- Create Vias (low MESH (M2) to top MESH (M7))

```
icc2_shell> set_pg_via_master_rule pgvia 2x4 -via_array_dimension {2 4}
Information: The command 'set_pg_via_master_rule' cleared the undo history. (UNDO-016)
Successfully set via def rule pgvia 2x4.
icc2_shell> set M2 M7 shape [get_shapes -filter "tag == low MESH 2 || tag == MESH 7"]
{PATH 23 89 PATH 23 90 PATH 23 91 PATH 23 92 PATH 23 93 PATH 23 94 PATH 23 95 PATH 23 96 PATH 23 97 PATH 23 98 PA
TH 23 99 PATH 23 100 PATH 23 101 PATH 23 102 PATH 23 103 PATH 23 104 PATH 23 105 PATH 23 106 PATH 23 107 PATH 23
108 PATH 23 109 PATH 23 110 PATH 23 111 PATH 23 112 PATH 23 113 PATH 23 114 PATH 23 115 PATH 23 116 PATH 23 117 P
ATH 23 118 PATH 23 119 PATH 23 120 PATH 23 121 PATH 23 122 PATH 23 123 PATH 23 124 PATH 23 125 PATH 23 126 PATH 2
3 127 PATH 23 128 PATH 23 129 PATH 23 130 PATH 23 131 PATH 23 132 PATH 23 133 PATH 23 134 PATH 23 135 PATH 23 136
PATH 23 137 PATH 23 138 PATH 23 139 PATH 23 140 PATH 23 141 PATH 23 142 PATH 23 143 PATH 23 144 PATH 23 145 PATH
23 146 PATH 23 147 PATH 23 148 PATH 23 149 PATH 23 150 PATH 23 151 PATH 23 152 PATH 23 153 PATH 23 154 PATH 23 1
55 PATH 23 156 PATH 23 157 PATH 23 158 PATH 23 159 PATH 23 160 PATH 23 161 PATH 23 162 PATH 23 163 PATH 23 164 PA
TH 23 165 PATH 23 166 PATH 23 167 PATH 23 168 PATH 23 169 PATH 23 170 PATH 23 171 PATH 23 172 PATH 23 173 PATH 23
174 PATH 23 175 PATH 23 176 PATH 23 177 PATH 23 178 PATH 23 179 PATH 23 180 PATH 23 181 PATH 23 182 PATH 23 183
PATH 23 184 PATH 23 185 PATH 23 186 PATH 23 187 PATH 23 188 ...}
icc2_shell> change_selection $M2 M7 shape
icc2_shell> create_pg_vias -nets {VDD VSS} -to_types {lib_cell_pin_connect stripe} -from_layers M2 \
-to_layers M7 -via_masters {pgvia 2x4} -tag {M2_7_via} -shapes [get_selection]
Set tag to M2_7_via
Query intersections and create vias.
Object query runtime: 0 seconds.
Determining intersections for 998 shapes starts
Intersection detection runtime: 0 seconds.
Instantiating vias for 17072 intersections starts
Via instantiation runtime: 0 seconds.
Starting via DRC checking ...
Checking DRC for 17072 stacked vias:0% 5% 10% 15% 20% 25% 30% 35% 40% 45% 50% 55% 60% 65% 70% 75% 80% 85% 90% 95%
100%
Overall via DRC checking runtime: 10 seconds.
Checking dangling/floating vias.
Start iteration 1: Checking 17072 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Start iteration 2: Checking 16892 stacked vias:0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
Committed 84460 vias.
Setting tag M2_7_via to committed wires.
Committing vias takes 0.00 seconds.
Overall runtime: 11 seconds.
Successfully created PG Vias.
1
```



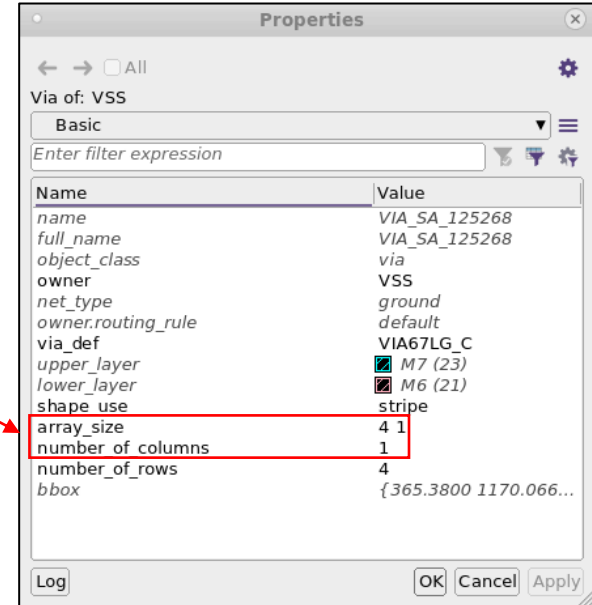
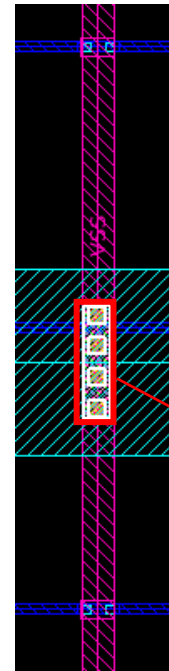
- Zoom in and check low MESH (M2) to top MESH (M7) stripes are correctly connected
- Open the properties window of the vias to check that (owner, array\_size, columns, rows) are correctly created

# Powerplan : Use ICC2 (IC Compiler II tool)

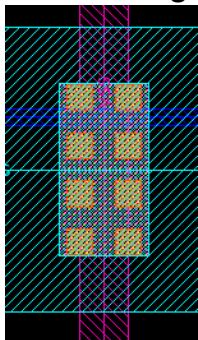
12. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/11_fix_DRC_and_create_via.tcl`

- Change Vias columns (low MESH (M2) to top MESH (M7) only VSS vias)

```
icc2_shell> set ch_M2 via [get vias -filter "tag == M2 7 via && owner.name == VSS"]
{VIA_SA_107234 VIA_SA_107235 VIA_SA_107236 VIA_SA_107237 VIA_SA_107238 VIA_SA_107239 VIA_SA_107240 VIA_SA_107241
VIA_SA_107242 VIA_SA_107243 VIA_SA_107244 VIA_SA_107245 VIA_SA_107246 VIA_SA_107247 VIA_SA_107248 VIA_SA_107249 V
IA_SA_107250 VIA_SA_107251 VIA_SA_107252 VIA_SA_107253 VIA_SA_107254 VIA_SA_107255 VIA_SA_107256 VIA_SA_107257 VI
A_SA_107258 VIA_SA_107259 VIA_SA_107260 VIA_SA_107261 VIA_SA_107262 VIA_SA_107263 VIA_SA_107264 VIA_SA_107265 VIA
SA_107266 VIA_SA_107267 VIA_SA_107268 VIA_SA_107269 VIA_SA_107270 VIA_SA_107271 VIA_SA_107272 VIA_SA_107273 VIA
SA_107274 VIA_SA_107275 VIA_SA_107276 VIA_SA_107277 VIA_SA_107278 VIA_SA_107279 VIA_SA_107280 VIA_SA_107281 VIA S
A_107282 VIA_SA_107283 VIA_SA_107284 VIA_SA_107285 VIA_SA_107286 VIA_SA_107287 VIA_SA_107288 VIA_SA_107289 VIA_SA
107290 VIA_SA_107291 VIA_SA_107292 VIA_SA_107293 VIA_SA_107294 VIA_SA_107295 VIA_SA_107296 VIA_SA_107297 VIA_SA
107298 VIA_SA_107299 VIA_SA_107300 VIA_SA_107301 VIA_SA_107302 VIA_SA_107303 VIA_SA_107304 VIA_SA_107305 VIA_SA_1
07306 VIA_SA_107307 VIA_SA_107308 VIA_SA_107309 VIA_SA_107310 VIA_SA_107311 VIA_SA_107312 VIA_SA_107313 VIA_SA_10
7314 VIA_SA_107315 VIA_SA_107316 VIA_SA_107317 VIA_SA_107318 VIA_SA_107319 VIA_SA_107320 VIA_SA_107321 VIA_SA_107
322 VIA_SA_107323 VIA_SA_107324 VIA_SA_107325 VIA_SA_107326 VIA_SA_107327 VIA_SA_107328 VIA_SA_107329 VIA_SA_1073
30 VIA_SA_107331 VIA_SA_107332 VIA_SA_107333 ...}
icc2_shell> change_selection $ch_M2 via
icc2_shell> set attribute -objects [get selection] -name number_of_columns -value 1
{VIA_SA_163403 VIA_SA_163402 VIA_SA_163401 VIA_SA_163400 VIA_SA_163399 VIA_SA_163398 VIA_SA_163397 VIA_SA_163396
VIA_SA_163395 VIA_SA_163394 VIA_SA_163393 VIA_SA_163392 VIA_SA_163391 VIA_SA_163390 VIA_SA_163389 VIA_SA_163388 V
IA_SA_163387 VIA_SA_163386 VIA_SA_163385 VIA_SA_163384 VIA_SA_163383 VIA_SA_163382 VIA_SA_163381 VIA_SA_163380 VI
A_SA_163379 VIA_SA_163378 VIA_SA_163377 VIA_SA_163376 VIA_SA_163375 VIA_SA_163374 VIA_SA_163373 VIA_SA_163372 VIA
SA_163371 VIA_SA_163370 VIA_SA_163369 VIA_SA_163368 VIA_SA_163367 VIA_SA_163366 VIA_SA_163365 VIA_SA_163364 VIA
SA_163363 VIA_SA_163362 VIA_SA_163361 VIA_SA_163360 VIA_SA_163359 VIA_SA_163358 VIA_SA_163357 VIA_SA_163356 VIA_S
A_163355 VIA_SA_163354 VIA_SA_163353 VIA_SA_163352 VIA_SA_163351 VIA_SA_163350 VIA_SA_163301 VIA_SA_163300 VIA_SA
163299 VIA_SA_163298 VIA_SA_163297 VIA_SA_163296 VIA_SA_163295 VIA_SA_163294 VIA_SA_163293 VIA_SA_163292 VIA_SA
163291 VIA_SA_163290 VIA_SA_163289 VIA_SA_163288 VIA_SA_163287 VIA_SA_163286 VIA_SA_163285 VIA_SA_163284 VIA_SA_1
63283 VIA_SA_163282 VIA_SA_163281 VIA_SA_163280 VIA_SA_163279 VIA_SA_163278 VIA_SA_163277 VIA_SA_163276 VIA_SA_16
3275 VIA_SA_163274 VIA_SA_163273 VIA_SA_163272 VIA_SA_163271 VIA_SA_163270 VIA_SA_163269 VIA_SA_163268 VIA_SA_163
267 VIA_SA_163266 VIA_SA_163265 VIA_SA_163264 VIA_SA_163263 VIA_SA_163262 VIA_SA_163261 VIA_SA_163260 VIA_SA_1632
59 VIA_SA_163258 VIA_SA_163257 VIA_SA_163256 ...}
```

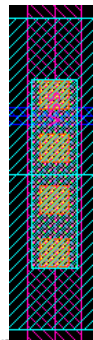


<Before changing>



Via size is larger than the width of the VSS stripe, the via size has been changed to fit the width

<After changing>



- Zoom in and check low MESH (M2) to top MESH (M7) VSS vias are correctly changed
- Open the properties window of the vias to check that (owner, array\_size, columns) are correctly changed

# Powerplan : Use ICC2 (IC Compiler II tool)

## 13. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/12\_create\_hard\_blockage.tcl

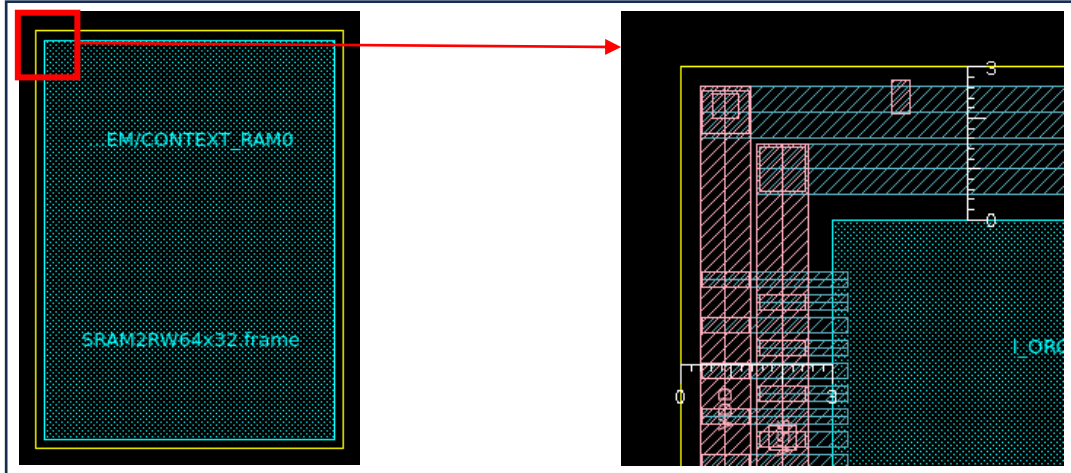
- Create hard placement blockage before placement

```
icc2_shell> remove_placement_blockage [get_placement_blockages -filter "blockage_type == hard"]
12
icc2_shell> set macro bbox [get_attr [get_cells $only_macros] bbox]
{{424.5430 881.7460} {521.4510 1015.1520}} {{550.4510 881.7460} {647.3590 1015.1520}} {{676.3590 881.7460} {773.2
670 1015.1520}} {{802.2670 881.7460} {899.1750 1015.1520}} {{877.7860 424.8460} {1015.4590 514.4760}} {{877.7860
532.1860} {1015.4590 621.8160}} {{424.5430 655.1680} {497.3820 788.5740}} {{589.4510 424.8460} {662.2900 558.2520
}} {{424.5430 424.8460} {521.4510 558.2520}} {{690.2900 424.8460} {787.1980 558.2520}}
icc2_shell> set i 0
0
icc2_shell>
icc2_shell> foreach x $macro_bbox {
? set llx [expr [lindex $x 0 0] - 2]
? set urx [expr [lindex $x 1 0] + 2]
? set lly [expr [lindex $x 0 1] - 2]
? set ury [expr [lindex $x 1 1] + 2]
? set new_bboxes "{$llx $lly} {$urx $ury}"
? create_placement_blockage -name "mc_blkq_hard_${i}" -type hard -boundary $new_bboxes
?
? incr i
? }
icc2_shell> set pll "I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD"
I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD
icc2_shell> set pll_bbox [get_attr [get_cells $pll] bbox]
{{899.2350 707.9240} {1016.4580 760.0320}} {{899.2350 638.8160} {1016.4580 690.9240}}
icc2_shell> set i 0
0
icc2_shell>
icc2_shell> foreach x $pll_bbox {
? set llx [expr [lindex $x 0 0] - 2]
? set urx [expr [lindex $x 1 0] + 2]
? set lly [expr [lindex $x 0 1] - 2]
? set ury [expr [lindex $x 1 1] + 2]
? set new_bboxes "{$llx $lly} {$urx $ury}"
? create_placement_blockage -name "pll_blkq_hard_${i}" -type hard -boundary $new_bboxes
?
? incr i
? }
```

# Powerplan : Use ICC2 (IC Compiler II tool)

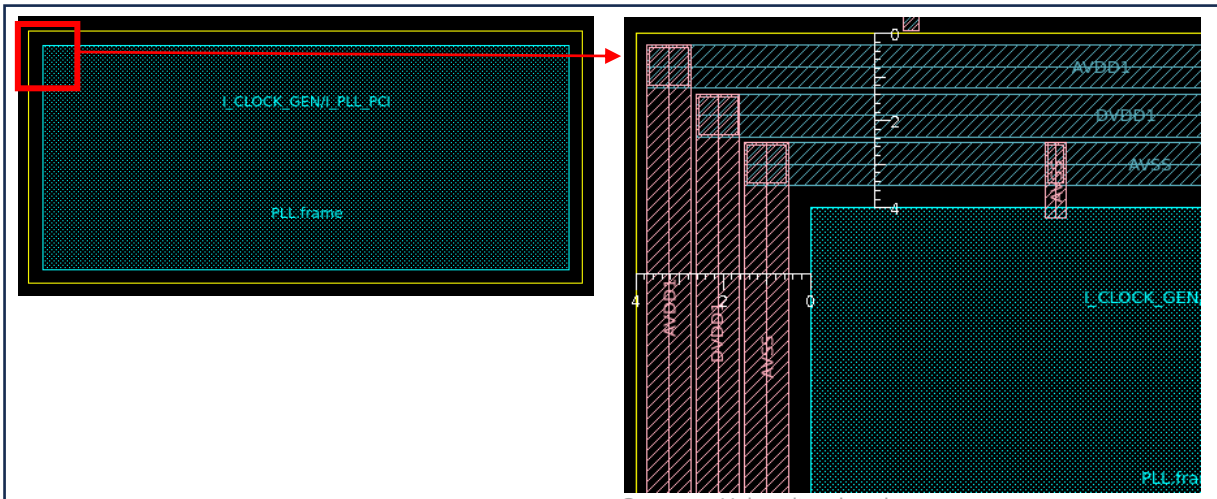
13. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/12_create_hard_blockage.tcl`

- Check hard placement blockage before placement
  - Macro Placement blockage



- Create blockages to prevent STD cells from being placed on the Macro & PLL boundary
- The placement blockage size is set to prevent STD cells from being placed in the Macro & PLL Ring

- PLL Placement blockage



# Powerplan : Use ICC2 (IC Compiler II tool)

14. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/13_final_check_and_save.tcl`

- `Check_pg_drc`

```
icc2_shell> check_pg_drc
Command check_pg_drc started at Wed Oct 4 15:05:11 2023
Command check_pg_drc finished at Wed Oct 4 15:05:17 2023
CPU usage for check_pg_drc: 5.77 seconds ( 0.00 hours)
Elapsed time for check_pg_drc: 6.08 seconds ( 0.00 hours)
No errors found.
```

Error Browser

File Errors Select Highlight Help

Show by: type

DRC Editor DRC

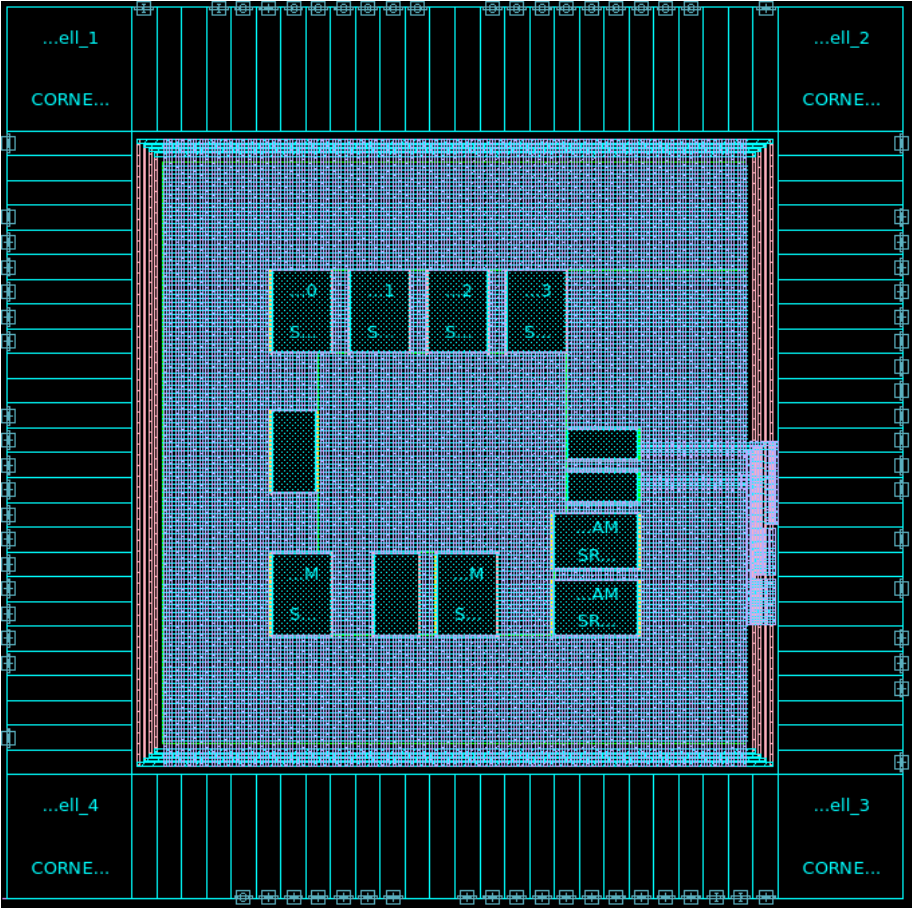
| ErrorSet        | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|-----------------|-------|---------|-------|---------|----------|----------|
| ORCA_powerpl... | 0     | 0       | 0     | 0       | 0        | 0        |
| └ DRC_report... | 0     | 0       | 0     | 0       | 0        | 0        |

# ID Status Color Type Layer Magnitude Error File Name Information Summary

Clear List Fixed

Show: all Follow: zoom 1.0 Dim

There are no violations



# Powerplan : Use ICC2 (IC Compiler II tool)

14. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/13_final_check_and_save.tcl`

- `Check_pg_missing_vias`

```
icc2_shell> check_pg_missing_vias
Check net AVDD vias...
Number of missing vias: 0
Checking net AVDD vias took 1 seconds.
Check net AVSS vias...
Number of missing vias: 0
Checking net AVSS vias took 0 seconds.
Check net DVDD vias...
Number of missing vias: 0
Checking net DVDD vias took 0 seconds.
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 0 seconds.
Check net VDDIO vias...
Number of missing vias: 0
Checking net VDDIO vias took 0 seconds.
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 1 seconds.
Check net VSSIO vias...
Number of missing vias: 0
Checking net VSSIO vias took 0 seconds.
Overall runtime: 2 seconds.
```

There are no missing vias (all cleared)

# Powerplan : Use ICC2 (IC Compiler II tool)

14. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/13_final_check_and_save.tcl`

- **Check\_pg\_connectivity** (exclude std cell – because std cells are not placed)

```
icc2_shell> check_pg_connectivity
Loading cell instances...
Number of Standard Cells: 8871
Number of Macro Cells: 12
Number of IO Pad Cells: 108
Number of Blocks: 0
Loading P/G wires and vias...
Number of AVDD Wires: 74
Number of AVDD Vias: 183
Number of AVDD Terminals: 1
*****Verify net AVDD connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 0
Number of floating hard macros: 0
Number of floating I/O pads: 2
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of AVSS Wires: 79
Number of AVSS Vias: 180
Number of AVSS Terminals: 1
*****Verify net AVSS connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 0
Number of floating hard macros: 0
Number of floating I/O pads: 2
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
```

```
Loading cell instances...
Loading P/G wires and vias...
Number of DVDD Wires: 69
Number of DVDD Vias: 164
Number of DVDD Terminals: 1
*****Verify net DVDD connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 0
Number of floating hard macros: 0
Number of floating I/O pads: 2
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VDD Wires: 2600
Number of VDD Vias: 56049
Number of VDD Terminals: 1
*****Verify net VDD connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 8871
Number of floating hard macros: 0
Number of floating I/O pads: 108
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VDDIO Wires: 0
Number of VDDIO Vias: 0
Number of VDDIO Terminals: 1
*****Verify net VDDIO connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 0
Number of floating hard macros: 0
Number of floating I/O pads: 106
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
```

## Ignore list

- ❖ (AVDD, AVSS, DVDD, VDD, VSS, VDDIO, VSSIO) Number of floating I/O pads
- ❖ STD cells (VDD, VSS pins)
- ❖ Number of floating terminals

# Powerplan : Use ICC2 (IC Compiler II tool)

14. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/13_final_check_and_save.tcl`

- **Check\_pg\_connectivity** (exclude std cell – because std cells are not placed)

```
Loading cell instances...
Loading P/G wires and vias...
Number of VDDIO Wires: 0
Number of VDDIO Vias: 0
Number of VDDIO Terminals: 1
*****Verify net VDDIO connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 0
Number of floating hard macros: 0
Number of floating I/O pads: 106
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 2811
Number of VSS Vias: 106873
Number of VSS Terminals: 1
*****Verify net VSS connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 8871
Number of floating hard macros: 0
Number of floating I/O pads: 106
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VSSIO Wires: 0
Number of VSSIO Vias: 0
Number of VSSIO Terminals: 1
*****Verify net VSSIO connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 0
Number of floating hard macros: 0
Number of floating I/O pads: 106
Number of floating terminals: 1
Number of floating hierarchical blocks: 0
*****
Overall runtime: 0 seconds.
```

## Ignore list

- ❖ (AVDD, AVSS, DVDD, VDD, VSS, VDDIO, VSSIO) Number of floating I/O pads
- ❖ STD cells (VDD, VSS pins)
- ❖ Number of floating terminals

# Powerplan : Use ICC2 (IC Compiler II tool)

14. `icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/03_powerplan/script/13_final_check_and_save.tcl`

- Check\_pg\_connectivity (exclude std cell – because std cells are not placed)

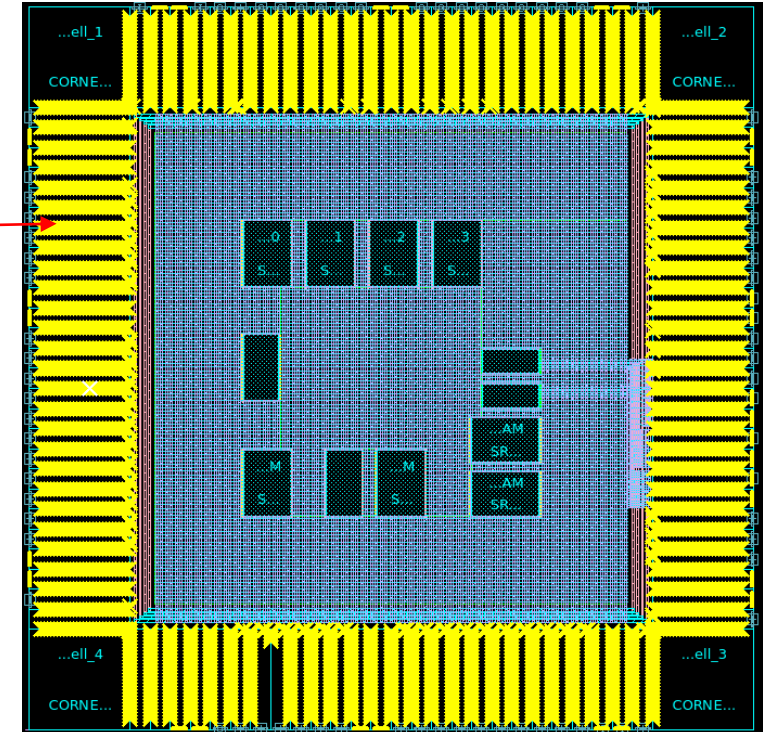
The Error Browser window displays a table of errors. The 'Floating ...' error set is highlighted with a red box, showing 24328 total errors, 24328 visible, and 7 detected. Below the table, a detailed view of error 12600 is shown, indicating a floating pin violation on layer M8 (25) for cell type IO. The error message states: 'Floating pin genblk1\_7\_pad\_iopad/VDDIO is not connected to net VDDIO'. The error ID is 12600 and the status is Error.

| ErrorSet        | Total | Visible | Fixed | Ignored | NULL Net | Detected |
|-----------------|-------|---------|-------|---------|----------|----------|
| ORCA_powerpl... | 42127 | 42127   | 0     | 0       | 0        | 42127    |
| ORCA_floati...  | 42127 | 42127   | 0     | 0       | 0        | 42127    |
| Floating ...    | 24328 | 24328   | 0     | 0       | 0        | 24328    |
| Floating ...    | 17792 | 17792   | 0     | 0       | 0        | 17792    |
| Floating ...    | 7     | 7       | 0     | 0       | 0        | 7        |

| #  | ID    | Status | Color | Type         | Layer   | Cell Type | Magnitude | Error File Name   | Information    | Summary      |
|----|-------|--------|-------|--------------|---------|-----------|-----------|-------------------|----------------|--------------|
| 0  | 12600 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 1  | 12601 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 2  | 12602 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 3  | 12603 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 4  | 12604 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 5  | 12605 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 6  | 12606 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 7  | 12607 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 8  | 12608 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 9  | 12609 |        |       | Floating ... | M8 (25) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 10 | 12610 |        |       | Floating ... | M9 (27) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |
| 11 | 12611 |        |       | Floating ... | M9 (27) | IO        |           | ORCA_floatingP... | Floating pi... | Status = ... |

0: Layer: M8 (25) Type: Floating pad pin violation  
Cell type: IO  
Type Summary : Status = error  
Obj Info : Floating pin genblk1\_7\_pad\_iopad/VDDIO is not connected to net VDDIO  
Cell: genblk1\_7\_pad\_iopad  
Error ID: 12600 Status: Error

Show: all Follow: zoom 1.0 Dim



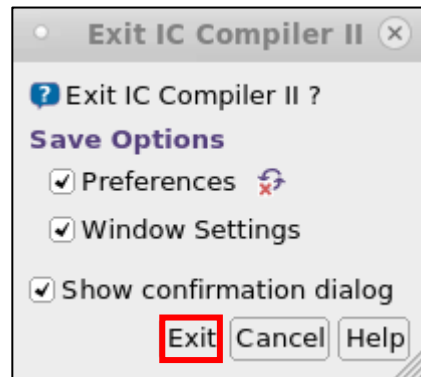
- Floating violations occurring in PADS & Terminals are ignored (Because of PAD ISSUE)
- All violations that occurred in the core area have been cleared

# Powerplan : Use ICC2 (IC Compiler II tool)

14. icc2\_shell> source \$WORK\_DIR/2\_BACKEND/1\_pnr/rundir/03\_powerplan/script/13\_final\_check\_and\_save.tcl

- Save & Exit

```
icc2_shell> save_lib -all  
Saving all libraries...  
1  
icc2_shell> exit
```



# WEEK 5 : Exercise for Powerplan

# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- In the "Exercise for floorplan, change the floorplan to proceed with powerplan, and check the results"
- **Using powerplan script & floorplan.nlib**
  - **powerplan script** - \${WORK\_DIR}/2\_BACKEND/1\_pnr/03\_powerplan/script/\*
  - **floorplan.nlib** - \${WORK\_DIR}/2\_BACKEND/1\_pnr/02\_floorplan/nlib/ORCA\_floorplan\_homework.nlib
- **Guidelines**
  - Find the change point when the powerplan runs after floorplan is changed
  - Check the powerplan script works correctly
  - Check that the Metal & Vias are correctly created according to the values specified in the script
  - Check that (DRC, connectivity, missing\_vias) are different from previous results after powerplan

# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- In the "Exercise for floorplan, change the floorplan to proceed with powerplan, and check the results"
  1. Find the change point when you run powerplan after change the floorplan
  2. Why is the location of the PLL fixed even if the floorplan is changed?
  3. Why we use vertical Layer of PLL Ring as M4?
  4. Check the width, offset of the Core Ring & Macro Ring
  5. Check the width, spacing, pitch, offset of the TOP MESH & Low MESH
  6. Check the width, rail\_shift of the STD Rail
  7. Check vias are correctly created as specified in the powerplan script
  8. Check that (DRC, connectivity, missing\_vias) are different from previous results after powerplan

# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- **solution**

- 1. Find the change point when you run powerplan after change the floorplan**

- (Macro Ring, TOP & Low MESH, STD Rail) are created in a different area due to macro position change

- 2. Why is the location of the PLL fixed even if the floorplan is changed?**

- Since signal & Power supply PADs to PLL are located on the right side, the position of PLL must be fixed from the beginning

- 3. Why we use vertical Layer of PLL Ring as M4?**

- Because the pins on the PLL are M6, M4 was used for the vertical layer of the PLL ring considering routing

- 4. Check the width, offset of the Core Ring & Macro Ring**

- **Core Ring**
  - width : vertical 1 horizontal 1
  - offset : vertical 0.5 horizontal 0.5
- **Macro Ring**
  - width : vertical 1 horizontal 1
  - offset : vertical 0.5 horizontal 0.5

# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- **solution**

## 5. Check the width, spacing, pitch, offset of the TOP MESH & Low MESH

- **TOP MESH (M8)**
  - width : 4.64
  - spacing : 5.088
  - pitch : 19.456
- **TOP MESH (M7)**
  - width : 1.104
  - spacing : 5.584
  - pitch : 13.376
- **Low MESH (M2)**
  - width : VDD (0.44) VSS (0.192)
  - spacing : VDD (2.274) VSS (3.456)
  - pitch : 9.728
  - offset : 1.216

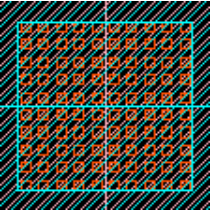
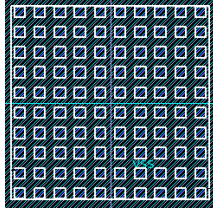
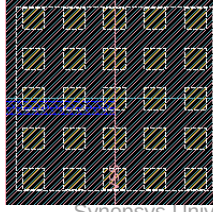
# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- solution

6. Check the width, rail\_shift of the STD Rail

- **STD Rail(M1)**
    - width : 0.06
    - rail\_shift : 3.344

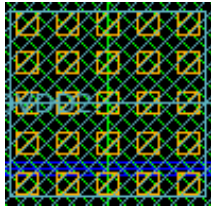
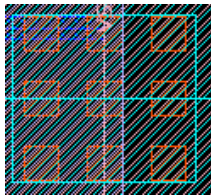
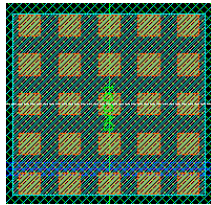
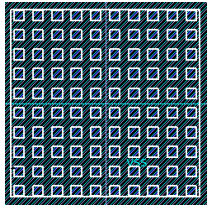
7. Check vias are correctly created as specified in the powerplan script

|                                                                                                                           |                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Core Ring (M6 to M7)</b> <ul style="list-style-type: none"><li>- columns : 10</li><li>- rows : 10</li></ul>            |    |
| <b>Core Ring (M7 to TOP MESH (M8))</b> <ul style="list-style-type: none"><li>- columns : 10</li><li>- rows : 10</li></ul> |   |
| <b>Macro Ring (M6 to M7)</b> <ul style="list-style-type: none"><li>- columns : 5</li><li>- rows : 5</li></ul>             |  |

# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- Solution

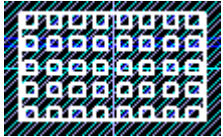
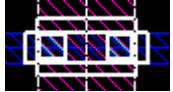
## 7. Check vias are correctly created as specified in the powerplan script

|                                                                       |                                                                                       |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>PLL Ring (M4 to M7)</b><br>- columns : 5<br>- rows : 5             |    |
| <b>Macro Ring (M6 to TOP MESH(M7))</b><br>- columns : 3<br>- rows : 3 |    |
| <b>PLL Ring (M4 to TOP MESH(M7))</b><br>- columns : 5<br>- rows : 5   |   |
| <b>PLL Manal Routing (M7 to M8)</b><br>- columns : 10<br>- rows : 10  |  |

# Powerplan : Use ICC2 (IC Compiler II tool) HomeWork

- **Solution**

## 7. Check vias are correctly created as specified in the powerplan script

|                                                           |                                                                                    |
|-----------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>TOP MESH (M7 to M8)</b><br>- cloumns : 8<br>- rows : 5 |  |
| <b>Low MESH (M1 to M2)</b><br>- cloumns : 2<br>- rows : 1 |   |
| <b>Low MESH (M2 to M7)</b><br>- cloumns : 1<br>- rows : 4 |   |

## 8. Check that (DRC, connectivity, missing\_vias) are different from previous results after powerplan

- Report & check check\_pg\_drc, check\_pg\_connectivity, check\_pg\_missing\_vias
- Check if it is the same as the violations that occurred in the powerplan in LAB (refer to 247p ~257p)

**Thank You**

